PHYSF202 Electromagnetism

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Preface

some more text

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1. Special Relativity

We start by discussing the theory of Special Relativity as one of the main pillars of electromagnetism. In fact, historically the laws of electromagnetism (in particular, propagation of electromagnetic waves) were the main motivation behind the Special Relativity.

1.1. Lorentz tramsformations

Relativity principle

• Motion of material bodies is described with respect to some *reference frame*. Among all the frames there is a special class called *inertial frames* which all move with constant velocities with respect to one another. In all inertial frames particles which are not subject to any forces move with constant velocities.

• The same may be formulated differently: the Newtonian mechanics respects the *relativity principle* which states that all the inertial frames are physically equivalent; the laws of physics (in particular, the laws of mechanics) have the same form in any of these frames. Relative motion of bodies confined in some space does not depend on the motion of this space as a whole if the latter has constant velocity.

In Newtonian mechanics this is often referred to as the Galilean relativity principle.

• In modern language, the relativity principle is a *symmetry* of mechanics. The identical form of the laws of mechanics in all inertial frames is a consequence of this symmetry. There are other fundamental symmetries of physics laws, like the invariance with respect to a shift of coordinates by a constant (in other words, change of a reference point) and with respect to rotation of the axes. These latter symmetries are referred to as **homogeneity** and **isotropy** of space.

Mathematically, a symmetry of mechanics is expressed as invariance of the Newton's laws under the corresponding *coordinate transformation*. Consider for instance the Newton's second law

$$m\frac{d^2\boldsymbol{x}}{dt^2} = \boldsymbol{F}. \tag{1.1}$$

$\boldsymbol{x}$ = position

m = mass

$\boldsymbol{F}$ = force

Here and after we use **bold face** to denote vectors.

If we change the reference point, the original coordinates $\boldsymbol{x}$ are related to the new ones $\boldsymbol{x}'$ as follows,

$$\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{x}_0,$$

where $\boldsymbol{x}_0$ is a constant vector. If the particle trajectory in the original frame was $\boldsymbol{x}(t)$, in the new frame it will be $\boldsymbol{x}'(t)$ satisfying

$$\boldsymbol{x}(t) = \boldsymbol{x}'(t) + \boldsymbol{x}_0.$$

The Newton's law in the new coordinaets will be expressed by the same equation (1.1) with the replacement of $\boldsymbol{x}$ by $\boldsymbol{x}'$. Likewise, if the frames $\boldsymbol{x}$ and $\boldsymbol{x}'$ are related to one another by a rotation, the Newton's equation

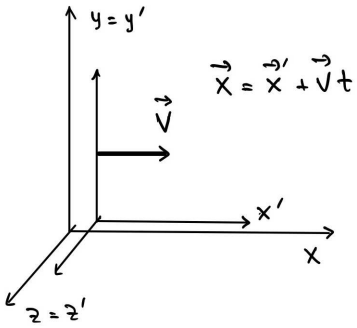
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preserves its form (1.1) in the rotated frame: both sides are vectors and the equaiton remains valid when multiplied by the same rotation matrix on both sides.

Finally, consider the case when one frame *moves* with respect to another with a constant velocity, i.e. the case of a transformation between inertial frames. Without loss of generality we may assume that the axes of the frame $\boldsymbol{x}'$ are parallell to those of the frame $\boldsymbol{x}$, and that the frame $\boldsymbol{x}'$ moves with respect to the frame $\boldsymbol{x}$ with the velocity $\boldsymbol{V}$ along the axis x, $\boldsymbol{V} = (V, 0, 0)$ in such a way that at $t = 0$ the axes of the two frames concide. Then we may write a transformation of the coordinates as follows:

Note that if this is not the case we can always use the rotations and translations to achieve this configuration.

$$\begin{aligned} x &= x' + Vt, \\ y &= y', \\ z &= z', \end{aligned}$$



or, in the vector form,

$$\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{V}t. \tag{1.2}$$

These coordinate transformations are known as the **Galilean transformations**. The laws of physics have to be invariant under these transformations in order to obey the relativity principle.

It is straightforward to see that eq. (1.1) is indeed invariant under these transformations. Differentiation of eq. (1.2) with respect to time gives:

$$\frac{d\boldsymbol{x}}{dt} = \frac{d\boldsymbol{x}'}{dt} + \boldsymbol{V},$$

$$\frac{d^2\boldsymbol{x}}{dt^2} = \frac{d^2\boldsymbol{x}'}{dt^2},$$

so that eq. (1.1) implies

$$m\frac{d^2\boldsymbol{x}'}{dt^2} = \boldsymbol{F}.$$

We see that the Newton's law has the same form in the new coordinates, and is therefore nvariant under the Galilean transformations.

• While the laws of mechanics respect the Galilean tramsformations and thus obey the relativity principle, *the laws of electromagnetism do not*. The complete set of the laws of electromagnetism is known as Maxwell's equations which we still have to derive. Anticipating, consider one consequence of these equations — the equation describing propagation of electromagnetic waves. It has the form

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}\right)\boldsymbol{E}(t, \boldsymbol{x}) = 0, \tag{1.3}$$

where $\boldsymbol{E}(t, \boldsymbol{x})$ is the electric field and $c = 2.998 \times 10^8 \mathrm{m/s}$ is the speed of light. It is easy to see that the Galilean transformation *does not* leave this equation invariant, and therefore it cannot leave the Maxwell's equations invariant either.

Thus, assuming the wave equation is correct, the Galilean relativity is not a symmetry of Nature. Note that translational and rotational symmetries still hold.

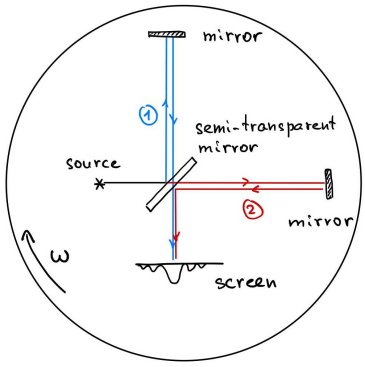
• It should be noted that it is not the wave equation (1.3) itself which poses a problem. These type of equations have been known way before the one for the electromagnetic waves — for instance, the propagation of sound waves is described by a similar equation with c replaced

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by the sound speed and E by the pressure or density. However, the sounds propagates in the medium — air, water, etc. For this reason the propagation of the sound waves indeed does not respect the relativity principle *if the corresponding medium is not included*: the reference frame where the medium is at rest and the one moving with respect to the medium are indeed not equivalent. As an illustration one may imagine making experiments with the propagation of the sound waves on an open train platform. But if the medium is included in the considered system, the equivalence principle is restored: experiments with sound waves in a closed compartment of a moving train should (and they do) give identical results to those carried out in a room provided other conditions (like temperature and pressure) are the same.

How do we know that the situation with the propagation of light is not similar to that of sound, in other words, that the light is not propagating in some hypothetical medium (dubbed "aether" at a time)? There is no way of knowing except by making an experiment.

Such an experiment was first carried out by Michelson and Morley in 1887 (and repeated many times in more sophisticated setups afterwards). The idea is to use the Earth rotation around the Sun with the velocity about 30 km/s to detect the "aether wind". The design is outlined on the side Figure. Two interferometer arms are mounted on a rotating table and tuned to have the same pathlength, so that, at a given orientation of the table, the two light beams 1 and 2 arrive to the detector in phase. The table is moving through the aether together with the Earth. After the beams are tuned, the table is rotated. The orientation of the arms with respect to the velocity of the aether changes periodically, the beams are not always in phase anymore and the interference fringes shift periodically with the period of rotation. Although the authors designed the experiment with the idea to detect the aether, they got a negative result, and the subsequent more precise experiments confirmed the absence of the effects due to the motion of the aether no matter what is the direction and velocity. The most plausible conclusion then was that there is no any aether.



• The failure to detect the "aether" inspired the development of alternative theories. Assuming Maxwell equations are correct, the relativity principle is not compatible with the Galilean transformations as they do not leave these equations invariant. A choice has to be made as to whether to keep the relativity *or* Galilean transformations. The choice in favor of relativity led Einstein to formulate two postulates:

1. *The laws of physics are the same in all inertial frames.*
2. *The speed of light is the same in all inertial frames and does not depend on the velocity of the source.*

It follows then that the Galilean transformations — and consequently, the laws of mechanics! — have to be modified. So, what are the correct transformations under which the laws of Nature are invariant? One can derive them from the above two postulates.

Interval

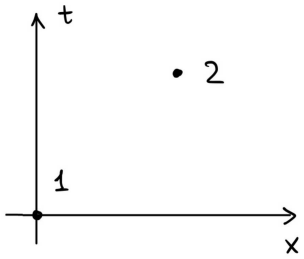
• It is useful to think in terms of "events", which are something that happens in a given point in space at a given moment of time and whose

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space extent and time duration is so short that it can be neglected in a given context. An example could be collision of particles, emission of a short flash of light, absorption of light, etc. Since events have no extent in space and time, they look the same in all reference frames. Events are represented by *points* on the space-time diagram, and are completely characterized by their 4 coordinates $(t, x, y, z) = (t, \boldsymbol{x})$.

For any two events 1 and 2 with the coordinates $(t_1, \boldsymbol{x}_1)$ and $(t_2, \boldsymbol{x}_2)$, respectively, one can define an **interval** between these events s_{12}^2 by the relation

$$s_{12}^2 \equiv c^2(t_2 - t_1)^2 - (\boldsymbol{x}_2 - \boldsymbol{x}_1)^2. \tag{1.4}$$



Note that if the two events happen in the same space-time point (and thus have the same coordinates) then $s_{12}^2 = 0$, but the opposite is not necessarily true. Note as well that, in general, s_{12}^2 can be both positive and negative; we will come back to these properties of the interval later.

• Making use of the Einstein postulates, and of the translational and rotational symmetry of space (homogeneity and isotropy), we will now show that the new transformations of coordinates which are to replace the Galilean transformations should *preserve the interval*.

Suppose that in some frame K where the coordinates are t and $\boldsymbol{x}$ we have two events:

event 1:	emission of light at $(0, \mathbf{0})$
event 2:	absorption of light at $(t, \boldsymbol{x})$

Because of the translational invariance, we did not lose generality by putting the first event at $(0, \mathbf{0})$.

Since the speed of light equals c we have

$$s_{12}^2 = c^2(t - 0)^2 - (\boldsymbol{x} - \mathbf{0})^2 = c^2t^2 - \boldsymbol{x}^2 = 0,$$

since ct is the distance traveled by the light in time t.

Consider now *the same two events* in the reference frame K' where the coordinates are denoted as t' and $\boldsymbol{x}'$ and which is moving with velocity $\boldsymbol{v}$ with respect to the original frame. By translational invariance, we may assume without loss of generality that the origins of the two frames coincide at the moment $t = 0$. Then in the frame K'

event 1:	emission of light at $(0, \mathbf{0})$
event 2:	absorption of light at $(t', \boldsymbol{x}')$

By the second postulate the velocity of light in the frame K' also equals c, and the same argument as before gives

$$s_{12}'^2 = c^2t'^2 - \boldsymbol{x}'^2 = 0.$$

It follows therefore that *for any two events such that the interval between them is zero in some inertial frame, it is also zero in any other inertial frame*. Indeed, whatever is the first event, we may imagine that in the same space-time point there is a flash of light emitted, and that this light is absorbed in the space-time point of the second event, so that the above argument applies.

• The next step is to show that the interval has the same value in all inertial frames even if $s_{12}^2 \neq 0$. From the translational invariance it follows

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that the transformation of coordinates should be linear. By rotational invariance, we may assume that the velocity of the frame K' is directed along the axis x, $\boldsymbol{v} = (v, 0, 0)$, in which case we may assume that the coordinates y and z are not affected by the transformation, $y = y'$ and $z = z'$. Thus, the coordinate transformation has the form

Indeed, changes of t and $\boldsymbol{x}$ by arbitrary constants should be possible to compensate by the changes of t' and $\boldsymbol{x}'$ by some other constants. Only linear transformations have this property.

$$\begin{aligned} ct &= \alpha ct' + \beta x', \\ x &= \gamma ct' + \delta x', \end{aligned} \tag{1.5}$$

with some constants α, β, γ and δ. In the matrix notations this is written as

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix}. \tag{1.6}$$

Here we put factors c for convenience in order that the coefficients α, β, γ and δ have no units. The interval in the frame K is then a quadratic form of ct' and x', which can always be written in the factorized form as follows:

$$s_{12}^2 \equiv c^2t^2 - \boldsymbol{x}^2 = A(ct' + Bx')(ct' + Dx'), \tag{1.7}$$

where A, B and D are some coefficients (they can, of course, be expressed in terms of α, β, γ and δ, but these expressions are of no interest at this point).

We may now use our previous result: for every value of ct' and x' for which the interval $s_{12}'^2 = (ct' - x')(ct' + x')$ is zero, the interval s_{12}^2 in eq. (1.7) must also be zero. Therefore, either $B = 1$, $D = -1$, or $B = -1$, $D = 1$. In either case we have

$$s_{12}^2 = A(c^2t'^2 - x'^2) = A\, s_{12}'^2. \tag{1.8}$$

We have thus established that the intervals between the same events in two different inertial frames are proportional to one another. The coefficient A may depend on the relative velocity of the two frames, but by rotational invariance it may only depend on the modulus $|v|$, so that $A = A(|\boldsymbol{v}|)$.

• The final step is to show that $A = 1$. To this end consider a third reference frame K'' which moves with respect to K with the velocity $\boldsymbol{v}_2$. For the uniformity of notations, denote the velocity of K' as $\boldsymbol{v} = \boldsymbol{v}_1$. In general, $\boldsymbol{v}_2$ is not parallel to $\boldsymbol{v}_1$. Applying eq. (1.8) we have

$$\begin{aligned} s^2 &= A(|\boldsymbol{v}_1|)\, s'^2, \\ s^2 &= A(|\boldsymbol{v}_2|)\, s''^2. \end{aligned}$$

We may also use eq. (1.8) to directly relate the intervals s'^2 and s''^2:

$$s''^2 = A(|\boldsymbol{v}_{12}|)\, s'^2,$$

where $\boldsymbol{v}_{12}$ is the relative velocity of the frames K' and K''. From these three relations we find

$$A(|\boldsymbol{v}_{12}|) = \frac{A(|\boldsymbol{v}_1|)}{A(|\boldsymbol{v}_2|)}. \tag{1.9}$$

We do not know the velocity composition law, but it is obvious that $|\boldsymbol{v}_{12}|$ depends on the relative orientation of $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$. It is clear then that

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eq. (1.9) can only be satisfied if A = const, because its r.h.s. does not depend on the angle between $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$, while the l.h.s. does unless A is a constant. But since in the limit $v \to 0$ we must have $A \to 1$ (the frames K and K' become identical), we conclude that this constant must be equal to 1,

$$A = 1.$$

It follows then from eq. (1.8) that

$$s_{12}^2 = s_{12}'^2. \tag{1.10}$$

Thus, in order to satisfy the Einstein postulates **the coordinate transformations must preserve the interval**.

Lorentz transformations

• We can now go back to the general linear transformations (1.5) and require that they preserve the interval, that is

$$c^2t'^2 - x'^2 = c^2t^2 - x^2.$$

Substituting ct and x from eq. (1.5) into the r.h.s. of this equation we find:

$$c^2t'^2 - x'^2 = (\alpha ct' + \beta x')^2 - (\gamma ct' + \delta x')^2 =$$
$$= (\alpha^2 - \gamma^2)c^2t'^2 + 2(\alpha\beta - \gamma\delta)ct'x' - (\delta^2 - \beta^2)x'^2.$$

Comparing the coefficients in front of three different terms t'^2, $t'x'$ and x'^2 we find three equations for the coefficients α, β, γ and δ,

$$\begin{aligned} \alpha^2 - \gamma^2 &= 1, \\ \alpha\beta - \gamma\delta &= 0, \\ \delta^2 - \beta^2 &= 1. \end{aligned}$$

We have got *three* equations for 4 unknowns. Thus, we can only determine three variables in terms of the fourth one, which remains a free parameter. This is exactly what we need for a *general* transformation which should depend on the relative velocity of the two frames.

We can rewrite these equations as follows:

$$\begin{aligned} \delta^2 &= \beta^2 + 1, \\ \frac{\gamma^2}{\beta^2} &= 1, \\ \alpha^2 &= 1 + \gamma^2. \end{aligned}$$

If we choose β as a free parameter, we can determine δ and γ from the first two equations and then α from the third one.

There is a nice way to parameterize the solution which also makes the resulting transformations easy to remember. Let us write the parameter as $\beta = \mathrm{sh}(\theta)$ where now the free parameter is θ (in particle physics it is called *rapidity*). Since the hyperbolic sine and cosine satisfy

$$\mathrm{ch}^2(\theta) - \mathrm{sh}^2(\theta) = 1,$$

Recall that:

$$\mathrm{ch}(\theta) \equiv \frac{1}{2}(e^{\theta} + e^{-\theta})$$

$$\mathrm{sh}(\theta) \equiv \frac{1}{2}(e^{\theta} - e^{-\theta})$$

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we have an obvious solution for the other coefficients

$$\begin{aligned} \delta &= \mathrm{ch}(\theta), \\ \gamma &= \mathrm{sh}(\theta), \\ \alpha &= \mathrm{ch}(\theta). \end{aligned}$$

There are other solutions that differ by signs; we will comment on them later. The transformation (1.5) then reads:

$$\begin{aligned} ct &= \mathrm{ch}(\theta)ct' + \mathrm{sh}(\theta)x', \\ x &= \mathrm{sh}(\theta)ct' + \mathrm{ch}(\theta)x'. \end{aligned} \tag{1.11}$$

One can also write this transformation in the matrix form,

$$\begin{pmatrix} ct \\ x \end{pmatrix} = \begin{pmatrix} \mathrm{ch}(\theta) & \mathrm{sh}(\theta) \\ \mathrm{sh}(\theta) & \mathrm{ch}(\theta) \end{pmatrix} \begin{pmatrix} ct' \\ x' \end{pmatrix} \tag{1.12}$$

This is the general form of the *Lorentz transformation* in the case when the relative velocity of the two frames is along the x-axis.

The Lorentz transformation written in this form strongly resembles usual space rotations, the difference is that trigonometric functions are replaced by the hyperbolic ones, and both $\mathrm{sh}(\theta)$-terms come with the same sign. This is not a coincidence: both transformations preserve a quadratic form — the interval in the case of Lorentz transformations, and the distance in the case of rotations.

Recall that for space rotations

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & cos\alpha \end{pmatrix} \begin{pmatrix} x' \\ y' \end{pmatrix}$$

• It remains to relate the parameter θ with the relative velocity of the two frames. To this end consider the case when the frame K' moves with the velocity $\boldsymbol{v} = (v, 0, 0)$ with respect to the frame K. Recall that we have assumed that at $t = t' = 0$ the origins of the two frames coincide. Let us focus on the motion of the origin of K', that is, of the point $x' = 0$, as seen from the frame K. By our assumption, this point is at $x = 0$ at the time $t = 0$ and moves with the velocity v in the positive direction of the axis x. Therefore, its coordinate x changes with time as $x = vt$.

Suppose an event happens in K' at time t' at the origin, i.e. at $x' = 0$. By eq. (1.11), its coordinates in the frame K are

$$\begin{aligned} ct &= \mathrm{ch}(\theta)\, ct' \\ x &= \mathrm{sh}(\theta)\, ct' \end{aligned}$$

where we have used $x' = 0$. But since this event happens at the origin of K', we know that $x = vt$ because this is the point where the origin of K' moved at time t. Dividing one equation by the other we get

$$\frac{v}{c} = \frac{\mathrm{sh}(\theta)}{\mathrm{ch}(\theta)} = \tanh(\theta). \tag{1.13}$$

This is the desired relation between the parameter θ and the relative velocity v of the frames.

• In order to rewrite the Lorentz transformation directly in terms of the velocity, we need to re-express $\mathrm{ch}(\theta)$ and $\mathrm{sh}(\theta)$ in terms of v. This is

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easily done by making use of the identity $\mathrm{ch}^2(\theta) - \mathrm{sh}^2(\theta) = 1$,

$$\frac{v^2}{c^2} = \frac{\mathrm{sh}^2(\theta)}{\mathrm{ch}^2(\theta)} = \frac{\mathrm{ch}^2(\theta) - 1}{\mathrm{ch}^2(\theta)} = 1 - \frac{1}{\mathrm{ch}^2(\theta)},$$

from where we can find $\mathrm{ch}(\theta)$, and then $\mathrm{sh}(\theta)$

$$\mathrm{ch}(\theta) = \frac{1}{\sqrt{1 - v^2/c^2}},$$

$$\mathrm{sh}(\theta) = \frac{v/c}{\sqrt{1 - v^2/c^2}}.$$

In terms of the velocity, the Lorentz transformations eq. (1.11) take the standard form

$$\begin{aligned} ct &= \frac{ct' + (v/c)x'}{\sqrt{1 - v^2/c^2}}, \\ x &= \frac{x' + vt'}{\sqrt{1 - v^2/c^2}}, \\ y &= y', \\ z &= z'. \end{aligned} \tag{1.14}$$

There is a standard notation for the inverse square root that appears always when Lorentz transformations are written in terms of velocity,

$$\gamma \equiv \frac{1}{\sqrt{1 - v^2/c^2}} = \mathrm{ch}(\theta)$$

which is called "gamma-factor". One also often denotes

$$\beta \equiv \frac{v}{c}.$$

The matrix of the Lorentz transformation is then

$$\begin{pmatrix} \gamma & \beta\gamma \\ \beta\gamma & \gamma \end{pmatrix}.$$

Here we have added for completeness the (trivial) transformation of y and z.

Recall now our original motivation that eq. (1.3) is not invariant under the Galilean transformations. It is straightforward to check, by changing variables in a standard way, that this equation *is invariant* under the transformation (1.11) or (1.14). However, under these new transformations the Newton's law is not invariant. If we want the laws of physics to be invariant under the Lorentz transformations we need to *modify the Newton's law*. It turns out that the required modification is such that the original Newton's law remains valid to a very good approximation at small velocities $v \ll c$, i.e. in the domain where it has been experimentally tested. The reason is that in the limit of small velocities the Lorentz transformations become the Galilean transformations. Indeed, at $v/c \ll 1$ from eqs. (1.14) we have

$$\begin{aligned} t &= t' + \ldots \\ x &= x' + vt + \ldots \end{aligned}$$

where dots denote terms suppressed by the powers of v/c. One may, therefore, expect that the relativistic dynamics becomes the Newtonian dynamics in this limit.

Consequences of Lorentz transformations

• *Causal structure and light cone.* Consider two events 1 and 2 with the coordinates $(t_1, \boldsymbol{x}_1)$ and $(t_2, \boldsymbol{x}_2)$. Let us choose the origin of our frame K coincident with the event 1, so that the event 2 has the coordinates $(t, \boldsymbol{x}) = (t_2 - t_1, \boldsymbol{x}_2 - \boldsymbol{x}_1)$. Does there exist a reference frame K' in which these events have the same space coordinate, that is, happen in the same point in space, $\boldsymbol{x}' = \boldsymbol{x}'_2 - \boldsymbol{x}'_1 = 0$?

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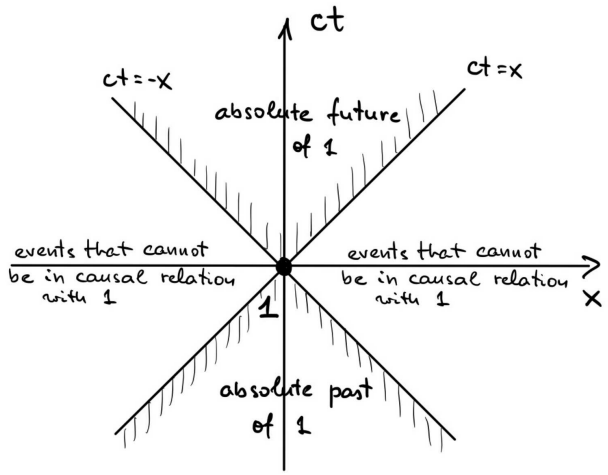


Figure 1.1.: Separation of space-time, with respect to a reference event 1, into three regions: absolute past of 1 (including part of the light cone $ct = -|\boldsymbol{x}|$), absolute future of 1 (including part of the light cone $ct = |\boldsymbol{x}|$), and the remaining region where events cannot be neither a cause nor a consequence of 1.

To answer this question, assume there is such a frame and consider the interval between the two events. Since the interval does not depend on the frame we have

$$c^2t^2 - \boldsymbol{x}^2 = c^2t'^2 - \underbrace{\boldsymbol{x}'^2}_{=0} = c^2t'^2 > 0.$$

We see that this is only possible if the interval between the two events is *positive*, otherwise we run into a contradiction. Positive intervals are called **time-like** for in this case the time contribution dominates.

If two events are separated by a time-like interval and $t_2 > t_1$, that is, the event 2 happens after the event 1, then this is true *in all reference frames*, $t'_2 > t'_1$. One may say in this case that the event 2 is in the future of the event 1, and the event 1 is in the past of the event 2 in all frames.

If the interval between the two events is *negative*, there exists a frame in which $t' = t'_2 - t'_1 = 0$, so that

$$c^2t^2 - \boldsymbol{x}^2 = -\boldsymbol{x}'^2 < 0.$$

Such intervals are called **space-like** as now the space contribution is dominant.

This leads to a striking consequence of the Lorentz transformations. The events separated by a space-like interval do not have an invariant chronological order: in some frames the event 1 may happen before the event 2 ($t' > 0$), in other frames the event 2 may happen before the event 1 ($t' < 0$). It is easy to show using eqs. (1.11) that for two events separated by a space-like interval both types of frames always exist. This has important physical implications: such events cannot be neither a cause, nor a consequence of one another.

Finally, the third possibility is that the interval between the events 1 and 2 is zero. Since the interval is invariant, this is also true in all other frames. Such intervals are called **light-like**.

This situation is summarized on Fig.1.1 which represents the causal structure of the space-time with respect to event 1. Taking the reference

In summary:

$s^2 > 0$: time-like
$s^2 < 0$: space-like
$s^2 = 0$: light-like

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event 1 at the origin, other events represented by space-time points are separated into three regions:

(i) *Absolute past of 1* — these are events with $ct \le -|\boldsymbol{x}|$; they happen before the event 1 in any reference frame. They can, potentially, be a cause of event 1. Note that this includes the past light cone $ct = -|\boldsymbol{x}|$.

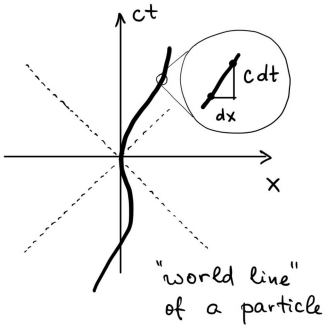
(ii) *Absolute future of 1* — these are events with $ct \ge |\boldsymbol{x}|$; they happen after the event 1 in any reference frame. They can, potentially, be a consequence of event 1. Note that this includes the future light cone $ct = |\boldsymbol{x}|$.

(iii) The remaining part of the space-time satisfies $|ct| < |\boldsymbol{x}|$. These events cannot be neither a cause nor a consequence of the event 1.

Importantly, *this division is preserved by the Lorentz transformations*; in other words, it does not depend on the reference frame.

• *Time dilation. Proper time.* Consider a particle moving with a time-dependent velocity $\boldsymbol{v}(t)$. On the space-time diagram this motion is represented by a line $\boldsymbol{x}(t)$ called a *world line* of the particle. Let the particle position at time t be $\boldsymbol{x}$. At an infinitesimally close time $t + dt$ the particle position will be $\boldsymbol{x} + d\boldsymbol{x}$, where $d\boldsymbol{x} = \boldsymbol{v}dt$ with $\boldsymbol{v}$ being the particle velocity at time t. The interval between these two infinitesimally close space-time points is

$$ds = \sqrt{c^2dt^2 - d\boldsymbol{x}^2} = c\,dt\sqrt{1 - v^2/c^2}.$$



Note that since $|\boldsymbol{v}| = v \le c$, all subsequent positions of the particle lie in the absolute future of its current position, and in particular $cdt \ge |d\boldsymbol{x}|$.

Now consider an inertial reference frame that moves with the (constant) velocity $\boldsymbol{v}$ coinciding with the particle velocity at the moment t, so the particle is at rest in this frame (for a brief moment of time). Call it a comoving frame at time t.

Denote the coordinates in the comoving frame as t' and $\boldsymbol{x}'$. In this frame the same two space-time points on the particle world line are separated by cdt' and $d\boldsymbol{x}' = 0$ because the particle is at rest. The interval between them is

$$ds' = \sqrt{c^2dt'^2 - d\boldsymbol{x}'^2} = c\,dt' \equiv cd\tau.$$

Here we have defined the **proper time** τ as the time that passes in the comoving frame. If our particle represents, say, a rocket, the proper time is the time that a clock in the rocket would measure.

Obviously, this definition only applies to infinitesimal intervals, since a given frame is, in general, comoving with the particle only for a short period of time.

Since the interval is the same in all frames, we have

$$ds = ds' \quad \Longrightarrow \quad dt\sqrt{1 - v^2/c^2} = d\tau.$$

This is the relation between the interval of the proper time $d\tau$ and the time interval dt in the frame in which the particle is in motion. For finite intervals, this relation gives

$$\tau_2 - \tau_1 = \int_{t_1}^{t_2} dt\sqrt{1 - v^2/c^2}$$

or, equivalently,

$$t_2 - t_1 = \int_{\tau_1}^{\tau_2} \frac{d\tau}{\sqrt{1 - v^2/c^2}}. \tag{1.15}$$

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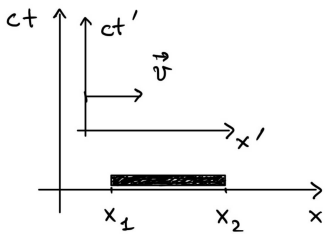
Since $\sqrt{1-v^2/c^2} < 1$, the *proper time is shorter than the time passing in any other frame*,

$$\tau_2 - \tau_1 < t_2 - t_1.$$

In other words, the time interval between two events happening with a moving object is shortest in the frame which is comoving with the object and in which the object is at rest. Obviously this is true for both finite and infinitesimal intervals.

This time dilation effect is of course only sizeable when the velocity of a body is comparable to the velocity of light. This is not the case for macroscopic bodies we have around in everyday life. However, in the world of elementary particles this is often the case — there the time dilation is a common phenomenon. For a concrete example, consider an unstable particle, say a μ-meson. The fact that the muon is unstable provides it with an internal "clock": after it is created, it decays in a certain period of time which is determined by some internal physics and is thus defined with respect to the muon proper frame. In other words, it counts the proper time. In case of a muon, the lifetime is $\tau_\mu = 2\times10^{-6}$ s — this is how long, on average, a muon at rest would live. Very high energy muons are copiously produced by cosmic rays in the atmosphere at an elevation of about 10 km. They have velocities very close to c. If these moving muons were also living 2×10^{-6} s, they would travel no further than $\sim$ 600 m. Meanwhile, they are observed at the sea level, i.e., often more than 10 km away from the creation point. This is possible because their decay time in the frame where they move with the velocity close to c is much longer than τ_μ in view of eq. (1.15).

• *Length contraction. Proper length.* In Galilean relativity the length of an object is the same in all inertial frames. This is no longer true in Lorentzian case. Consider first an object at rest, say, a bar oriented along the axis x. Suppose the end points of the bar have coordinates x_1 and x_2. Then the length of the bar is defined as



$$L = x_2 - x_1.$$

The length of an object in the frame where it is at rest is called **proper length**.

The same bar can be viewed in the frame that moves with the velocity $\boldsymbol{v} = (v, 0, 0)$ with respect to the original frame. As usual, we will assume that at $t = 0$ the origins of the two frames coincide. With respect tot he new frame the bar is moving. How do we determine the length of a moving object? The answer is rather obvious: we measure the coordinates of the bar endpoints x_1' and x_2' at *the same* moment of time t' and subtract the two,

$$L' = x_2' - x_1'.$$

Any moment of time t' is as good as any other, the result should not depend on it. But pay attention: it is important that the positions of the endpoints are both measured at *the same* time t'.

We may now use Lorentz transformations to find the relation between L

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and L'. From eq. (1.14) we have

$$\begin{aligned} x_1 &= \frac{x_1' + vt'}{\sqrt{1 - v^2/c^2}} \\ x_2 &= \frac{x_2' + vt'}{\sqrt{1 - v^2/c^2}} \end{aligned}$$

Subtracting these equations we find

$$x_2 - x_1 = \frac{x_2' - x_1'}{\sqrt{1 - v^2/c^2}}$$

or

$$L = \frac{L'}{\sqrt{1 - v^2/c^2}}. \tag{1.16}$$

Since $\sqrt{1 - v^2/c^2} < 1$, the *proper length L is larger than the length L′ in any moving frame.* The length in the moving frame is correspondingly shorter than the proper length — this is called Lorentz contraction. Again, this effect is only sizeable when v is comparable to c.

• *Velocity composition law.* Consider a frame K with coordinates t, $\boldsymbol{x}$, and another frame K' with coordinates t', $\boldsymbol{x}'$, which moves with respect to the first one with a constant velocity $\boldsymbol{V} = (V, 0, 0)$ in the x-direction. Consider a moving particle. Its velocity in the two frames is

$$\begin{aligned} \boldsymbol{v} &= \frac{d\boldsymbol{x}}{dt} \qquad \text{in the frame } K \\ \boldsymbol{v}' &= \frac{d\boldsymbol{x}'}{dt'} \qquad \text{in the frame } K' \end{aligned}$$

How $\boldsymbol{v}$ and $\boldsymbol{v}'$ are related? In the non-relativistic mechanics we have $\boldsymbol{v} = \boldsymbol{v}' + \boldsymbol{V}$. This is no longer true in the relativistic case.

In order to derive the relation between $\boldsymbol{v}$ and $\boldsymbol{v}'$ we use the Lorentz transformations (1.14), which we can write directly for the differentials:

$$\begin{aligned} dx &= \frac{dx' + V dt'}{\sqrt{1 - V^2/c^2}} \\ dy &= dy' \\ dz &= dz' \\ dt &= \frac{dt' + (V/c^2) dx'}{\sqrt{1 - V^2/c^2}} \end{aligned}$$

Dividing the first equation by the fourth one we get:

$$v_x = \frac{dx}{dt} = \frac{dx' + V dt'}{dt' + (V/c^2) dx'} = \frac{dx'/dt' + V}{1 + (V/c^2)\, dx'/dt'}.$$

Thus, we have expressed v_x in terms of $v_x' = dx'/dt'$. Repeating the same

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calculations for the second and third equation we find:

$$v_x = \frac{v'_x + V}{1 + Vv'_x/c^2} \tag{1.17}$$

$$v_y = \frac{v'_y}{1 + Vv'_x/c^2}\sqrt{1 - V^2/c^2} \tag{1.18}$$

$$v_z = \frac{v'_z}{1 + Vv'_x/c^2}\sqrt{1 - V^2/c^2} \tag{1.19}$$

Note that, unlike in the Galilean case, the velocity components v_y and v_z perpendicular to $\boldsymbol{V}$ also change, and that they change in a way different from the parallel component v_x.

This is the relativistic law of velocity composition.

Remarks. Our result for the relativistic velocity composition is complicated and non-linear, so let us discuss it in some more detail. First we note that the Galilean velocity composition is recovered in the limit $V/c \ll 1$. Indeed, in this limit the second terms in denominators of eqs. (1.17–1.19) can be neglected, as well as the second term under the square root. Thus, in this limit we have

The non-linearity of the velocity transformation is due to the fact that the velocity is a ratio of dx and dt both of which change upon the change of the frame.

$$\begin{aligned} v_x &= v'_x + V + \dots \\ v_y &= v'_y + \dots \\ v_z &= v'_z + \dots, \end{aligned}$$

where dots stand for the terms suppressed by powers of V/c. This, of course, is obvious from the fact that the Lorentz transformations themselves reduce to the Galilean ones in this limit.

Second, and less trivial property of this composition law is that if v' and V are both smaller than c, the velocity v is also always smaller than c. Consider an extreme case when both V and v' are close to c, so that $V = (1-\epsilon)c$ and $v'_x = (1-\epsilon)c$, where ϵ is a small dimensionless parameter. In the Galilean case we would have in this situation $v_x \simeq 2c$, which would be a disaster as it would not be consistent with our causality arguments. Instead, from eq. (1.17) we have in this case

$$v_x = \frac{2(1-\epsilon)}{1 + (1-\epsilon)^2}c = \frac{2(1-\epsilon)}{2(1-\epsilon) + \epsilon^2}c.$$

The factor in front of c in the last expression is obviously smaller than 1 as its denominator is larger than the numerator. We have therefore $v_x < c$.

Finally, let us check that the relativistic velocity composition preserves the speed of light. We will assume for simplicity that the velocity $\boldsymbol{v}'$ is parallel to $\boldsymbol{V}$, but the statement is true in general. So assume $\boldsymbol{v}' = (c, 0, 0)$. Eq. (1.17) gives if this case:

$$v_x = \frac{c + V}{1 + V/c} = c\frac{1 + V/c}{1 + V/c} = c,$$

and thus the velocity is c in both frames whatever their relative velocity V.

• *Graphical representation.* In practical problems it is often convenient to represent Lorentz transformation on a space-time diagram. For instance, one may ask a question where on a space-time diagram are the events that are simultaneous (or have the same coordinate x) in some other frame? So, consider two reference frames: K with coordinates ct, x and

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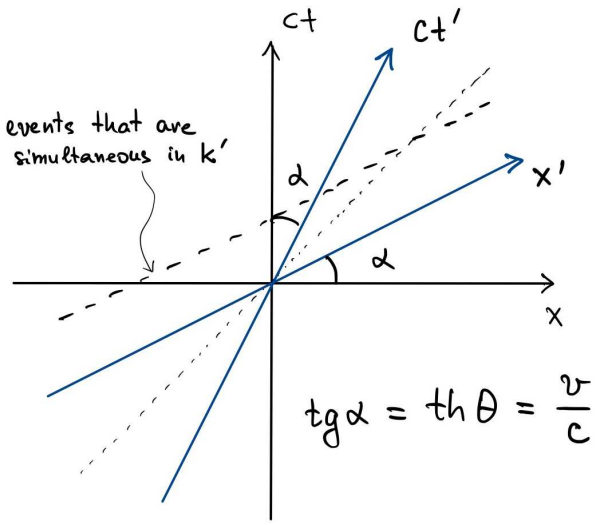


Figure 1.2.: Axes of a frame K' as viewed from the frame K. All events that lie on a line parallel to the x'-axis (for instance, on a dashed line) are simultaneous in the frame K'.

K' with coordinates ct', x'. The coordinates are related by the Lorentz transformation (1.11).

- First we note that, because the Lorentz transformation is linear, straight lines on the space-time diagram K' are transformed into straight lines on K and vice versa. Suppose we have a straight line in K' given parametrically as

$$\begin{aligned} x' &= A\lambda + x'_0, \\ ct' &= B\lambda + ct'_0, \end{aligned} \tag{1.20}$$

  where λ is a parameter, B/A is the slope and x'_0 and ct'_0 are some constants. Making use of eq. (1.11), this line in the frame K has the following equation:

$$\begin{aligned} x &= (B\mathrm{sh}\theta + A\mathrm{ch}\theta)\lambda + x_0, \\ ct &= (B\mathrm{ch}\theta + A\mathrm{sh}\theta)\lambda + ct_0, \end{aligned} \tag{1.21}$$

  where x_0 and ct_0 are some other constants of no interest to us. This line has the slope

$$\frac{B\mathrm{ch}\theta + A\mathrm{sh}\theta}{B\mathrm{sh}\theta + A\mathrm{ch}\theta} = \frac{(B/A)\mathrm{ch}\theta + \mathrm{sh}\theta}{(B/A)\mathrm{sh}\theta + \mathrm{ch}\theta} = F(B/A). \tag{1.22}$$

  We see that two lines in K' that have the same slope B/A, also have the same slope $F(B/A)$ in K. Therefore, *parallel lines are transformed into parallel lines*.
- Consider what happens to the axes. In K' the x'-axis is described by eqs. (1.20) with $A = 1$, $B = 0$ and $x'_0 = ct'_0 = 0$. Its counterpart in K is the straight line with the slope $\mathrm{sh}\theta/\mathrm{ch}\theta$, cf. eq. (1.22).
  Likewise, the ct'-axis in K' is described by eqs. (1.20) with $A = 0$, $B = 1$ and $x'_0 = ct'_0 = 0$. Its counterpart in K is the straight line with the slope $\mathrm{ch}\theta/\mathrm{sh}\theta$.
  This is summarized in Fig. 1.2 Events that happen at $t' = 0$ in K' are situated at the x'-axis. Events that happen at the same time t' form a line parallel to the x'-axis. Similarly, events that happen at the same point x' in the frame K' form a line parallel to the ct'-axis.

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1.2. Formalism of 4-vectors

Before we move further and discuss the relativistic dynamics, let us develop the formalism that is suited for working with the relativistic quantities. A useful analogy is suggested by ordinary vectors. The laws of physics are invariant under space rotations. For this reason it is convenient to use the formalism that operates with quantities having clear transformation properties under rotations — the vectors. In this way it is easy to ensure that physical relations are valid in any frame. For instance, in the Newton's law $\boldsymbol{F} = m\boldsymbol{a}$ both sides of the equation change in the same way when the axes are rotated, so that the equation remains valid.

The Lorentz transformations are similar to rotations from mathematical point of view (we have already pointed out this analogy before). We have learned in the previous section that the laws of physics must be invariant under Lorentz transformations. So it is often convenient to work with the generalization of vectors, the *4-vectors*.

The analogy is very close, so it is instructive to start with the normal space vectors and recall some of their properties before going into 4-dimensional case. This will also let us introduce some useful notations and calculational techniques that will be extensively used in the rest of these lectures.

3-vectors and rotations

• In ordinary 3-dimensional space points are associated with vectors that denote their position. In a given frame, the position vector $\boldsymbol{x}$ is characterized by its three components (x, y, z),

$$\boldsymbol{x} \equiv \vec{x} = (x, y, z) = (x_1, x_2, x_3)$$

are different ways to write the same vector $\boldsymbol{x}$. We will often refer to these three components as

$$\boldsymbol{x} = \{x_i\}, \qquad i = 1, 2, 3.$$

These index notations will come very handy in calculations, particularly in parts involving vector analysis.

The length $|\boldsymbol{x}|$ of a vector $\boldsymbol{x}$ (the distance of the corresponding point from the origin) is found from the Pythagoras theorem,

$$|\boldsymbol{x}|^2 = x^2 + y^2 + z^2 = (x_1)^2 + (x_2)^2 + (x_3)^2 = \sum_{i=1,2,3} x_i x_i. \tag{1.23}$$

We can also write this as follows,

$$|\boldsymbol{x}|^2 = \sum_{i,j=1,2,3} \delta_{ij} x_j x_i \tag{1.24}$$

where the matrix $\{\delta_{ij}\}$ is

In other words,

$$\delta_{ij} = \begin{cases} 1 & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases}$$

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$$\{\delta_{ij}\} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

is a 3×3 unit matrix called the *metric*. This form is a starting point for generalization to the case of non-orthonormal frames or curved coordinate systems.

• Rotations are linear transformations that preserve the distance (1.23). If x_i and $\tilde{x}_j$ are three coordinates of the same vector $\boldsymbol{x}$ in the original and rotated frame, they are related as follows:

For instance, in the case of rotation around the z-axis this matrix has the form

$$R = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$x_i = \sum_{j=1,2,3} R_{ij}\tilde{x}_j, \qquad \text{matrix form: } x = R\tilde{x}, \tag{1.25}$$

where R is a 3×3 matrix of rotation. This matrix must satisfy the relation

$$R^T R = 1, \tag{1.26}$$

that is, must be an orthogonal matrix. Indeed, for any vector $\boldsymbol{x}$ we must have

$$\sum_{i,j} \delta_{ij} x_j x_i = \sum_{i,j} \delta_{ij} \left(\sum_q R_{jq}\tilde{x}_q \right) \left(\sum_p R_{ip}\tilde{x}_p \right) =$$

$$= \sum_{p,q} \underbrace{\left[\sum_{i,j} \delta_{ij} R_{jq} R_{ip} \right]}_{\text{must be equal to } \delta_{pq}} \tilde{x}_p \tilde{x}_q = \sum_{p,q} \delta_{pq} \tilde{x}_p \tilde{x}_q .$$

Since the last equality must hold for any $\tilde{x}_i$, the quantity in the square brackets must be equal to δ_{pq},

In the matrix form: $R^T R = 1$

$$\Longrightarrow \quad \sum_{i,j} \delta_{ij} R_{jq} R_{ip} = \sum_i R_{iq} R_{ip} = \sum_i (R^T)_{qi} R_{ip} = \delta_{pq}$$

which is equivalent to eq. (1.26).

• The basic vector algebra may be rewritten in this "component" form. For two different vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ one defines a scalar product

$$\boldsymbol{ab} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \sum_i a_i b_i . \tag{1.27}$$

This scalar product is automatically invariant under rotations (1.25) due to the property of the rotation matrix (1.26).

Summation convention. Since in physics one deals with laws that are invariant under rotations, the product, say, of the first components of two vectors like $a_1 b_1$ (almost) never appears except in the combination $a_1 b_1 + a_2 b_2 + a_3 b_3 = \sum_i a_i b_i$. It is therefore a common practice not to write explicitely the sign of the sum when there are *two repeating indices in the product*. In these cases the summation is assumed (unless explicitely stated otherwise), but not shown explicitely, for example

$$\sum_i a_i b_i \quad \text{one writes as} \quad a_i b_i .$$

We will follow this convention from now on. We will therefore write the scalar product of two vectors as $\boldsymbol{ab} = a_i b_i = \delta_{ij} a_i b_j$ (the summation over

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both i and j is assumed in the last expression).

• To write the vector product in components we need one more object, the absolute antisymmetric symbol ε_{ijk} defined by the following relations:

$$\varepsilon_{ijk} = -\varepsilon_{jik} = -\varepsilon_{ikj},$$

$$\varepsilon_{123} = 1.$$

Because ε_{ijk} is totally antisymmetric, it is zero if any of the two indices repeat: $\varepsilon_{112} = \varepsilon_{331} = \varepsilon_{232} = 0$, etc. The non-zero components (those which have all three indices different) are equal 1 or -1 depending on the parity of permutation of indices compared to 123. In components, the vector product of two vectors is written as

$$(\boldsymbol{a} \times \boldsymbol{b})_i = \varepsilon_{ijk} a_j b_k. \tag{1.28}$$

Note that there is an implicit summation over repeating j and k on the r.h.s. of eq. (1.28). Note also that the antisymmetry of ε_{ijk} automatically implies that $\boldsymbol{a} \times \boldsymbol{b} = -\boldsymbol{b} \times \boldsymbol{a}$:

$$(\boldsymbol{a} \times \boldsymbol{b})_i = \varepsilon_{ijk} a_j b_k = -\varepsilon_{ikj} a_j b_k = -\varepsilon_{ikj} b_k a_j = -(\boldsymbol{b} \times \boldsymbol{a})_i.$$

• *Properties of δ_{ij} and ε_{ijk}.* All the formulas of vector algebra, as well as of vector analysis as we will see later, follow from a few the identities involving δ_{ij} and ε_{ijk}. Here they are:

$$\delta_{ij} A_{jkp\ldots} = A_{ikp\ldots} \qquad \text{for any } A_{jkp\ldots} \tag{1.29}$$

$$\delta_{ii} = 3 \tag{1.30}$$

$$\varepsilon_{ijk} S_{jkq\ldots} = 0 \qquad \text{for any symmetric } S_{jkq\ldots} = S_{kjq\ldots} \tag{1.31}$$

$$\varepsilon_{ijk}\varepsilon_{ijk} = 6 \tag{1.32}$$

$$\varepsilon_{ijk}\varepsilon_{ijp} = 2\delta_{kp} \tag{1.33}$$

$$\varepsilon_{kij}\varepsilon_{kpq} = \delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp} \tag{1.34}$$

The demonstration is straightforward. The first four identities are obvious. To show (1.33) we use the fact that δ_{ij} and ε_{ijk} are the only two rotationally invariant tensors. Therefore, the r.h.s. must be some combination of them. By symmetry, this can only be δ_{ij} with some yet unknown coefficient,

$$\varepsilon_{ijk}\varepsilon_{ijp} = A\delta_{kp}.$$

If we now take $p = k$ and sum both sides over k, we get $6 = 3A$, and therefore $A = 2$. The same arguments work for eq. (1.34) as well.

Note that the summation over the repeating indices is assumed.

For an illustration, let us work out the double vector product $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$. The result is a vector; its i-th component is

$$[\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})]_i = \varepsilon_{ijk} a_j (\varepsilon_{kpq} b_p c_q) = \varepsilon_{kij}\varepsilon_{kpq} a_j b_p c_q =$$

$$= (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}) a_j b_p c_q = b_i a_j c_j - c_i a_j b_j = [\boldsymbol{b}(\boldsymbol{a}\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\boldsymbol{b})]_i.$$

We have, therefore,

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\boldsymbol{b}).$$

Another example: consider a scalar product of $\boldsymbol{a}$ and $\boldsymbol{b} \times \boldsymbol{c}$,

$$\boldsymbol{a}(\boldsymbol{b} \times \boldsymbol{c}) = a_i \varepsilon_{ijk} b_j c_k = b_j \varepsilon_{jki} c_k a_i = \boldsymbol{b}(\boldsymbol{c} \times \boldsymbol{a}).$$

In a similar way all other identities of the vector algebra can be reproduced.

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4-vectors and Lorentz transformations

• Similar to the 3d case, one labels points of space-time (that is, events) by 4 coordinates arranged into a **contravariant 4-vector**

$$\{x^\mu\} \equiv (x^0, x^1, x^2, x^3) \equiv (ct, x, y, z) = (ct, \boldsymbol{x}).$$

We used here $x^0 = ct$ as the time coordinate so that all four coordinates have the same units of length.

It is a tradition to use the Greek letters for 4-vector indices. In case of 4-vectors one makes a difference between upper and lower indices, more on that below. Vectors with an upper index, like x^μ, are called "contravariant".

• In space-time the analog of the distance (invariant under rotations) is the interval (invariant under Lorentz transformations). We can write the interval as follows:

$$s^2 = c^2t^2 - \boldsymbol{x}^2 = \sum_{\mu,\nu=0,1,2,3} \eta_{\mu\nu} x^\mu x^\nu,$$

where the **Minkowski metric** $\eta_{\mu\nu}$ is the diagonal matrix

$$\eta = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

cf. eq. (1.24). It is customary to use the *summation convention* in the 4d case as well: a pair of repeating indices in a product, one upper and one lower, is assumed to be summed over. The interval is then written as $s^2 = \eta_{\mu\nu} x^\mu x^\nu$ where the sum over both μ and ν is assumed, but not written explicitely.

• Making use of the metric, one defines the **covariant vector** $\{x_\mu\}$ as

$$x_\mu = \eta_{\mu\nu} x^\nu \qquad \text{(summation over } \nu \text{ is implicit).}$$

In mathematics contravariant and covariant vectors are parts of a more formal construction and are objects of a different nature. Here we do not have to make this distinction and will consider x^μ and x_μ as different forms of the same physical quantity.

The components of the covariant vector x_μ are

$$\{x_\mu\} = (ct, -x, -y, -z) = (ct, -\boldsymbol{x}).$$

The interval can be written as

$$s^2 = \eta_{\mu\nu} x^\nu x^\mu = x_\mu x^\mu. \tag{1.35}$$

One defines the *inverse metric* $\{\eta^{\mu\nu}$ as the matrix inverse to $\{\eta_{\mu\nu}\}$, so that

$$\eta^{\mu\nu}\eta_{\nu\lambda} = \delta^\mu_\lambda\,,$$

where δ^μ_λ is a unit matrix. Clearly,

$$\{\eta^{\mu\nu}\} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

the same matrix as $\eta_{\mu\nu}$. Making use of $\eta_{\mu\nu}$ and $\eta^{\mu\nu}$ one may raise and lower the indices of vectors,

$$x_\mu = \eta_{\mu\nu} x^\nu, \qquad x^\mu = \eta^{\mu\nu} x_\nu.$$

Lowering and then raising the index of a vector one recovers its starting form:

$$\eta^{\mu\nu}(\eta_{\nu\lambda}x^\lambda) = (\eta^{\mu\nu}\eta_{\nu\lambda})x^\lambda = \delta^\mu_\lambda x^\lambda = x^\mu.$$

• A general **Lorentz transformation** is defined as *a linear transformation preserving the interval*. We can write this transformation as

$$x^\mu = \Lambda^\mu{}_\nu x'^\nu, \tag{1.36}$$

where the 4×4 matrix of the transformation $\{\Lambda^\mu{}_\nu\}$ has to satisfy certain conditions in order to preserve the interval. To find these conditions we require, similar to what we have done in the case of rotations, that the interval (1.35) is the same when written in transformed and non-transformed coordinates,

$$\eta_{\mu\nu}x^\mu x^\nu = \eta_{\mu\nu}\left(\Lambda^\mu{}_\rho x'^\rho\right)\left(\Lambda^\nu{}_\sigma x'^\sigma\right) = \left(\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\right)x'^\rho x'^\sigma = \eta_{\rho\sigma}x'^\rho x'^\sigma.$$

From the last equality which must be valid for any x'^μ we conclude that

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}.$$

All quantities here are 4×4-matrices, and one may recognize the matrix multiplication, so that in the matrix notations this equality reads

$$\Lambda^T\eta\Lambda = \eta. \tag{1.37}$$

This condition is equivalent to the requirement that the transformation (1.36) preserves the interval; one may use this relation as an equivalent definition of a general Lorentz transformation.

As an example, consider the Lorentz transformation that we have derived in Sect. 1.1. The matrix Λ in this case has the form

$$\Lambda = \begin{pmatrix} \mathrm{ch}\theta & \mathrm{sh}\theta & 0 & 0 \\ \mathrm{sh}\theta & \mathrm{ch}\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.38}$$

It is straightforward to check that this matrix satisfies eq. (1.37). Calculating $\Lambda^T\eta\Lambda$ we find:

$$\Lambda^T\eta\Lambda = \begin{pmatrix} \mathrm{ch}^2\theta - \mathrm{sh}^2\theta & \mathrm{sh}\theta\mathrm{ch}\theta - \mathrm{ch}\theta\mathrm{sh}\theta & 0 & 0 \\ \mathrm{sh}\theta\mathrm{ch}\theta - \mathrm{ch}\theta\mathrm{sh}\theta & -\mathrm{ch}^2\theta + \mathrm{sh}^2\theta & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \eta.$$

• From the transformation (1.36) and the definition of the covariant vector we may immediately derive the transformation law for the covariant vectors:

$$x_\mu = \eta_{\mu\nu}x^\nu = \eta_{\mu\nu}\Lambda^\nu{}_\rho x'^\rho = \left(\eta_{\mu\nu}\Lambda^\nu{}_\rho\eta^{\rho\sigma}\right)x'_\sigma. \tag{1.39}$$

The matrix in parenthesis transforms the covariant vectors. It is in fact the inverse of Λ^T as follows from eq. (1.37).

• From two 4-vectors with components x^μ and y^μ and Minkowskian

To make correspondence with the standard matrix notations one may think of the contravariant x^μ (denote it x for brevity) as a column and covariant x_μ as a row. In these matrix notations eq. (1.36) is then simply $x = \Lambda x'$, while the expression for the interval (1.35) is $s^2 = x^T\eta x$. The condition that the Lorentz transformation preserves the interval becomes, in these notations

$$x^T\eta x = (\Lambda x')^T\eta\Lambda x' =$$
$$= x'^T(\Lambda^T\eta\Lambda)x' = x'^T\eta x'$$

from where we conclude that

$$\Lambda^T\eta\Lambda = \eta.$$

This is another way to derive eq. (1.37).

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metric one may form a scalar product according to

$$(x \cdot y) = \eta_{\mu\nu} x^\mu y^\nu .$$

Note the analogy with the 3d scalar product (1.27). This scalar product is *invariant under the Lorentz transformations* (1.36). Indeed,

$$\eta_{\mu\nu} x^\mu y^\nu = \eta_{\mu\nu} \Lambda^\mu{}_\rho x'^\rho \Lambda^\nu{}_\sigma y'^\sigma = \eta_{\rho\sigma} x'^\rho y'^\sigma .$$

The last equality holds due to the property of the Lorentz matrix (1.37). Therefore, the scalar product of two 4-vectors $(x \cdot y)$ can be calculated in any reference frame. There is no analog of the vector product of two 4-vectors.

• **Lorentz tensors** are defined as objects with some number of contravariant and some number of covariant indices that transform as the corresponding product of 4-vectors, one per each index. For example, the tensor $t^{\mu\nu}$ transforms under the Lorentz transformation as the product $x^\mu x^\nu$,

$$t^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma t'^{\rho\sigma}, \tag{1.40}$$

where $t'^{\rho\sigma}$ are components of the same tensor in another reference frame. A similar rule applies in the case of more than two, or other type of indices: upon Lorentz transformation, the tensor is multiplied by the matrix Λ per each contravariant (upper) index, and by the matrix $\eta\Lambda\eta$ from eq. (1.39) per each covariant (lower) index.

• The Minkowski metric $\eta_{\mu\nu}$, or its inverse $\eta^{\mu\nu}$, may itself be regarded as a *constant tensor*. This tensor is *invariant* under the Lorentz transformations. This is easy to show; consider the inverse metric as an example. Applying the rule (1.40) we have

$$\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \eta^{\rho\sigma} = \eta^{\mu\nu}$$

as follows from the basic property of the Lorentz matrices, eq. (1.37).

There is another constant invariant tensor — totally antisymmetric tensor $\epsilon^{\mu\nu\rho\sigma}$ defined by the relations

$$\epsilon^{\mu\nu\rho\sigma} = -\epsilon^{\nu\mu\rho\sigma} = -\epsilon^{\mu\rho\nu\sigma} = -\epsilon^{\mu\nu\sigma\rho},$$

$$\epsilon^{0123} = 1.$$

To show the invariance let us apply the tensor transformation rules. We have to multiply by Λ-matrices 4 times as there are 4 indices. Denote the result of the transformation as $A^{\alpha\beta\gamma\delta}$:

$$A^{\alpha\beta\gamma\delta} \equiv \Lambda^\alpha{}_\mu \Lambda^\beta{}_\nu \Lambda^\gamma{}_\rho \Lambda^\delta{}_\sigma \epsilon^{\mu\nu\rho\sigma}.$$

Our goal is to show that $A^{\alpha\beta\gamma\delta} = \epsilon^{\alpha\beta\gamma\delta}$.

First let us demonstrate that $A^{\alpha\beta\gamma\delta}$ is totally antisymmetric. Consider, e.g., the first pair of indices. We have

$$A^{\beta\alpha\gamma\delta} = \Lambda^\beta{}_\mu \Lambda^\alpha{}_\nu \Lambda^\gamma{}_\rho \Lambda^\delta{}_\sigma \epsilon^{\mu\nu\rho\sigma} =$$

...rename the summation indices $\mu \to \nu$, $\nu \to \mu$

$$= \Lambda^\beta{}_\nu \Lambda^\alpha{}_\mu \Lambda^\gamma{}_\rho \Lambda^\delta{}_\sigma \epsilon^{\nu\mu\rho\sigma} =$$

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...and permute the first two indices of $\epsilon^{\nu\mu\rho\sigma}$. Because of its antisymmetry this gives a minus sign,

$$= -\Lambda^{\beta}{}_{\nu}\Lambda^{\alpha}{}_{\mu}\Lambda^{\gamma}{}_{\rho}\Lambda^{\delta}{}_{\sigma}\epsilon^{\mu\nu\rho\sigma} = -A^{\alpha\beta\gamma\delta}.$$

So, we have shown that

$$A^{\beta\alpha\gamma\delta} = -A^{\alpha\beta\gamma\delta}.$$

The same argument applies to any other pair of adjacent indices.

Since $A^{\alpha\beta\gamma\delta}$ is totally antisymmetric, it is completely determined by a single component, say $A^{0123} = A$ — other non-zero components are obtained by permutations. We can therefore write

$$A^{\alpha\beta\gamma\delta} = A\epsilon^{\alpha\beta\gamma\delta}.$$

To find the constant A, let us calculate

$$A = A^{0123} = \Lambda^{0}{}_{\mu}\Lambda^{1}{}_{\nu}\Lambda^{2}{}_{\rho}\Lambda^{3}{}_{\sigma}\epsilon^{\mu\nu\rho\sigma}.$$

One may recognize in this expression the definition of the determinant of the matrix Λ. Thus,

$$A = \det(\Lambda).$$

Finally, to find the determinant of the matrix Λ we take the determinant of both sides of eq. (1.37). Using properties of the determinant we get

$$\det(\Lambda)\det(\eta)\det(\Lambda) = \det(\eta),$$

and therefore

$$\det(\Lambda) = \pm 1.$$

The Lorentz transformations that we have considered so far have the property that they become the identity transformation in the limit when the relative velocity of the two frames goes to zero. By continuity, all these transformations have $\det(\Lambda) = 1$, and therefore $\epsilon^{\nu\mu\rho\sigma}$ is invariant under these transformations.

A simple example of a matrix Λ that satisfies eq. (1.37) and has the determinant equal to -1 is a diagonal matrix $\Lambda = \mathrm{diag}(1, -1, 1, 1)$. This matrix performs a reflection of the x-axis, $x \to -x$.

• From mathematical point of view, the set of all the matrices Λ satisfying eq. (1.37) has a structure of a group called *Lorentz group*. The demonstration is straightforward by using the definition of a group. This set of matrices includes, in particular:

- Boosts or proper Lorentz transformations — the ones that we have derived above.
- Rotations — these are matrices of the form

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & & & \\ 0 & & R_{ij} & \\ 0 & & & \end{pmatrix},$$

  where R_{ij} is a 3×3 rotation matrix. For these matrices the condition (1.37) becomes the orthogonality condition for R_{ij}.
- Discrete transformations — reflections of the axes.

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1.3. Relativistic mechanics

4-velocity

Motivated by electromagnetism and the relativity principle, we are constructing a theory invariant under the Lorentz transformations. As it has already been discussed, the conventional Newton's laws are not invariant, so we need a replacement. We are looking therefore for new mechanical laws which are Lorentz-invariant and at the same time reproduce the conventional Newton's laws in the non-relativistic limit $v/c \to 0$. This way we will be able to reconcile relativity principle with both electromagnetism and (experimentally verified) Newton's mechanics.

The laws of any relativistic theory should be possible to formulate entirely in terms of Lorentz scalars, vectors, tensors etc. We have seen that time and space coordinates form a 4-vector x^μ. To formulate dynamical laws we also need velocity and acceleration. So, let us start by discussing the relativistic generalization of velocity.

As we have seen in sect. 1.1, the usual velocity of a particle $\boldsymbol{v} = d\boldsymbol{x}/dt$ has very complicated and non-linear transformation rules from one frame to the other. It is evident that it cannot be a space part of any 4-vector. There is only one natural candidate for a replacement. The differentials of the space-time coordinates dx^μ form a Lorentz vector, while the infinitesimal interval ds is a Lorentz scalar, so if we divide one by the other we get a Lorentz vector

$$u^\mu \equiv c\frac{dx^\mu}{ds}, \tag{1.41}$$

where we have added a factor c to give u^μ the units of velocity. This quantity is called **4-velocity**. It transforms under the Lorentz transformation in the same way as x^μ or any other 4-vector: $u^\mu = \Lambda^\mu{}_\nu u'^\nu$. In geometrical terms, u^μ is a tangent vector to the world line of a particle.

Let us see what it equals to in conventional terms. The numerator in eq. (1.41) is a 4-vector with the components $dx^\mu = (cdt, d\boldsymbol{x})$. The denominator is

$$ds = \sqrt{dx_\mu dx^\mu} = \sqrt{c^2dt^2 - d\boldsymbol{x}^2} = cdt\sqrt{1 - \boldsymbol{v}^2/c^2}.$$

Dividing one by the other we find

$$u^\mu = \left(\frac{c}{\sqrt{1 - \boldsymbol{v}^2/c^2}}, \frac{\boldsymbol{v}}{\sqrt{1 - \boldsymbol{v}^2/c^2}}\right). \tag{1.42}$$

In the non-relativistic limit $v/c \ll 1$ the 4-velocity u^μ becomes

$$u^\mu \simeq (c, \boldsymbol{v}). \tag{1.43}$$

Its space part thus becomes the usual velocity in this limit.

Since the 4 components of u^μ depend on three components of $\boldsymbol{v}$ they are not all independent and there is a relation among them. Indeed, if we

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calculate the Minkowskian scalar product $u_\mu u^\mu$ we find

$$u_\mu u^\mu = c^2 \frac{dx_\mu}{ds}\frac{dx^\mu}{ds} = c^2 \frac{dx_\mu dx^\mu}{ds^2} = c^2 \tag{1.44}$$

regardless of the value of $\boldsymbol{v}$.

Energy and momentum

In the non-relativistic mechanics very important quantities are energy and momentum, because these quantities are conserved in the evolution of an isolated system. It is not a coincidence that the number of these conserved quantities (4 in total: energy and 3 components of momentum) is the same as the number of components of a 4-vector. In fact, the conservation of energy and momentum can be traced back to the symmetry with respect to translations of the 4 coordinates. This suggests that the energy and momentum somehow form a 4-vector. We will see below that this is indeed the case.

The non-relativistic momentum is defined as $\boldsymbol{P} = m\boldsymbol{v}$. Having in mind the non-relativistic limit (1.43) of u^μ, a good candidate for this 4-vector is

$$P^\mu = mu^\mu. \tag{1.45}$$

Indeed, we will now show that this is the only expression compatible with the Lorentz symmetry, conservation law and having correct non-relativistic limit.

• We start from general expressions for the energy and momentum. Since the space has no preferred direction, the momentum must have the same direction as $\boldsymbol{v}$. Apart from that, both energy and momentum are some functions of $v = |\boldsymbol{v}|$. We can write them as

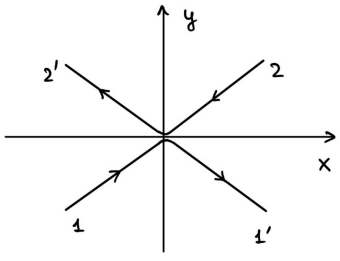
$$\boldsymbol{P} = \mathscr{M}(v)\boldsymbol{v}, \tag{1.46}$$
$$E = E(v), \tag{1.47}$$

where $\mathscr{M}(v)$ and $E(v)$ are two functions to be determined. These functions must be such that at small velocities the standard Newtonian expressions are recovered, and therefore

$$\mathscr{M}(v) = m + \ldots \tag{1.48}$$
$$E(v) = \text{const} + \frac{1}{2}mv^2 + \ldots \tag{1.49}$$

where dots stand for terms suppressed by powers of v/c.

• Let us start with the momentum. Consider the collision of two *identical* particles. Without loss of generality we may assume that it happens in the plane (xy), where we can choose the axes in such a way that the collision happens symmetrically as shown on the picture. The particles head towards each other along the lines 1 and 2, collide at the origin and then part along the lines 1′ and 2′.



We will now consider this process in a particular frame K and write the equations of momentum conservation. Let us choose this frame K in such a way that the particle 1 has zero velocity in the x-direction as shown on Fig. 1.3, left plot. (The particle 1 is arriving and leaving along the y-axis;

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its trajectory on the plot is slightly displaced from the axis for visibility.) The momenta of particles are in the plane (xy) both before and after collision, so we only have to consider their projections on the axes x and y. The conservation of momentum gives the following two equations,

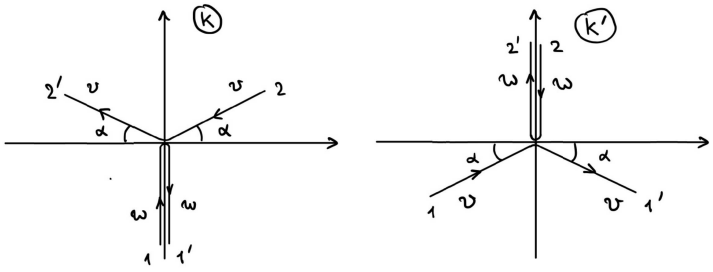


Figure 1.3.: Collision of identical particles in two different frames K and K'.

$$\begin{aligned} &\text{axis } x\text{:} && 0 + \mathcal{M}(v)v\cos\alpha = 0 + \mathcal{M}(v)v\cos\alpha, \\ &\text{axis } y\text{:} && \mathcal{M}(w)w - \mathcal{M}(v)v\sin\alpha = \mathcal{M}(v)v\sin\alpha - \mathcal{M}(w)w. \end{aligned}$$

The first of these equations is identically satisfied, while the second gives

$$\mathcal{M}(w)w = \mathcal{M}(v)v\sin\alpha. \tag{1.50}$$

This is all we can extract from the frame K alone.

To get additional equations, consider the same process in a different frame which we choose in such a way that now the particle 2 has zero velocity in the x-direction, as show on the right plot of Fig. 1.3. Because the particles are identical, and because of the symmetry between the two pictures, the angle α and the velocities v and w *are the same on both plots*. On the other hand, the two pictures are related by a Lorentz transformation. The relative velocity of the two frames is clearly in the x direction and equals $V = -v\cos\alpha$. Recall that under the Lorentz transformation both parallel and perpendicular components of velocity change, so we get two equations. Let us concentrate on the particle 2 (the equations for the particle 1 must be identical because of the symmetry). Applying eqs. (1.17) and (1.18) with $v'_x = 0$ (due to our choice of the frame K') we get

$$\begin{aligned} v\cos\alpha &= -V, \\ v\sin\alpha &= w\sqrt{1-\frac{V^2}{c^2}} = w\sqrt{1-\cos^2\alpha\frac{v^2}{c^2}}. \end{aligned} \tag{1.51}$$

The first of these equations we already know, so it gives nothing new. The second equation can be used to substitute $v\sin\alpha$ in eq. (1.50). The velocity w then cancels and we get

$$\mathcal{M}(w) = \mathcal{M}(v)\sqrt{1-\cos^2\alpha\frac{v^2}{c^2}}.$$

This equation still has too many unknowns. We can get rid of them by going to the limit $\alpha \to 0$, in which this equation still has to be valid. In this limit obviously $\cos\alpha \to 1$, but also $w \to 0$ as follows from eq. (1.51), so that we find

$$\mathcal{M}(0) = \mathcal{M}(v)\sqrt{1-\boldsymbol{v}^2/c^2}$$

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or, recalling that $\mathcal{M}(0) = m$ in view of eq. (1.48),

$$\mathcal{M}(v) = \frac{m}{\sqrt{1 - \boldsymbol{v}^2/c^2}}. \tag{1.52}$$

From this expression and eq. (1.46) we finally get

$$\boldsymbol{P} = \frac{m\boldsymbol{v}}{\sqrt{1 - \boldsymbol{v}^2/c^2}}. \tag{1.53}$$

By our derivation, this is the only expression for the momentum that is compatible with the momentum conservation and Lorentz transformations.

• This expression for the relativistic momentum is quite remarkable. First, we note that the relativistic momentum $\boldsymbol{P}$ can be arbitrarily large even though the particle velocity is limited by the speed of light c. Indeed, when v approaches c the denominator goes to zero and correspondingly $\boldsymbol{P}$ goes to infinity.

Second, comparing it with the 4-velocity u^μ, eq. (1.42), we see that the relativistic momentum indeed can be written as

$$P^i = mu^i.$$

We have thus established the space part of the equation (1.45) which we anticipated before.

• Let us now derive the relativistic expression for the energy of a particle. In principle, it is possible to do so by considering again the scattering of identical particles in a manner similar to what we have done in the case of momentum, and requiring the conservation and Lorentz invariance. There is however a simpler way — we can guess the right expression and then show that it satisfies all the required properties.

Let us have a closer look at the *time* component of eq. (1.45),

$$P^0 = mu^0 = \frac{mc}{\sqrt{1 - \boldsymbol{v}^2/c^2}}.$$

If we expand this quantity in powers of v/c, in the non-relativistic limit $v/c \ll 1$ we find

$$P^0 = mc(1 - v^2/c^2)^{-1/2} \simeq mc\left(1 + \frac{1}{2}\frac{v^2}{c^2} + \dots\right) = mc + \frac{1}{2c}mv^2 + \dots$$

and therefore

$$cP^0 \simeq mc^2 + \frac{1}{2}mv^2 + \dots$$

Comparing to eq. (1.49) we see that cP^0 has precisely the non-relativistic limit that we expect for the energy.

Let us show that P^0 (and therefore cP^0) is conserved in particle collisions. Consider a collision in which the total initial momentum is $\boldsymbol{P}_{\text{in}}$ and the total final momentum $\boldsymbol{P}_{\text{out}}$. The momentum is conserved, in particular its x-component is the same before and after collision,

$$P^x_{\text{in}} = P^x_{\text{out}}.$$

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If we consider the same collision in a frame that moves with the velocity $\boldsymbol{V} = (V, 0, 0)$ with respect to the original one, the momentum in the new frame is also conserved, so that $P'^x_{\text{in}} = P'^x_{\text{out}}$. Now recall that all 4-vectors transform in the same way, so that making use of the Lorentz transformation from the original frame to the new one we can write this equation as

$$\frac{P^x_{\text{in}} - \frac{V}{c}P^0_{\text{in}}}{\sqrt{1 - V^2/c^2}} = \frac{P^x_{\text{fin}} - \frac{V}{c}P^0_{\text{fin}}}{\sqrt{1 - V^2/c^2}}.$$

With the account of the conservation of the x-component $P^x_{\text{in}} = P^x_{\text{out}}$ this equations implies

$$P^0_{\text{in}} = P^0_{\text{out}}.$$

Since the quantity cP^0 is conserved and has the correct non-relativistic limit, we identify it with the energy, and thus

$$E = cP^0 = \frac{mc^2}{\sqrt{1 - \boldsymbol{v}^2/c^2}}. \tag{1.54}$$

The energy and momentum are, therefore, time and space parts of *the same 4-vector*,

$$P^\mu = (E/c, \boldsymbol{P}). \tag{1.55}$$

• It is instructive to re-derive the expression for the relativistic energy (1.54) in a different, less formal way. In Newtonian mechanics the energy of a system is related to the work performed on the system by an external force. For a particle, this work is equal to the change of its energy and may be expressed in terms of the change of its momentum as follows,

$$dE = \boldsymbol{F}d\boldsymbol{x} = \frac{d\boldsymbol{P}}{dt}d\boldsymbol{x} = d\boldsymbol{P}\frac{d\boldsymbol{x}}{dt} = \boldsymbol{v}d\boldsymbol{P}.$$

We can now use the relativistic expression for the momentum (1.53) to rewrite this expression,

$$dE = \boldsymbol{v}d\left(\frac{m\boldsymbol{v}}{\sqrt{1 - \boldsymbol{v}^2/c^2}}\right) = \frac{m(\boldsymbol{v}d\boldsymbol{v})}{\sqrt{1 - \boldsymbol{v}^2/c^2}} + mv^2\frac{(\boldsymbol{v}d\boldsymbol{v})/c^2}{(1 - v^2/c^2)^{3/2}} =$$

$$= m(\boldsymbol{v}d\boldsymbol{v})\frac{1 - v^2/c^2 + v^2/c^2}{(1 - v^2/c^2)^{3/2}} = \frac{m(\boldsymbol{v}d\boldsymbol{v})}{(1 - v^2/c^2)^{3/2}} = d\left(\frac{mc^2}{\sqrt{1 - \boldsymbol{v}^2/c^2}}\right),$$

Here we are using

$$d(v^2) = d(\boldsymbol{v}\boldsymbol{v}) = 2\boldsymbol{v}d\boldsymbol{v},$$

or, using the index notations,

$$d(v^2) = d(v_i v_i) = 2v_i dv_i.$$

so that

$$dE = d\left(\frac{mc^2}{\sqrt{1 - \boldsymbol{v}^2/c^2}}\right).$$

Integrating this expression we get

$$E = \frac{mc^2}{\sqrt{1 - \boldsymbol{v}^2/c^2}},$$

the same expression as before. Thus, we see that the quantity cP^0 that we have associated with energy on formal grounds is indeed related to the work performed on the particle.

• Several important conclusions can be drawn from these results. In summary:

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- The energy and momentum are time and space parts of the same 4-vector $P^\mu = mu^\mu$ according to eq. (1.55). Therefore, they transform under the change of the reference frame as any 4-vector. To write the transformation rules explicitely in the case of the relative velocity of the two frames in the x-direction we just have to take the Lorentz transformations of coordinates (1.11) and replace ct with E/c and $\boldsymbol{x}$ with $\boldsymbol{P}$,

$$\begin{aligned} \frac{E}{c} &= \mathrm{ch}\theta\, \frac{E'}{c} + \mathrm{sh}\theta\, P'^x \\ P^x &= \mathrm{sh}\theta\, \frac{E'}{c} + \mathrm{ch}\theta\, P'^x \\ P^y &= P'^y \\ P^z &= P'^z \end{aligned}$$

  where θ is the parameter determined by the relative velocity of the two frames, see Sect. 1.1.
- Both energy and momentum grow infinitely when the velocity approaches the velocity of light $v \to c$. Thus, to accelerate a (massive) particle to the speed of light *an infinite amount of work* would be required.
- It follows from the explicit expression for the energy (1.54) that when the velocity tends to zero the energy goes to a *finite constant*,

$$E \to mc^2.$$

  This is called a *rest energy* of a particle. This is the minimum energy a particle can have; the minimum is reached when the velocity is zero, cf. eq. (1.54).

  Because the speed of light is very large, the rest energy of macroscopic bodies is enormous. To get an idea of numbers, the energy stored in 1 gramm of matter is

$$(1\mathrm{g})\, c^2 = 10^{-3}(3\cdot 10^8)^2\ \mathrm{J} = 10^{14}\ \mathrm{J}.$$

  For comparison, burning of 1 kg of wood produces about 10^7 J, so that the rest energy stored in 1 g of matter is equivalent to the one obtained by burning of 10^4 tons of wood!

  This energy is real in the sense that it can, in principle, be converted into other forms of energy. This happens, for instance, when a massive particle M decays into two lighter ones m_1 and m_2 whose sum of masses is $m_1 + m_2 < M$. By energy conservation, after the decay the two particles together must still have the total energy equal Mc^2. Their smallest possible energy is $m_1c^2 + m_2c^2 < Mc^2$, so it is clear that the difference $(M - m_1 - m_2)c^2$ is their combined kinetic energy that can be extracted by stopping these particles. This is precisely what happens in nuclear reactions when an uranium nucleus decays into lighter debris. In this case $\sim 10^{-3}$ of the uranium rest energy is liberated as the kinetic energy of the debris and is converted to heat.
- The energy of a particle may be expressed in terms of its momentum. To do so recall that the Lorentz square of any 4-vector is a Lorentz invariant (e.g., for the position 4-vector x^μ this is the interval). For the 4-vector of energy-momentum we have

$$P_\mu P^\mu = \frac{E^2}{c^2} - \boldsymbol{P}^2 = \frac{1}{1 - v^2/c^2}\left(\frac{1}{c^2}m^2c^4 - m^2v^2\right) = \\ = m^2c^2, \tag{1.56}$$

  Such relations are called *dispersion relations*.

  An even easier way to obtain this relation is by eq. (1.44),

$$P_\mu P^\mu = m^2 u_\mu u^\mu = m^2c^2.$$

  so that

$$E^2 - c^2\boldsymbol{P}^2 = m^2c^4. \tag{1.57}$$

  From this equation we can express the particle energy as a function

of its momentum,

$$E = \sqrt{m^2c^4 + c^2\boldsymbol{P}^2}. \tag{1.58}$$

- Comparing the explicit expressions (1.53) and (1.54) for the energy and momentum in terms of particle velocity, we can write another relation,

$$\boldsymbol{P} = \frac{E}{c^2}\boldsymbol{v}, \tag{1.59}$$

which is useful if we want to express the particle velocity in terms of its energy and momentum.

- Finally, consider an *ultra-relativistic* particle which has velocity $v \to c$. In this case from eqs. (1.53) and (1.54) both E and $\boldsymbol{P}$ become very large,

$$E \gg mc^2, \tag{1.60}$$

$$|\boldsymbol{P}| \gg mc. \tag{1.61}$$

In this limit we can neglect mass terms in eqs. (1.57) and (1.58) which then give

$$E = c|\boldsymbol{P}|. \tag{1.62}$$

We can turn this argument in a different way. Assume that the momentum is fixed and *the mass m goes to zero*, and look at eq. (1.58). When $m \to 0$ we can neglect the mass term under the square root. The energy then goes to a finite limit given by eq. (1.62) which becomes exact for $m = 0$. So, despite zero mass a particle still can have energy and momentum, which are related by eq. (1.62). It follows then from eq. (1.59) by taking modulus of both sides that

$$|\boldsymbol{P}| = \frac{E}{c^2}|\boldsymbol{v}| = \frac{c|\boldsymbol{P}|}{c^2}|\boldsymbol{v}| = \frac{|\boldsymbol{v}|}{c}|\boldsymbol{P}|,$$

and thus

$$|\boldsymbol{v}| = c.$$

We conclude that massless particles, if they exist, must have the velocity equal to the velocity of light. In fact, the light itself consists of massless particles — photons.

From the above expressions it is clear that massless and very light (and therefore ultra-relativistic) particles have very similar behavior. In other words, the limit $m \to 0$ is smooth.

The question becomes more subtle in the case of particles that carry internal angular momentum called *spin*. In this case the smoothness of the limit $m \to 0$ is not so straightforward to establish. We do not have the tools to address this question here.

Relativistic version of the Newton's law

Now that we have derived the expressions for the energy and momentum, it is not difficult to re-write the Newton's law in a manifestly relativistic form, that is, in terms of 4-vectors. Such an equation must reproduce the conventional non-relativistic Newton's law in the small velocity limit, because we know from experiment that it is correct. Following the same logic as for the 4-velocity, a natural candidate is

$$mc\frac{du^\mu}{ds} = f^\mu. \tag{1.63}$$

On the left we have the 4-acceleration cdu^μ/ds; it is a 4-vector constructed from the 4-velocity in the same manner the 4-velocity itself was con-

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structed from the position x^μ in eq. (1.41). On the right we have a 4-force f^μ which we still have to determine. Before doing that we note that not all the components of this 4-vector are independent. Indeed, if we multiply both sides of eq. (1.63) by u_μ and sum over μ we get

$$mcu_\mu \frac{du^\mu}{ds} = u_\mu f^\mu.$$

The l.h.s. of this equation is identically zero because

$$u_\mu \frac{du^\mu}{ds} = \frac{1}{2}\frac{d}{ds}(u_\mu u^\mu) = \frac{1}{2}\frac{d}{ds}c^2 = 0.$$

Thus, the 4-force must satisfy

$$u_\mu f^\mu = 0.$$

As a consequence, we will see below that the 4 components of f^μ are expressed in term of 3 components of the usual force $\boldsymbol{F}$.

First, we remark that when deriving the expression for the relativistic energy through the work of a force we used the equation $\boldsymbol{F} = d\boldsymbol{P}/dt$ where $\boldsymbol{P}$ is the *relativistic* momentum. So let us rewrite eq. (1.63) in terms of the 4-momentum $P^\mu = mu^\mu$,

Here we use again $ds = c\,dt\sqrt{1-\boldsymbol{v}^2/c^2}$.

$$f^\mu = c\frac{dP^\mu}{ds} = \frac{1}{\sqrt{1-\boldsymbol{v}^2/c^2}}\frac{dP^\mu}{dt}.$$

The space components of this equation give

$$f^i = \frac{1}{\sqrt{1-\boldsymbol{v}^2/c^2}}\frac{dP^i}{dt} = \frac{F^i}{\sqrt{1-\boldsymbol{v}^2/c^2}}.$$

Let us have a look at the time component. Differentiating the energy,

$$f^0 = \frac{1}{\sqrt{1-\boldsymbol{v}^2/c^2}}\frac{dP^0}{dt} = \frac{mc}{\sqrt{1-\boldsymbol{v}^2/c^2}}\frac{d}{dt}\left(\frac{1}{\sqrt{1-\boldsymbol{v}^2/c^2}}\right) =$$

$$= \frac{m}{c\sqrt{1-\boldsymbol{v}^2/c^2}}\frac{(\boldsymbol{v}\dot{\boldsymbol{v}})}{(1-v^2/c^2)^{3/2}}.$$

On the other hand, we have

$$\boldsymbol{F} = \frac{d\boldsymbol{P}}{dt} = \frac{d}{dt}\frac{m\boldsymbol{v}}{\sqrt{1-\boldsymbol{v}^2/c^2}} = \frac{m}{\sqrt{1-\boldsymbol{v}^2/c^2}}\left(\dot{\boldsymbol{v}} + \frac{\boldsymbol{v}(\boldsymbol{v}\dot{\boldsymbol{v}})/c^2}{(1-v^2/c^2)^{3/2}}\right),$$

and therefore

$$(\boldsymbol{v}\boldsymbol{F}) = \frac{m(\boldsymbol{v}\dot{\boldsymbol{v}})}{\sqrt{1-\boldsymbol{v}^2/c^2}}\left(1 + \frac{v^2/c^2}{1-v^2/c^2}\right) = \frac{m(\boldsymbol{v}\dot{\boldsymbol{v}})}{(1-v^2/c^2)^{3/2}}.$$

Comparing this to the expression for f^0 we find

$$f^0 = \frac{(\boldsymbol{v}\boldsymbol{F})}{c\sqrt{1-\boldsymbol{v}^2/c^2}}.$$

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Finally, putting the two results together,

$$f^\mu = \left(\frac{(\boldsymbol{v}\boldsymbol{F})/c}{\sqrt{1-\boldsymbol{v}^2/c^2}}, \frac{F^i}{\sqrt{1-\boldsymbol{v}^2/c^2}} \right). \tag{1.64}$$

We see that indeed the 4-force is completely determined by the standard force $\boldsymbol{F}$, the zero-th component being proportional to the work of this force per unit time, $\boldsymbol{v}\boldsymbol{F} = (\boldsymbol{F}d\boldsymbol{x})/dt$.

• *To summarize:* the relativistic Newton's law can be written in a manifestly 4-dimensional form (1.63). However, out of 4 equations contained in eq. (1.63) the three space component are equivalent to the equation

$$\frac{d\boldsymbol{P}}{dt} = \boldsymbol{F} \tag{1.65}$$

which formally has the same form as its non-relativistic analog, except here the momentum is a *relativistic momentum* (1.53). The fourth equation, the 0-th component of (1.63), contains nothing new, it simply expresses the energy conservation corresponding to eq. (1.64):

$$\frac{c}{c\sqrt{1-\boldsymbol{v}^2/c^2}} \frac{dP^0}{dt} = \frac{1}{\sqrt{1-\boldsymbol{v}^2/c^2}} \frac{d(E/c)}{dt} = \frac{(\boldsymbol{v}\boldsymbol{F})/c}{\sqrt{1-\boldsymbol{v}^2/c^2}}$$

or

$$\frac{dE}{dt} = \boldsymbol{v}\boldsymbol{F} = \frac{(\boldsymbol{F}d\boldsymbol{x})}{dt}.$$

This equation is a consequence of eq. (1.65). In practice, therefore, it is sufficient to solve eq. (1.65). We will use this equation later when studying the motion of a relativistic particle in electric and magnetic fields in Chapter. 3.

Kinematics of particle decay

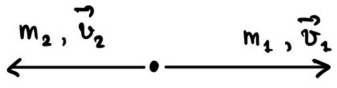
The conservation of energy and momentum can be written together as follows

$$\sum_{\text{in}} P^\mu_{\text{in}} = \sum_{\text{fin}} P^\mu_{\text{fin}} \tag{1.66}$$

where *in* and *fin* label initial and final particles. This equation imposes constraints on what a possible outcome of a given process could be. In some cases this is enough to determine some or even all the characteristics of the final state. Below we consider two examples: decay of a particle into two lighter ones (the rest of this section), and the so-called elastic scattering of particles on each other (next section).

• Consider a particle of mass M which decays into two particles of masses m_1 and m_2. In a relativistic theory we may choose any frame to describe this process, so let us choose the frame where the *total momentum is zero*, $\boldsymbol{P} = 0$. This is called **center-of-mass frame**. Obviously, since the momentum is conserved, the total momentum is zero both before and after decay. In this frame, before the decay we have

$$P^\mu_{\text{in}} = (Mc, \boldsymbol{0}).$$



Center-of-mass frame $\Leftrightarrow \boldsymbol{P}_{\text{tot}} = 0$.

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Recall that the energy of a particle is $E = \sqrt{m^2c^4 + c^2\boldsymbol{P}^2}$, and since $\boldsymbol{P} = 0$ we have $E = mc^2$, the rest energy. Then $P^0 = E/c = Mc$.

After the decay we must also have the total momentum equal to zero, which means that momenta of the two particles resulting from the decay are opposite in direction and equal in modulus. Without loss of generality let us assume that they are both aligned with the x-axis.

• Let us put all this information into equations. The conservation of the energy and momentum gives

$$\begin{aligned} Mc^2 &= E_1 + E_2, && (1.67)\\ 0 &= P_1 - P_2 && (1.68) \end{aligned}$$

where we have denoted P_1 and P_2 the absolute values of momenta $P_{1,2} = |\boldsymbol{P}_{1,2}|$ of particles 1 and 2, respectively. In addition, we have two dispersion relations for two final particles,

$$\begin{aligned} E_1^2 - c^2P_1^2 &= m_1^2c^4, && (1.69)\\ E_2^2 - c^2P_2^2 &= m_2^2c^4. && (1.70) \end{aligned}$$

We have 4 equations here for 4 unknowns, so we expect being able to determine all the variables.

We could have done this counting from the start. Each of the two particles in the final state is described by its own 4-momentum; this makes $4 \cdot 2 = 8$ variables in total. We have 4 conservation equations (1 for energy and 3 for momentum), and two dispersion relations. So it looks like we are two equations short. However, because two of the variables (the angles describing the orientation of the momentum of, say, particle 1) do not enter the equations, we have 6 equations for 6 remaining variables. The two angles describing the orientation of the momentum of particle 1 (and therefore of particle 2, since $\boldsymbol{P}_1 = -\boldsymbol{P}_2$) remain arbitrary.

• It is not difficult to solve this system of equations. First we note that from eq. (1.68) we have

$$P_1^2 = P_2^2.$$

Then we can subtract equations (1.69) and (1.70) from one another and get

$$E_1^2 - E_2^2 = (E_1 - E_2)(E_1 + E_2) = (E_1 - E_2)Mc^2 = c^4(m_1^2 - m_2^2).$$

At the second step we have used eq. (1.67). Together with eq. (1.67) we now have two linear equations for two unknown energies,

$$\begin{aligned} E_1 - E_2 &= c^2\frac{m_1^2 - m_2^2}{M},\\ E_1 + E_2 &= c^2M, \end{aligned}$$

from where we find

$$\begin{aligned} E_1 &= \frac{c^2}{2M}(M^2 + m_1^2 - m_2^2), && (1.71)\\ E_2 &= \frac{c^2}{2M}(M^2 - m_1^2 + m_2^2). && (1.72) \end{aligned}$$

Once the energies are found, the momenta P_1 and P_2 can be determined form eqs. (1.69) and (1.70).

• Let us have a closer look at these equations. First, from the energy conservation it is clear that the decay is not possible unless $M > m_1 + m_2$. Indeed, eq. (1.67) together with the dispersion relations gives

$$Mc^2 = \sqrt{m_1^2c^4 + c^2P_1^2} + \sqrt{m_2^2c^4 + c^2P_2^2}.$$

This equation has no solutions if $M < m_1 + m_2$ because the energy of

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each final particle is at least as large as $m_i c^2$.

• From our solution we can find the *kinetic* energies of particles. For instance, for the particle 1 we have

$$K_1 \equiv E_1 - c^2 m_1 = \frac{c^2}{2M}(M^2 + m_1^2 - m_2^2 - 2Mm_1) =$$

$$= \frac{c^2}{2M}(M - m_1 - m_2)(M - m_1 + m_2). \tag{1.73}$$

This is positive as long as $M > m_1 + m_2$.

The kinetic energy of final particles is the part of energy that can be extracted and converted into other forms, for instance, into heat. The total energy $\Delta = K_1 + K_2$ liberated in the decay is thus

$$\Delta = Mc^2 - m_1 c^2 - m_2 c^2.$$

This energy is shared between the particles 1 and 2 in a proportion that follows from eq. (1.73):

$$\begin{aligned} K_1 = E_1 - c^2 m_1 &= \frac{\Delta}{2}\frac{M - m_1 + m_2}{M}, \\ K_2 = E_2 - c^2 m_2 &= \frac{\Delta}{2}\frac{M - m_2 + m_1}{M}. \end{aligned}$$

• Consider some limiting cases.

- If both particles are very light or massless, $m_{1,2} \ll M$, then we have

  $$\Delta \simeq Mc^2,$$
  $$K_1 \simeq K_2 \simeq \frac{\Delta}{2}.$$

  The liberated energy is shared equally between the final particles.
- In the opposite limit $\Delta \ll Mc^2$ the sum of final masses is close to the initial mass. For simplicity, assume also that $\Delta \ll m_{1,2}c^2$. In this case we can rewrite the kinetic energies as

  $$\begin{aligned} K_1 &= \frac{\Delta}{2}\frac{(\Delta + 2m_2c^2)}{Mc^2} \simeq \Delta\frac{m_2}{M}, \\ K_2 &= \frac{\Delta}{2}\frac{(\Delta + 2m_1c^2)}{Mc^2} \simeq \Delta\frac{m_1}{M}. \end{aligned}$$

  In this regime the lighter the final particle, the larger fraction of the liberated energy in carries.

Cherenkov radiation

Consider an example that has numerous applications in experimental particle physics — the Cherenkov radiation. This phenomenon may occur when a charged particle moves through a medium with very high velocity. We will see in the second part of the course that the velocity of light in media v is generically smaller than c, $v < c$ (for instance, in water it is $v \sim 0.75c$). If a charged particle moves through such a medium with the velocity u exceeding v, that is $v < u < c$, it emits electromagnetic radiation. This radiation is named after its discoverer

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Pavel Cherenkov. The Cherenkov radiation is emitted into the forward cone with the opening angle θ_c around the particle trajectory; θ_c is called the Cherenkov angle. This phenomenon is used in many modern detectors of relativistic charged particles. We are not prepared to discuss the full theory of Cherenkov radiation, however, the formalism that we have already developed allows one to determine the Cherenkov angle.

We may view the Cherenkov radiation as a collection of photons, and its emission as a sequential emission of individual photons. An elementary process would look like in Fig. 1.4. The momentum of a photon is typically very small, so we can neglect the change in the trajectory of the charged particle due to the emission, in which case we approximately have $\boldsymbol{P}_1 \parallel \boldsymbol{P}_2$.

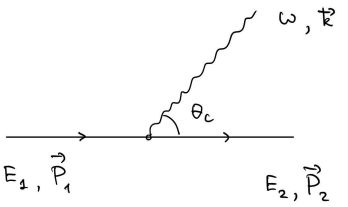


Figure 1.4.: Emission of a Cherenkov photon.

A moving charged particle in the vacuum cannot emit a photon because this is forbidden by the energy-momentum conservation. To see this it is sufficient to go to the rest frame of a particle, where its energy is at the absolute minimum and cannot decrease any further, so there is no energy available to pass to the emitted photon. In the medium the situation is different because the Lorentz invariance is lost and the argument does not apply.

For simplicity let us assume that the charged particle is ultra-relativistic, $E \gg mc^2$, so that its energy and the absolute value of momentum are related by

$$E = cP.$$

The dispersion relation of a photon in the medium is different from that in the vacuum; we will see in the second part of the course that the in a medium the photon energy ω and momentum $k = |\boldsymbol{k}|$ are related as

$$\omega = vk,$$

where v is the speed of light in the medium. Note that for $v = c$ (vacuum) we recover the standard relation $\omega = ck$.

Let us write the energy-momentum conservation and see what it gives. From energy conservation we have

$$E_1 = E_2 + \omega. \tag{1.74}$$

From the momentum projection onto the x-axis

$$P_1 = P_2 + k\cos\theta_c.$$

Note: the projection of $\boldsymbol{k}$ onto the y-axis can be balanced by a small change of the direction of $\boldsymbol{P}_2$, so we do not need to worry about it.

We now use the dispersion relations to replace momenta in this equation by the energies,

$$\frac{E_1}{c} = \frac{E_2}{c} + \frac{\omega}{v}\cos\theta_c,$$

from where, multiplying by c,

$$E_1 = E_2 + \frac{c}{v}\cos\theta_c\,\omega. \tag{1.75}$$

Comparing this equation to eq. (1.74) we see that $(c/v)\cos\theta_c$ must be equal to 1, or

$$\cos\theta_c = \frac{v}{c}.$$

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In water, for instance, $\cos\theta_c = 0.75$, so that $\theta_c \simeq 41^\circ$.

Kinematics of particle collisions

Consider now how the conservation of energy and momentum works in the case of particle collisions. In general, when two particles collide various other particles may be produced. However, at sufficiently low energy it often happens that no extra particles are produced, so that the result of the collision is that the initial particles change their energies and momenta. This is called *elastic collision*. Regardless of the energy, the elastic collision is always possible with some non-zero probability. So, consider this case.

• The elastic collision is simplest to describe in the center-of-mass (CM) frame in which the total spatial momentum is zero. Such a collision is shown in Fig. 1.5. Two particles of masses m_1 and m_2 approach each

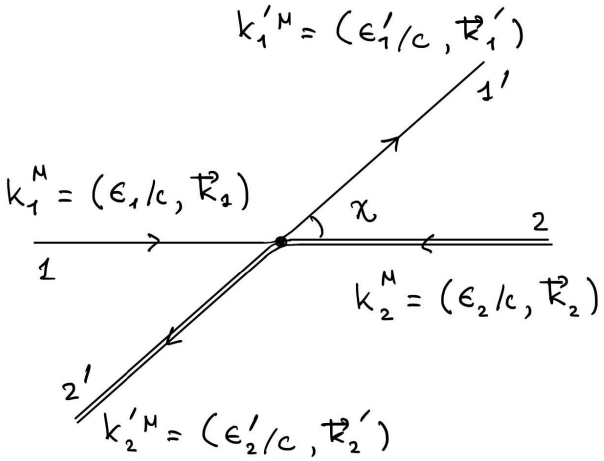


Figure 1.5.: Collision of two particles in the CM frame. Indices 1 and 2 refer to particles, primes mark quantities after the collision.

other along the x-axis, collide and leave back to back at an angle χ to the original direction.

Let us show that energies and absolute values of momenta in the elastic collision of two particles *do not change* when considered in the CM frame. Denoting $k_1 = |\boldsymbol{k}_1|$ and similarly for other momenta, we can write the energy and momentum conservation, together with the CM condition, as follows:

$$\begin{aligned} \epsilon_1(k_1) + \epsilon_2(k_2) &= \epsilon_1(k_1') + \epsilon_2(k_2'), \\ k_1 - k_2 &= k_1' - k_2' = 0. \end{aligned} \tag{1.76}$$

Here

$$\epsilon_1(k) = \sqrt{m_1^2c^4 + c^2k^2}, \tag{1.77}$$

$$\epsilon_2(k) = \sqrt{m_2^2c^4 + c^2k^2}. \tag{1.78}$$

Pay attention that we have used in eq. (1.76) the fact that the nature of particles does not change, so these are the same functions ϵ_1 and ϵ_2 on the l.h.s. and on the r.h.s., only of different arguments.

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Now use the momentum conservation eqs. (1.77) to rewrite this equation as follows

$$\epsilon_1(k_1) + \epsilon_2(k_1) = \epsilon_1(k_1') + \epsilon_2(k_1').$$

The l.h.s. of this equation is considered as known; we have to determine the momentum after the collision k_1'. One solution to this equation is obvious,

$$k_1' = k_1.$$

In fact, this is the only solution. Indeed, the function $f(k) = \epsilon_1(k) + \epsilon_2(k)$ is a *monotonic* function, so it takes a given value $f(k_1)$ only once. The solution for k_1' is thus unique.

• It is clear from our derivation that the energy and momentum conservation can be satisfied at any angle χ which, by consequence, cannot be determined from these arguments. Extra information is needed to fix the scattering angle χ. To understand what is happening consider the particle trajectories close to the collision point in more detail. In real world the collision does not happen in one point. Particles interact through the *interaction potential* $U(R)$, R being the distance between the particles (for example, the Coulomb potential in the case of electric charges). At large distances the potential can be neglected, and the particles move freely so that their trajectories are straight lines. In the CM frame these lines are parallel, but are in general displaced by some distance ρ called *impact parameter*. It is clear from the figure that this is the impact parameter that determines at which angle the particles will scatter. The general purpose of the collision experiments is to measure the dependence $\chi(\rho)$ and determine from that the interaction potential $U(R)$.

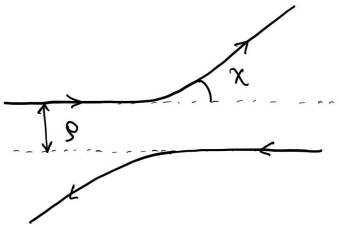


Figure 1.6.: Realistic picture of a collision in the CM frame.

• The scattering experiments are often performed on a *fixed target* and therefore not in the CM frame. A typical setup is shown in Fig. 1.7; call it a Lab frame. The particle 2 before the collision is at rest. One measures the energies and momenta of the final particles (including the angles θ_1 and θ_2) as a function of the impact parameter, and wants to reconstruct the interaction potential between particles. For this, however, one needs to determine the scattering angle χ in the CM frame. One may, of course, perform explicitely the Lorentz transformation to the CM frame. Here we would like to use this example to illustrate a different technique which uses the properties of the 4-vector of momentum.

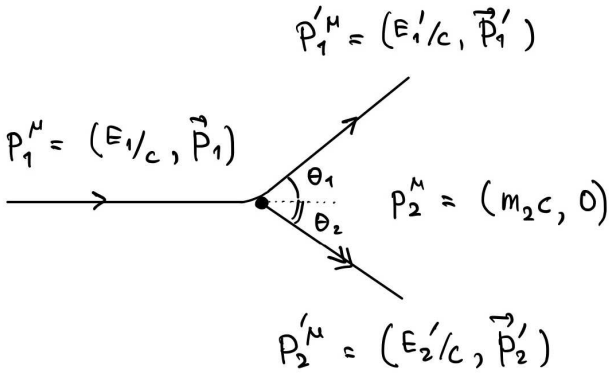


Figure 1.7.: Collision on a fixed target (particle 2 before the collision is at rest).

We would like to derive equations that relate the angle χ with measurable quantities in the Lab frame. Let us start by writing the conservation of

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the energy and momentum in the 4-vector form,

$$p_1^\mu + p_2^\mu = p_1'^\mu + p_2'^\mu.$$

Moving, say, the momentum $p_1'^\mu$ to the left we get

$$p_1^\mu + p_2^\mu - p_1'^\mu = p_2'^\mu.$$

We now take the square of each side (the Lorentz scalar product with itself) and get

$$(p_1 \cdot p_1) + (p_2 \cdot p_2) + (p_1' \cdot p_1') + 2(p_1 \cdot p_2) - 2(p_1 \cdot p_1') - 2(p_2 \cdot p_1') = (p_2' \cdot p_2'),$$

where $(a \cdot b) \equiv a_\mu b^\mu$ denotes the 4d scalar product of two vectors.

Many terms in this expression can be simplified. First we note that $(p_1 \cdot p_1) = m_1^2 c^2$ by means of eq. (1.56) and similarly for squares of other momenta, so we have

$$m_1^2 c^2 + m_2^2 c^2 + m_1^2 c^2 + 2(p_1 \cdot p_2) - 2(p_1 \cdot p_1') - 2(p_2 \cdot p_1') = m_2^2 c^2,$$

or, simplifying,

$$m_1^2 c^2 + (p_1 \cdot p_2) - (p_1 \cdot p_1') - (p_2 \cdot p_1') = 0. \tag{1.79}$$

Now let us calculate the scalar products. The simplest are two products containing $p_2^\mu = (m_2 c, 0)$ (recall that the particle 2 is initially at rest). They are

$$(p_1 \cdot p_2) = E_1/c \; m_2 c = E_1 m_2,$$
$$(p_2 \cdot p_1') = E_1'/c \; m_2 c = E_1' m_2.$$

To calculate $(p_1 \cdot p_1')$ we may use the fact that the scalar product is invariant under Lorentz transformations, so it can be calculated in any frame. Let us use this fact and calculate it *in the CM frame*. From Fig. 1.5

$$(p_1 \cdot p_1') = (k_1 \cdot k_1') = \frac{1}{c^2}\epsilon_1 \epsilon_1' - k_1 k_1' \cos\chi.$$

Now recall that we have proven that in the CM frame $k_1' = k_1$ and therefore $\epsilon_1' = \epsilon_1$. Therefore we can write

$$(p_1 \cdot p_1') = \frac{1}{c^2}\epsilon_1^2 - k_1^2 \cos\chi = m_1^2 + k_1^2(1 - \cos\chi),$$

where in the last equation we have expressed the energy ϵ_1^2 in terms of the momentum k_1 using eq. (1.57). We can now rewrite eq. (1.79) as

$$E_1 - E_1' = \frac{k_1^2}{m_2}(1 - \cos\chi). \tag{1.80}$$

We are not done yet because we need to express k_1^2 in terms of quantities measured in the Lab frame. To do this we may use the Lorentz invariance of another scalar product, $(p_1 \cdot p_2) = (k_1 \cdot k_2)$. Explicitely,

$$m_1 E_1 = \frac{1}{c^2}\epsilon_1 \epsilon_2 - \boldsymbol{k}_1 \boldsymbol{k}_2 = \frac{1}{c^2}\epsilon_1 \epsilon_2 + k_1^2,$$

where we have used $\boldsymbol{k}_2 = -\boldsymbol{k}_1$ in the CM frame. We now solve this

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equation for k_1:

$$m_1 E_1 - k_1^2 = \sqrt{(c^2 m_1^2 + k_1^2)(c^2 m_2^2 + k_1^2)}.$$

Taking square and simplifying (note that k_1^4 cancels), we find

$$k_1^2 = \frac{m_2^2 (E_1^2 - c^4 m_1^2)}{c^2 (m_1^2 + m_2^2) + 2 m_2 E_1}.$$

Now everything is expressed in terms of the energies in the Lab frame and the angle χ, so that we can find the latter from the equation

$$E_1' = E_1 - \frac{m_2 (E_1^2 - c^4 m_1^2)}{c^2 (m_1^2 + m_2^2) + 2 m_2 E_1} (1 - \cos\chi). \tag{1.81}$$

This solves the problem of expressing the CM scattering angle χ in terms of quantities measured in the Lab frame.

As a by-product of our calculation, we get the expression for the maximum transferred energy in the Lab frame. Using eq. (1.81) and the energy conservation we may write

$$\begin{aligned} E_1' &= E_1 - \Delta E, \\ E_2' &= E_2 + \Delta E, \end{aligned}$$

where

$$\Delta E = \frac{m_2 (E_1^2 - c^4 m_1^2)}{c^2 (m_1^2 + m_2^2) + 2 m_2 E_1} (1 - \cos\chi).$$

The transferred energy ΔE varies between 0 and $\Delta E_{\max}$ where

$$\Delta E_{\max} = \frac{2 m_2 (E_1^2 - c^4 m_1^2)}{c^2 (m_1^2 + m_2^2) + 2 m_2 E_1}$$

corresponds to the scattering angle $\chi = \pi$ (the head-on collision). In the case of particles of equal masses $m_1 = m_2$ and the head-on collision, the incident particle 1 gets stopped, and the particle 2 gets all of its kinetic energy, just like in the non-relativistic case.

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2. Maxwell equations

We will now leave relativistic physics aside for a while and revise the laws of electromagnetism. The purpose is to re-formulate them in the form of differential equations, so that at the end we will arrive at a complete set of the laws of electromagnetism formulated in terms of differential equations relating continuous variables, known as Maxwell's equations.

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2.1. Electrostatics

Equations of electrostatics

• *Coulomb law*. Our starting point is the Coulomb law, which expresses the force acting on a charge q_1 from another charge q_2,

$$\boldsymbol{F} = \frac{1}{4\pi\epsilon_0} q_1 q_2 \frac{\boldsymbol{x}_1 - \boldsymbol{x}_2}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^3}, \tag{2.1}$$

where the coefficient

$$\frac{1}{4\pi\epsilon_0} = 8.988 \times 10^9 \, \frac{\mathrm{N\,m^2}}{\mathrm{C^2}}. \tag{2.2}$$

If we focus on the charge q_1 we see that the force acting on this charge is proportional to q_1, which we can write as

$$\boldsymbol{F} = q_1 \boldsymbol{E}. \tag{2.3}$$

Here we have introduced the *electric field* $\boldsymbol{E}$ at the point $\boldsymbol{x}_1$,

$$\boldsymbol{E}(\boldsymbol{x}_1) = \frac{1}{4\pi\epsilon_0} q_2 \frac{\boldsymbol{x}_1 - \boldsymbol{x}_2}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|^3}. \tag{2.4}$$

We interpret this equation as the field $\boldsymbol{E}$ created at $\boldsymbol{x}_1$ by the charge q_2 located in the point $\boldsymbol{x}_2$. Note that this interpretation implicitly assumes that the electric field $\boldsymbol{E}$ is more than the proportionality coefficient in eq. (2.3) — rather it is a dynamical quantity by itself, a material object that produces a force on the charge q_1 and thus must carry energy and momentum. We will see later that this interpretation is correct.

• Forces acting on a charge q due to several other charges add like vectors, and so do electric fields. Thus, the field created by several charges q_a in the positions $\boldsymbol{x}_a$ is

$$\boldsymbol{E}(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0} \sum_a q_a \frac{\boldsymbol{x} - \boldsymbol{x}_a}{|\boldsymbol{x} - \boldsymbol{x}_a|^3}. \tag{2.5}$$

• In the limit when the charges are very numerous and fill some part of the space, like, e.g., atoms of a medium, one may describe their distribution by the **charge density** $\rho(\boldsymbol{x})$ which, in general, depends on coordinates.

By definition, the total charge dq contained in an infinitesimal volume $dV = dx^1 dx^2 dx^3 \equiv d^3x$ equals

$$dq = \rho(x)dV = \rho(x)d^3x. \tag{2.6}$$

The electric field created by a continuous charge distribution is obviously written as

$$\boldsymbol{E}(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\int d^3x'\rho(\boldsymbol{x}')\frac{\boldsymbol{x}-\boldsymbol{x}'}{|\boldsymbol{x}-\boldsymbol{x}'|^3}. \tag{2.7}$$

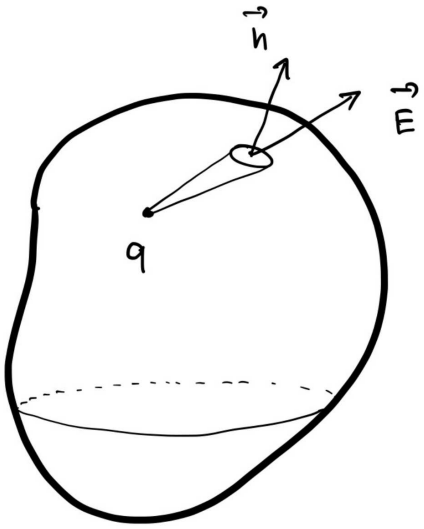


Figure 2.1.: Derivation of the Gauss' law. Charge q is inside the closed surface S; $\boldsymbol{E}$ is the electric field; $\boldsymbol{n}$ is the unit vector normal to the surface; $d\boldsymbol{s} = \boldsymbol{n}ds$ is the surface element.

• *Gauss' law*. Consider a closed surface S containing a charge q somewhere inside, see Fig. 2.1. Let us calculate the total flux of the electric field across the surface S. At each point of the surface we take the infinitesimal area element ds and form the vector called the surface element, $d\boldsymbol{s} = \boldsymbol{n}ds$, where $\boldsymbol{n}$ is the unit vector in the direction of the external normal to the surface. The flux of the electric field across this area element is then written as the scalar product $d\boldsymbol{s}\boldsymbol{E}$. The total flux F through the whole surface S is, therefore,

$$F = \int_S d\boldsymbol{s}\boldsymbol{E}. \tag{2.8}$$

Let us rewrite this integral in a different way. First, we write the scalar product as

$$d\boldsymbol{s}\boldsymbol{E} = dsE\cos\theta,$$

where θ is the angle between $\boldsymbol{n}$ and $\boldsymbol{E}$. The area element ds defines a (infinitsimally thin) conus centered on the charge and passing through the boundary of ds, see Fig. 2.2. The product $ds\cos\theta$ is *the area* which this conus cuts out of the sphere of radius r centered on the charge, where r is the distance of the element ds from the change. This area is proportional to r^2. One may use the definition of the solid angle Ω to write this as

$$ds\cos\theta = r^2 d\Omega,$$

where $d\Omega$ is the solid angle of the conus passing through ds.

We can now evaluate the integral in eq. (2.8). Replacing the modulus of

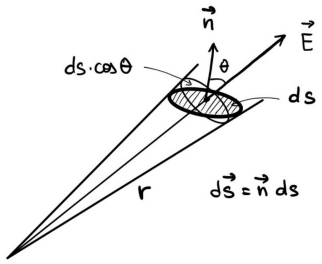


Figure 2.2.: Calculation of the flux.

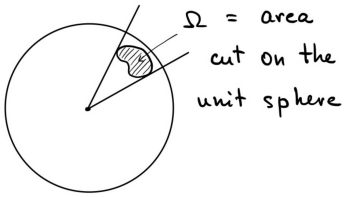


Figure 2.3.: Solid angle Ω of a conus is the *area* cut by the conus from the *unit sphere*. The shape of this area does not matter. Obviously, Ω varies from 0 to 4π.

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the electric field by its expression from eq. (2.4) we get

$$F = \int_S dsE\cos\theta = \int_S r^2 d\Omega \frac{q}{4\pi\epsilon_0}\frac{1}{r^2} = \frac{q}{4\pi\epsilon_0}\int d\Omega = \frac{q}{\epsilon_0}$$

because the integral over the solid angle gives 4π. We have therefore established the relation

$$\int_S d\boldsymbol{s}\boldsymbol{E} = \frac{q}{\epsilon_0}. \tag{2.9}$$

This relation is known as the *Gauss' law*.

If there are several charges inside S, their electric fields add. One may apply eq. (2.9) to each of these fields individully and sum the results. One will find, therefore, that the flux of the electric field through S is determined by the sum of the charges.

If the charge is *outside* of the surface S, the flux it produces across S is *zero*. Indeed, in this case the infinitesimal cones that we used in the demonstration above will cross the surface *twice* giving each time two contributions: one with $d\boldsymbol{s}\boldsymbol{E} > 0$ and one with $d\boldsymbol{s}\boldsymbol{E} < 0$. These contributions exactly exactly cancel each other for every cone, so the total integral will be zero. Therefore, the flux of electric field through a closed surface is determined by the *total charge inside* this surface. In the case of a continuous charge distribution characterized by the density $\rho(\boldsymbol{x})$ this gives

$$\int_S d\boldsymbol{s}\boldsymbol{E} = \frac{1}{\epsilon_0}\int_V d^3x\rho(\boldsymbol{x}), \tag{2.10}$$

where V is the volume bounded by the surface S.

From this equaion we can derive a local relation between the electric field and the charge density. Making use of the Srokes theorem we may rewrite the l.h.s. of eq. (2.10) as

$$\int_S d\boldsymbol{s}\boldsymbol{E} = \int_V d^3x\boldsymbol{\nabla}\boldsymbol{E}.$$

From eq. (2.10) we have then

$$\int_V d^3x\boldsymbol{\nabla}\boldsymbol{E}(\boldsymbol{x}) = \frac{1}{\epsilon_0}\int_V d^3x\rho(\boldsymbol{x}).$$

Since this equation is valid for any volume, it implies the local equation

$$\boldsymbol{\nabla}\boldsymbol{E}(\boldsymbol{x}) = \frac{\rho(\boldsymbol{x})}{\epsilon_0}. \tag{2.11}$$

This is first of the four Maxwell equaitions.

• *Electrostatic potential*. As a consequence of the Coulomb law, the components of the electric field are not independent and can be expressed in terms of a single scalar function. Indeed, consider eq. (2.7) and use the identity

$$\frac{\boldsymbol{x}-\boldsymbol{x}'}{|\boldsymbol{x}-\boldsymbol{x}'|^3} = -\boldsymbol{\nabla}\frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|}.$$

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To prove this identity let us calculate the r.h.s.:

$$-\partial_i \frac{1}{\sqrt{(x-x')_j (x-x')_j}} = \frac{1}{2} \frac{2(x-x')_i}{[(x-x')_j (x-x')_j]^{3/2}} = \frac{(x-x')_i}{|\boldsymbol{x}-\boldsymbol{x}'|^3}.$$

When we plug this expression into eq. (2.7) we may take the gradient operator outside of the integral, because it acts on $\boldsymbol{x}$ while the integration is over $\boldsymbol{x}'$. Then we find

$$\boldsymbol{E}(\boldsymbol{x}) = -\boldsymbol{\nabla} \left\{ \frac{1}{4\pi\epsilon_0} \int \frac{d^3x' \rho(x')}{|\boldsymbol{x}-\boldsymbol{x}'|} \right\} \equiv -\boldsymbol{\nabla}\phi(\boldsymbol{x}) \tag{2.12}$$

where

$$\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0} \int \frac{d^3x' \rho(x')}{|\boldsymbol{x}-\boldsymbol{x}'|} \tag{2.13}$$

is called the electrostatic potential, or the *scalar potential*.

• The fact that the vector of the electric field $\boldsymbol{E}$ can be written as the gradient of a scalar function implies another equation for the electric field. Taking the curl of both sides of eq. (2.12) and making use of the fact that curl of a gradient is identically zero, we find

$$\boldsymbol{\nabla} \times \boldsymbol{E} = 0. \tag{2.14}$$

This is the second Maxwell's equation (to be more precise, its static version).

• When one has to find the electric field of a given charge distribution $\rho(\boldsymbol{x})$, it is often simpler to first calculate the scalar potential $\phi(x)$ and then determine the electric field from eq. (2.12). One may obtain a closed equation for ϕ by combining eqs. (2.11) and (2.12),

$$\boldsymbol{\nabla}\boldsymbol{\nabla}\phi(\boldsymbol{x}) \equiv \Delta\phi(\boldsymbol{x}) = -\frac{\rho(\boldsymbol{x})}{\epsilon_0}. \tag{2.15}$$

This is known as the **Poisson equation**. Its a particular case when the charges are absent, $\rho = 0$, it reduces to the **Laplace equation**:

$$\Delta\phi(\boldsymbol{x}) = 0. \tag{2.16}$$

The operator

$$\boldsymbol{\nabla}\boldsymbol{\nabla} = \Delta = \partial_i \partial_i = \partial_1^2 + \partial_2^2 + \partial_3^2$$

is called the **Laplacian**.

• *Surface chargs.* In practical problems one often has to deal with interfaces between different media, in which case there may arise charge distributions that are concentrated along the interface rather than being distributed in a volume. Such distributions may be characterized by the *surface density* of charges, $\sigma(\boldsymbol{x})$, which in general may depend on the position along the surface. The meaning of $\sigma(\boldsymbol{x})$ is that a small (infinitesimal) area ds of a surface carries total charge dq which is

$$dq = \sigma(\boldsymbol{x}) ds.$$

When the surface density is constant, this relation works, of course, for arbitrarily large pieces of the surface.

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In a similar way one may define a line charge density in cases when charges are distribted aling a line. In practice, such distributions arise only in rare cases.

• *Continuity conditions.* If an interface between two media carries a surface charge density, the normal component of the electric field is discontinuous across this surface, and the value of the discontinuity is related to the surface density of charge.

In order to derive the continuity conditions consider first an infinitesimal flat rectangular volume (box) with thickness much smaller than the other two dimensions, oriented in such a way that its flat side is parallel to the interface, and the box is cut in halfs by the interface, see Fig.2.4. The interface has the surface charge density σ which we may assume to be constant inside the box.

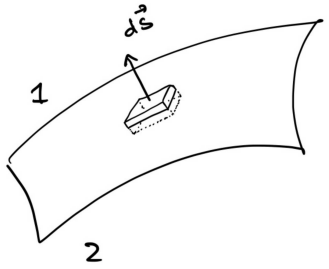


Figure 2.4.: The volume used to derive the jump of the normal component, 3d view.

Let us use the Gauss' law for this infinitesimal volume. Denoting the area of the flat side by ds, the total charge inside the volume is σds, where σ is the surface charge density at the position of the box. The electric field is, in general, different on two sides of the interface. For a sufficiently small box we can neglect its variation along the surface, so the field on the side 1 is a constant vector $\boldsymbol{E}_1$, and on the side 2 a different constant vector $\boldsymbol{E}_2$ as in Fig. 2.5. There are three contributions to the flux of the electric field across the surfae of the box: from the two flat sides parallel to the interface, and from the side perpendicular to it. The last contribution can be neglected if we take the box sufficiently thin. Retaining only the two contributions from the two flat sides we have from the Gauss' law

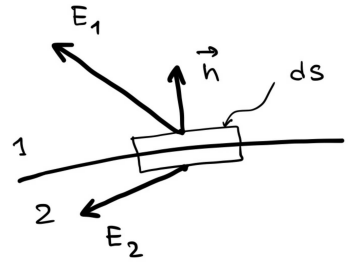


Figure 2.5.: The volume used to derive the jump of the normal component, cut across the interface.

$$\boldsymbol{n}\boldsymbol{E}_1 ds - \boldsymbol{n}\boldsymbol{E}_2 ds = \frac{\sigma ds}{\epsilon_0},$$

or, simplifying by ds

$$\boldsymbol{n}(\boldsymbol{E}_1 - \boldsymbol{E}_2) = \frac{\sigma}{\epsilon_0}. \tag{2.17}$$

In other words, the electric field component normal to the interface is discontinuous, with the jump equal σ/ϵ_0.

Consider now the conditions on the parallel components. To derive them consider this time a (infinitesimal) closed rectangular contour that goes parallel to the sirface for some distance $d\boldsymbol{l}$, crosses it and comes back, see Fig. 2.6. We will assume that the parts of this contour that are parallel to the interface are much longer than the perpendicular ones.

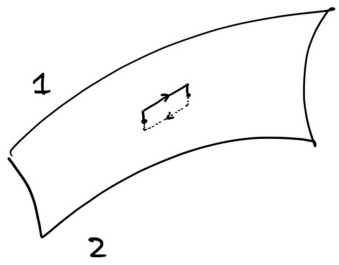


Figure 2.6.: The contour used to derive the continuity of the parallel components, 3d view.

The contour is a boundary ∂S of a rectangular surface S. Let us integrate eq. (2.14) over this surface. We get

$$\int_S \nabla \times \boldsymbol{E} = 0.$$

By the Stokes theorem the integral over the surface S can be transformed into the integral over its boundary, i.e., along our contour:

$$\int_S \nabla \times \boldsymbol{E} = \int_{\partial S} \boldsymbol{E} d\boldsymbol{l} = 0.$$

Approximating as before the electric fields on two sides of the interface by constant vestors $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$, and neglecting the parts of the contour

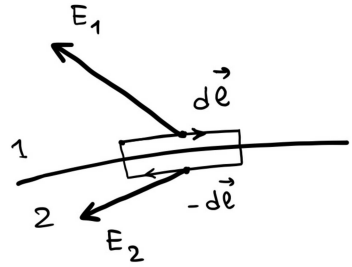


Figure 2.7.: The contour used to derive the continuity of the parallel components, cut across the interface.

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perpendicular to the interface we find (cf. Fig.2.7):

$$d\boldsymbol{l}(\boldsymbol{E}_1 - \boldsymbol{E}_2) = 0.$$

There are two possible independent orientations of $d\boldsymbol{l}$ parallel to the interface; correspondingly, there are two continuity equations for two components of $\boldsymbol{E}$ parallel to the surface. One may write them together as follows,

$$\boldsymbol{n} \times (\boldsymbol{E}_1 - \boldsymbol{E}_2) = 0, \tag{2.18}$$

where as before $\boldsymbol{n}$ is a unit vector normal to the interface.

Eqs. (2.17) and (2.18) give three continuity conditions for three components of the electric field at the interface of two media. Note that they follow directly from equations (2.11) and (2.14). Note also that in the basence of the charge density at the interface, that is if $\sigma = 0$, the electric field is continuous.

As an application of these equations consider a charged conductor (e.g., a piece of metal). In conductors charges can move freely. Because of that the electric field inside a conductor cannot be non-zero: if there was a non-zero field inside it would produce a force acting on charges, and since the charges can move, the equilibrium is only possible when the take such a configuration that the internal electric field is zero. The charges will always concentrate on the boundaries; the previous argument does not apply to the boundaries because the charges cannot leave the conductor.

Since the field inside the conductor is zero, its components parallel to the surface are also zero. By the continuity condition (2.18) the electric field just outside the conductor also must have zero components parallel to the surface. This field is therefore *perpendicular to the surface of the conductor*.

• *Interpretation of the scalar potential* ϕ. Imagine we have an electric field $\boldsymbol{E}(\boldsymbol{x})$ and we slowly move a charge q in this field from point A to point B. At each point of the trajectory the charge is subject to the force $\boldsymbol{F} = q\boldsymbol{E}$. To displace the charge by $d\boldsymbol{l}$ we have to do the work $\boldsymbol{F}d\boldsymbol{l} = q\boldsymbol{E}d\boldsymbol{l}$. The total work we have to perform to move the charge from A to B is thus

$$W = -\int_A^B d\boldsymbol{l}\,\boldsymbol{F} = -q\int_A^B d\boldsymbol{l}\,\boldsymbol{E} = q\int_A^B d\boldsymbol{l}\,\boldsymbol{\nabla}\phi.$$

In the last integral, the scalar product is

$$d\boldsymbol{l}\,\boldsymbol{\nabla}\phi = dl_1\frac{\partial\phi}{\partial x_1} + dl_2\frac{\partial\phi}{\partial x_2} + dl_3\frac{\partial\phi}{\partial x_3} = d\phi$$

We have therefore

$$W = q\int_A^B d\phi = q(\phi_B - \phi_A),$$

so that the change of particle energy when moved from point A to point B is

$$q\phi_B - q\phi_A.$$

We see that the quantity $q\phi$ is the *potential energy of the charge* q *in the electric field*.

Strictly speaking, the Stokes theorem only applies to differentable functions, while the components of the electric field are not even necessarily continuous across the interface surface. So, before using the Stokes theorem in the above argument we first have to imagine that the surface has a finite thickness and the components of the electric field are smoothly connected across this layer. After the application of the Stokes theorem the thickness of the interface is sent to 0.

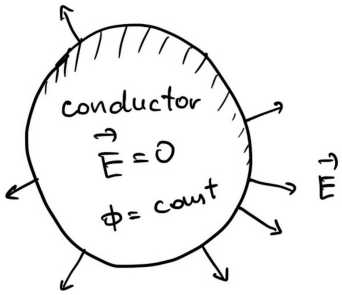


Figure 2.8.: The electric field is normal to the surface of a conductor.

Note that when we displace a particle in the direction of the force we have $d\boldsymbol{l}\,\boldsymbol{F} > 0$. But in this case we perform a *negative* work; this is why there is a minus sign in this equation.

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The scalar potential is defined by the relation (2.12) up to an additive constant. This constant is usually choosen so as to have ϕ vanishing at infinity, far from all charges.

• *Energy of electric field*. Let us calculate the total potential energy stored in a system of charges. Consider first n point charges q_a, $a = 1, ..., n$ which are placed in the points $\boldsymbol{x}_a$.

To calculate the energy of this system let us imagine that we are bringing the charges from infinity one by one. The energy of the system is the sum of the works we have to do at each step.

To bring the first charge to its position $\boldsymbol{x}_1$ costs no work, $w_1 = 0$. To bring the second charge costs the work equal to the potential energy of the second charge in the field created by the first charge,

$$w_2 = \frac{1}{4\pi\epsilon_0}\frac{q_1 q_2}{|\boldsymbol{x}_1 - \boldsymbol{x}_2|}.$$

In general, bringing the charge a when $a-1$ charges are already in place costs the work

$$w_a = \frac{q_a}{4\pi\epsilon_0}\sum_{b=1}^{a-1}\frac{q_b}{|\boldsymbol{x}_b - \boldsymbol{x}_a|}.$$

So, the total work, and therefore the total enery of our system of charges is

$$W = \sum_{a=1}^{n} w_a = \frac{1}{4\pi\epsilon_0}\sum_{a=1}^{n}\sum_{b=1}^{a-1}\frac{q_a q_b}{|\boldsymbol{x}_b - \boldsymbol{x}_a|} = \frac{1}{8\pi\epsilon_0}\sum_{b\neq a}\frac{q_a q_b}{|\boldsymbol{x}_b - \boldsymbol{x}_a|}.$$

In the last sum the two contributions with $a < b$ and $a > b$ are identical, which is compenated by the extra factor $1/2$.

It is straightforwarg to generalize the last expression to the case of a continuous charge distribution. We have to change $q_a \to \rho(\boldsymbol{x})d^3x$ and replace the sum by the integral. We then have

$$W = \frac{1}{8\pi\epsilon_0}\int d^3x d^3x' \frac{\rho(\boldsymbol{x})\rho(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} = \frac{1}{2}\int d^3x \rho(\boldsymbol{x})\phi(\boldsymbol{x}). \tag{2.19}$$

If the charge density is localized in some region of space, the integral in this expression only needs to be evaluated over this region because outside of it the integrand is zero. We will assume, however, that the integration here is extended to the whole space. This is not important here, but will be important later when rewriting this expression in order to drop the surface term when integrating by parts.

In the last equation we have used eq. (2.13).

It is instructive to rewrite this equation in a different way. We can express the charge density from the Poisson equation (2.15),

$$\rho(\boldsymbol{x}) = -\epsilon_0\Delta\phi(\boldsymbol{x})$$

and substitute it into eq. (2.19). This gives

$$W = -\frac{\epsilon_0}{2}\int d^3x \phi(\boldsymbol{x})\Delta\phi(\boldsymbol{x}) = -\frac{\epsilon_0}{2}\int d^3x \phi(\boldsymbol{x})\boldsymbol{\nabla}\boldsymbol{\nabla}\phi(\boldsymbol{x}) =$$

$$= \frac{\epsilon_0}{2}\int d^3x (\boldsymbol{\nabla}\phi(\boldsymbol{x}))^2 = \frac{\epsilon_0}{2}\int d^3x \boldsymbol{E}^2.$$

To go from the first to the second line in this expression we have integrated by parts using the identity

$$\phi(\boldsymbol{x})\boldsymbol{\nabla}\boldsymbol{\nabla}\phi(\boldsymbol{x}) = \boldsymbol{\nabla}[\phi(\boldsymbol{x})\boldsymbol{\nabla}\phi(\boldsymbol{x})] - (\boldsymbol{\nabla}\phi(\boldsymbol{x}))^2$$

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and set to zero the boundary term arising fromt the integration of $\nabla[\phi(\boldsymbol{x})\nabla\phi(\boldsymbol{x})]$ by the Stokes theorem.

We see that the energy of the continuous distribution of charges may be viewed as arising from the electric field it creates, the enery density w_E in the electric field being

$$w_E = \frac{\epsilon_0}{2}|\boldsymbol{E}|^2. \tag{2.20}$$

We will see below that this is a correct interpretation — the electric field indeed carries energy.

Multipole expansion

The electric field of a system of charges is given by the vector sum of fields of individual charges, eqs. (2.5) and (2.7). For systems of many charges calculation of the field may be a technically complicated task. If we are interested in the field at distances R much larger than the characteristic size of the system r, we may want to use an approximate expression expanding the field in powers of a small parameter r/R.

Consider a system of charges q_a at the positions $\boldsymbol{x}^a$ which are localized within a region of the characteristic size r. If we choose the origin of the coordinates somewhere in this region, then $|\boldsymbol{x}^a| \lesssim r$. Let us calculate the electric field of these charges at the point $\boldsymbol{R}$ far away from the system, such that $R = |\boldsymbol{R}| \gg r$.

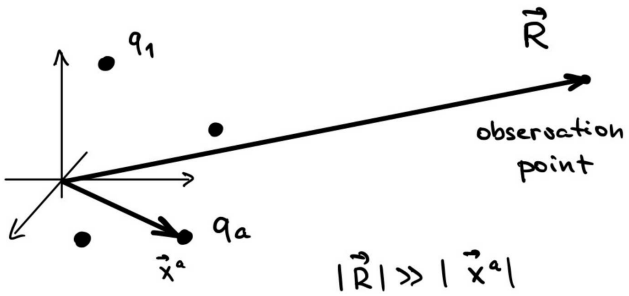


Figure 2.9.: Localized system of charges q_a in positions $\boldsymbol{x}^a$.

It is convenient to calculate first the scalar potential $\phi(\boldsymbol{R})$. So we start by eq. (2.13) which in the case of several point charges reads

$$\phi(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\sum_a \frac{q_a}{|\boldsymbol{R}-\boldsymbol{x}^a|}. \tag{2.21}$$

The idea now is to expand this function in Taylor series in powers of $\boldsymbol{x}^a$ around the point $\boldsymbol{x}^a = 0$.

For any function $f(\boldsymbol{R})$ we can write its Taylor expansion as follows,

$$f(\boldsymbol{R}-\boldsymbol{x}) = f(\boldsymbol{R}) - x_i\frac{\partial f}{\partial R_i} + \frac{1}{2}x_i x_j \frac{\partial^2 f}{\partial R_i \partial R_j} + \dots$$

In our case $f(\boldsymbol{R}) = 1/|\boldsymbol{R}|$. Let us first calculate the derivatives:

$$\frac{\partial}{\partial R_i}\frac{1}{|\boldsymbol{R}|} = -\frac{R_i}{R^3},$$

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$$\frac{\partial^2}{\partial R_i \partial R_j}\frac{1}{|\boldsymbol{R}|} = \frac{\partial}{\partial R_j}\left(-\frac{R_i}{|\boldsymbol{R}|^3}\right) = -\frac{\delta_{ij}}{R^3} + 3\frac{R_i R_j}{R^5} = \frac{3R_i R_j - R^2\delta_{ij}}{R^5},$$

where $R \equiv |\boldsymbol{R}|$. Therefore,

$$\frac{1}{|\boldsymbol{R}-\boldsymbol{x}|} = \frac{1}{R} + \frac{(\boldsymbol{x}\boldsymbol{R})}{R^3} + \frac{3(\boldsymbol{x}\boldsymbol{R})(\boldsymbol{x}\boldsymbol{R}) - R^2x^2}{2R^5} + \dots \tag{2.22}$$

Note that in this expression the second term is smaller than the first one by x/R, the third is smaller by $(x/R)^2$, etc., so on the right we have the expansion of the left side in powers of the small parameter x/R.

Let us now use this expansion in eq. (2.21). The potential ϕ then becomes a series

$$\phi(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\sum_a q_a\left\{\frac{1}{R} + \frac{(\boldsymbol{x}^a\boldsymbol{R})}{R^3} + \frac{3(\boldsymbol{x}^a\boldsymbol{R})(\boldsymbol{x}^a\boldsymbol{R}) - R^2\boldsymbol{x}^a\boldsymbol{x}^a}{2R^5} + \dots\right\} =$$

$$= \phi_0(\boldsymbol{R}) + \phi_1(\boldsymbol{R}) + \phi_2(\boldsymbol{R}) + \dots$$

where each subsequent term is suppressed by the extra power of a small parameter r/R. The leading contribution is

$$\phi_0(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{1}{R}\sum_a q_a = \frac{1}{4\pi\epsilon_0}\frac{Q}{R} \tag{2.23}$$

where

$$Q = \sum_a q_a \tag{2.24}$$

is the *total charge* of our system. We see that in the leading approximation the potential is just the Coulomb potential of the point charge Q equal to the total charge or the system.

Consider the first correction $\phi_1(\boldsymbol{R})$. It comes from the second term on the r.h.s. of eq. (2.22). We have

$$\phi_1(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{1}{R^3}\sum_a q_a(\boldsymbol{x}^a\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{R}}{R^3}\sum_a q_a\boldsymbol{x}^a.$$

The sum in this expression is a vector that does not depend on $\boldsymbol{R}$ and is an internal characteristics of our system of charges. It is called the **dipole moment** $\boldsymbol{d}$ of the system,

$$\boldsymbol{d} \equiv \sum_a q_a\boldsymbol{x}^a. \tag{2.25}$$

In terms of the dipole moment, the first correction to the potential reads

$$\phi_1(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{\boldsymbol{R}\boldsymbol{d}}{R^3}. \tag{2.26}$$

note that while the leading term ϕ_0 is proportional to $1/R$, the first correction ϕ_1 is proportional to $1/R^2$. The larger the distance R, the smaller its relative contribution.

Consider now the second correction $\phi_2(\boldsymbol{R})$ which corresponds to the

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third term in the expansion (2.22). For this correction we have

$$\phi_2(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{1}{R^5}\frac{1}{2}\sum_a q_a[3(\boldsymbol{x}^a\boldsymbol{R})(\boldsymbol{x}^a\boldsymbol{R}) - \boldsymbol{x}^{a\,2}R^2].$$

Both terms in square brackets contain $\boldsymbol{R}$ to the second power, so we can factor out R_iR_j,

$$\phi_2(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\frac{1}{2}\frac{R_iR_j}{R^5}\sum_a q_a[3x_i^a x_j^a - \delta_{ij}x^{a\,2}].$$

As in the case of the dipole, the sum does not depend on $\boldsymbol{R}$ and is a characteristics of the system of charges, which is called the **quadrupole moment** D_{ij} where

$$D_{ij} \equiv \sum_a q_a[3x_i^a x_j^a - \delta_{ij}(\boldsymbol{x}^a)^2]. \tag{2.27}$$

This is a matrix wit two indices (a tensor with respect to rotations). The matrix D_{ij} is symmetric and has zero trace $D_{ii} = 0$, so that it has 5 independent components.

Assembling all the contributions we may write the scalar potential at large distances from our system as follows:

$$\phi(\boldsymbol{R}) = \frac{1}{4\pi\epsilon_0}\left\{\frac{Q}{R} + \frac{\boldsymbol{d}\boldsymbol{R}}{R^3} + \frac{1}{2}\frac{R_iR_jD_{ij}}{R^5} + \dots\right\} \tag{2.28}$$

This series is called the *multipole expansion* of the potential. The first term (monopole) is proportional to $1/R$, the second (dipole) is proportional to $1/R^2$, the third (quadrupole) to $1/R^3$ etc. Clearly we could continue this expansion to an arbitrarily high order, with each subsequent term being smaller by an extra power of $1/R$ than the previous one. In practice, the terms that we have calculated are usually sufficient.

• The multipole expansion (2.28) expresses the potential created by a system of charges in terms of the total charge, dipole, quadrupole and higher multipoles of the system. But because each power of R appears only once, this equation may also be used in the opposite direction: if one knows the potential $\phi(\boldsymbol{R})$ of a system, one may expand it in powers of $1/R$ and read off the charge, dipole moment, quadrupole moment, etc. of this system from the coefficients in front of powers of $1/R$.

• The multipoles of a given system of charges are defined with respect to a chosen coordinate frame. A natural question is whether they really depend on the choice of the frame? Obviously, the total charge (2.24) does not. What about the dipole moment? Suppose we shift the origin of the coordinates by a constant vector $\boldsymbol{b}$, so that $\boldsymbol{x} = \boldsymbol{x}' + \boldsymbol{b}$ where $\boldsymbol{x}'$ are the new coordinates. From eq. (2.25) we have

$$\boldsymbol{d} = \sum_a q_a\boldsymbol{x}^a = \sum_a q_a(\boldsymbol{x}'^a + \boldsymbol{b}) = \boldsymbol{d}' + Q\boldsymbol{b}.$$

We see that the dipole moment does depend on the frame unless $Q = 0$, in which case it does not. Similar calculations show that the quadrupole moment is independent of the frame provided that $Q = 0$ and $\boldsymbol{d} = 0$. In general, the lowest non-zero multipole of any system is independent of the reference frame.

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• Finally, let us calculate the electric field by evaluating the gradient of eq. (2.28),

$$\boldsymbol{E}(\boldsymbol{x}) = -\boldsymbol{\nabla}\phi(\boldsymbol{x}) = -\frac{1}{4\pi\epsilon_0}\boldsymbol{\nabla}\left\{\frac{Q}{x} + \frac{\boldsymbol{x}\boldsymbol{d}}{x^3} + \dots\right\}$$

where we limited ourselves to the first two contributions. The leading term produces the electric field of a point charge Q,

$$-\frac{1}{4\pi\epsilon_0}\boldsymbol{\nabla}\frac{Q}{x} = \frac{Q}{4\pi\epsilon_0}\frac{\boldsymbol{x}}{x^3}.$$

For the dipole contribution we have to calculate the derivative

$$\partial_i\left(\frac{\boldsymbol{x}\boldsymbol{d}}{x^3}\right) = \partial_i\left(\frac{d_j x_j}{x^3}\right) = d_j\partial_i\left(\frac{x_j}{x^3}\right) = d_j\left(\frac{\delta_{ij}}{x^3} - 3\frac{x_i x_j}{x^5}\right).$$

Mkang use of this expression we find

$$\boldsymbol{E}(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\left\{\frac{Q\boldsymbol{x}}{x^3} + \frac{3\boldsymbol{x}(\boldsymbol{x}\boldsymbol{d}) - x^2\boldsymbol{d}}{x^5} + \dots\right\}. \tag{2.29}$$

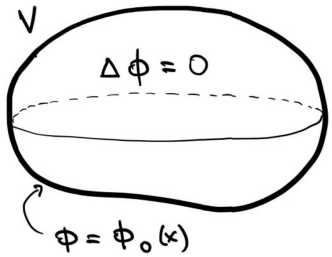
The second term — the dipole contribution — is proportional to $1/x^3$ at large x; it is therefore subdominant to the first one whch is proportional to $1/x^2$.

2.2. Poisson and Laplace equations

The Poisson equation (2.15) and Laplace equation (2.16) are very common in physics. They arise in many different contexts, not necessarily related to electromagnetism. So let us discuss a few general methods to solve these equations.

Uniqueness of the solution

First let us demonstrate that, given the appropriate boundary conditions, the solution to the Poisson (and, by consequence, of the Laplace) equation is unique. This theorem is very useful in practice.



Consider a closed volume V bounded by the surface ∂V. Suppose the scalar function $\phi(\boldsymbol{x})$ satisfies the Poisson equation inside this volume with a given function $\rho(\boldsymbol{x})$, and takes a given value $\phi_0(\boldsymbol{x})$ at the boundary,

$$\Delta\phi(\boldsymbol{x}) = -\frac{\rho(\boldsymbol{x})}{\epsilon_0}, \tag{2.30}$$

$$\phi(\boldsymbol{x})|_{\partial V} = \phi_0(\boldsymbol{x}). \tag{2.31}$$

We want to show that any other function satisfying the same equations (2.30) and (2.31) equals $\phi(\boldsymbol{x})$.

Let us assume there are two solutions $\phi_1(\boldsymbol{x})$ and $\phi_2(\boldsymbol{x})$ both satisfying eqs. (2.30) and (2.31). Consider their difference

$$u(\boldsymbol{x}) = \phi_2(\boldsymbol{x}) - \phi_1(\boldsymbol{x}).$$

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Clearly the function $u(\boldsymbol{x})$ satisfies the Laplace equation inside V and takes zero value at the boundary,

$$\Delta u(\boldsymbol{x}) = 0, \tag{2.32}$$
$$u(\boldsymbol{x})|_{\partial V} = 0. \tag{2.33}$$

We will show now that these equations imply $u(\boldsymbol{x}) = 0$.

Let us multiply eq. (2.32) by $u(\boldsymbol{x})$ and integrate over the volume V,

$$\int_V d^3x\, u(\boldsymbol{x})\Delta u(\boldsymbol{x}) = 0.$$

The integral on the l.h.s. can be re-written as

$$\int_V d^3x\, u(\boldsymbol{x})\Delta u(\boldsymbol{x}) = \int_V d^3x\, \left(\boldsymbol{\nabla}(u\boldsymbol{\nabla}u) - (\boldsymbol{\nabla}u)^2\right).$$

We have therefore

$$\int_V d^3x\, \boldsymbol{\nabla}(u\boldsymbol{\nabla}u) - \int_V d^3x\, (\boldsymbol{\nabla}u)^2 = 0.$$

The first integral can be transformed into an integral over the surface ∂V by the Stokes theorem,

$$\int_V d^3x\, \boldsymbol{\nabla}(u\boldsymbol{\nabla}u) = \int_{\partial V} d\boldsymbol{s}\, u\boldsymbol{\nabla}u = 0$$

where and last integral equals zero because at the boundary ∂V the function $u(\boldsymbol{x})|_{\partial V} = 0$. Thus,

$$\int_V d^3x\, (\boldsymbol{\nabla}u)^2 = 0.$$

Since the integrand is non-negative, this equation implies

$$\boldsymbol{\nabla}u(\boldsymbol{x}) = 0 \quad \Longrightarrow \quad u = \text{const}.$$

However, since at the boundary $u = 0$ this constant must be zero, and we finally get

$$u(\boldsymbol{x}) = 0$$

which completes the proof.

Dirac δ-function

We have derived the Poison equation (2.15) for the scalar potential by combining eqs. (2.11) and (2.12) which both follow from the Coulomb law. The Coulomb law also gave us its solution, eq. (2.13). It is not evident, however, *why* eq. (2.13) is a solution. In other words, how check this by the direct substitution of (2.13) into the Poisson equation? Clarifying this question leads to a general method of solving equations similar to the Poisson equation, called the method of Green's functions.

To formulate this method we first need to introduce the notion of the Dirac **delta-function**. At the intuitive level the Dirac delta-function $\delta(x)$

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in one dimension is defined by the relation

$$\int dx\, \delta(x - x_0) f(x) = f(x_0) \tag{2.34}$$

for any continuous function $f(x)$. It is clear that to satisfy this property one must have

$$\delta(x) = 0 \quad \text{for any } x \neq 0,$$

so formally this function must be zero everywhere except for one point. This however contradicts to eq. (2.34) because one can prove that any function that is zero everywhere except one point will give zero when integrated with a continuous function $f(x)$. So, in the strict mathematical sense the δ-function either does not exist, of is not a function.

To give mathematical meaning to eq. (2.34) let us define the "regularized version" of the δ-function, $\delta_\varepsilon(x)$ which is a normal function that is zero everywhere except the vicinity of the point $x = 0$ where it has a peak of width ε and height $1/\varepsilon$, so that the integral of this function is 1. There are many ways to define such a function — it may be a rectangular peak, or a smooth function, for instance,

$$\delta_\varepsilon(x) = \frac{1}{\sqrt{2\pi}\varepsilon} \mathrm{e}^{-x^2/2\varepsilon^2}.$$

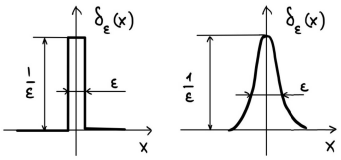


Figure 2.10.: The examples of a regularized δ-function $\delta_\varepsilon(x)$.

When ε is very small, the function $\delta_\varepsilon(x)$ has approximately the desired properties,

$$\int dx \delta_\varepsilon(x - x_0) f(x) \simeq f(x_0),$$

the smaller the ε, the better this equation is satisfied.

Formally, if we take a limit $\varepsilon \to 0$ we get a "function" that is zero everywhere except one point, and we recover the mathematical inconsistency discussed above. However, if we *first calculate the integral, and then take the limit* $\varepsilon \to 0$ everything works just fine and we get an exact relation

$$\lim_{\varepsilon \to 0} \int dx\, \delta_\varepsilon(x - x_0) f(x) = f(x_0).$$

Therefore, we can use the Dirac δ-function as a normal function if we keep in mind that in principle we *first have to regularize it*, perform the integral and remove the regularization *after* the calculation is done. In most cases of practical interest there is no need to do it explicitely.

These ideas can be formulated in a mathematically strict way, known as the theory of distributions. One defines a space of test functions that are continuous piecewise differentiable functions decaying at infinity faster than any power. A *distribution* is a linear mapping from the space of test functions into real numbers. Every piece-wise continuous function $f(x)$ defines a distribution that maps any test function into the value of its integral with $f(x)$. The regularized functions $\delta_\varepsilon(x - x_0)$ define in this way distributions parameterized by ε; in the limit $\varepsilon \to 0$ they all converge to a *distribution* that has the properties of the δ-function: it maps any test function $\phi(x)$ into its value $\phi(x_0)$ in the point x_0.

• As an illustration of these concepts consider the integral of a delta function of an argument which itself is a function of x,

$$\int dx\, \delta(g(x)) f(x),$$

where $f(x)$ and $g(x)$ are two smooth functions. The contributions to this integral only come from vicinities of the points where $g(x) = 0$. The function $g(x)$ may have several zeros $\{x_i\}$ such that

$$g(x_i) = 0.$$

We will assume that all the zeros are isolated and of the first order. Let us regularize the δ-function and take ε so small that the ε-vicinity of the

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zero x_i contains no other zeros. The contribution of this region is then

$$\int_{x_i-\varepsilon}^{x_i-\varepsilon} dx\, \delta_\varepsilon(g(x))f(x).$$

To evaluate this integral we make a change of variables

$$g(x) = y.$$

This defines x as a function of y, $x = x(y)$ such that $x(0) = x_i$. Differentiating we also find

$$dx g'(x) = dy \quad \Longrightarrow \quad dx = \frac{dy}{g'(x(y))}.$$

Thus, the above contribution equals

$$\int_{x_i-\varepsilon}^{x_i-\varepsilon} dx\, \delta_\varepsilon(g(x))f(x) = \int dy\, \frac{f(x(y))}{g'(x(y))}\delta_\varepsilon(y).$$

Now that we have the δ-function of a simple argument we can remove the regularization. We do not need to know what precisely are the limits of integration, only whether the lower limit is smaller or larger than the upper limit. In the case of positive derivative $g'(x_i) > 0$ we get from the above expression

$$\int_{x_i-\varepsilon}^{x_i-\varepsilon} dx\, \delta_\varepsilon(g(x))f(x) = \frac{f(x_i)}{g'(x_i)}.$$

while for negative derivative $g'(x_i) < 0$

$$\int_{x_i-\varepsilon}^{x_i-\varepsilon} dx\, \delta_\varepsilon(g(x))f(x) = -\frac{f(x_i)}{g'(x_i)}.$$

Together these two expressions may be written as

$$\int_{x_i-\varepsilon}^{x_i-\varepsilon} dx\, \delta_\varepsilon(g(x))f(x) = \frac{f(x_i)}{|g'(x_i)|}.$$

It only remains to sum over all zeros of the function $g(x)$ and we find

$$\int dx\, \delta(g(x))f(x) = \sum_i \frac{f(x_i)}{|g'(x_i)|}, \quad \text{for } \{x_i\} \text{ such that } g(x_i) = 0.$$

Method of Green's function

• *Laplacian of* $1/r$. Now that we have adequate tools, let us calculate the Laplacian of $1/r$, where $r = |\boldsymbol{x}|$. We will show that

$$\Delta\frac{1}{r} = -4\pi\delta^3(\boldsymbol{x}), \tag{2.35}$$

where $\delta^3(\boldsymbol{x}) \equiv \delta(x)\delta(y)\delta(z)$ is the 3-dimensional δ-function which has the same property as its 1-dimensional analog:

$$\int d^3x f(\boldsymbol{x})\delta^3(\boldsymbol{x}) = f(0)$$

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for any continuous $f(\boldsymbol{x})$.

Let us first consider the points away from the origin, that is $r \neq 0$. At these points the function $1/r$ is differentiable. We get

$$\partial_i \frac{1}{r} = -\frac{\partial_i r}{r^2} = -\frac{1}{r^2}\frac{x_i}{r} = -\frac{x_i}{r^3},$$

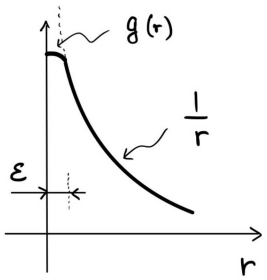
$$\partial_i \partial_i \frac{1}{r} = -\partial_i \left(\frac{x_i}{r^3}\right) = -\frac{\delta_{ii}}{r^3} + 3\frac{x_i x_i}{r^5} = -\frac{3}{r^3} + \frac{3}{r^3} = 0.$$

We see that away from the point $r = 0$ the Laplacian of $1/r$ is zero, just as we need for eq. (2.35).

To figure out what happens at the point $r = 0$ where the function $1/r$ is singular, let us regularize it by replacing with the function $f_\varepsilon(r)$ such that

$$f_\varepsilon(r) = \begin{cases} 1/r & \text{at } r > \varepsilon \\ g(r) & \text{at } r < \varepsilon \end{cases}$$

where $g(r)$ is some twice differentiable finite function defined in the range $r < \varepsilon$ and smoothly matching into $1/r$ at $r = \varepsilon$, see the plot. Its precise shape will not needed as we will see below.



Consider the integral of the Laplacian $\Delta f_\varepsilon(r)$ with some continuous function $F(\boldsymbol{x})$,

$$\int d^3x\, F(\boldsymbol{x}) \Delta f_\varepsilon(r).$$

Since by construction $f_\varepsilon(r) = 1/r$ for $r > \varepsilon$ and the Laplacian of $1/r$ is zero, we can limit the integration to the spherical volume $|\boldsymbol{x}| \leq \varepsilon$,

$$\int d^3x\, F(\boldsymbol{x}) \Delta f_\varepsilon(r) = \int_{|\boldsymbol{x}|\leq\varepsilon} d^3x\, F(\boldsymbol{x}) \Delta f_\varepsilon(r).$$

If ε is sufficiently small, we can replace $F(\boldsymbol{x})$ by its value at the origin $F(0)$ and take it out of the integral,

$$\int_{|\boldsymbol{x}|\leq\varepsilon} d^3x\, F(\boldsymbol{x}) \Delta f_\varepsilon(r) = F(0) \int_{|\boldsymbol{x}|\leq\varepsilon} d^3x\, \Delta f_\varepsilon(r) + \mathcal{O}(\varepsilon).$$

Since the function $f_\varepsilon(r)$ is smooth everywhere, we can use the Stokes theorem to transform this integral into a surface integral,

$$\int_{|\boldsymbol{x}|\leq\varepsilon} d^3x\, \Delta f_\varepsilon(r) = \int_{|\boldsymbol{x}|\leq\varepsilon} d^3x\, \boldsymbol{\nabla}\boldsymbol{\nabla} f_\varepsilon(r) = \int_{|\boldsymbol{x}|=\varepsilon} d\boldsymbol{s}\, \boldsymbol{\nabla} f_\varepsilon(r).$$

The integral is taken over the sphere of radius ε where the function $f_\varepsilon(r) = 1/r$, so we can evaluate it. Denoting $\boldsymbol{n}$ a unit radial vector $\boldsymbol{n} = \boldsymbol{x}/|\boldsymbol{x}|$ we have

$$\boldsymbol{\nabla} f_\varepsilon(r)|_{r=\varepsilon} = -\frac{\boldsymbol{n}}{\varepsilon^2}.$$

Therefore,

$$\int_{|\boldsymbol{x}|=\varepsilon} d\boldsymbol{s}\, \boldsymbol{\nabla} f_\varepsilon(r) = -\frac{1}{\varepsilon^2} \int_{|\boldsymbol{x}|=\varepsilon} d\boldsymbol{s}\, \boldsymbol{n} = -\frac{1}{\varepsilon^2}\varepsilon^2 \int d\Omega = -4\pi.$$

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Here we have used $d\boldsymbol{s} = \varepsilon^2 \boldsymbol{n} d\Omega$. Assembling the factors we finally fine

$$\int d^3x\, F(\boldsymbol{x}) \Delta f_\varepsilon(r) = -4\pi F(0) + \mathcal{O}(\varepsilon).$$

When we take the limit $\varepsilon \to 0$ the correction on the r.h.s. disappears, the function $f_\varepsilon(r)$ becomes $1/r$ and we see from the above equation that $\Delta(1/r) = \delta^3(x)$ by definition of the latter. We have therefore proven eq. (2.35).

• We have now the tools to show that the potential (2.13) indeed satisfies eq. (2.30). Acting on $\phi(\boldsymbol{x})$ by the Laplacian we get

$$\Delta\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0} \Delta \int \frac{d^3x' \rho(x')}{|\boldsymbol{x} - \boldsymbol{x}'|}.$$

The Laplacian acts on the coordinates $\boldsymbol{x}$ while the integration is over $\boldsymbol{x}'$, so we can take the Laplacian inside the integral. Thus

$$\Delta\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \rho(x') \Delta \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} =$$

$$= \frac{1}{4\pi\epsilon_0} \int d^3x' \rho(x') [-4\pi\delta^3(\boldsymbol{x} - \boldsymbol{x}')] = -\frac{\rho(\boldsymbol{x})}{\epsilon_0},$$

so we have verified eq. (2.15).

• *Green's function*. The above line of reasoning can be turned into a general method of solving second order differential equations similar to Poisson equation (2.15), known as the method of Green's function. Suppose we have to solve the equation

$$L^{(2)} f(\boldsymbol{x}) = g(\boldsymbol{x}), \tag{2.36}$$

where $L^{(2)}$ is some second order differential operator acting on coordinates $\boldsymbol{x}$ (Laplacian in the case of the Poisson equation) and $g(\boldsymbol{x})$ is a given function. This is an inhomogeneous linear equation; the inhomogeneous term $g(\boldsymbol{x})$ is often called the source term.

One may simplify the problem by using the linearity of this equation which implies that if the source term $g(\boldsymbol{x})$ is a sum of two contributions, one may solve the equation separately for these contributions and add the solutions. Thus, one may first solve the equation (2.36) for a *point source*, and then combine the solutions. This is precisely what we have done in Sect. 2.1 when we generalized the Coulomb law for a point charge to a continuous charge distribution.

In mathematical terms, this is done as follows. Assume we have found a solution $G(\boldsymbol{x}, \boldsymbol{x}')$ of the following equation instead of (2.36),

$$L^{(2)} G(\boldsymbol{x}, \boldsymbol{x}') = \delta^3(\boldsymbol{x} - \boldsymbol{x}'). \tag{2.37}$$

The r.h.s. of this equation represents a point source ("point charge") in the position $\boldsymbol{x}'$. This solution $G(\boldsymbol{x}, \boldsymbol{x}')$ is called *Green's function* of the operator $L^{(2)}$. Eq. (2.37) is simpler than the original equation because it does not contain an arbitrary function $g(\boldsymbol{x})$. Moreover, there exist standard methods to construct $G(\boldsymbol{x}, \boldsymbol{x}')$ from the solutions of the *homogeneous* version of eq. (2.36), that is, the one with $g(\boldsymbol{x}) = 0$.

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If we have found the Green's function $G(\boldsymbol{x}, \boldsymbol{x}')$, we can write down the solution to eq. (2.36) as follows:

$$f(\boldsymbol{x}) = \int d^3\boldsymbol{x}'\, G(\boldsymbol{x}, \boldsymbol{x}') g(\boldsymbol{x}'). \tag{2.38}$$

Indeed, if we act by the operator $L^{(2)}$ on both sides of this equation, then by the definition of $G(\boldsymbol{x}, \boldsymbol{x}')$ we have

$$L^{(2)} f(\boldsymbol{x}) = L^{(2)} \int d^3\boldsymbol{x}'\, G(\boldsymbol{x}, \boldsymbol{x}') g(\boldsymbol{x}') = \int d^3\boldsymbol{x}'\, L^{(2)} G(\boldsymbol{x}, \boldsymbol{x}') g(\boldsymbol{x}') =$$

$$= \int d^3\boldsymbol{x}'\, \delta^3(\boldsymbol{x} - \boldsymbol{x}') g(\boldsymbol{x}') = g(\boldsymbol{x}),$$

so that $f(\boldsymbol{x})$ given by eq. (2.38) is indeed a solution to eq. (2.36).

Coming back to the case of the Poisson equation we see that the function

$$G(\boldsymbol{x}, \boldsymbol{x}') = -\frac{1}{4\pi} \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|}$$

is the Green's function of the Laplace operator Δ, as it follows from eq. (2.35):

$$\Delta \left\{ -\frac{1}{4\pi} \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} \right\} = \delta^3(\boldsymbol{x} - \boldsymbol{x}').$$

In order to get a *unique* solution of the equation (2.36) one has to specify the *boundary conditions*, cf. the discussion of the Poisson and Laplace equations in the beginning of this section. How do we satisfy the boundary conditions with the solution (2.38)? To do so recall that the equation for the Green's function (2.37) is a linear inhomogeneous equation whose general solution is a sum of a particular solution to the inhomogeneous equation (e.g., given by eq. (2.38)) *and a general solution of the homogeneous equation*. In other words, if $G(\boldsymbol{x}, \boldsymbol{x}')$ is a solution to eq. (2.37), then the sum

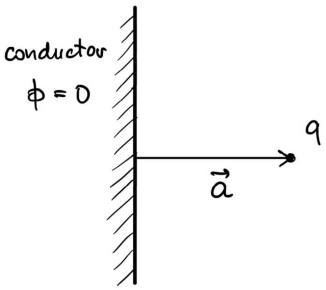
$$\tilde{G}(\boldsymbol{x}, \boldsymbol{x}') = G(\boldsymbol{x}, \boldsymbol{x}') + F(\boldsymbol{x}, \boldsymbol{x}')$$

where

$$L^{(2)} F(\boldsymbol{x}, \boldsymbol{x}') = 0$$

is *also a solution*. We may use the freedom in the function $F(\boldsymbol{x}, \boldsymbol{x}')$ to satisfy the required boundary conditions.

• **Method of images**. As an illustration consider a method to find the electrostatic potential in presence of conductors known as the *method of images*. Suppose we have a plane conductor occupying the region $x \le 0$ and a charge q at a distance a from it so that the charge is at the position $\boldsymbol{a} = (a, 0, 0)$, see the figure. We want to find the electric field everywhere in the region $x > 0$.



We have argued above in Sect. 2.1 that the electric field inside the conductor (that is, at $x \le 0$) is zero, and therefore the potential is ϕ = const. Let us choose this constant to be zero, so $\phi(x \le 0) = 0$. We also know that the electric field close to the conductor must be perpendicular to its surface.

The charge q produces an electric field that is directed radially from its position. Near the surface of the conductor and away from the origin,

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this field is not perpendicular to the conductor, so this cannot be the correct answer; there must be other charges contributing to the total field which make it perpendicular to the surface of the conductor close to it. These charges must be concentrated on the surface of the conductor, and their distribution along this surface must be such that the total electric field is normal to the surface. How can we find the distribution of these charges?

Let us approach the problem formally and find the potential $\phi(\boldsymbol{x})$ that satisfies eq. (2.15) everywhere in the region $x > 0$ and the boundary condition $\phi = 0$ on the plane $x = 0$. The potential satisfying eq. (2.15) is given by eq. (2.13) with the charge density $\rho(\boldsymbol{x}') = q\delta^3(\boldsymbol{x}' - \boldsymbol{a})$, which gives simply the potential of the point charge q,

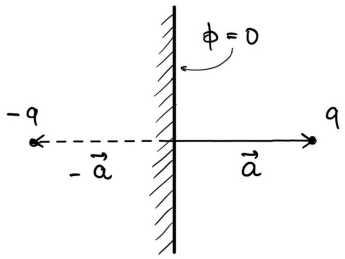
$$\phi_1(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\frac{q}{|\boldsymbol{x} - \boldsymbol{a}|}.$$

This potential alone does not satisfy the boundary condition because on the plane $x = 0$ it is non-zero:

$$\phi_1(0, y, z) = \frac{1}{4\pi\epsilon_0}\frac{q}{\sqrt{y^2 + z^2 + a^2}}.$$

We need to add to this potential a function that satisfies the *homogeneous* equation in the region $x > 0$ and such that its sum with ϕ_1 is zero at $x = 0$.

We have seen above that the potential created by a charge satisfies the Laplace equation $\Delta\phi = 0$ everywhere except at the position of the charge. Therefore, any potential which is created by charges at $x < 0$ will have zero Laplacian at $x > 0$ and its sum with ϕ_1 will satisfy the Poisson equation at $x > 0$. To find the potential which will cancel the contribution of ϕ_1 at the boundary of conductor, let us forget the conductor for a moment and put the (imaginary) charge $-q$ in the point $-\boldsymbol{a}$, see the plot. It will create the potential



$$\phi_2(\boldsymbol{x}) = -\frac{1}{4\pi\epsilon_0}\frac{q}{|\boldsymbol{x} + \boldsymbol{a}|}.$$

In the region $x > 0$ the sum

$$\phi(\boldsymbol{x}) = \phi_1(\boldsymbol{x}) + \phi_2(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\left\{\frac{q}{|\boldsymbol{x} - \boldsymbol{a}|} - \frac{q}{|\boldsymbol{x} + \boldsymbol{a}|}\right\} \tag{2.39}$$

satisfies the Poisson equation with the charge q as a source. At the boundary $x = 0$ the vector $\boldsymbol{x}$ is orthogonal to $\boldsymbol{a}$, so that

$$|\boldsymbol{x} - \boldsymbol{a}| = |\boldsymbol{x} + \boldsymbol{a}|$$

and thus

$$\phi(0, y, z) = \frac{1}{4\pi\epsilon_0}\left\{\frac{q}{\sqrt{y^2 + z^2 + a^2}} - \frac{q}{\sqrt{y^2 + z^2 + a^2}}\right\} = 0.$$

We see therefore that the potential $\phi(\boldsymbol{x})$ in eq. (2.39) satisfies all the conditions, so it is the solution to our problem.

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In reality, of course, there is no charge at the point $\boldsymbol{x} = -\boldsymbol{a}$ inside the conductor. Forgetting the conductor and putting an imaginary charge there was just a mathematical trick to find the good function ϕ_2 which would cancel the potential at the plane $x = 0$. In reality, the potential ϕ_2 at $x > 0$ is created by the surface charges distributed at the boundary of the conductor.

To find this distribution we first calculate the electric field by eq. (2.12),

$$\boldsymbol{E}(\boldsymbol{x}) = -\nabla\phi(\boldsymbol{x}) = \frac{q}{4\pi\epsilon_0}\left\{\frac{\boldsymbol{x}-\boldsymbol{a}}{|\boldsymbol{x}-\boldsymbol{a}|^3} - \frac{\boldsymbol{x}+\boldsymbol{a}}{|\boldsymbol{x}+\boldsymbol{a}|^3}\right\}. \tag{2.40}$$

At $x = 0$ this expression becomes

$$\boldsymbol{E}(0, y, z) = -\frac{q}{4\pi\epsilon_0}\frac{2\boldsymbol{a}}{|\boldsymbol{x}-\boldsymbol{a}|^3}.$$

We see that the electric field on the border of the conductor is indeed normal to the surface as it has the direction parallel to the vector $\boldsymbol{a}$. We can now use the field continuity conditions to find the surface charge density $\sigma(\boldsymbol{x})$. Applying eq. (2.17) we find

$$\sigma(y, z) = -\frac{qa}{2\pi(y^2 + z^2 + a^2)^{3/2}}.$$

This surface charge has the sign opposite of q. Integrating over the whole plane we find the total surface charge on the border of the conductor,

$$Q = \int dy\, dz\, \sigma(y, z) = -qa \int d\theta\, r dr \frac{1}{2\pi(r^2 + a^2)^{3/2}} = -q.$$

This result is in fact evident from the Gauss law applied to a very large sphere around our system. Indeed, imagine a very large sphere, $|\boldsymbol{x}| \to \infty$, over which we have to integrate $\boldsymbol{En}$, $\boldsymbol{n}$ being the unit radial vector. The field at negative x is zero, so that part of the sphere does not contribute. The field at $x > 0$ is given by eq. (2.40). This field behaves like $1/|\boldsymbol{x}|^3$ at large distances: each of the two terms behaves like $1/|\boldsymbol{x}|^2$, but due to the minus sign the leading $1/|\boldsymbol{x}|^2$ contribution cancels out. So, at large distances the integral behaves like $|\boldsymbol{x}|^2/|\boldsymbol{x}|^3 = 1/|\boldsymbol{x}|$ and goes to zero as $|\boldsymbol{x}| \to \infty$. By the Gauss law the total charge in a very large sphere is zero, and therefore the charge on the surface must be equal to $-q$.

Separation of variables

Another powerful method to solve partial differential equations is *separation of variables*. This method works differently in different coordinate systems, and only works in some of them. For the Laplace equation, we will consider Cartesian, polar and spherical coordinates.

• *Cartesian coordinates.* In the Cartesian coordinates the Laplace equation reads

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0. \tag{2.41}$$

The key idea is to look for solutions that have a particular form, namely, the product of three functions, each depending on only one variable. So

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let us look for the solutions of the following form,

$$\phi(x, y, z) = U(x)V(y)W(z).$$

If we substitute this into the Laplace equation we get

$$\frac{d^2U(x)}{dx^2}V(y)W(z) + U(x)\frac{d^2V(y)}{dy^2}W(z) + U(x)V(y)\frac{d^2W(z)}{dz^2} = 0.$$

Dividing by $\phi = UVW$ we get the following equation:

$$\frac{1}{U(x)}\frac{d^2U(x)}{dx^2} + \frac{1}{V(y)}\frac{d^2V(y)}{dy^2} + \frac{1}{W(z)}\frac{d^2W(z)}{dz^2} = 0. \tag{2.42}$$

A remarkable feature of this equation is that every term depends on *only one variable*. The only way to satisfy this equation is to require that *every term is a constant* and the sum of these constants is zero.

Among these three constants there must be at least one positive, and at least one negative. The remaining third one may be either positive or negative. Any choice satisfying these conditions is possible. Let us assume that the first two constants are negative, and the third one positive. Then we get the following three *ordinary* differential equations for three functions U, V and W,

$$\frac{1}{U(x)}\frac{d^2U(x)}{dx^2} = -\alpha^2 \tag{2.43}$$

$$\frac{1}{V(y)}\frac{d^2V(y)}{dy^2} = -\beta^2 \tag{2.44}$$

$$\frac{1}{W(z)}\frac{d^2W(z)}{dz^2} = \gamma^2. \tag{2.45}$$

The original equation is separated into three equations, each for its own variable, hence the name of the method. Eq. (2.42) is satisfied as long as

$$\gamma^2 = \alpha^2 + \beta^2. \tag{2.46}$$

The coefficients α and β are arbitrary at this point (we will see later that they are determined by the boundary conditions).

The resulting ordinary differential equations can be easily solved. Eq. (2.43) reads

$$\frac{d^2U(x)}{dx^2} + \alpha^2 U(x) = 0.$$

This is the equation of a harmonic oscillator. The general solution is

$$U(x) = Ae^{i\alpha x} + A^*e^{-i\alpha x},$$

where A is an arbitrary complex constant. Equivalently, one may write this solution as a linear combination of $\sin(\alpha x)$ and $\cos(\alpha x)$) with two arbitrary real coefficients.

Eq. (2.44) is the same as for U except for the names of variables. The solution for V is therefore

$$V(y) = Be^{i\beta y} + B^*e^{-i\beta y},$$

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where B is another arbitrary complex constant.

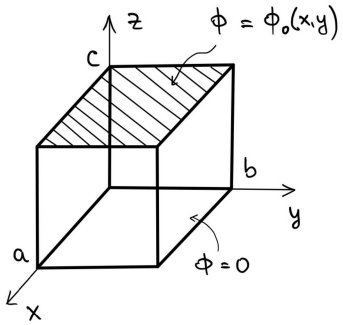
Finally, eq. (2.45) reads

$$\frac{d^2W(z)}{dz^2} - \gamma^2 W(z) = 0.$$

The solution to this equation is

$$W(x) = Ce^{\gamma z} + De^{-\gamma z},$$

where again C and D are two arbitrary real constants. Note that at least one of the constants in eqs. (2.43)–(2.45) must be positive, and the corresponding function will have exponential rather than oscillatory behavior.



• *Boundary conditions.* So far we have found many particular solutions to eq. (2.41): for any constants α, β and any amplitudes A, B, C and D the product $U(x)V(y)W(z)$ satisfies this equation. Let us see that this freedom is enough to satisfy boundary conditions. As an illustration, consider the problem of solving eq. (2.41) in a rectangular region $0 \le x \le a, 0 \le y \le b$, $0 \le z \le c$ with boundary conditions $\phi = 0$ everywhere except on the side $z = c$ where we impose the boundary condition $\phi = \phi_0(x, y)$, see the plot.

Let us start with the solution for $U(x)$. To satisfy the boundary condition at $x = 0$ we have to choose the constants in such a way that $\phi(0, y, z) = 0$. This is achieved by taking

$$U(x) = \sin(\alpha x)$$

which corresponds to the purely imaginary complex amplitude A. We still have an arbitrary overall amplitude (the value of the imaginary part of A), but at this point it is irrelevant because in the product $U(x)V(y)W(z)$ only one common amplitude matters.

Now we have the second boundary condition at $x = a$, $\phi(a, y, z) = 0$. However, there are no more arbitrary amplitudes which we can tune to ensure it. The only choice is to tune the constant α in such a way that $\sin(\alpha a) = 0$, which requires that α takes *discrete values*

$$\alpha_n = \frac{\pi n}{a}, \qquad n = 1, 2, \ldots$$

For any natural n the function $U(x) = \sin(\alpha_n x)$ is zero both at $x = 0$ and at $x = a$. We have therefore satisfied the boundary conditions at two sides of our region that are perpendicular to the axis x.

Equation for $V(y)$ is the same as the one for $U(x)$. Repeating the arguments we find that β has to take discrete values

$$\beta_m = \frac{\pi m}{b}, \qquad m = 1, 2, \ldots$$

and

$$V(y) = \sin(\beta_m y).$$

This ensures the boundary conditions $\phi = 0$ at the sides $y = 0$ and $y = b$.

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Let us now turn to the equation for $W(z)$ and the remaining two conditions. At $z = 0$ we have the zero boundary condition, $\phi(x, y, 0) = 0$. So, we must ensure that $W(0) = 0$. This is achieved by taking $D = -C$, which gives, up to a normalization constant,

$$W(z) = \sinh(\gamma z).$$

Unlike previous cases, here we do not have any freedom to chose γ because we must have $\gamma = \sqrt{\alpha^2 + \beta^2}$.

To summarize, we have constructed a discrete set of solutions to eq. (2.41) labeled by two integers $n, m = 1, 2, \ldots$

$$\phi_{nm}(x, y, z) = \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} z)$$

where

$$\gamma_{nm} = \sqrt{\alpha_n^2 + \beta_m^2} = \pi \sqrt{\frac{n^2}{a^2} + \frac{m^2}{b^2}}.$$

Each of these solutions satisfies the zero boundary conditions on 5 sides, but does not satisfy the one on the side $z = c$.

We have no more freedom left except for combining these solutions with arbitrary coefficients A_{nm},

$$\phi(x, y, z) = \sum_{n,m=1}^{\infty} A_{nm} \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} z).$$

Note that any such combination satisfies, by construction, the Laplace equation and 5 zero boundary conditions. Now we have to choose the coefficients A_{nm} in order to satisfy the remaining condition at $z = c$, $\phi(x, y, c) = \phi_0(x, y)$. This gives the following equation for the coefficients A_{nm}:

$$\phi_0(x, y) = \sum_{n,m=1}^{\infty} A_{nm} \sin(\alpha_n x) \sin(\beta_m y) \sinh(\gamma_{nm} c). \tag{2.47}$$

One may recognize the double Fourier series for the function $\phi_0(x, y)$ with unknown coefficients A_{nm} which we now have to determine.

In order to solve this equation for the coefficients A_{nm} one may use the following orthogonality properties of the functions $\sin(\alpha_n x)$:

$$\int_0^a dx \sin(\alpha_n x) \sin(\alpha_k x) = \frac{a}{2} \delta_{nk},$$

and similarly for $\sin(\beta_m y)$. If we multiply eq. (2.47) by $\sin(\alpha_k x) \sin(\beta_q y)$ and integrate, we get

$$\int_0^a dx \int_0^b dy \sin(\alpha_k x) \sin(\beta_q y) \phi_0(x, y) =$$

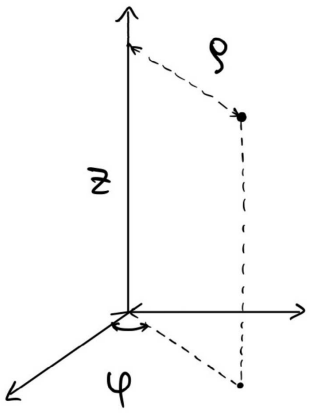
$$= \sum_{n,m=1}^{\infty} A_{nm} \frac{ab}{4} \delta_{nk} \delta_{mq} \sinh(\gamma_{nm} c) = \frac{ab}{4} A_{kq} \sinh(\gamma_{kq} c),$$

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so that

$$A_{kq} = \frac{4}{ab\sinh(\gamma_{kq}c)} \int_0^a dx \int_0^b dy \sin(\alpha_k x)\sin(\beta_q y)\phi_0(x,y).$$

This gives the coefficients A_{nm} in terms of the function $\phi_0(x,y)$.



• *Cylindrical coordinates.* The separation of variables in the Laplace equation can also be performed in the cylindrical coordinates. In the cylindrical coordinates a given point is characterized by its coordinate z, the distance from the z-axis ρ and the angle φ which the projection onto the plane xy makes with the x-axis, see the plot. The relations with Cartesian coordinates are

$$\begin{aligned} x &= \rho\cos\varphi, \\ y &= \rho\sin\varphi. \end{aligned}$$

In order to write down the Laplace equation in these coordinates we need the expression for the Laplace operator. To find this expression we need the rewrite derivatives ∂_x and ∂_y in terms of the derivatives ∂_ρ and ∂_φ.

The simplest way to do it is by using differentials. We have

$$\begin{aligned} dx &= d\rho\cos\varphi - \rho\sin\varphi d\varphi, \\ dy &= d\rho\sin\varphi + \rho\cos\varphi d\varphi. \end{aligned}$$

These relations can be inverted to give

$$\begin{aligned} d\rho &= \cos\varphi dx + \sin\varphi dy \equiv \frac{\partial\rho}{\partial x}dx + \frac{\partial\rho}{\partial y}dy, \\ d\varphi &= \frac{1}{\rho}\sin\varphi dx - \frac{1}{\rho}\cos\varphi dy \equiv \frac{\partial\varphi}{\partial x}dx + \frac{\partial\varphi}{\partial y}dy. \end{aligned}$$

From these expressions one may read off the partial derivatives $\partial\rho/\partial x = \cos\varphi$ etc. Therefore,

$$\begin{aligned} \partial_x &= \frac{\partial\rho}{\partial x}\partial_\rho + \frac{\partial\varphi}{\partial x}\partial_\varphi = \cos\varphi\partial_\rho - \frac{1}{\rho}\sin\varphi\partial_\varphi, \\ \partial_y &= \frac{\partial\rho}{\partial y}\partial_\rho + \frac{\partial\varphi}{\partial y}\partial_\varphi = \sin\varphi\partial_\rho + \frac{1}{\rho}\cos\varphi\partial_\varphi. \end{aligned}$$

From these expressions one finds, after some calculations,

$$\partial_x^2 + \partial_y^2 = \partial_\rho^2 + \frac{1}{\rho}\partial_\rho + \frac{1}{\rho^2}\partial_\varphi^2 = \frac{1}{\rho}\partial_\rho(\rho\partial_\rho) + \frac{1}{\rho^2}\partial_\varphi^2.$$

Therefore, in cylindrical coordinates the Laplace equation reads

$$\frac{1}{\rho}\partial_\rho(\rho\partial_\rho\phi) + \frac{1}{\rho^2}\partial_\varphi^2\phi + \partial_z^2\phi = 0 \tag{2.48}$$

where the potential ϕ is a function of cylindrical coordinates, $\phi = \phi(\rho,\varphi,z)$.

In order to separate the variables in this equation, let us look again for

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the solutions in the form of a product of three functions

$$\phi(\rho, \varphi, z) = R(\rho)\Psi(\varphi)W(z).$$

When we substitute this into eq. (2.48) and divide by ϕ we get

$$\frac{1}{R\rho}\partial_\rho(\rho\partial_\rho R) + \frac{1}{\rho^2\Psi}\partial_\varphi^2\Psi + \frac{1}{W}\partial_z^2 W = 0 \tag{2.49}$$

The only term in this equation that depends on z is the last term. To satisfy the equation we must therefore require that this term is a constant. Let us choose this constant to be positive,

$$\frac{1}{W}\partial_z^2 W = k^2.$$

We get therefore the following equation for W,

$$\partial_z^2 W - k^2 W = 0, \tag{2.50}$$

which is the same as in the Cartesian case.

Once we satisfy the equation for W we have for the two remaining variables the following equation:

$$\frac{1}{R\rho}\partial_\rho(\rho\partial_\rho R) + \frac{1}{\rho^2\Psi}\partial_\varphi^2\Psi + k^2 = 0.$$

Consider now the dependence on φ. It only appears in the second term of this equation. In order to satisfy the equation we must require that the second term does not depend on φ, which gives

$$\frac{1}{\Psi}\partial_\varphi^2\Psi = -n^2.$$

Here we have chosen the negative constant. Unlike in the previous case where we could choose any sign, here have to impose an extra condition: the function $\Psi(\varphi)$ must be periodic in φ, since $\varphi = 0$ and $\varphi = 2\pi$ correspond to the same space point. We therefore cannot have exponential behavior, so the sign must be negative. Moreover, periodicity also requires that n is an *integer*. The solutions to this equation is, therefore, a linear combination

$$\Psi(\varphi) = A\sin(n\varphi) + B\cos(n\varphi).$$

For any A and B and integer n this function is periodic in φ.

When this solution is substituted back into eq. (2.49) we finally get a closed equation for R,

$$\frac{1}{R\rho}\partial_\rho(\rho\partial_\rho R) - \frac{n^2}{\rho^2} + k^2 = 0,$$

which we can rewrite in a more conventional way as follows,

$$\partial_\rho^2 R + \frac{1}{\rho}\partial_\rho R + \left(k^2 - \frac{n^2}{\rho^2}\right)R = 0.$$

• *Bessel equaiton.* First let us bring this equation into its canonical form.

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One may notice that the constant k can be rescaled out. Indeed, dividing this equation by k^2,

$$\frac{1}{k^2}\partial_\rho^2 R + \frac{1}{k\rho}\frac{1}{k}\partial_\rho R + \left(1 - \frac{n^2}{k^2\rho^2}\right) R = 0,$$

one may notice that k enters only in the combination $k\rho$. Defining a new variable $x = k\rho$ we get

$$\partial_x^2 R + \frac{1}{x}\partial_x R + \left(1 - \frac{n^2}{x^2}\right) R = 0. \tag{2.51}$$

This equation is known as the **Bessel equation**.

This is a second order linear differential equation, so it has two linearly independent solutions. To understand the behavior of these solutions consider two asymptotic cases: small $x \ll 1$ and large $x \gg 1$.

In the limit $x \ll 1$ the second term in the brackets is much larger than the first one,

$$\frac{n^2}{x^2} \gg 1.$$

If we neglect the small term the equation simplifies and can be solved explicitely. We have in this approximation

$$R'' + \frac{1}{x}R' - \frac{n^2}{x^2}R = 0,$$

where prime denotes the derivative with respect to x.

One may notice that the resulting equation is *homogeneous* in x — that is, every term scales as $1/x^2$. We therefore look for solutions in terms of a power of x,

$$R = x^\alpha,$$
$$R' = \alpha x^{\alpha-1},$$
$$R'' = \alpha(\alpha-1)x^{\alpha-2}.$$

Substituting this into the equation we get an algebraic (quadratic) equation for α,

$$\alpha(\alpha-1) + \alpha - n^2 = 0.$$

The solutions are obviously

$$\alpha = \pm n.$$

So, in the limit of small x the solutions to the Bessel equation behave like

$$R \propto x^n \qquad \text{or} \qquad R \propto \frac{1}{x^n}.$$

The first solution is regular at $x \to 0$, the second one diverges in this limit.

Consider now the asymptotic behavior at large $x \gg 1$. Before making approximations let us make a change of variables in order to get rid of the first derivative. To this end let us make a substitution

$$R = x^\beta \xi(x),$$

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$$R' = \beta x^{\beta-1}\xi + x^\beta \xi',$$
$$R'' = \beta(\beta-1)x^{\beta-2}\xi + 2\beta x^{\beta-1}\xi' + x^\beta \xi''.$$

When we substitute this into the Bessel equation we see that the term proportional to ξ' is

$$(2\beta+1)x^{\beta-1}\xi'.$$

So, if we chose $\beta = -1/2$, the resulting equation will not contain the first derivative ξ'. The equation for ξ then reads

$$\xi'' + \left(1 - \frac{n^2 - 1/4}{x^2}\right)\xi = 0. \tag{2.52}$$

This equation is still equivalent to the Bessel equation, but is more convenient to make an approximation at large x. Indeed, in this limit the second term in the brackets can be neglected, and we get a very simple equation

$$\xi'' + \xi = 0, \tag{2.53}$$

which has solutions

$$\xi = \sin x \qquad \text{or} \qquad \xi = \cos x.$$

Now recalling that for $\beta = -1/2$ we have $R = \xi/\sqrt{x}$, we finally find the large-x asymptotics

$$R = \frac{1}{\sqrt{x}}\sin x \qquad \text{or} \qquad R = \frac{1}{\sqrt{x}}\cos x. \tag{2.54}$$

As a by-product of this exercise we notice that if we consider eq. (2.51) from a mathematical point of view we do not have to assume that n is integer. For $n = 1/2$ we see that the Bessel equation can be solved *exactly*. In this case eq. (2.52) simplifies to eq. (2.53) without any approximations, so the solutions (2.54) are exact solutions.

At $x \sim 1$ the solutions of the Bessel equation cannot be expressed in terms of elementary functions. These solutions are called **Bessel functions**. The solution which is regular at the origin $x \to 0$ (the one that behaves like x^n) is denoted $J_n(x)$; the one that is singular is denoted $N_n(x)$ and is sometimes called Neumann function. These functions are defined for a general n, not necessarily integer. In particular,

$$\begin{aligned} J_{1/2}(x) &= \sqrt{\frac{2}{\pi x}}\sin x, \\ N_{1/2}(x) &= -\sqrt{\frac{2}{\pi x}}\cos x. \end{aligned} \tag{2.55}$$

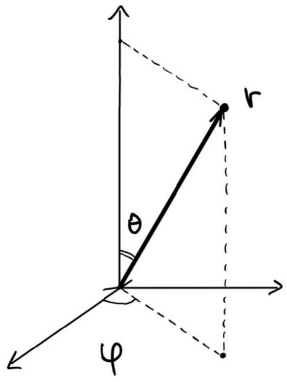
Coming back to our problem of solving the Laplace equation, we can write the solutions which are regular at the origin as the product $R\Psi W$,

$$\phi_{kn}(\rho,\varphi,z) = R\Psi W \propto J_n(k\rho)\mathrm{e}^{\pm in\varphi}\mathrm{e}^{\pm kz}.$$

These products solve the Laplace equation for any choice of k, any integer n and any choice of signs in the exponentials. The boundary conditions must be satisfied by choosing an appropriate linear combination of all these solutions, similarly to what we have done in the Cartesian case.

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• *Spherical coordinates.* The last coordinate system which we will consider and in which the variables may be separated is spherical coordinates. In spherical coordinates the space points are characterized by their distance from the center r and two angles θ and φ. The correspondence with Cartesian coordinates is as follows:



$$\begin{aligned} x &= r\sin\theta\cos\varphi, \\ y &= r\sin\theta\sin\varphi, \\ z &= r\cos\theta. \end{aligned} \tag{2.56}$$

In these coordinates the Laplace operator has the form

$$\Delta = \frac{1}{r^2}\partial_r r^2\partial_r + \frac{1}{r^2\sin\theta}\partial_\theta\sin\theta\partial_\theta + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2. \tag{2.57}$$

Correspondingly, the Laplace equation reads

$$\frac{1}{r^2}\partial_r(r^2\partial_r\phi) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta\phi) + \frac{1}{r^2\sin^2\theta}\partial_\varphi^2\phi = 0. \tag{2.58}$$

Following the general logic of the method, let us look for solutions in the following form,

$$\phi(r,\theta,\varphi) = R(r)P(\theta)\Phi(\varphi).$$

Substituting this into eq. (2.58) and dividing by ϕ we get

$$\frac{1}{r^2R}\partial_r(r^2\partial_r R) + \frac{1}{r^2\sin\theta}\frac{1}{P}\partial_\theta(\sin\theta\partial_\theta P) + \frac{1}{r^2\sin^2\theta}\frac{1}{\Phi}\partial_\varphi^2\Phi = 0. \tag{2.59}$$

Let us first look at the φ-dependent third term. Since this is the only term that contains φ, it must not depend on it. Thus, we must have

$$\frac{1}{\Phi}\partial_\varphi^2\Phi = -m^2.$$

Similarly to the cylindrical coordinates, the periodicity of the solution in φ implies that the constant must be negative and m must be integer. From this equation we find two linearly independent solutions

$$\Phi(\varphi) = \mathrm{e}^{\pm im\varphi}.$$

Eq. (2.59) now becomes

$$\frac{1}{r^2R}\partial_r(r^2\partial_r R) + \frac{1}{r^2\sin\theta}\frac{1}{P}\partial_\theta(\sin\theta\partial_\theta P) - \frac{m^2}{r^2\sin^2\theta} =$$

$$= \frac{1}{r^2R}\partial_r(r^2\partial_r R) + \frac{1}{r^2}\left[\frac{1}{\sin\theta}\frac{1}{P}\partial_\theta(\sin\theta\partial_\theta P) - \frac{m^2}{\sin^2\theta}\right] = 0. \tag{2.60}$$

Only the last two terms depend on θ, and this dependence comes through an overall factor in square brackets that does not contain other variables. Therefore, this factor must be a constant. The variable θ is also an angular variable, so regularity of the solution at $\theta = 0$ and $\theta = \pi$ imposes constraints on this constant. One can show that this constant must have the form $-l(l+1)$ where l is an integer $l \geq |m|$. We have therefore the

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following equation for $P(\theta)$,

$$\frac{1}{\sin\theta}\partial_\theta(\sin\theta\partial_\theta P) + \left[l(l+1) - \frac{m^2}{\sin^2\theta}\right]P = 0. \tag{2.61}$$

Finally, when the square bracket in eq. (2.60) is replaced by $-l(l+1)$ we get the equation for R,

$$\frac{1}{r^2}\partial_r(r^2\partial_r R) - \frac{l(l+1)}{r^2}R = 0.$$

We have therefore separated the variables, as now we have three ordinary differential equations instead of the original partial differential equation.

The equation for R is easy to solve as it is homogeneous in r. We look for the solutions in the form $R = r^\alpha$. When this is substituted into the equation for R we get a quadratic equation for α,

$$\alpha(\alpha+1) - l(l+1) = 0.$$

The two solutions are

$$\begin{aligned}\alpha &= l,\\ \alpha &= -l-1.\end{aligned}$$

Therefore,

$$R = r^l \qquad \text{or} \qquad R = r^{-l-1}.$$

As in the cylindrical case one solution is regular at $r \to 0$, and another one is singular.

• *Legendre polynomials.* Finally, consider eq. (2.61). This equation is brought to the standard form by the change of variables

$$x = \cos\theta$$

which implies

$$\begin{aligned}\sin^2\theta &= 1 - x^2,\\ \partial_x &= -\frac{1}{\sin\theta}\partial_\theta.\end{aligned}$$

In terms of the variable x eq. (2.61) becomes

$$\frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right] + \left[l(l+1) - \frac{m^2}{1-x^2}\right]P = 0. \tag{2.62}$$

This is **generalized Legendre equation**. For the case $m = 0$ it reduces to the ordinary **Legendre equation**

$$\frac{d}{dx}\left[(1-x^2)\frac{dP}{dx}\right] + l(l+1)P = 0. \tag{2.63}$$

We are interested in solutions that are finite and continuous in the interval $-1 \le x \le 1$ corresponding to $0 \le \theta \le \pi$. The solutions to eq. (2.63) having these properties are called **Legendre polynomials**. Explicitely

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they are

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \frac{1}{2}(3x^2-1) \\ P_3(x) &= \frac{1}{2}(5x^3-3x) \\ &\dots \\ P_l(x) &= \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l. \end{aligned}$$

Note that $P_l(x)$ is a polynomial of degree l.

The solutions to the generalized Legendre equation (2.62) with the required properties exist for $m = \pm 1, \mp 2, \pm 3, \dots \pm l$; they are called **associated Legendre functions** $P_l^m(x)$. They can be written in the general case as follows,

$$P_l^m(x) = \frac{(-1)^m}{2^l l!}(1-x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l.$$

The solutions for Φ and P are labeled by the two discrete indices l and m. They are usually combined into a single function $Y_{lm}(\theta,\varphi)$ called **spherical harmonics**. Explicitely, these functions are defined as follows:

$$\begin{aligned} Y_{lm}(\theta,\varphi) &= \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}}P_l^m(\cos\theta)\mathrm{e}^{im\varphi} \qquad (2.64) \\ Y_{l,-m} &= (-1)^m Y_{lm}^*. \end{aligned}$$

The few lowest spherical harmonics are

$$\begin{aligned} &\underline{l=0:} \\ &\qquad Y_{00} = \frac{1}{\sqrt{4\pi}} \\ &\underline{l=1:} \\ &\qquad Y_{10} = \frac{3}{\sqrt{4\pi}}\cos\theta \\ &\qquad Y_{11} = -\frac{3}{\sqrt{8\pi}}\sin\theta\mathrm{e}^{i\varphi} \\ &\qquad Y_{1,-1} = \frac{3}{\sqrt{8\pi}}\sin\theta\mathrm{e}^{-i\varphi} \\ &\dots \end{aligned}$$

These functions are mutually orthogonal and normalized to 1 when integrated over the unit sphere.

For any natural $l = 0, 1, 2, \dots$ and integer $m = 0, \mp 1, \pm 2, \dots, \pm l$ the products

$$r^l Y_{lm}(\theta,\varphi) \qquad \text{and} \qquad \frac{1}{r^{l+1}}Y_{lm}(\theta,\varphi) \qquad (2.65)$$

are solutions to the Laplace equation. The first are regular at the origin $r = 0$, the second are singular. Conversely, the first are growing at $r \to \infty$ while the second are decaying.

There is a correspondence between the decaying solutions (2.65) and terms in the multipole expansion constructed in Sect. 2.1. The electrostatic potential outside of the localized system of charges satisfies the Laplace equation and decays at infinity. Since in this expansion each power of r appears only once, it is clear that *each term individually* must satisfy the Laplace equation. Each multipole contribution, therefore, must be a linear combination of the decaying solutions (2.65) with the corresponding power of r. This is indeed the case. The monopole term Q/r is the same as the solution with $l = 0$ which is $1/r$ up to a normalization constant.

The dipole term (2.26) is proportional to $1/r^2$,

$$\phi_1 \propto \frac{(\boldsymbol{dn})}{r^2},$$

where $\boldsymbol{n}$ is a unit radial vector with the components which one can read from eq. (2.56),

$$\boldsymbol{n} = (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta).$$

So, ϕ_1 is a linear combination of three functions: n_1/r^2, n_2/r^2 and n_3/r^2. These functions are, up to a normalization, the three solutions with $l = 1$

$$\begin{aligned}
\frac{n_1}{r^2} &= \frac{\sin\theta\cos\varphi}{r^2} \propto \frac{Y_{11} - Y_{1,-1}}{r^2}, \\
\frac{n_2}{r^2} &= \frac{\sin\theta\sin\varphi}{r^2} \propto \frac{Y_{11} + Y_{1,-1}}{ir^2}, \\
\frac{n_3}{r^2} &= \frac{\cos\theta}{r^2} \propto \frac{Y_{10}}{r^2}.
\end{aligned}$$

In the next order, the quadrupole contribution ϕ_2 is a combination of 5 functions composed of the products $n_i n_j$ divided by r^3,

$$\phi_2 \propto D_{ij}\frac{n_i n_j}{r^3}.$$

These 5 functions (recall that D_{ij} is a symmetric traceless matrix which has 5 independent components) can be represented as linear combinations of the 5 solutions with $l = 2$, namely Y_{2m}/r^3, $m = 0, \pm1, \pm2$. And similarly for all higher multipoles.

2.3. Magnetostatics

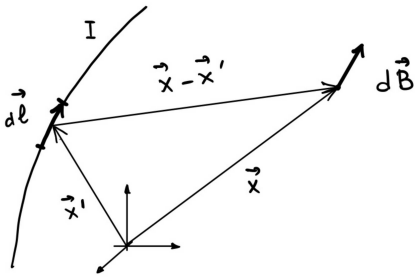
Biot-Savart law

• Magnetic fields are created by electric currents wich in turn are produced by moving charges. Clearly, currents cannot be point-like; the simplest current is a linear current — that is, a current along a line (one may imagine an infinitely-thin wire). Such currents are characterized by a single quantity I which is the charge passing in a unit time across a section of the wire, $I = dQ/dt$. In a stationary case conservation of the electric charge implies that I is constant along the wire.

The magnetic field produced by a current gets contributions from all parts of the wire; these contributions sum as vectors. A contribution of an

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infinitesimal piece of the wire $d\boldsymbol{l}$ located at $\boldsymbol{x}'$ and carrying the current I is given by the **Biot-Savart law**,



$$d\boldsymbol{B}(x) = \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{l} \times (\boldsymbol{x} - \boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|^3}, \tag{2.66}$$

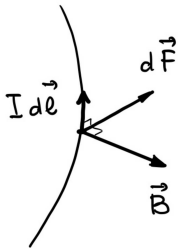
where the constant is

$$\frac{\mu_0}{4\pi} = 10^{-7} \frac{\mathrm{N}}{\mathrm{A}^2}.$$

This is the analog of the Coulomb law of electrostatics. The unit of the magnetic field is **Tesla** (T) or **Gauss** (G = 10^{-4}T),

$$\mathrm{T} = \frac{\mathrm{N}}{\mathrm{A} \cdot \mathrm{m}}.$$

Note that the magnetic field of a current element $d\boldsymbol{l}$ decreases as $1/x^2$ with the distance x from the current element, similarly to the electric field of a point charge.

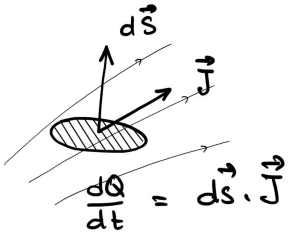


• An element of a current $Id\boldsymbol{l}$ in a magnetic field $\boldsymbol{B}$ is subject to a force

$$d\boldsymbol{F} = Id\boldsymbol{l} \times \boldsymbol{B}. \tag{2.67}$$

This force is perpendicular to both the current and the magnetic field.

Continuous current distribution



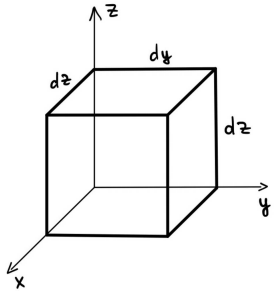
• *Current density.* In many realistic applications currents cannot be approximated by infinitely thin lines. Currents that are distributed over space can be characterized by the **current density** $\boldsymbol{J}(\boldsymbol{x})$. This is a *vector* whose direction is the direction of the current and the magitude is the *current per unit area* perpendicular to $\boldsymbol{J}$. It follows from this definition that the current dI passing through an infinitesimal area $d\boldsymbol{s}$ is given by the scalar product

$$dI = d\boldsymbol{s}\, \boldsymbol{J}.$$

The units of the current density are A/m^2.

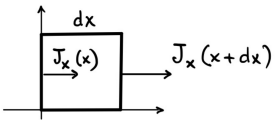
• *Conservation equation*. In general, the current density may depend on both the spatial coordinates $\boldsymbol{x}$ and time. This dependence, however, cannot be completely arbitrary because of the conservation of electric charge: charges can move around, but they can neither be created nor destroyed. Let us derive the equation that expresses this property mathematically.

Consider an infinitesimal rectangular volume at the point $\boldsymbol{x}$ of the size $dx \times dy \times dz$, see the plot. The charge inside this volume is $\rho(x)dxdydz$. By the charge conservation, its change over time must be the difference between the charges flowing in and out of the volume. These latter are given by the currents through the sides of the box. We have, therefore:



$$\begin{aligned} dxdydz \frac{\partial \rho}{\partial t} &= dydz[J_x(x) - J_x(x+dx)] \\ &+ dxdz[J_y(y) - J_y(y+dy)] \\ &+ dxdy[J_z(z) - J_z(z+dz)]. \end{aligned}$$

Note that in this expression we can neglect the variation of J_x over the side of the box because this variation is of a higher order in differentials,



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and similarly for other components. Therefore, in the expression below we only show one argument in each case. Expanding to the leading order in differentials we get

$$dxdydz\frac{\partial\rho}{\partial t} = -dydz\frac{\partial J_x}{\partial x}dx - dxdz\frac{\partial J_y}{\partial y}dy - dxdy\frac{\partial J_z}{\partial z}dz$$

or, simplifying by the volume of the box,

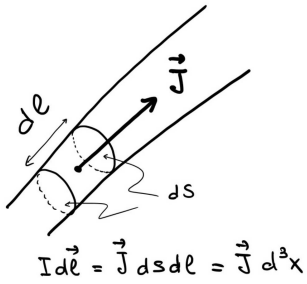
$$\frac{\partial\rho}{\partial t} + \nabla\boldsymbol{J} = 0. \tag{2.68}$$

This is known as the **continuity equation**. In the stationary case when all time dependencies vanish, this equaiton reduces to

$$\nabla\boldsymbol{J} = 0. \tag{2.69}$$

Because of this equation, the statioanry current distribution can be represented by lines of current which never cross nor end.

• The Biot-Savart law can be generalized to the case of continuous current distribution as follows. Such a distribution can be thought of as consisting of thin tubes of current, such that together they fill the whole space without gaps, do not intersect and have a constant current along each tube. An infinitesimal piece of such a tube is a cylinder of the section ds and the length dl, aligned with the local direction of the current $\boldsymbol{J}(\boldsymbol{x})$. The current flowing in the tube is $I = Jds$. The contribution of such a piece of tube into the magnetic field in the Biot-Savart law is expressed in terms of $Id\boldsymbol{l}$ which can be written as



$$Id\boldsymbol{l} = \boldsymbol{J}dsdl = \boldsymbol{J}d^3x.$$

Here we have used the fact that $d\boldsymbol{l}$ and $\boldsymbol{J}$ are aligned. We see, therefore, that summation over individual pieces of all the current tubes amounts to integration over the whole space once we replace

$$Id\boldsymbol{l} \to \boldsymbol{J}d^3x$$

in the Biot-Savart law. Thus, the magnetic field created by a continuous current distribution $\boldsymbol{J}$ is

$$\boldsymbol{B}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\int d^3x'\boldsymbol{J}(\boldsymbol{x}')\times\frac{(\boldsymbol{x}-\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|^3}. \tag{2.70}$$

• This equation may be rewritten is a different way by noting, as in the case of Coulomb law, that

$$\frac{\boldsymbol{x}-\boldsymbol{x}'}{|\boldsymbol{x}-\boldsymbol{x}'|^3} = -\nabla\frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|}.$$

As the gradient acts on the coordinates $\boldsymbol{x}$ and has nothing to do with the integration, we can take it outside of the integral. Then

$$\boldsymbol{B}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\nabla\times\int d^3x'\frac{\boldsymbol{J}(\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|}. \tag{2.71}$$

We see from this expression that the magnetic field is represented as

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a curl of a vector. We will come back to this fact shortly. For now we conclude that

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0. \tag{2.72}$$

This is the third Maxwell equation.

• To derive the fourth Maxwell equaiton let us calculate the curl of the magnetic field. In components:

$$(\boldsymbol{\nabla} \times \boldsymbol{B})_i = \epsilon_{ijk}\partial_j \left[\frac{\mu_0}{4\pi} \epsilon_{kpq} \partial_p \int d^3x' \frac{J_q(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} \right] =$$

$$= \frac{\mu_0}{4\pi} (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}) \partial_j \partial_p \int d^3x' \frac{J_q(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} =$$

$$= \frac{\mu_0}{4\pi} \partial_i \partial_q \int d^3x' \frac{J_q(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} - \frac{\mu_0}{4\pi} \partial_j \partial_j \int d^3x' \frac{J_i(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|}.$$

Consider the second term. The derivatives act on $\boldsymbol{x}$. Taking them inside the integral we have

$$\partial_j \partial_j \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} = \Delta \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} = -4\pi\delta^3(\boldsymbol{x} - \boldsymbol{x}').$$

The factors (-4π) cancel and the second term becomes

$$\mu_0 \int d^3x' J_i(\boldsymbol{x}') \delta^3(\boldsymbol{x} - \boldsymbol{x}') = \mu_0 J_i(\boldsymbol{x}).$$

The first term is in fact zero. To see this let us transform it as follows:

$$\frac{\mu_0}{4\pi} \partial_i \int d^3x' J_q(\boldsymbol{x}') \partial_q \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} = -\frac{\mu_0}{4\pi} \partial_i \int d^3x' J_q(\boldsymbol{x}') \partial'_q \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|}$$

where ∂'_q is the derivative with respect to $\boldsymbol{x}'$. The advantage of having this derivative is that now we can integrate by parts. At large distances the current distribution $J_i(\boldsymbol{x})$ is zero, so we can drop the boundary term. We get therefore

$$\frac{\mu_0}{4\pi} \partial_i \int d^3x' [\partial'_q J_q(\boldsymbol{x}')] \frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|}.$$

In the stationary case when the charge density does not depend on time, the current has zero divergence $\partial_i J_i = 0$. This term is therefore zero. We have finally

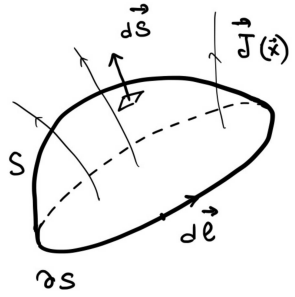
$$(\boldsymbol{\nabla} \times \boldsymbol{B})_i = \mu_0 J_i(\boldsymbol{x}),$$

or in vector notations

$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0 \boldsymbol{J}. \tag{2.73}$$

This is the fourth Maxwell eqaution. Note that to derive it we have explicitely used the equation $\boldsymbol{\nabla J} = 0$ which is only valid in the static case.

Ampere's law.



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• Consider a surface S with the border ∂S, and let us inregrate eq. (2.73) over this surface,

$$\int_S d\boldsymbol{s}\boldsymbol{\nabla}\times\boldsymbol{B} = \mu_0 \int_S d\boldsymbol{s}\boldsymbol{J}.$$

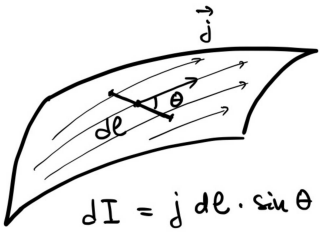
On the left we may use the Stokes theorem and transform the integral over the surface into an integral over the boundary of the surface,

$$\int_{\partial S} d\boldsymbol{l}\boldsymbol{B} = \mu_0 \int_S d\boldsymbol{s}\boldsymbol{J}. \tag{2.74}$$

This relation is known as the Ampere's law. The quantity in the l.h.s. is called the *circulation* of the field $\boldsymbol{B}$ along the contour ∂S. On the right, the integral is the total current flowing through the contour ∂S.

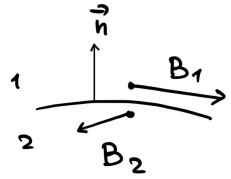
Continuity conditions

• Simlar to the electrostatic case, the equations of magnetostatics (2.72) and (2.73) determine the continuity conditions of the field $\boldsymbol{B}$ across the interface of two media. Such an interface may carry a current that is localized at the surface, called the *surface current*. It is characterized by the *surface current density* $\boldsymbol{j}$. Similarly to the volume current density, the direction of $\boldsymbol{j}$ shows the direction of the current while the modulus $j = |\boldsymbol{j}|$ characterizes its intensity: the total current dI passing across an infinitesimal line element dl lying in the surface and perpendicular to $\boldsymbol{j}$ is $dI = jdl$. For the element $d\boldsymbol{l}$ making an angle θ with the local direction of $\boldsymbol{j}$ (see the plot) the current passing through it is



$$dI = jdl\sin\theta.$$

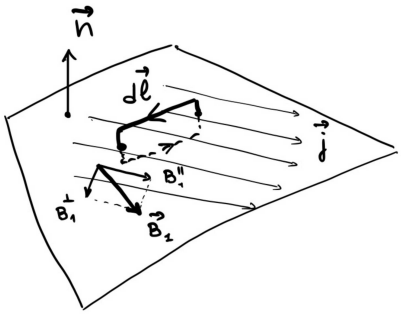
• Consider first the condition that follows from eq. (2.72). This equation is the same as the electrostatic equation in the absence of charges. Therefore, the same arguments that were used to derive the continuity condition for the electric field give the condition



$$\boldsymbol{n}(\boldsymbol{B}_1 - \boldsymbol{B}_2) = 0. \tag{2.75}$$

Here $\boldsymbol{n}$ is the unit vector normal to the interface, and $\boldsymbol{B}_1$ and $\boldsymbol{B}_1$ are magnetic fields on the two sides of it, see the plot. Thus, the component of the magnetic field perpendicular to the interface is continuous across the interface.

• In order to derive conditions on the two components parallel to the surface consider eq. (2.73). Following the same method as in the electrostatic case, let us choose the infinitesimal contour of the length dl that is half on one side of the surface, half on the other side. We now apply the Ampere's law to this contour,



$$\int d\boldsymbol{l}\boldsymbol{B} = \mu_0 \int d\boldsymbol{s}\boldsymbol{j}.$$

On the right we just have the total current flowing through the contour wich equals $jdl\sin\theta$. On the left we neglect the vertical side of the

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contour and get $d\boldsymbol{l}(\boldsymbol{B}_1 - \boldsymbol{B}_2)$. Thus, using the definition of the surface current we get the equation

$$d\boldsymbol{l}(\boldsymbol{B}_1 - \boldsymbol{B}_2) = \mu_0 j dl \sin\theta.$$

Let us decompose the magnetic field parallel to the surface into the sum of two components: $\boldsymbol{B}^{\parallel}$ which is parallel to the current $\boldsymbol{j}$, and $\boldsymbol{B}^{\perp}$ which is perpendicular to it, both component beign parallel to the surface.

Consider the two cases separately. For the parallel component we can choose $d\boldsymbol{l}$ parallel to $\boldsymbol{j}$ and $\boldsymbol{B}^{\parallel}$. Then on the right we will have just the product $dl(B_1^{\parallel} - B_2^{\parallel})$, while on the left we will have zero because $\sin\theta = 0$. Thus,

$$B_1^{\parallel} - B_2^{\parallel} = 0$$

i.e. the parallel component is continuous.

Second, consider the perpendicular component $\boldsymbol{B}^{\perp}$. In this case we choose $d\boldsymbol{l}$ parallel to $\boldsymbol{B}^{\perp}$ and perpendicular to $\boldsymbol{j}$. Then $\sin\theta = 1$ and we have

$$B_1^{\perp} - B_2^{\perp} = \mu_0 j.$$

These two conditions can be written together as a single vector equation:

$$\boldsymbol{n} \times (\boldsymbol{B}_1 - \boldsymbol{B}_2) = \mu_0 \boldsymbol{j}. \tag{2.76}$$

This gives the continuity condition for the components parallel to the interface: the one parallel to the current is continuous, the one perpendicular to it has a discontinuoty at the surface.

Vector potential

• The fact that $\boldsymbol{B}$ has zero divergence implies that it can be written as a curl of some vector field,

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}. \tag{2.77}$$

The field $\boldsymbol{A}$ is called **vector potential**. We can read off the vector potential $\boldsymbol{A}$ from eq. (2.71). However, there is a subtlety: equation (2.77) does not fix $\boldsymbol{A}$ uniquely. There are many different vector potentials that lead to the same magnetic field $\boldsymbol{B}$. Indeed, a gradient of an arbitrary scalar function has zero curl, so it can be added to $\boldsymbol{A}$ without changing the magnetic field. Thus, form eq. (2.71) we have

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\boldsymbol{J}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} + \nabla\Psi(\boldsymbol{x}).$$

Here the last term containing an arbitrary function $\Psi(\boldsymbol{x})$ reflects the freedom in the choice of $\boldsymbol{A}$. This freedom is called **gauge freedom**. Changing $\boldsymbol{A}$ by a gradient of a scalar function,

$$\boldsymbol{A}(\boldsymbol{x}) \to \boldsymbol{A}(\boldsymbol{x}) + \nabla\Psi(\boldsymbol{x})$$

is called the **gauge transformation**.

• Let us derive a closed equaiton for the vector potential. If we express the magnetic field in terms of the vector potential by means of eq. (2.77), the first equation of magnetostatics is automatically satisfied. Therefore, the

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equation for $\boldsymbol{A}$ must come from the second equation of magnetostatics (2.73). Substituting eq. (2.77) into this equation and using the identity

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \boldsymbol{A}) - \Delta \boldsymbol{A}$$

we find

$$\boldsymbol{\nabla}(\boldsymbol{\nabla} \boldsymbol{A}) - \Delta \boldsymbol{A} = \mu_0 \boldsymbol{J}(\boldsymbol{x}).$$

Let us now use the gauge freedom to set $\boldsymbol{\nabla} \boldsymbol{A} = 0$. This is called the *Coulomb gauge*. Note that we have one arbitrary function, so that we can satisfy one scalar condition.

With this condition the eqaution for $\boldsymbol{A}$ becomes

$$-\Delta \boldsymbol{A} = \mu_0 \boldsymbol{J}(\boldsymbol{x}). \tag{2.78}$$

We see that in this gauge all the components of the vector potential independently satisfy the Poisson equation.

We have seen this equation before — this is the same Poisson equation that is satisfied by the scalar potential ϕ, eq. (2.15), except for a different source term. By analogy, we may write the solution,

$$A(\boldsymbol{x}) = \frac{\mu_0}{4\pi} \int d^3x' \frac{\boldsymbol{J}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|}. \tag{2.79}$$

This is the same expression as the one following from eqs. (2.71) and (2.77).

Multipole expansion

• In many physical problems one needs to find a magnetic field of a compact distribution of currents at large distances from it. In this case one may develop an approximation similar to the one in the electrostatic case (cf. sect. 2.1), expanding in inverse powers of the distance. More precisely, the expansion is in powers of the small parameter d/x, where d is the characteristic size of the current distribution, and $x = |\boldsymbol{x}|$ is the distance from it to the observation point, see Fig. 5.2.

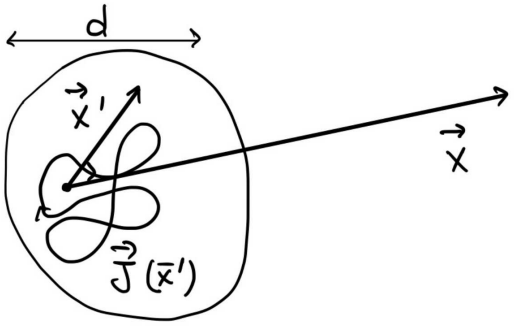


Figure 2.11.: Localized system of currents $\boldsymbol{J}(\boldsymbol{x}')$. We want to determine the magnetic field $\boldsymbol{B}(\boldsymbol{x})$ in the point $\boldsymbol{x}$ such that $|\boldsymbol{x}| \gg |\boldsymbol{x}'| \sim d$.

Like in the electrostatic case we will first calculate the vector potential $\boldsymbol{A}(\boldsymbol{x})$ from eq. (2.79) and then determine the magnetic field from eq. (2.77). To build the expansion we use eq. (2.22) in which this time we keep only

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two first terms,

$$\frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|} = \frac{1}{x} + \frac{(\boldsymbol{x}\boldsymbol{x}')}{x^3} + \dots$$

Upon substitution into eq. (2.79) this gives

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\int d^3x'\boldsymbol{J}(\boldsymbol{x}')\left\{\frac{1}{x} + \frac{(\boldsymbol{x}\boldsymbol{x}')}{x^3} + \dots\right\} =$$

$$= \frac{\mu_0}{4\pi}\left\{\frac{1}{x}\int d^3x'\boldsymbol{J}(\boldsymbol{x}') + \frac{x_j}{x^3}\int d^3x'\boldsymbol{J}(\boldsymbol{x}')x'_j + \dots\right\} \tag{2.80}$$

where we have taken out of the integral all the x'-independent factors. Note that each subsequent term is suppressed by an extra power of $1/x$.

Consider the first integral, and let us show that it is zero. The i-th component of the current can be written as

$$J_i(\boldsymbol{x}') = J_k(\boldsymbol{x}')\partial'_k x'_i$$

because $\partial'_k x'_i = \delta_{ki}$. Substituting it into the first integral we have

$$\int d^3x' J_i(\boldsymbol{x}') = \int d^3x' J_k(\boldsymbol{x}')\partial'_k x'_i = -\int d^3x'\partial'_k J_k(\boldsymbol{x}')x'_i =$$

$$= -\int d^3x'\boldsymbol{\nabla}'\boldsymbol{J}(\boldsymbol{x}')x'_i.$$

When integrating by parts we have set to zero the surface term because for a localized distribution the current vanishes outside of the system. As we have discussed before, for a stationary distribution of currents the divergence of the current is zero, cf. eq. (2.69). Therefore,

$$\int d^3x'\boldsymbol{J}(\boldsymbol{x}') = 0,$$

and the leading term in the vector potential vanishes.

Consider now the second term. In component notations the integral is

$$\int d^3x' J_i(\boldsymbol{x}')x'_j.$$

A similar trick as before may be used to show that this expression is in fact antisymemtric in its indices. Indeed, consider an identity

$$\partial_k(J_k x_i x_j) = \delta_{ki}J_k x_j + \delta_{kj}J_k x_i + x_i x_j \underbrace{\partial_k J_k}_{=0} = J_i x_j + J_j x_i$$

Integrating this equation we get on the l.h.s. by the Stokes theorem:

$$\int d^3x\partial_k(J_k(\boldsymbol{x})x_i x_j) = \int_{\partial V} ds_k J_k(\boldsymbol{x})x_i x_j = 0$$

because the current is zero outside of the localized system. Therefore,

$$0 = \int d^3x J_j(\boldsymbol{x})x_i + \int d^3x J_i(\boldsymbol{x})x_j,$$

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that is, the integral is antisymmetric with respect to i and j.

Making use of this antisymmetry, we can transform eq. (2.80) as follows. Since the first term is zero, let us start with the integral in the second term. In components,

$$\int d^3x' J_i(\boldsymbol{x}')x'_j = \frac{1}{2}\int d^3x' \left(J_i(\boldsymbol{x}')x'_j - J_j(\boldsymbol{x}')x'_i =\right)$$

$$= \frac{1}{2}\epsilon_{ijk}\epsilon_{kpq}\int d^3x' J_p(\boldsymbol{x}')x'_q.$$

The last equality is proven by rewriting the product of two ϵ-symbols as a combination of deltas. One may recognize the double vector product in this expression. Let us define the vector $\boldsymbol{m}$ as follows,

$$m_k = -\frac{1}{2}\epsilon_{kpq}\int d^3x' J_p(\boldsymbol{x}')x'_q$$

or, equivalenty, in vector notations

$$\boldsymbol{m} = \frac{1}{2}\int d^3x' \boldsymbol{x}' \times \boldsymbol{J}(\boldsymbol{x}'). \tag{2.81}$$

This quantity is called the **magnetic dipole moment** of our system of currents. Then we have

$$\int d^3x' J_i(\boldsymbol{x}')x'_j = -\epsilon_{ijk}m_k.$$

When we substitute this into eq. (2.80), we get

$$A_i(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\frac{1}{x^3}\epsilon_{ikj}m_k x_j.$$

In vector notations this equation reads,

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\frac{\boldsymbol{m}\times\boldsymbol{x}}{x^3} + \dots, \tag{2.82}$$

where dots stand for terms that are suppressed by higher powers of $1/x$. If we compare this expansion with that for the scalar potential, eq. (2.29) we see that the first term (naturally called the "monopole term") is missing. There are no magnetic charges (or, equivalently, magnetic monopoles).

• *Magnetic field of a dipole.* The magnetic field at large distances can be found from eq. (2.77). The calculation is simpler in componemts:

$$B_i(\boldsymbol{x}) = \epsilon_{ijk}\partial_j A_k = \frac{\mu_0}{4\pi}\epsilon_{ijk}\epsilon_{kpq}m_p\partial_j\left(\frac{x_q}{x^3}\right) =$$

$$= \frac{\mu_0}{4\pi}(\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp})m_p\left(\frac{\delta_{jq}}{x^3} - 3x_q\frac{x_j}{x^5}\right)$$

$$= \frac{\mu_0}{4\pi}\left(m_i\frac{3}{x^3} - m_i\frac{3}{x^3} - m_i\frac{1}{x^3} + 3x_i\frac{(\boldsymbol{x}\boldsymbol{m})}{x^5}\right).$$

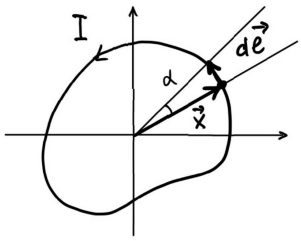
The first two terms cancel, while the last two are written in vector notations as follows:

$$\boldsymbol{B}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\frac{3\boldsymbol{x}(\boldsymbol{x}\boldsymbol{m}) - \boldsymbol{m}x^2}{x^5} + \dots, \tag{2.83}$$

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where dots denote higher order terms. It is interesting to compare this expression with the dipole contribution into the electric field, eq. (2.29). We see that, apart from the numerical coefficient and substitution of the electric dipole with the magnetic dipole, they are the same.

• *Magnetic dipole.* As is seen from eq. (2.81), the units of the magnetic dipole $\boldsymbol{m}$ are (current) × (area). This relation is more transparent in a particular case of a flat loop of current. Suppose we have a linear current I that makes a flat loop. In this case eq. (2.81) reduces to a linear integral



$$\boldsymbol{m} = \frac{I}{2}\oint \boldsymbol{x}\times d\boldsymbol{l}.$$

To evaluate this integral we choose the origin of coordinates somewhere inside the loop. For a given piece of the loop $d\boldsymbol{l}$ its contribution to the integral is

$$|\boldsymbol{x}\times d\boldsymbol{l}| = |\boldsymbol{x}|\,|d\boldsymbol{l}|\sin\alpha = 2\times(\text{area of the triangle}),$$

while the direction of this contribution is always along the z-axis. When we sum all these contributions the triangles sum into the **total area** A of the loop, so that we get the vector with the components

$$\boldsymbol{m} = (0,0,m)$$

where

$$m = I\,A.$$

This gives an idea how the magnetic dipole depends on the parameters of the localized system of currents.

• Magnetic moment of moving charges. Suppose we have a system of charges which move in a localized region like, for instance, electrons in an atom. The trajectory of the charge a is described by its time-dependent position vector $\boldsymbol{x}_a(t)$. The currents these charges produce are then $\boldsymbol{J}_a(\boldsymbol{x}) = q_a\boldsymbol{v}_a\delta(\boldsymbol{x}-\boldsymbol{x}_a(t))$, where $\boldsymbol{v}_a$ are the velocities of charges. The total current is therefore

$$\boldsymbol{J}(\boldsymbol{x}) = \sum_a q_a\boldsymbol{v}_a\delta(\boldsymbol{x}-\boldsymbol{x}_a(t)).$$

From eq. (2.81), the magnetic moment of this system is

$$\boldsymbol{m} = \frac{1}{2}\int d^3x'\boldsymbol{x}'\times\sum_a q_a\boldsymbol{v}_a\delta(\boldsymbol{x}-\boldsymbol{x}_a(t)) = \frac{1}{2}\sum_a q_a(\boldsymbol{x}_a\times\boldsymbol{v}_a).$$

If the charges are non-relativistic, $|\boldsymbol{v}_a| \ll c$, the last expression can be re-written in terms of the angular momentum. For one particle, the angular momentum is

$$\boldsymbol{L}_a = M_a\boldsymbol{x}_a\times\boldsymbol{v}_a,$$

where M_a is the mass of the a-th particle. Thus,

$$\boldsymbol{m} = \sum_a \frac{q_a}{2M_a}\boldsymbol{L}_a.$$

If all the moving particles have the same charge-to-mass ratio $q_a/M_a =$

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q/M, it can be taken out of the sum and we get

$$\boldsymbol{m} = \frac{q}{2M}\sum_a \boldsymbol{L}_a = \frac{q}{2M}\boldsymbol{L}_{\text{tot}}.$$

This is a well-known classical relation between the angular momentum and the magnetic moment. This relation is violated by the *intrinsic* magnetic moments of particles. For instance, for an electron the magnetic moment is almost exactly

$$\mu_B = \frac{e}{2m_e}\hbar,$$

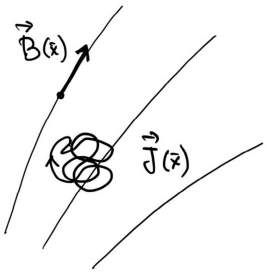
where $\hbar$ is the Planck constant, while its intrinsic angular momentum (spin) is $\hbar/2$. The ratio is then e/m_e.

Forces acting on a localized system of currents

• *Force.* If a localized system of currents is placed in an external magnetic field, it experiences a force. In the case when the magnetic field varies slowly with coordinates, that is when

$$\frac{d}{B}|\partial_i B_k| \ll 1, \tag{2.84}$$

where d is the size of the system, this force can be expressed (approximately) through the magnetic moment of the system.



Let us put the origin of coordinates somewhere in our system, and expand the magnetic field in Taylor series around this point. For the k-th component we have

$$B_k(\boldsymbol{x}) = B_k(0) + x_i\partial_i B_k(0) + \dots$$

Note that by the condition (2.84) the second term is much smaller than the first one, because $|\boldsymbol{x}| \lesssim d$.

For the continuous current distribution in a magnetic field the force (2.67) is obviously generalized as follows,

$$\boldsymbol{F} = \int d^3x \boldsymbol{J}(x)\times\boldsymbol{B}(x).$$

The i-th component of the force is thus

$$F_i = \epsilon_{ijk}\int d^3x J_j(\boldsymbol{x})\{B_k(0) + x_m\partial_m B_k(0) + \dots\}$$

In the first term we can take $B_k(0)$ outside of the integral. What remains is $\int d^3x J_j(\boldsymbol{x}) = 0$ as we have shown before. The first term is therefore zero.

In the second term we can again factor out the derivative of B so that we have

$$F_i = \epsilon_{ijk}\partial_m B_k(0)\int d^3x J_j(\boldsymbol{x})x_m + \dots$$

The integral in this expression we have also seen before while considering the multipole expanson. Following the same arguments we rewrite it as

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follows,

$$\int d^3x J_j(\boldsymbol{x}) x_m = \frac{1}{2}\int d^3x (J_j x_m - J_m x_j) =$$
$$= \epsilon_{jms}\epsilon_{spq}\frac{1}{2}\int d^3x J_p x_q = \epsilon_{mjs} m_s,$$

where we have used the definition (2.81) of the magnetic dipole moment.

Substituting this into expression for the force we find

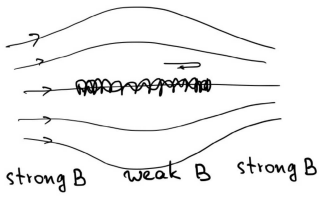
$$F_i = \epsilon_{ijk}\partial_m B_k \epsilon_{mjs} m_s = \partial_m B_k m_s(\delta_{im}\delta_{ks} - \delta_{is}\delta_{km}) = \partial_i B_k m_k$$

where the second term is zero because $\partial_k B_k = 0$. In vector form this equation reads

$$\boldsymbol{F} = \boldsymbol{\nabla}(\boldsymbol{mB}). \tag{2.85}$$

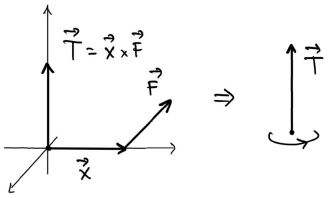
Thus, a magnetic dipole in a varying magnetic field experiences a force depending on its orientation with respect to the gradient of the field.

This force leads to an interesting phenomenon known as *magnetic trapping*. We will see later in Sect.?? that a moving charged particle makes circles around the magnetic field lines. Such a circular motion, when averaged over many periods, can be viewed as a circular current which produces a magnetic moment. This moment is *anti-aligned* with the magnetic field, $\boldsymbol{mB} < 0$. It is clear then from eq. (2.85) that the particle will be *repelled* from the regions of a stronger field. One may organize the magnetic field in such a way that the particle gets trapped.



• *Torque.* The external magnetic field also produces a *moment of force*, or torque, which tries to rotate the system of currents. In mechanics, the torque $\boldsymbol{T}$ is defined as

$$\boldsymbol{T} = \sum_a \boldsymbol{x}_a \times \boldsymbol{F}_a.$$



A small volume d^3x with a current density $\boldsymbol{J}(\boldsymbol{x})$ in a magnetic field is subject to the force

$$d\boldsymbol{F} = d^3x \boldsymbol{J} \times \boldsymbol{B}.$$

Therefore, its contribution to the torque is

$$d\boldsymbol{T} = \boldsymbol{x} \times (d^3x \boldsymbol{J} \times \boldsymbol{B}).$$

Integrating, we find

$$d\boldsymbol{T} = \int d^3x\, \boldsymbol{x} \times (\boldsymbol{J} \times \boldsymbol{B}).$$

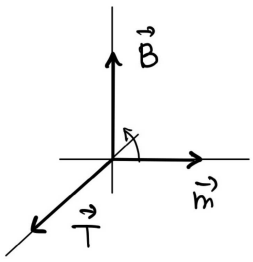
For the torque already the leading term in the Taylor expansion of $\boldsymbol{B}$ gives a non-zero contribution. Keeping only this term we have

$$T_i = \epsilon_{ijk}\epsilon_{kpq} B_q(0) \int d^3x\, x_j J_p(\boldsymbol{x}) = (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}) B_q \epsilon_{jps} m_s$$

where as before we have expressed the integral in terms of the magnetic dipole moment of the system. The second term does not contribute as $\delta_{jp}\epsilon_{jps} = 0$, so we get

$$T_i = \epsilon_{isq} m_s B_q$$

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which in the vector notations reads

$$\boldsymbol{T} = \boldsymbol{m} \times \boldsymbol{B}.$$

The torque is trying to *align* the directions of $\boldsymbol{m}$ and $\boldsymbol{B}$.

Faraday induction law

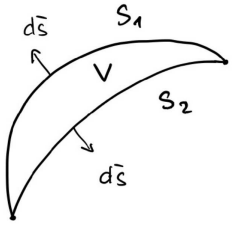
• Up to now we have considered static distributions of charges and currents. The equations that we have derived — two equations of electrostatics and two equations of magnetostatics — only apply in this case. When the charge or current distribution changes with time these equations may have to be modified. Some of these modifications follow from the *Faraday induction law*.

To formulate the Faraday induction law we first define the *flux* Φ of the magnetic field through a contour. Suppose we have some magnetic field in space, and consider a closed contour C. This contour defines a surface S of which it is a boundary. There are many such surfaces; consider any one of them. The magnetic flux through the contour is defined as

$$\Phi = \int_S d\boldsymbol{s}\boldsymbol{B}. \tag{2.86}$$

It is easy to see that the flux Φ does not depend on our choice of the surface. Indeed, suppose we have two surfaces S_1 and S_2, and corresponding fluxes Φ_1 and Φ_2. The difference $\Phi_1 - \Phi_2$ is the integral of the magnetic field through the closed surface $S_1 + S_2$,

$$\Phi_1 - \Phi_2 = \int_{S_1} d\boldsymbol{s}\boldsymbol{B} - \int_{S_2} d\boldsymbol{s}\boldsymbol{B} = \int_{S_1+S_2} d\boldsymbol{s}\boldsymbol{B} = \int_V d^3x \nabla\boldsymbol{B} = 0,$$



where V is the volume between surfaces S_1 and S_2, and we have used eq. (2.72). Note that when we integrate over $S_1 + S_2$ we have to change the sign of the normal to the surface as compared to the integral over S_2 in Φ_2.

The Faraday induction law states that the change of the magnetic flux through the contour C is related to the "electromotive force" $\mathscr{E}$ induced in the contour,

$$\mathscr{E} = -\frac{d\Phi}{dt}.$$

The "electromotive force" $\mathscr{E}$ is not actually a force; this is a work that one would have to do to move a unit charge around the contour C. Only electric field can do a work on the charge, so $\mathscr{E}$ must be a line integral of the electric field over the contour,

$$\mathscr{E} = \int_C \boldsymbol{E} d\boldsymbol{l}.$$

We therefore can write the Faraday law as follows,

$$\int_C \boldsymbol{E} d\boldsymbol{l} = -\frac{d}{dt} \int_S d\boldsymbol{s}\boldsymbol{B}.$$

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On the right we transform the integral using the Stokes theorem,

$$\int_C \boldsymbol{E} d\boldsymbol{l} = \int_S d\boldsymbol{s}(\boldsymbol{\nabla} \times \boldsymbol{E}).$$

With both terms on one side, the Faraday law becomes

$$\int_S d\boldsymbol{s} \left\{ (\boldsymbol{\nabla} \times \boldsymbol{E}) + \frac{\partial \boldsymbol{B}}{\partial t} \right\} = 0.$$

Finally, since the surface S is arbitrary, we conclude that

$$\boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = 0. \tag{2.87}$$

This equation replaces, in the case of time-dependent charge and current distributions, the second equation of electrostatics.

• *Energy of magnetic field.* Let us calculate the energy associated to the localized system of currents. We will assume that these currents are distributed in space and are described by the density of current $\boldsymbol{J}(\boldsymbol{x})$ which occupies some finite region. For the stationary current distribution the current $\boldsymbol{J}$ has zero divergence, $\boldsymbol{\nabla J} = 0$. This implies that the region occupied by the current can be decomposed into non-intersecting infinitesimally thin *current loops* that fill the whole region, such that the current flowing along each loop is constant along its length (the thickness of a given loop will *not* be, in general, constant along its length). One may view the distribution $\boldsymbol{J}(\boldsymbol{x})$ as a collection of such loops of current.

In order to calculate the energy of this system of currents suppose that we are assembling this distribution by gradually powering up these loops. For a given loop, raising its current costs no energy (if there is no resistance). However, the change of the current implies the change of magnetic field it creates, which in turn changes the magnetic flux through all loops. When the flux through a loop is changing, by Faraday law one needs to do a work to maintain the current. For a given loop with the current I, this work can be written as follows,

$$\delta w = I \delta \Phi,$$

where Φ is the magnetic flux through the loop.

To understand why assume the change of $\delta\Phi$ happens over the time δt. In this time the charge flowing through a given section of the loop is δQ such that $I = \delta Q/\delta t$. The current I is thus equivalent to taking the charge δQ along the whole loop in time δt. By Faraday law we can write $\delta\Phi = -\mathscr{E}\delta t$. By the definition of $\mathscr{E}$ the work on the system δW is

$$\delta w = -\mathscr{E}\delta Q = -\mathscr{E}\delta t \frac{\delta Q}{\delta t} = I\delta\Phi.$$

Consider one of these loops and let us rewrite this work in a different way,

$$\delta w = I \delta \Phi = I \int_S \delta \boldsymbol{B} d\boldsymbol{s},$$

where S is some surface bound by the loop. The change of magnetic field $\delta \boldsymbol{B}$ in turn can be written in terms of the change of the vector potential $\delta \boldsymbol{A}$,

$$\delta w = I \int_S d\boldsymbol{s}(\boldsymbol{\nabla} \times \delta \boldsymbol{A}) = I \int_{\partial S} d\boldsymbol{l} \delta \boldsymbol{A}.$$

In the last equality we have used the Stokes theorem, and the integral is now over the length of the loop. Finally, let us express I in terms of the current density $\boldsymbol{J}$. If we denote σ the cross section of the loop, then $I = J\sigma$. While I is constant along the loop, both $\boldsymbol{J}$ and σ change, so before

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using this equation we have to put I inside the integral,

$$\delta w = \int_{\partial S} d\boldsymbol{l} J \sigma \delta \boldsymbol{A} = \int_{\partial S} dl \sigma \boldsymbol{J} \delta \boldsymbol{A} = \int_{\text{loop volume}} d^3x \boldsymbol{J} \delta \boldsymbol{A},$$

because obviously $dl\ \sigma = d^3x$.

This gives a work associated with a single loop. But it is now trivial to sum contributions of all loops: it is sufficient to integrate over the whole space instead of the loop volume. Thus, the total work on all loops δW is

$$\delta W = \int d^3x \boldsymbol{J} \delta \boldsymbol{A}. \tag{2.88}$$

Let us transform this equation by making use of the second equation of magnetostatics (2.73),

$$\delta W = \frac{1}{\mu_0} \int d^3x \delta \boldsymbol{A} (\boldsymbol{\nabla} \times \boldsymbol{B}).$$

In order to express this result in terms of the magnetic field we need to integrate this expression by parts. Let us firt rewrite the integrand:

$$\delta \boldsymbol{A} (\boldsymbol{\nabla} \times \boldsymbol{B}) = \delta A_i \epsilon_{ijk} \partial_j B_k = \epsilon_{ijk} \left(\partial_j (\delta A_i B_k) - B_k \partial_j \delta A_i \right).$$

The integral of the first term is transformed into a surface integral by the Stokes theorem; since our current distribution is localized, this term vanishes when the surface is taken to infinity. In the second term we may recognize the curl of $\delta \boldsymbol{A}$. Thus,

$$\delta W = \frac{1}{\mu_0} \int d^3x \boldsymbol{B} (\boldsymbol{\nabla} \times \delta \boldsymbol{A}) = \frac{1}{\mu_0} \int d^3x \boldsymbol{B} \delta \boldsymbol{B}.$$

Furthermore, we can write

$$\delta W = \delta \left\{ \frac{1}{2\mu_0} \int d^3x \boldsymbol{B}^2. \right\}$$

We see therefore that the quantity in the brackets is the energy E of the system of currents, so that

$$W = \frac{1}{2\mu_0} \int d^3x \boldsymbol{B}^2. \tag{2.89}$$

This energy is, in fact, associated with the magnetic fields; the integrand

$$\frac{1}{2\mu_0} \boldsymbol{B}^2$$

is the magnetic field energy density.

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Maxwell equations

• Let us summarize the equations that we have derived so far from the Coulomb law, the Biot-Savart law and the Faraday induction law:

$$\begin{aligned} \nabla \boldsymbol{E} &= \frac{\rho}{\epsilon_0}, && \text{(Coulomb law)} && (2.90)\\ \nabla \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} &= 0, && \text{(Faraday law)} && (2.91)\\ \nabla \boldsymbol{B} &= 0, && \text{(Biot-Savart law)} && (2.92)\\ \nabla \times \boldsymbol{B} &= \mu_0 \boldsymbol{J}. && \text{(Biot-Savart law)} && (2.93) \end{aligned}$$

The second equation is the only one that is derived with the account of time-dependence. In addition, there is a charge conservation equation

$$\frac{\partial \rho}{\partial t} + \nabla \boldsymbol{J} = 0 \tag{2.94}$$

which is correct in general, not only in the static case.

It is easy to see that these 5 equations contradict to each other in the case of time-dependent charge distribution. Indeed, taking the divergence of eq. (2.93) we find

$$\nabla(\nabla \times \boldsymbol{B}) = 0 = \mu_0 \nabla \boldsymbol{J},$$

which contradicts to eq. (2.94). Thus, these equations have to be modified.

The right modification has been proposed by Maxwell in 1865. He noted that the current conservation equation implies that a combination of the current and the time derivative of the electric field is divergence-free:

$$\frac{\partial \rho}{\partial t} + \nabla \boldsymbol{J} = \frac{\partial}{\partial t} \epsilon_0 \nabla \boldsymbol{E} + \nabla \boldsymbol{J} = \nabla \left\{ \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} + \boldsymbol{J} \right\} = 0.$$

Therefore, if one replaces the current in the r.h.s. of eq. (2.93) by this combination,

$$\nabla \times \boldsymbol{B} = \mu_0 \boldsymbol{J} \quad \rightarrow \quad \nabla \times \boldsymbol{B} = \mu_0 \left\{ \boldsymbol{J} + \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right\},$$

the mathematical consistency is recovered and the resulting equations are compatible with each other.

The resulting equations are called the **Maxwell equations**. They read:

$$\begin{aligned} \nabla \boldsymbol{E} &= \frac{\rho}{\epsilon_0}, && (2.95)\\ \nabla \times \boldsymbol{B} - \mu_0 \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} &= \mu_0 \boldsymbol{J}, && (2.96)\\ \nabla \boldsymbol{B} &= 0, && (2.97)\\ \nabla \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} &= 0. && (2.98) \end{aligned}$$

These equations incorporate all the known laws of electrostatics and magnetostatics, as well as the Faraday induction law. By itself this does not guarantee that they correctly describe the nature, which is an *experimental question*. It turns out that these equations predict a number of new

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phenomena, notably the existence of *electromagnetic waves*. All known consequences of these equations are by now confirmed experimentally.

From mathematical point of view the Maxwell equations are linear inhomogeneous equations. The homogeneous part of these equations (that is, when $\rho = \boldsymbol{J} = 0$) depends on a single constant $\mu_0\epsilon_0$. The inverse of this constant has units of velocity squared, and equals c^2,

$$\frac{1}{\mu_0\epsilon_0} = 8.988 \cdot 10^9 \cdot 10^7 \frac{\mathrm{Nm}^2}{\mathrm{C}^2}\frac{\mathrm{C}^2}{\mathrm{Ns}^2} = \left(3 \cdot 10^8 \frac{\mathrm{m}}{\mathrm{s}}\right)^2 = c^2.$$

The solutions to the homogeneous equations describe electromagnetic waves that propagate with the speed of light.

Relativistic form of Maxwell eqautions

• When we introduced Lorentz transformaitons we were motivated by the fact that electromagnetism (in particular, the wave equation) is not compatible with the Galilean relativity principle. We have constructed Lorentz transformations in such a way that the wave equation is invariant under them. This suggests that the full set of Maxwell equations, of which the wave equation is a consequence, can also be written in the relativistic form — that is, in the form of 4-vectors. This is indeed the case.

To rewrite the Maxwell equations in the 4-vector notations, let us start with the current conservation. This equation must be valid in any reference frame, so it must be a Lorentz scalar (in other words, it must be invariant under the Lorentz transformations). We can rewrite this equation as follows,

$$0 = \frac{\partial\rho}{\partial t} + \frac{\partial}{\partial x^i}J^i = \frac{\partial(c\rho)}{\partial(ct)} + \frac{\partial}{\partial x^i}J^i = \partial_\mu J^\mu$$

where we have defined the quantity

$$J^\mu = (c\rho, \boldsymbol{J})$$

which must, therefore, transform as a 4-vector to make the conservation equation Lorentz-invariant. Thus, the charge density and the current density are temporal and spatial components of the same 4-vector called the *4-vector of current*.

To progress further we need to reconsider the relation between the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ and the scalar and vector potentials in order to include the time-dependent case. Since the equation $\boldsymbol{\nabla B} = 0$ is still valid in the time-dependent case, we may define the vector potential as before,

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}.$$

However, the equation $\boldsymbol{\nabla} \times \boldsymbol{E} = 0$ is not valid in the non-static case, so $\boldsymbol{E}$ cannot be represented as a gradient of a scalar finction. We have instead

$$\boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times \left\{\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t}\right\} = 0.$$

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Therefore, the scalar potential should be defined as

$$\boldsymbol{E} + \frac{\partial \boldsymbol{A}}{\partial t} = -\boldsymbol{\nabla}\phi,$$

which leads to

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}. \tag{2.99}$$

This is the generalization of $\boldsymbol{E} = -\boldsymbol{\nabla}\phi$ for the case of time-dependent fields.

• *Gauge transformations.* In the non-static case there is a larger freedom in the definition of ϕ and $\boldsymbol{A}$. In the static case, the transformation that leaves both $\boldsymbol{E}$ and $\boldsymbol{B}$ unchanged was

$$\begin{aligned} \phi &\rightarrow \phi \\ A^i &\rightarrow A^i + \partial^i \lambda(\boldsymbol{x}) \end{aligned}$$

with an arbitrary function of space coordinates $\lambda(\boldsymbol{x})$. In the non-static case the following, more general, transformation

$$\begin{aligned} \phi &\rightarrow \phi + \frac{\partial}{\partial t}\lambda(\boldsymbol{x}, t) \\ \boldsymbol{A} &\rightarrow \boldsymbol{A} - \boldsymbol{\nabla}\lambda(\boldsymbol{x}, t) \end{aligned} \tag{2.100}$$

with an arbitrary function $\lambda(\boldsymbol{x}, t)$ *of space-time* coordinates does not change either $\boldsymbol{B}$ or $\boldsymbol{E}$. For $\boldsymbol{B}$ this is obvious, while for $\boldsymbol{E}$ we have from eq. (2.99):

$$\boldsymbol{E} = -\boldsymbol{\nabla}\left(\phi + \frac{\partial}{\partial t}\lambda(\boldsymbol{x}, t)\right) - \frac{\partial}{\partial t}\left(\boldsymbol{A} - \boldsymbol{\nabla}\lambda(\boldsymbol{x}, t)\right) = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}.$$

• Consider now the second Maxwell equation (2.96) and substitute the expressions for $\boldsymbol{B}$ and $\boldsymbol{E}$ in terms of the potentials:

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) - \frac{1}{c^2}\frac{\partial}{\partial t}\left(-\boldsymbol{\nabla}\phi \frac{\partial \boldsymbol{A}}{\partial t}\right) = \mu_0 \boldsymbol{J}.$$

Expanding the double curl and regrouping the terms we can rewrite this equation as follows,

$$\boldsymbol{\nabla}\left(\boldsymbol{\nabla}\boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t}\right) + \left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{A} = \mu_0 \boldsymbol{J}.$$

The r.h.s. of this equation is a space component of a 4-vector, and so must be the l.h.s. On the left, the second bracket is a Lorentz-invariant operator $\partial_\mu \partial^\mu$. Therefore, $\boldsymbol{A}$ must be a space component of a 4-vector. In the first term, the gradient is also a space component of a 4-vector, so the first bracket must be a scalar. Thus, the scalar and vector potentials must be parts of the same 4-vector,

$$\mathcal{A}^\mu = (\phi, c\boldsymbol{A}).$$

Then the first bracket is

$$\boldsymbol{\nabla}\boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = \frac{1}{c}\partial_\mu \mathcal{A}^\mu$$

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which is obviously a scalar.

• Both $\boldsymbol{B}$ and $\boldsymbol{E}$ are linear in potentials and in derivatives. They are in fact parts of an antisymmetric tensor

$$F^{\mu\nu} \equiv \partial^\mu \mathcal{A}^\nu - \partial^\nu \mathcal{A}^\mu$$

called *field stress tensor*. Since $F^{\mu\nu}$ is antisymmetric, it has 6 independent components which are 3 components of $\boldsymbol{E}$ plus 3 components of $\boldsymbol{B}$. Indeed, the non-zero components of $F^{\mu\nu}$ are

$$F^{i0} = \partial^i \mathcal{A}^0 - \partial^0 \mathcal{A}^i = -\partial_i \mathcal{A}^0 - \partial^0 \mathcal{A}^i = -\partial_i \phi - \frac{\partial}{\partial t} A_i = E_i$$

and

$$F^{ij} = \partial^i \mathcal{A}^j - \partial^j \mathcal{A}^i = -\partial_i \mathcal{A}^j + \partial_j \mathcal{A}^i = -c\epsilon_{ijk}\epsilon_{kpq}\partial_p A_q = -c\epsilon_{ijk} B_k$$

The inverse relations read

$$B_i = -\frac{1}{2c}\epsilon_{ijk} F^{jk}.$$

In the matrix form we may represent $F^{\mu\nu}$ as follows:

$$\{F^{\mu\nu}\} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -cB_3 & cB_2 \\ E_2 & cB_3 & 0 & -cB_1 \\ E_3 & -cB_2 & cB_1 & 0 \end{pmatrix}$$

• The Maxwell equations can be rewritten entirely in terms of the field stress tensor $F^{\mu\nu}$. We first note that the two homogeneous equations are automatically satisfied once we express the electric and magnetic fields in terms of the potentials. For the inhomogeneous equations we have

$$\boldsymbol{\nabla E} = \partial_i F^{i0} = \partial_\mu F^{\mu 0}$$

since $F^{00} = 0$ because of antisymmetry of $F^{\mu\nu}$. Thus, eq. (2.95) becomes

$$\partial_\mu F^{\mu 0} = \frac{1}{c\epsilon_0} J^0.$$

Both sides of this equation are time components of a 4-vector.

Now, using the expression of B_i and E_i in terms of $F^{\mu\nu}$ we have

$$(\boldsymbol{\nabla} \times \boldsymbol{B})_i - \frac{1}{c^2}\frac{\partial E_i}{\partial t} = \epsilon_{ijk}\partial_j B_k - \frac{1}{c^2}\frac{\partial E_i}{\partial t} = -\frac{1}{c}\partial_j F^{ij} - \frac{1}{c}\partial_0 F^{i0} =$$

$$= -\frac{1}{c}\partial_\mu F^{i\mu} = \frac{1}{c}\partial_\mu F^{\mu i}.$$

Then eq. (2.96) becomes

$$\frac{1}{c}\partial_\mu F^{\mu i} = \mu_0 J^i.$$

Therefore, both equations together are written as a single 4-vector equa-

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tion

$$\partial_\mu F^{\mu\nu} = \frac{1}{c\epsilon_0} J^\nu.$$

This is the relativistic form of the Maxwell equations.

Field invariants

•. The fact that the electromagnetic fields constitute parts of the field stress tensor $F^{\mu\nu}$ gives their transformation law upon the change of the reference frame:

$$F^{\mu\nu} = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma F'^{\rho\sigma},$$

where $\Lambda^\mu{}_\rho$ is the matrix of a Lorentz transformation. This transformation law is more complicated than for the vectors, and we will not discuss its explicit form here.

A simpler, but still very important, question is what are combinations of fields that are *invariant*. In the 4-vector formalism it is easy to construct invariants: one has to build products of field strength $F^{\mu\nu}$ and two invariant tensors, $\eta_{\mu\nu}$ and $\varepsilon^{\mu\nu\lambda\rho}$ in all possible combinations such that all the indices are contracted in a Lorentzian way — an upper index with a lower one.

It is easy to see that there are no invariants which are linear in the fields. Indeed, If one tries to contract the indices of $F^{\mu\nu}$ with the metric the result is zero,

$$F^{\mu\nu}\eta_{\mu\nu} = 0,$$

as for any contraction of the symmetric tensor with an antisymmetric one. Adding $\varepsilon^{\mu\nu\lambda\rho}$ obviously does not change this.

At the quadratic level we have two non-zero combinations with all the indices contracted, $F^{\mu\nu}F_{\mu\nu}$ and $F_{\mu\nu}F_{\lambda\rho}\varepsilon^{\mu\nu\lambda\rho}$. Let us express them in terms of the fields $\boldsymbol{E}$ and $\boldsymbol{B}$. For the first one we have

$$F^{\mu\nu}F_{\mu\nu} = F^{0i}F_{0i} + F^{i0}F_{i0} + F^{ij}F_{ij} = -2F^{i0}F^{i0} + F^{ij}F^{ij} =$$

$$= -2E_iE_i + c^2\epsilon_{ijk}B_k\epsilon_{ijp}B_p = -2(\boldsymbol{E}^2 - c^2\boldsymbol{B}^2).$$

So, the first combination of the fields that is Lorentz-invariant is

$$I_1 \equiv \boldsymbol{E}^2 - c^2\boldsymbol{B}^2. \tag{2.101}$$

To calculate the second combination we note that because of the antisymmetry of $\varepsilon^{\mu\nu\lambda\rho}$ one of its indices must be 0. It is easy to see that all four possibilities give the same answer, so that

$$F_{\mu\nu}F_{\lambda\rho}\varepsilon^{\mu\nu\lambda\rho} = 4F_{0i}F_{jk}\epsilon^{ijk} = -4F^{i0}2cB_i = -8c\boldsymbol{E}\boldsymbol{B}$$

Therefore, the second Lorentz-invariant combination of the fields is

$$I_2 \equiv \boldsymbol{E}\boldsymbol{B}.$$

One can show that all the invariants constructed with the higher powers of $F^{\mu\nu}$ can be expressed in terms of these two, so they add nothing new.

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2.4. Electromagnetic fields in media

So far we have considered electric and magnetic fields in the vacuum. This is often not a good approximation. In fact, most of the everyday applications of electromagnetism deal with fields in media: air, water, solid materials. All these media are, to a good approximation, electrically neutral, but at the microscopic level they are made of electrically charged particles — atomic nuclei and electrons which carry positive and negative charges, respectively. They may, therefore, have non-trivial electromagnetic properties despite the overall neutrality. We have already discussed one of the examples — conductors, in which charges can move freely and compensate completely external electric fields.

The overall electric neutrality of materials is ensured by enormous electric forces between macroscopic charges. To get a quantitative idea assume one somehow managed to separate positive and negative charges (protons and electrons) of 1 g of hydrogen and put them at a distance of 1 m. The protons of 1 g of hydrogen have a total positive charge of $Q = eN_A = 1.6 \cdot 10^{-19}\mathrm{C} \cdot 6 \cdot 10^{23} \sim 10^5$ C. The electrons have equal and opposite charge $-Q$. At a distance of $r = 1$ m the attraction force between these two charges would be



$$F = \frac{1}{4\pi\epsilon_0}\frac{Q^2}{r^2} = 9 \cdot 10^9 \frac{\mathrm{Nm}^2}{\mathrm{C}^2} \cdot 10^{10} \frac{\mathrm{C}^2}{\mathrm{m}^2} \sim 10^{20}\mathrm{N}.$$

This is an enormous force surpassing by many orders of magnitude all the forces we observe in everyday life. In fact, this force equals, within a factor 2, to the force holding the Moon on its orbit around the Earth!

The electromagnetic properties of a given medium depend both on the properties of its atoms and on the way they are arranged together. There is an enormous variety of different media, which react in a different way to external electric and magnetic fields. Apart from the properties of the medium itself, the electromagnetic response depends on how fast the applied fields vary with time. In these lectures we will only consider basic notions applicable to (relatively) slowly varying fields and to simplest — but very common — types of media.

Maxwell equations which we have derived in the previous section are, of course, valid in the case of media. But in order to apply them we would need to know the positions of all the charges (electrons and nuclei), which is clearly impractical. Even if it was technically possible, the microscopic electric and magnetic fields with all their variations in the interatomic space is not that we typically want to know. We need to develop instead some kind of averaged, macroscopic description.

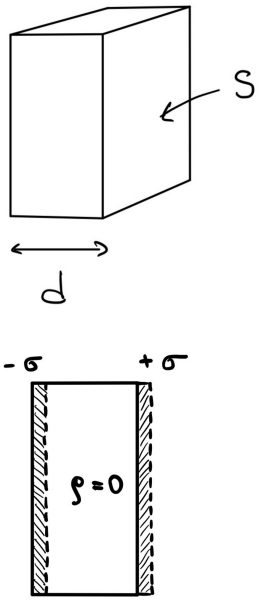
Electrostatics in media

We will start our discussion with electrostatic properties of *dielectrics*. In dielectrics electrons are bound to their molecules and cannot move freely. The atoms are electrically neutral; we will also assume that they have no dipole moment in the absence of external electric fields. Such a medium does not produce a macroscopic electric field by itself.

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When an external electric field is applied to an atom of such a medium, the positive charge (nucleus) is pulled in the direction of the field, while negative charges in the opposite direction. The atom becomes *polarized*: the positive and negative charges get displaced with respect to each other, so the dipole moment is no longer zero. In a medium subject to an external electric field all the atoms are polarized coherently in the same direction parallel to the field. We want to describe the effect of such a polarization.

• *Polarization vector.* Consider first a small slab of matter of thickness d and area S, so that the volume is $S \cdot d$. We will assume that the polarization (displacement of charges) is in the direction perpendicular to the slab. We do not need to know at this point what was the reason for this polarization. It could have been the external field as discussed above, or the result of an external mechanical force like in a piezoelectric crystal, or any other cause. They are all described in the same way.



In the middle of the slab nothing changes: the charge density is still equal to zero because the positive and negative charges still compensate each other on average. However, at the sides a net charge density is formed. The thickness of the charged layer is of atomic size, so from macroscopic point of view it is described by the surface charge density $\pm\sigma$. While the total charge of the slab is zero, because of the appearance of the surface charges the slab will have a *dipole moment*.

Let us calculate this dipole moment. Clearly, it is given by the product of the total charge on one side, σS, and the distance between the sides d,

$$D_{\text{slab}} = \sigma S d = \sigma V,$$

where $V = Sd$ is the volume of the slab. Note that the dipole moment does not depend on d and S separately, but only on their product which is the volume of the slab. We see therefore that the state of polarization can be characterized by a local quantity — the *polarization vector* $\boldsymbol{P}$. By definition, the polarization vector is the dipole moment *per unit volume*,

$\boldsymbol{P}$ = dipole moment per unit volume

$$|\boldsymbol{P}| = \frac{D_{\text{slab}}}{V}.$$

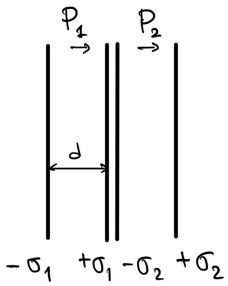
As a by-product we see from this equation that the surface charge density at the border perpendicular to the polarization vector is given by its modulus, $\sigma = |\boldsymbol{P}|$.

• *Polarization charge density.* The polarization vector is a local quantity and may vary from point to point, $\boldsymbol{P} = \boldsymbol{P}(\boldsymbol{x})$. In this case the charge density called *polarization charge density* ρ_P will be induced within the volume of the dielectric.

To understand why this happens consider a particular case when the polarization vector is directed along the x-axis and varies as a function of x, $\boldsymbol{P}(\boldsymbol{x}) = (P(x), 0, 0)$. Let us approximate this situation by a collection of thin slices perpendicular to x-axis, each of thickness d. The polarization vector is constant in each slice, but varies from slice to slice. Consider two neighboring slices with polarizations P_1 and P_2. If $P_1 = P_2$, the surface charges in the middle exactly compensate each other, and the only net charge is at the external sides. However, if $P_1 \neq P_2$, the charges in the middle *do not compensate each other*, so there will be a net charge

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accumulated between the slices 1 and 2. In the continuum limit $d \to 0$ this charge corresponds to



$$\Delta Q = S(\sigma_1 - \sigma_2) = S(P_1 - P_2) = Sd\frac{P_1 - P_2}{d} = V\frac{P_1 - P_2}{d}.$$

Thus, the polarization charge density ρ_P is

$$\rho_P = \frac{\Delta Q}{V} = \frac{P_1 - P_2}{d} \to -\frac{\partial P}{\partial x} \quad \text{at } d \to 0.$$

It is straightforward to generalize the above arguments to a general case when the polarization vector $\boldsymbol{P}$ changes both in value and in direction as a function of all coordinates, and the dielectric boundary is not necessarily perpendicular to $\boldsymbol{P}$. One finds:

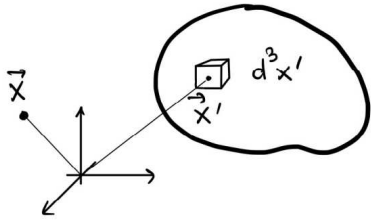
$$\rho_P = -\boldsymbol{\nabla P}, \tag{2.102}$$

$$\sigma_P = \boldsymbol{nP}. \tag{2.103}$$

Here $\boldsymbol{n}$ is a unit vector normal to the surface of dielectric, and we have added the subscript P to the σ_P to indicate that this is the surface charge density resulting from the polarization.

• *Formal derivation*. Let us re-derive these relations in a more formal way. Consider a polarized dielectric body characterized by the polarization vector $\boldsymbol{P}(\boldsymbol{x})$. In general, in addition to polarization, this body may carry some volume charge density $\rho_{\text{ext}}(\boldsymbol{x})$ and surface charge density $\sigma_{\text{ext}}(\boldsymbol{x})$. The subscript "ext" indicates that these are charges not related to polarization and, in this sense, external to the medium. Let us calculate the electrostatic potential $\phi(\boldsymbol{x})$ created by this body at a point $\boldsymbol{x}$.

There are three contributions to the potential at the point $\boldsymbol{x}$: from the external charges $\rho_{\text{ext}}(\boldsymbol{x})$, from $\sigma_{\text{ext}}(\boldsymbol{x})$ and from the polarization. Consider a point $\boldsymbol{x'}$ and an infinitesimal volume around this point d^3x'. If this volume is internal, it produces two contributions to the potential at the point $\boldsymbol{x}$. The first is due to the net external charge of the volume, $\rho_{\text{ext}}(\boldsymbol{x'})d^3x'$. The second contribution results from the dipole moment of the volume. By definition of the polarization vector, this dipole moment is $\boldsymbol{P}d^3x'$, while its contribution to the potential is given by eq. (2.28). If the volume is at the boundary, there is a third contribution from the surface charge $ds\,\sigma_{\text{ext}}(\boldsymbol{x'})$. So, in general,



$$d\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\left\{\frac{\rho_{\text{ext}}(\boldsymbol{x'})d^3x'}{|\boldsymbol{x}-\boldsymbol{x'}|} + \frac{(\boldsymbol{x}-\boldsymbol{x'})\boldsymbol{P}d^3x'}{|\boldsymbol{x}-\boldsymbol{x'}|^3} + \frac{\sigma_{\text{ext}}(\boldsymbol{x'})ds}{|\boldsymbol{x}-\boldsymbol{x'}|}\right\}.$$

It remains to integrate this expression over the body,

$$\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\left[\int d^3x' \left\{\frac{\rho_{\text{ext}}(\boldsymbol{x'})}{|\boldsymbol{x}-\boldsymbol{x'}|} + \frac{(\boldsymbol{x}-\boldsymbol{x'})\boldsymbol{P}(\boldsymbol{x'})}{|\boldsymbol{x}-\boldsymbol{x'}|^3}\right\} + \int ds\,\frac{\sigma_{\text{ext}}(\boldsymbol{x'})}{|\boldsymbol{x}-\boldsymbol{x'}|}\right]$$

Let us transform the term containing the polarization vector as follows,

$$\int d^3x' \frac{(\boldsymbol{x}-\boldsymbol{x'})\boldsymbol{P}(\boldsymbol{x'})}{|\boldsymbol{x}-\boldsymbol{x'}|^3} = \int d^3x' \boldsymbol{P}(\boldsymbol{x'})\boldsymbol{\nabla'}\frac{1}{|\boldsymbol{x}-\boldsymbol{x'}|} =$$

$$= \int d^3x' \frac{-\boldsymbol{\nabla'}\boldsymbol{P}(\boldsymbol{x'})}{|\boldsymbol{x}-\boldsymbol{x'}|} + \int ds\,\frac{\boldsymbol{nP}}{|\boldsymbol{x}-\boldsymbol{x'}|}.$$

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Here the second term is the surface term which resulted from integration by parts, and we have used $d\boldsymbol{s} = \boldsymbol{n}\, ds$ for the surface element. Combining all the contributions we thus have

$$\phi(\boldsymbol{x}) = \frac{1}{4\pi\epsilon_0}\left[\int d^3x' \frac{\rho_{\text{ext}}(\boldsymbol{x}') - \boldsymbol{\nabla}'\boldsymbol{P}(\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|} + \int ds \frac{\sigma_{\text{ext}}(\boldsymbol{x}') + \boldsymbol{n}\boldsymbol{P}}{|\boldsymbol{x}-\boldsymbol{x}'|}\right].$$

We see that polarization produces additional charge densities, both in the volume and on the surface, such that the total densities are

$$\rho_{\text{tot}} = \rho_{\text{ext}} + \rho_P,$$

$$\sigma_{\text{tot}} = \sigma_{\text{ext}} + \sigma_P,$$

where

$$\rho_P(\boldsymbol{x}) = -\boldsymbol{\nabla}\boldsymbol{P}(\boldsymbol{x}),$$

$$\sigma_P(\boldsymbol{x}) = \boldsymbol{n}\boldsymbol{P}(\boldsymbol{x}).$$

We have therefore established eqs. (2.102) and (2.103) .

• *Consistency check*. By their nature, the polarization charges ρ_P and σ_P are both due to the *displacement* of microscopic charges in an otherwise neutral medium. Therefore, in an arbitrary compact piece of dielectric material, polarized in an arbitrary way, the total sum of polarization charges must be zero. Let us show that eqs. (2.102) and (2.103) guarantee it.

Consider a piece of dielectric occupying volume V. Its total polarization charge is

$$Q_P = \int_V d^3x \rho_P(\boldsymbol{x}) + \int_{\partial V} ds \sigma_P.$$

Here ds is an infinitesimal area of the surface of the dielectric. Making use of eqs. (2.102) and (2.103) we can rewrite this as follows:

$$Q_P = -\int_V d^3x \boldsymbol{\nabla}\boldsymbol{P} + \int_{\partial V} ds \boldsymbol{n}\boldsymbol{P} = -\int_{\partial V} d\boldsymbol{s}\boldsymbol{P} + \int_{\partial V} d\boldsymbol{s}\boldsymbol{P} = 0.$$

In the first integral we have used the Stokes theorem to rewrite it as a an integral over the surface of the dielectric with the surface element $d\boldsymbol{s}$, while in the second we used the relation $d\boldsymbol{s} = ds\boldsymbol{n}$ with $\boldsymbol{n}$ being a unit normal vector to the surface. We see therefore that the two contributions cancel.

• *Displacement vector*. On macroscopic scales the polarization results in the appearance of polarization charges in addition to the external charges that might have been present. The total charge density is therefore

$$\rho(\boldsymbol{x}) = \rho_{\text{ext}}(\boldsymbol{x}) + \rho_P(\boldsymbol{x}) = \rho_{\text{ext}}(\boldsymbol{x}) - \boldsymbol{\nabla}\boldsymbol{P}(\boldsymbol{x}).$$

The total charge density determines the divergence of the electric field by the first Maxwell equation,

$$\boldsymbol{\nabla}\boldsymbol{E} = \frac{\rho(\boldsymbol{x})}{\epsilon_0} = \frac{1}{\epsilon_0}(\rho_{\text{ext}} - \boldsymbol{\nabla}\boldsymbol{P}).$$

One may rewrite this equation as

$$\boldsymbol{\nabla}(\boldsymbol{P} + \epsilon_0\boldsymbol{E}) = \rho_{\text{ext}}.$$

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The vector that appears in the r.h.s.,

$$\boldsymbol{D} = \epsilon_0 \boldsymbol{E} + \boldsymbol{P} \tag{2.104}$$

is called the **electric displacement** vector. The first Maxwell equation thus becomes

$$\boldsymbol{\nabla} \boldsymbol{D} = \rho_{\text{ext}}. \tag{2.105}$$

• *Relation between* $\boldsymbol{P}$ *and* $\boldsymbol{E}$. So far we did not specify the origin of the polarization. If the polarization vector $\boldsymbol{P}(\boldsymbol{x})$ is given, then one may use equations (2.105), (2.104) and the second equation of electrostatics $\boldsymbol{\nabla} \times \boldsymbol{E} = 0$ to determine $\boldsymbol{E}$ from the density of external charges ρ_{ext}. Indeed, one may calculate the density of polarization charges $\rho_P(\boldsymbol{x})$ from eq. (2.102), add it to the external density, and the problem of finding $\boldsymbol{E}(\boldsymbol{x})$ reduces (mathematically) to the one of electrostatics in the vacuum with the density of charges $\rho = \rho_{\text{ext}} + \rho_P$.

In many cases, however, the polarization is created by the electric field itself, and in order to obtain a closed system of equations one need to specify a relation between $\boldsymbol{P}$ and $\boldsymbol{E}$. In general, this relation can be very complicated. However, if the external electric field is sufficiently small, one can expand the polarization vector in powers of $\boldsymbol{E}$ and keep only the leading term. In this case one may write

$$P_i = \sum_j \alpha_{ij} E_j,$$

where the coefficients α_{ij} do not depend on $\boldsymbol{E}$. In general, the directions of $\boldsymbol{P}$ and $\boldsymbol{E}$ do not coincide. If we assume further that the medium is *isotropic*, i.e., does not have a preferred direction, then the matrix α_{ij} is proportional to the unit matrix, $\alpha_{ij} = \alpha \delta_{ij}$, and the directions of $\boldsymbol{P}$ and $\boldsymbol{E}$ are the same. In this case they are proportional to one another, and one may write

$$\boldsymbol{P} = \epsilon_0 \chi_e \boldsymbol{E}, \tag{2.106}$$

where we have written the proportionality coefficient as $\epsilon_0 \chi_e$. The coefficient χ_e is called *electric susceptibility* of the medium. Comparing to eq. (2.104) we see that it has no units. Since the positive charges are displaced in the direction of the electric field, this coefficient is positive, $\chi_e > 0$.

χ_e = electric susceptibility

• If we substitute eq. (2.106) into the definition of $\boldsymbol{D}$ we get the relation between $\boldsymbol{D}$ and $\boldsymbol{E}$,

$$\boldsymbol{D} = \epsilon \boldsymbol{E}, \tag{2.107}$$

where the coefficient

$$\epsilon = \epsilon_0 (1 + \chi_e)$$

is called *electric permittivity*. The ratio $\epsilon/\epsilon_0 = 1 + \chi_e \geq 1$ is called the *dielectric constant* or relative electric permittivity.

ϵ = electric permittivity
ϵ/ϵ_0 = dielectric constant

• *Continuity conditions.* At the boundary of a dielectric, or at the interface between two dielectrics with different dielectric constants, the fields $\boldsymbol{E}$ and $\boldsymbol{D}$ must satisfy continuity conditions. We have seen in the vacuum case that these conditions follow directly from the electrostatic equations (2.11)

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and (2.14). In the presence of dielectric we have instead the equations

$$\nabla \boldsymbol{D} = \rho_{\text{ext}}, \tag{2.108}$$

$$\nabla \times \boldsymbol{E} = 0. \tag{2.109}$$

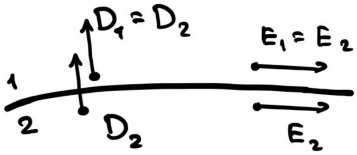
Note that here ρ_{ext} represents only the *external* charge density not including the polarization charges.

The second equation remains unchanged compared to the vacuum case. Therefore, the continuity conditions that follow from it are the same as in the vacuum. Denoting, as before, the unit vector normal to the interface by $\boldsymbol{n}$ we may write, as in the vacuum case,

$$\boldsymbol{n} \times (\boldsymbol{E}_2 - \boldsymbol{E}_1) = 0, \tag{2.110}$$

where $\boldsymbol{E}_1$ and $\boldsymbol{E}_2$ are the electric fields on two sides of the interface. Thus, the components of the electric field parallel to the interface are continuous.

The continuity condition for the perpendicular component follows from the divergence equation. Since this equation has (mathematically) the same form as in the vacuum, we may directly write the condition it implies by analogy with the vacuum case:



$$\boldsymbol{n} \cdot (\boldsymbol{D}_2 - \boldsymbol{D}_1) = \sigma_{\text{ext}}, \tag{2.111}$$

where σ_{ext} is the surface density of *external* charges not including the polarization charge density. In particular, in the absence of the external surface charge at the boundary the normal component of $\boldsymbol{D}$ is continuous.

• For as isotropic homogeneous dielectric the divergence equation (2.105) can be written as

$$\nabla \boldsymbol{E} = \frac{\rho_{\text{ext}}}{\epsilon}.$$

Comparing this to the case of vacuum we see that it only differs by the replacement $\epsilon_0 \to \epsilon$, so the same must be true for the solutions. In particular, the electric field of an external charge q_{ext} in an isotropic dielectric with the permittivity ϵ is

$$\boldsymbol{E}(\boldsymbol{x}) = \frac{q_{\text{ext}}}{4\pi\epsilon} \frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}. \tag{2.112}$$

Since $\epsilon > \epsilon_0$, this field is *weaker* than it would be in the vacuum. This is what one should expect on physical grounds: the charges in the dielectric are displaced by the electric field so that the charges of opposite sign are attracted to q_{ext} and (partially) compensate it. The external charge is *screened* by the polarization charges of the medium.

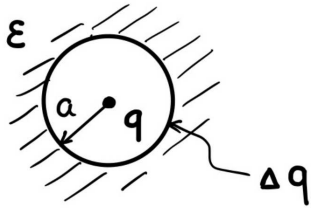
Likewise, in a capacitor with a dielectric filling the space between the plates the electric field is weaker compared to what it would be in the vacuum by the factor ϵ_0/ϵ,

$$E = \frac{\sigma}{\epsilon},$$

where σ is the surface charge density on the plates. Since the field is weaker, the potential difference $U = \int Edx$ is smaller by the same factor, while the capacity $C = Q/U$ is larger.

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• In the capacitor the electric field is weaker because the polarization charges which arise at the boundary of dielectric next to the capacitor plates partially compensate the charges on the plates. Where are the polarization charges in the case of a point charge embedded in a dielectric? If we consider a large sphere centered on our charge and apply the Gauss' theorem we will find from eq. (2.112) that the total charge inside the sphere is $q_{\text{ext}}\epsilon_0/\epsilon < q_{\text{ext}}$. Where is the extra charge?



To understand what is going on let us assume the external charge is put in a very small spherical cavity of radius a. The field outside of the cavity is given by eq. (2.112). Knowing $\boldsymbol{E}$, from eq. (2.134) we can determine $\boldsymbol{D}$ in this region, and then $\boldsymbol{P}$ from eq. (2.104):

$$\boldsymbol{D}(\boldsymbol{x}) = \frac{q_{\text{ext}}}{4\pi}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3},$$

$$\boldsymbol{P}(\boldsymbol{x}) = (\epsilon - \epsilon_0)\frac{q_{\text{ext}}}{4\pi\epsilon}\frac{\boldsymbol{x}}{|\boldsymbol{x}|^3}.$$

Note that the polarization vector satisfies $\boldsymbol{P}(\boldsymbol{x}) = \epsilon_0\chi_e\boldsymbol{E}(\boldsymbol{x})$ everywhere outside the cavity.

The polarization vector tells us where the polarization charges are. Its divergence determines the volume polarization charge density through eq. (2.102). Since the polarization $\boldsymbol{P}(\boldsymbol{x})$ has the same *form* as the electric field of a point charge in the vacuum, its divergence is zero, and therefore $\rho_P(\boldsymbol{x}) = 0$. There are no polarization charges in the volume of the dielectric.

At the boundary of the cavity the value of the polarization vector determines the surface density of polarization charges by eq. (2.103),

$$\sigma_P = \boldsymbol{n}\boldsymbol{P} = -P = -q_{\text{ext}}\frac{\epsilon - \epsilon_0}{\epsilon}\frac{1}{4\pi a^2}.$$

The minus sign here is due to opposite directions of $\boldsymbol{P}$ and the external normal to the dielectric $\boldsymbol{n}$ which points inwards. Multiplying by the area of the sphere $4\pi a^2$ we find from here the total polarization charge Δq that arises at the boundary of the cavity,

$$\Delta q = \frac{\epsilon_0 - \epsilon}{\epsilon}q_{\text{ext}}.$$

This must be the extra charge that has changed the flux of the electric field in the Gauss law. From this expression, the total charge inside a large sphere around q_{ext} is

$$q_{\text{tot}} = q_{\text{ext}} + \Delta q = \frac{\epsilon_0}{\epsilon}q_{\text{ext}},$$

in agreement with the Gauss' theorem.

• In practice, the continuity conditions are often the most convenient tool to solve problems. Consider an example shown in Fig. 5.2: a slice of dielectric with permittivity ϵ is embedded into a uniform electric field perpendicular to the slice. The field outside of the dielectric is $\boldsymbol{E}_0$. One needs to find the electric field $\boldsymbol{E}$, the polarization $\boldsymbol{P}$ and the displacement vector inside. There are two equivalent ways to approach the problem. We will discuss both of them one after the other.

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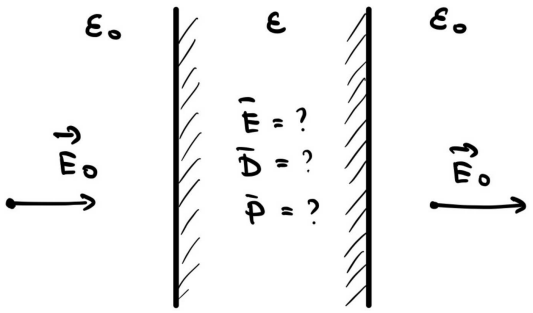


Figure 2.12.: Slice of dielectric of permittivity ϵ is embedded into a uniform electric field $\boldsymbol{E}_0$ perpendicular to the slice.

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Approach 1: formal using continuity conditions. The easiest way to solve this problem is to use the continuity conditions. It is clear from the symmetry that all three vectors have the same direction as the external electric field $\boldsymbol{E}_0$. One may therefore choose the x-axis in the direction of the external field and focus on the x-components of vectors.

Let us start from the exterior region to the left. In the vacuum, the displacement vector and the electric field are related by ϵ_0,

$$D_0 = \epsilon_0 E_0,$$

and, obviously, in this region $P_0 = 0$.

We may now use the continuity condition (2.111) to determine the displacement vector inside the dielectric. Since there are no external charges, we simply have

$$D = D_0 = \epsilon_0 E_0. \tag{2.113}$$

Once we know the displacement, we may use eq. (2.134) to determine the electric field,

$$E = \frac{D}{\epsilon} = \frac{\epsilon_0}{\epsilon} E_0. \tag{2.114}$$

Finally, for the polarization we have

$$P = D - \epsilon_0 E = (\epsilon - \epsilon_0)E = \frac{\epsilon_0}{\epsilon}(\epsilon - \epsilon_0)E_0. \tag{2.115}$$

This solves the problem.

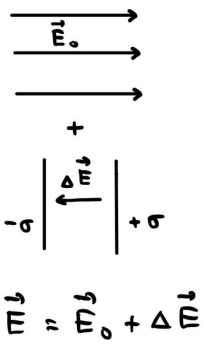
Approach 2: what is actually happening. Let us have a look at the same problem from physics perspective and try to understand what is happening. When we apply the electric field E_0 to the dielectric, its molecules get polarized in a coherent way — this is what we have described above by the polarization vector $\boldsymbol{P}$. The polarization vector is proportional to the electric field, but this is not the external field $\boldsymbol{E}_0$ outside of the dielectric — the atoms feel the local electric field $\boldsymbol{E}$ right where they are! So we may write

$$P = \epsilon_0 \chi_e E = (\epsilon - \epsilon_0)E.$$

But this is not a solution to the problem, because P and E are both unknown.

We need a second equation for these variables. The polarization P creates surface polarization charges at the boundaries of the dielectric, positive and negative on the right and left boundary, respectively. These charge

densities act like plates of a capacitor; they produce an electric field ΔE directed from the positive to the negative plate, that is, opposite to E_0. The value of this field is



$$\Delta E = -\frac{\sigma}{\epsilon_0} = -\frac{P}{\epsilon_0}.$$

The field inside the dielectric is a sum of the external field E_0 and the field created by the polarization charges,

$$E = E_0 + \Delta E = E_0 - \frac{P}{\epsilon_0}.$$

This is another equation that relates unknown E and P. We can now solve these two equations and find

$$\begin{aligned} E &= \frac{\epsilon_0}{\epsilon} E_0, \\ P &= \frac{\epsilon_0}{\epsilon}(\epsilon - \epsilon_0) E_0, \end{aligned}$$

which are the same expressions as before.

• How different are permittivities ϵ_0 and ϵ in practice? In other words, how big is χ_e? Before giving a table with a few values let us make a rough estimate from first principles, to see what we should expect.

Let us assume that in the absence of the external electric field each atom has zero charge and zero dipole moment. When the field is applied it acquires a dipole moment $\boldsymbol{d}$ which is aligned with the field. The polarization vector $\boldsymbol{P}$ is then

$$\boldsymbol{P} = n\boldsymbol{d},$$

where n is the number of atoms per unit volume. The dipole moment of each atom is in turn estimated as

$$d = |\boldsymbol{d}| \sim e \cdot \Delta x,$$

where e is the electron charge and Δx is the displacement of positive and negative charges under the action of the external electric field. Therefore, we need to estimate Δx.

In the absence of the external field atoms are in a stable equilibrium. If perturbed, they would oscillate around the equilibrium position. We may therefore model an atom as a harmonic oscillator satisfying the equation

$$m\ddot{x} = -kx + eE,$$

where $x = 0$ is the equilibrium position at $E = 0$ and the last term is the force due to the external electric field. Dividing by m we rewrite this equation as

$$\ddot{x} = -\omega^2 x + \frac{eE}{m},$$

where the $k/m = \omega^2$ is the oscillation frequency squared. We are not interested in oscillations, but rather in a new equilibrium position at $E \neq 0$. In the equilibrium $\ddot{x} = 0$, so the new equilibrium position is

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obtained by setting the r.h.s. of the previous equation to 0. This gives

$$x = \frac{eE}{m\omega^2} = \Delta x$$

since the original equilibrium was at $x = 0$. Combining the above expressions we have

$$P \sim n\frac{e^2 E}{m\omega^2},$$

and since by definition $P = \epsilon_0 \chi_e E$ we find

$$\chi_e = \frac{ne^2}{\epsilon_0 m\omega^2}.$$

To estimate the numerical value of χ_e we need to know ω. The frequencies of atomic oscillations are in correspondence with the frequencies of photons these atoms emit, which in turn can be written in terms of the photon wavelength λ,

$$\omega = \frac{2\pi c}{\lambda}.$$

If we take a typical value $\lambda \sim 1000\text{Å} = 10^{-7}$ m we get $\omega = 2 \cdot 10^{16}\,\text{s}^{-1}$. We can now substitute the numerical values:

$$\chi_e \sim 4\pi \frac{1}{4\pi\epsilon_0}\frac{N_A}{1\text{cm}^3}\frac{e^2}{m_e\omega^2} =$$

$$= 4\pi\, 10^{10}\frac{\text{N m}^2}{\text{C}^2}\cdot\frac{6\cdot 10^{23}}{10^{-6}\text{m}^3}\cdot(1.6\cdot 10^{-19}\text{C})^2\frac{1}{10^{-30}\text{kg}}\cdot\frac{10^{-32}}{4}\text{s}^2 \sim 5.$$

Here we have assumed N_A atoms per cm^3 which is characteristic for condensed media. In this case we have $\chi_e \sim \mathcal{O}(1)$. For gases at normal conditions one expects rather $\chi_e \sim \mathcal{O}(10^{-3})$.

Substance	**Dielectric constant** ϵ/ϵ_0
Vacuum	1.0
Air	≈ 1.0006
Water (20°C)	≈ 80.1
Glass	4 – 10
Silicon (Si)	11.7
Polyethylene	≈ 2.25
Teflon (PTFE)	≈ 2.1
Paper	3.0 – 4.0
Wood (dry)	2 – 6
Wood (wet)	10 – 30
Diamond	≈ 5.7
Rubber (natural)	2.5 – 3.0
Rubber (synthetic)	2.7 – 4.0

Table 2.1.: Dielectric constants of common materials

• *Electrostatic energy in a medium.* We start as in Sect. 2.1 by calculating the potential energy of a given charge distribution $\rho_{\text{ext}}(\boldsymbol{x})$. We may follow the same logic as in the case of vacuum, except now as we bring charges from infinity to assemble the distribution they polarize the medium, so the expression for the work changes. We do not need to repeat all the

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steps and may start directly from the equation

$$\delta W = \int d^3x \delta\rho_{\text{ext}}(\boldsymbol{x})\phi(\boldsymbol{x}).$$

Here $\phi(\boldsymbol{x})$ is the full electrostatic potential created by *all* the charges, including the polarization charges. It gradient gives the full electric field, $\boldsymbol{E} = -\boldsymbol{\nabla}\phi$.

We may now use the equation $\boldsymbol{\nabla D} = \rho_{\text{ext}}$ to express the variation of the charge density in terms of the variation of $\boldsymbol{D}$. Replacing $\delta\rho_{\text{ext}}$ by $\boldsymbol{\nabla}\delta\boldsymbol{D}$ we have

$$\delta W = \int d^3x \boldsymbol{\nabla}\delta\boldsymbol{D}(\boldsymbol{x})\phi(\boldsymbol{x}) = \int d^3x \delta\boldsymbol{D}(\boldsymbol{x})[-\boldsymbol{\nabla}\phi(\boldsymbol{x})] = \int d^3x\, \delta\boldsymbol{D}\cdot\boldsymbol{E}.$$

In order to advance further we need to specify the relation between $\boldsymbol{D}$ and $\boldsymbol{E}$. In the simplest case of a linear isotropic medium we have $\boldsymbol{D} = \epsilon\boldsymbol{E}$. In this case we may integrate the equation for δW which gives

$$W = \frac{1}{2}\int d^3x \boldsymbol{ED} = \frac{1}{2}\int d^3x \epsilon\boldsymbol{E}^2.$$

Note that ϵ may be a function of the coordinates, so in general it cannot be taken outside of the integral. We see that in this simple case the energy of electric field in medium can be obtained from the vacuum expression by substituting $\epsilon_0 \to \epsilon$, but in general this may not be the case.

• *Energy of a dielectric in an external field.* When a dielectric is placed in an external field, the polarization of medium partially compensates (screens) the field and thus reduces the energy stored in the electric field compared to its value without the dielectric. Thus, dielectrics are attracted into the regions of larger field. Formally, this can be demonstrated as follows. Suppose we have a medium with an electric permittivity ϵ_1, not necessarily constant in space, and an electric field $\boldsymbol{E}_1$ created by some external charges. We will consider these charges to be fixed in value and position. Let the corresponding displacement vector be $\boldsymbol{D}_1$.

Suppose we introduce in this medium a piece of dielectric with permittivity ϵ_2 but do not change the external charges. Denote the electric field and displacement in the presence of the piece of dielectric by $\boldsymbol{E}_2$ and $\boldsymbol{D}_2$, respectively. They are in general different from $\boldsymbol{E}_1$ and $\boldsymbol{D}_1$ both inside and outside of the dielectric piece.

Let us calculate the change in the electrostatic energy of the system due to the presence of the dielectric. It equals

$$\Delta W = \frac{1}{2}\int d^3x(\boldsymbol{D}_2\boldsymbol{E}_2 - \boldsymbol{D}_1\boldsymbol{E}_1).$$

This difference can be rewritten as follows:

$$\Delta W = \frac{1}{2}\int d^3x(\boldsymbol{D}_1\boldsymbol{E}_2 - \boldsymbol{D}_2\boldsymbol{E}_1) + \frac{1}{2}\int d^3x(\boldsymbol{E}_2 + \boldsymbol{E}_1)(\boldsymbol{D}_2 - \boldsymbol{D}_1).$$

The second integral is in fact zero. To show it we write

$$\boldsymbol{E}_2 + \boldsymbol{E}_1 = -\boldsymbol{\nabla}(\phi_1 + \phi_2)$$

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where ϕ_1 and ϕ_2 are electrostatic potentials before and after insertion of the dielectric. Substituting this in the second integral and integrating by parts we find

$$\frac{1}{2}\int d^3x(\phi_1+\phi_2)\boldsymbol{\nabla}(\boldsymbol{D}_2-\boldsymbol{D}_1)$$

which is zero because divergencies of $\boldsymbol{D}_2$ and $\boldsymbol{D}_1$ are given by the same external charges. We have therefore

$$\Delta W=\frac{1}{2}\int d^3x(\boldsymbol{D}_1\boldsymbol{E}_2-\boldsymbol{D}_2\boldsymbol{E}_1)$$

Assuming the linear relations between the polarization and the applied field, we may rewrite this energy difference as follows,

$$\Delta W=-\frac{1}{2}\int d^3x(\epsilon_2-\epsilon_1)\boldsymbol{E}_1\boldsymbol{E}_2.$$

The integral now is over the volume V of the dielectric piece.

If $\epsilon_2>\epsilon_1$ the difference in energy ΔW is negative. This means that dielectrics are *attracted* into the regions of stronger electric field. The physical reason is, of course, that this lowers the energy of the system.

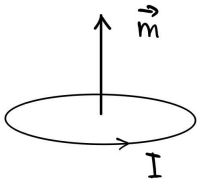
Magnetostatics in media

Being composed of atoms, the matter always contains charges in motion. The electrons are moving around nuclei, protons are moving inside nuclei and even protons and electrons themselves have internal angular momentum called *spin*. These moving charges create magnetic moments which, in turn, produce microscopic magnetic fields. These microscopic fields may or may not average to zero. However, even if magnetic moments of atoms are absent at zero external field, they may be induced when the field is applied.

• *Magnetization vector.* Similarly to the electric case, the magnetic properties of a medium can be characterized by the *vector of magnetization* $\boldsymbol{M}(\boldsymbol{x})$. To understand the nature of this quantity consider a magnetized medium in which microscopic magnetic moments of atoms do not average to zero. Consider a small volume such that it is much larger than the atomic size, but small enough to be considered homogeneous. The magnetic moment of this volume is produced by a closed surface current flowing along the sides parallel to $\boldsymbol{M}$ a shown on see Fig. 2.13. If we denote this surface current j, the total current, circulating around our volume is $I=jL$ where L is the height of the box. We may therefore write the total magnetic moment of the box as follows:

$$M_{\text{box}}=S\cdot I=SLj=Vj$$

where V is the volume of the box.



Recall:

$$\boldsymbol{m}=\frac{1}{2}\int d^3x\boldsymbol{x}\times\boldsymbol{J}(\boldsymbol{x})$$

and for a plane loop of area S

$$|\boldsymbol{m}|=IS.$$

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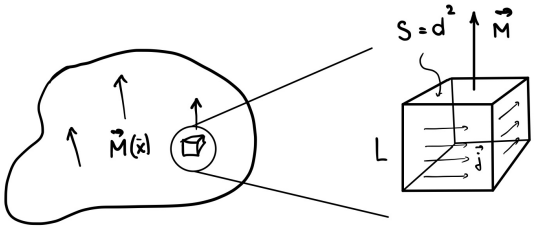


Figure 2.13.: Rectangular piece of a magnetized medium.

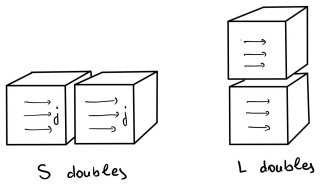
One can easily see by combining several boxes in different ways that the total magnetic moment is the sum of the magnetic moments of the individual boxes and thus is proportional to their total volume independently of the shape. This means that one can characterize the magnetized medium by the *density* of the magnetic moment, or in other words, the magnetic moment per volume,

$$\boldsymbol{M} = \frac{\boldsymbol{M}_{\text{box}}}{V}.$$

In the last equation it is assumed that the volume is small, $V \to 0$, so that $\boldsymbol{M} = \boldsymbol{M}(\boldsymbol{x})$ can be considered constant across the box, even though it may change on larger scales. We also have for the absolute values

$$M = \frac{M_{\text{box}}}{V} = j,$$

so the magnetization vector determines the value of the surface current.



• *Magnetization current.* When the magnetization $\boldsymbol{M}$ varies in space, there may arise a net current density in the *volume* of the medium. This current is called the *magnetization current* $\boldsymbol{J}_M$. Consider the case when the magnetization is in the direction z and varies as a function of y. We may approximate this situation by taking small volumes with the magnetization that is constant across each volume but different from box to box, see Fig. 2.14. If the magnetizations in the nearby boxes are different, the surface currents are also different and do not compensate each other any more when the boxes are put together. Thus, there will be net current flowing along the common side of the two boxes which equals

$$I = L(j_2 - j_1) = Ld\frac{j_2 - j_1}{d} = Ld\frac{M_2 - M_1}{d}.$$

This current is perpendicular to the (yz) plane and is associated with the

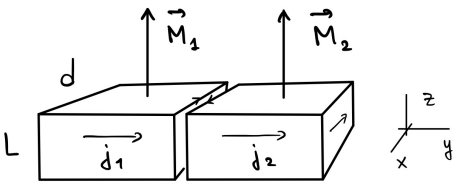


Figure 2.14.: Approximate representation of the magnetization $\boldsymbol{M} = (0, 0, M(y))$ varying as a function of y.

area Ld: if one imagines n blocks put together, there will be $n-1$ currents while the area will be nLd, so the area per current will be $\approx Ld$. If we

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divide the current by the area perpendicular to it we get, by definition, the current density. Thus,

$$\frac{I}{Ld} = J_x = \frac{M_2 - M_1}{d}.$$

In the continuum limit this becomes

$$J_x = \frac{\partial M_z}{\partial y}.$$

For arbitrary $\boldsymbol{M}(\boldsymbol{x})$ this relation generalizes to

$$\boldsymbol{J}_M(\boldsymbol{x}) = \boldsymbol{\nabla} \times \boldsymbol{M}(\boldsymbol{x}). \tag{2.116}$$

It is easy to see that eq. (2.116) indeed reduces to $J_x = \partial M_z/\partial y$ in our particular case.

• Let us re-derive this relation in a formal way. Suppose we have a piece of matter carrying some external currents $\boldsymbol{J}_{\text{ext}}(\boldsymbol{x})$ and having a non-zero magnetization $\boldsymbol{M}(\boldsymbol{x})$. Each volume d^3x' of this matter makes two contributions to the vector potential $\boldsymbol{A}(\boldsymbol{x})$ in the point $\boldsymbol{x}$. The first one is due to the external current,

$$\delta\boldsymbol{A}_1(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\frac{\boldsymbol{J}_{\text{ext}}(\boldsymbol{x}')d^3x'}{|\boldsymbol{x} - \boldsymbol{x}'|}.$$

The second contribution arises because this volume carries the magnetic moment $\boldsymbol{M}(\boldsymbol{x})d^3x'$, so from eq. (2.82) we have

$$\delta\boldsymbol{A}_2(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\frac{\boldsymbol{M}(\boldsymbol{x}')d^3x' \times (\boldsymbol{x} - \boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|^3}.$$

Adding the two contributions and integrating over d^3x' we obtain the total vector potential,

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\int d^3x' \left\{\frac{\boldsymbol{J}_{\text{ext}}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} + \frac{\boldsymbol{M}(\boldsymbol{x}') \times (\boldsymbol{x} - \boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|^3}\right\}.$$

The last term can be transformed by using the identity

$$\frac{\boldsymbol{x} - \boldsymbol{x}'}{|\boldsymbol{x} - \boldsymbol{x}'|^3} = \boldsymbol{\nabla}'\frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|}$$

and integrating by parts. When integrating by parts, we will assume that the magnetized material occupies a finite volume, and will take into account the boundary term. We have

$$\int d^3x' \frac{\boldsymbol{M}(\boldsymbol{x}') \times (\boldsymbol{x} - \boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|^3} = \int d^3x' \boldsymbol{M}(\boldsymbol{x}') \times \boldsymbol{\nabla}'\frac{1}{|\boldsymbol{x} - \boldsymbol{x}'|} =$$

$$= \int d^3x' \frac{\boldsymbol{\nabla}' \times \boldsymbol{M}(\boldsymbol{x}')}{|\boldsymbol{x} - \boldsymbol{x}'|} + \int ds \frac{\boldsymbol{M}(\boldsymbol{x}') \times \boldsymbol{n}}{|\boldsymbol{x} - \boldsymbol{x}'|}.$$

The second contribution is the surface term; the integration is over the boundary of our volume where $\boldsymbol{n}ds$ represents the area element, $\boldsymbol{n}$ being the unit vector normal to this surface. We now substitute this expression

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into the vector potential and get

$$\boldsymbol{A}(\boldsymbol{x}) = \frac{\mu_0}{4\pi}\int d^3x' \frac{\boldsymbol{J}_{\rm ext}(\boldsymbol{x}') + \boldsymbol{\nabla}'\times\boldsymbol{M}(\boldsymbol{x}')}{|\boldsymbol{x}-\boldsymbol{x}'|} + \frac{\mu_0}{4\pi}\int ds \frac{\boldsymbol{M}(\boldsymbol{x}')\times\boldsymbol{n}}{|\boldsymbol{x}-\boldsymbol{x}'|}.$$

Comparing the $\boldsymbol{M}$-dependent contributions to that of $\boldsymbol{J}_{\rm ext}$ we see that they are produced by two currents: the volume current $\boldsymbol{\nabla}'\times\boldsymbol{M}(\boldsymbol{x}')$ and the surface current $\boldsymbol{M}(\boldsymbol{x}')\times\boldsymbol{n}$. The total volume current is

$$\boldsymbol{J}_{\rm tot}(\boldsymbol{x}) = \boldsymbol{J}_{\rm ext}(\boldsymbol{x}) + \boldsymbol{\nabla}\times\boldsymbol{M}(\boldsymbol{x}) \equiv \boldsymbol{J}_{\rm ext}(\boldsymbol{x}) + \boldsymbol{J}_M(\boldsymbol{x}). \tag{2.117}$$

Thus, we have recovered eq. (2.116). We have also derived the surface magnetization current in terms of the magnetization $\boldsymbol{M}$,

$$\boldsymbol{j}_M = \boldsymbol{M}(\boldsymbol{x}')\times\boldsymbol{n}. \tag{2.118}$$

This relation is the magnetic analog of electric $\sigma_P = \boldsymbol{P}\boldsymbol{n}$. Comparing to Fig. 2.13 we see that it gives the correct direction of $\boldsymbol{j}_M$.

• *Magnetic field and magnetic induction.* The macroscopic magnetic field is produced by the total current (2.117) which includes the external current and the current due to magnetization. Therefore, the equation of magnetostatics reads

$$\boldsymbol{\nabla}\times\boldsymbol{B} = \mu_0(\boldsymbol{J}_{\rm ext} + \boldsymbol{\nabla}\times\boldsymbol{M}).$$

Moving the second term to the left side we can write this equation as

$$\boldsymbol{\nabla}\times\boldsymbol{H} = \boldsymbol{J}_{\rm ext}$$

where

$$\boldsymbol{H} \equiv \frac{1}{\mu_0}\boldsymbol{B} - \boldsymbol{M}. \tag{2.119}$$

Note that this expression is analogous to eq. (2.104) of the electric case.

Historically, the vector $\boldsymbol{H}$ is called the magnetic field and the vector $\boldsymbol{B}$ the *magnetic induction*. Important point to note, however, is that $\boldsymbol{B}$ is a fundamental quantity, while $\boldsymbol{H}$ is a combination of $\boldsymbol{B}$ and magnetization $\boldsymbol{M}$ defined in eq. (2.119) which is introduced for convenience. For instance, forces acting on charged particles are determined by $\boldsymbol{B}$ and not by $\boldsymbol{H}$.

• *Relation between $\boldsymbol{B}$ and $\boldsymbol{H}$.* In order to be able to determine the magnetic field from the distribution of the external currents we need to provide the relation between $\boldsymbol{B}$ and $\boldsymbol{H}$. In general, such relation may be quite complicated.

There are in general three types of media: *diamagnetic*, *paramagnetic* and *ferromagnetic*. The first two are characterized by the property that $\boldsymbol{M} = 0$ in the absence of the external field. In case of sufficiently small external field the response of diamagnetics and paramagnetics is linear. For an isotropic medium we may write in this case

$$\boldsymbol{B} = \mu\boldsymbol{H} \tag{2.120}$$

The constant μ called *magnetic permeability* has the same units as its vacuum analog μ_0 (cf. eq. (2.119)) and can be larger or smaller than μ_0 :

$\mu > \mu_0$ for *paramagnetics*,

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$\mu < \mu_0$ for *diamagnetics*.

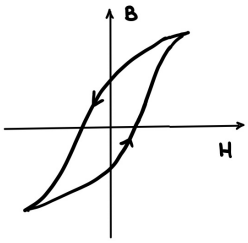
In either case the value of μ is very close to μ_0; typically

$$\frac{|\mu - \mu_0|}{\mu_0} \sim 10^{-5}.$$

For the *ferromagnetic* substances the relation between $\boldsymbol{B}$ and $\boldsymbol{B}$ is non-linear and can be written as

$$\boldsymbol{B} = \boldsymbol{F}(\boldsymbol{H}).$$

Moreover, there is a phenomenon of *hysteresis*, i.e., $\boldsymbol{B}$ is not a single-valued function of $\boldsymbol{H}$ and depends also on the history of its changes. The permeability in this case is defined as



$$\mu(H) = \frac{\partial B}{\partial H}.$$

The value of $\mu(H)/\mu_0$ can be as high as 10^6.

•. *Continuity conditions.* The continuity conditions for $\boldsymbol{B}$ and $\boldsymbol{H}$ across the boundary of two media follow directly from the equations for these fields. The latter read

$$\nabla \boldsymbol{B} = 0, \tag{2.121}$$

$$\nabla \times \boldsymbol{H} = \boldsymbol{J}_{\text{ext}}. \tag{2.122}$$

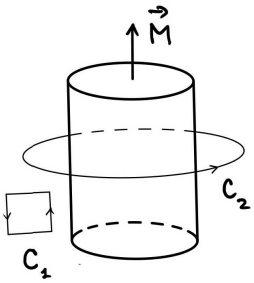
If we compare these equations to the magnetostatic equations in the vacuum we may immediately generalize the vacuum continuity conditions to the case of a medium,

$$\boldsymbol{n}(\boldsymbol{B}_2 - \boldsymbol{B}_1) = 0, \tag{2.123}$$

$$\boldsymbol{n} \times (\boldsymbol{H}_2 - \boldsymbol{H}_1) = \boldsymbol{j}_{\text{ext}}. \tag{2.124}$$

Here $\boldsymbol{B}_{1,2}$ and $\boldsymbol{H}_{1,2}$ are the fields on sides 1 and 2 with respect to the boundary, while $\boldsymbol{j}_{\text{ext}}$ is the *external* surface current on the boundary.

• *Example: magnetized cylinder.* Consider an infinite (very long) cylinder which is magnetized with a constant magnetization $\boldsymbol{M}$ parallel to its axis, see the plot. What is the value of $\boldsymbol{B}$ and $\boldsymbol{H}$ inside and outside of the cylinder?



First, it is straightforward to argue that the field $\boldsymbol{B}$ outside of the cylinder must be zero. Indeed, by the symmetry of the problem the field must be axially symmetric and independent of z, so that $\boldsymbol{B} = \boldsymbol{B}(r)$. This field cannot have a radial component, because then its flux through a cylindrical surface enclosing the magnetized cylinder will not be zero, which contradicts to $\nabla \boldsymbol{B} = 0$. Neither can it have an azimuthal component. If there was one, then the linear integral of the field along the contour C_2 would not be zero. Since in the vacuum $\boldsymbol{B}$ and $\boldsymbol{H}$ are proportional, the same applies to $\boldsymbol{H}$. But this contradicts to eq. (2.122) in the absence of external currents.

The only remaining option is $\boldsymbol{B} = (0, 0, B(r))$. But this is not possible either. Since $\nabla \times \boldsymbol{B} = 0$, the line integral of the field over the contour C_1 which lies in a plane passing through the cylinder axis must be zero, and therefore $B(r)$ must be constant. But since the field must decay at

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infinity, this constant must be zero. Therefore, outside of the cylinder $\boldsymbol{B} = \boldsymbol{H} = 0$.

Inside the cylinder obviously both $\boldsymbol{B}$ and $\boldsymbol{H}$ are parallel to $\boldsymbol{M}$. Since there is no azimuthal components and we have $\boldsymbol{\nabla} \times \boldsymbol{H} = 0$ inside the cylinder, we may apply to $\boldsymbol{H}$ the same argument with the contour C_1 as above, but now with C_1 being inside the cylinder. This way we may show that $\boldsymbol{H}$ is independent of r and thus is constant across the section of the cylinder.

We are now ready to use the continuity conditions. Since there are no field components perpendicular to the surface of the cylinder, eq. (2.123) is trivially satisfied. The other equation (2.124) tells us that the components of $\boldsymbol{H}$ parallel to the surface are continuous. Since outside $\boldsymbol{H} = 0$, we conclude that inside the cylinder $\boldsymbol{H} = 0$ as well. One may now use eq. (2.119) to determine the magnetic field inside the cylinder,

$$\boldsymbol{B} = \mu_0 \boldsymbol{M}.$$

From physics point of view what is happening is the following. We have seen above that the magnetization produces a surface magnetization current. In our example this is the current that flows along the circle in the azimuthal direction. The cylinder is thus equivalent to a solenoid with the current j per unit length. From the Ampere theorem the magnetic field inside this solenoid is given precisely by the above expression.

• *Energy of the magnetic field in medium.* The arguments that we have used in the vacuum to derive the energy of magnetic field are still applicable in the presence of a medium. We do not need to repeat them in full and may start instead from the analog of eq. (2.88) where we have to replace $\boldsymbol{J}$ by $\boldsymbol{J}_{\text{ext}}$,

$$\delta W = \int d^3x \boldsymbol{J}_{\text{ext}}\, \delta \boldsymbol{A}.$$

We now express $\boldsymbol{J}_{\text{ext}}$ through eq. (2.122) as a curl of $\boldsymbol{H}$, so that

$$\delta W = \int d^3x (\boldsymbol{\nabla} \times \boldsymbol{H})\, \delta \boldsymbol{A}$$

Now we may use the identity

$$\boldsymbol{\nabla} \cdot (\delta \boldsymbol{A} \times \boldsymbol{H}) = \boldsymbol{H} \cdot (\boldsymbol{\nabla} \times \delta \boldsymbol{A}) - \delta \boldsymbol{A} \cdot (\boldsymbol{\nabla} \times \boldsymbol{H})$$

to write this as

$$\delta W = \int d^3x \left[\boldsymbol{H} \cdot (\boldsymbol{\nabla} \times \delta \boldsymbol{A}) - \boldsymbol{\nabla} \cdot (\delta \boldsymbol{A} \times \boldsymbol{H})\right]$$

The second term is converted to a surface integral that vanishes, while in the first term we use $\delta \boldsymbol{B} = \boldsymbol{\nabla} \times \delta \boldsymbol{A}$ and obtain

$$\delta W = \int d^3x \boldsymbol{H} \cdot \delta \boldsymbol{B}.$$

If the relation between $\boldsymbol{B}$ and $\boldsymbol{H}$ is linear this equation can be integrated explicitely and gives

$$W = \frac{1}{2} \int d^3x\, \boldsymbol{H}\boldsymbol{B} \tag{2.125}$$

This is the analog of eq. (2.89) in a medium.

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Maxwell equations in medium

Let us start with the summary of the static equations in medium:

$$\begin{aligned} \boldsymbol{\nabla} \boldsymbol{D} &= \rho_{\text{ext}}, && (2.126)\\ \boldsymbol{\nabla} \times \boldsymbol{E} &= 0, && (2.127)\\ \boldsymbol{\nabla} \boldsymbol{B} &= 0, && (2.128)\\ \boldsymbol{\nabla} \times \boldsymbol{H} &= \boldsymbol{J}_{\text{ext}} && (2.129) \end{aligned}$$

In addition, we also have the conservation of the *external* current,

$$\frac{\partial \rho_{\text{ext}}}{\partial t} + \boldsymbol{\nabla} \boldsymbol{J}_{\text{ext}} = 0$$

Note that the external current is conserved separately.

We may use the same logic as in the vacuum to generalize these equations to the time-dependent case. In a medium, the Faraday induction law still remains valid, so the second equation has to be changed as follows,

$$\boldsymbol{\nabla} \times \boldsymbol{E} = 0 \qquad \rightarrow \qquad \boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = 0.$$

However, as in the vacuum case, eq. (2.129) is not compatible with the external current conservation: taking the divergence of both sides we find $\boldsymbol{\nabla} \boldsymbol{J}_{\text{ext}} = 0$ which is not true in general, only in the static case when ρ_{ext} does not depend on time.

From the vacuum case, we know how to modify this equation to make it compatible with the remaining ones. If we take the external charge density from eq. (2.126) and substitute into the current conservation equation we find

$$\frac{\partial \boldsymbol{\nabla} \boldsymbol{D}}{\partial t} + \boldsymbol{\nabla} \boldsymbol{J}_{\text{ext}} = \boldsymbol{\nabla} \left(\frac{\partial \boldsymbol{D}}{\partial t} + \boldsymbol{J}_{\text{ext}} \right) = 0,$$

i.e., the combination in the brackets has zero divergence. We may put this combination instead of $\boldsymbol{J}_{\text{ext}}$ in the r.h.s. of eq. (2.129). Then if we take the divergence of the resulting equation both sides give zero, and there is no contradiction any more. Thus, we have to change the last equation as follows,

$$\boldsymbol{\nabla} \times \boldsymbol{H} = \boldsymbol{J}_{\text{ext}} \qquad \rightarrow \qquad \boldsymbol{\nabla} \times \boldsymbol{H} - \frac{\partial \boldsymbol{D}}{\partial t} = \boldsymbol{J}_{\text{ext}}.$$

With these changes we get a fully consistent set of equations,

$$\begin{aligned} \boldsymbol{\nabla} \boldsymbol{D} &= \rho_{\text{ext}}, && (2.130)\\ \boldsymbol{\nabla} \times \boldsymbol{H} - \frac{\partial \boldsymbol{D}}{\partial t} &= \boldsymbol{J}_{\text{ext}}, && (2.131)\\ \boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} &= 0, && (2.132)\\ \boldsymbol{\nabla} \boldsymbol{B} &= 0. && (2.133) \end{aligned}$$

Unlike the Maxwell equations in the vacuum, these equations are not sufficient to find the electromagnetic field created by a given distribution of external charges and currents. They have to be supplemented by the

relations between $\boldsymbol{E}$ and $\boldsymbol{D}$, and between $\boldsymbol{B}$ and $\boldsymbol{H}$. If the medium is linear and isotropic, these relations can be written as

$$\boldsymbol{D} = \epsilon \boldsymbol{E} \tag{2.134}$$

$$\boldsymbol{H} = \frac{1}{\mu}\boldsymbol{B}. \tag{2.135}$$

One may then substitute $\boldsymbol{D}$ and $\boldsymbol{H}$ into the first two equations and find

$$\begin{aligned} \boldsymbol{\nabla} \boldsymbol{E} &= \frac{\rho_{\text{ext}}}{\epsilon}, \\ \boldsymbol{\nabla} \times \boldsymbol{B} - \mu\epsilon \frac{\partial \boldsymbol{E}}{\partial t} &= \mu \boldsymbol{J}_{\text{ext}}. \end{aligned}$$

These equations are formally the same as in the vacuum except for the substitution $\epsilon_0, \mu_0 \to \epsilon, \mu$. Remarkably, in the absence of the external charges and currents the whole set of Maxwell equations depends on a single constant $\mu\epsilon$ which has units $\mathrm{s}^2/\mathrm{m}^2$. This constant determines the speed of propagation v of electromagnetic waves in the medium,

$$v^2 = \frac{1}{\mu\epsilon}. \tag{2.136}$$

Since μ is very close to μ_0 and $\epsilon > \epsilon_0$ this speed is smaller than in the vacuum, $v < c$.

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Charged particle in electromagnetic fields 3.

Up to now we have addressed the question of how to find the electric and magnetic fields given the distribution of charges and currents. We have solved this problem in the static case, and also derived the complete set of equations (the Maxwell's equations) that determine the electromagnetic fields in a general situation when the charges and currents are not static.

Let us now consider the opposite question: given the electric and magnetic fields, how do electric charges move? To keep things simple we will limit ourselves to particles that are completely characterized by their electric charge e and mass m. This is not always the case; for instance, electrons and protons also have internal angular momenta (spins) and magnetic moments. However, in many situations these complications may be neglected, at least in the first approximation. The starting point is the equation of motion.

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3.1. Equation of motion.

• *Lorentz force.* In the non-relativistic case, the behavior of a charged particle with charge e in the electromagnetic field is governed by the Newton's law $m\boldsymbol{a} = \boldsymbol{F}$, where the force $\boldsymbol{F}$ is the *Lorentz force*

$$\boldsymbol{F} = e[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}],$$

$\boldsymbol{v}$ being the velocity of the particle. Both contributions to the force, electric and magnetic, are proportional to the particle charge. The magnetic force also depends on the particle velocity. So, the non-relativistic equation of motion is

$$m\frac{d\boldsymbol{v}}{dt} = e[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}].$$

m = particle mass
e = particle charge

In many applications, however, the non-relativistic approximation is not sufficient

• *Relativistic equation.* We have seen above in Sect. 1.3 that the relativistic equation can be obtained from a non-relativistic one by rewriting the l.h.s. in terms of the particle momentum as $m\,d\boldsymbol{v}/dt = d\boldsymbol{p}/dt$, and replacing the momentum $\boldsymbol{p}$ by its relativistic expression,

$$\boldsymbol{p} = m\boldsymbol{v} \quad \rightarrow \quad \boldsymbol{p} = \frac{m\boldsymbol{v}}{\sqrt{1 - v^2/c^2}} = m\boldsymbol{v}\gamma.$$

The equation of motion then becomes

$$\frac{d\boldsymbol{p}}{dt} = e[\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}]. \tag{3.1}$$

Given the electric and magnetic fields, this equation completely determines the particle motion. Note that in the limit $v \ll c$ this equation reduces to the non-relativistic one because in this limit $\gamma \rightarrow 1$.

• *Equation of motion in the 4-vector form.* Before discussing solutions to eq. (3.1), it is instructive to check that, despite its 3-dimensional

appearance, this is a relativistic equation. In other words, if a solution to this equation is transformed into a different frame, it remains a solution to the transformed equation.

The easiest way to do it is to rewrite this equation in the form of 4-vectors — then its validity in any reference frame will be automatic. From Sect. 1.3 we know that the l.h.s. of the relativistic equation has to be of the form

$$c\frac{dp^\mu}{ds} = \dots$$

The spatial part of this expression corresponding to $\mu = i$ will then be

$$\frac{1}{\sqrt{1-v^2/c^2}}\frac{dp^i}{dt} = \dots$$

which is what we need to recover eq. (3.1), apart from the extra factor $1/\sqrt{1-v^2/c^2}$.

Let us try to guess the right hand side. It has to be a 4-vector, and it has to be linear in velocity and linear in electromagnetic field. The 4-vector containing velocity is the 4-velocity u^μ, eqs. (1.41), (1.42). The electric and magnetic fields are contained in the field stress tensor $F^{\mu\nu}$ which is anti-symmetric with respect to μ and ν. So the r.h.s. of our equation must be proportional to the product of $F^{\mu\nu}$ and u^μ. Therefore, we can try the relativistic equation of the form

$$c\frac{dp^\mu}{ds} = \alpha F^{\mu\nu}u_\nu,$$

where α is an unknown coefficient. If our guess is correct, its value will follow from comparison with eq. (3.1).

Recall that

$$\begin{aligned} u^0 &= \frac{c}{\sqrt{1-v^2/c^2}} \\ u^i &= \frac{v_i}{\sqrt{1-v^2/c^2}} \\ F^{i0} &= E_i \\ F^{ij} &= -c\epsilon_{ijk}B_k \end{aligned}$$

Consider the space part of this equation, $\mu = i$. We have

$$\frac{1}{\sqrt{1-v^2/c^2}}\frac{dp^i}{dt} = \alpha(F^{i0}u_0 + F^{ij}u_j) =$$

$$= \alpha\left(E_i\frac{c}{\sqrt{1-v^2/c^2}} + c\epsilon_{ijk}B_k\frac{v_j}{\sqrt{1-v^2/c^2}}\right) =$$

$$= \frac{\alpha c}{\sqrt{1-v^2/c^2}}(E_i + \epsilon_{ijk}v_jB_k).$$

Note that

$$\begin{aligned} u_0 &= u^0 \\ u_i &= -u^i \end{aligned}$$

The second term in the brackets is precisely the i-th component of the vector product of $\boldsymbol{v}$ and $\boldsymbol{B}$. We see therefore that we recover eq. (3.1) by setting $\alpha = e/c$. Thus, eq. (3.1) is the space part of the 4-vector equation

$$\frac{dp^\mu}{ds} = \frac{e}{c^2}F^{\mu\nu}u_\nu. \tag{3.2}$$

The question remains: what is the extra equation, the time component of eq. (3.2) with $\mu = 0$? As we have seen in Sect. 1.3, this equation should not be independent and should express the energy conservation which is a consequence of the equations with $\mu = i$. Let us check if this is indeed

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the case. Setting $\mu = 0$ in eq. (3.2) gives

$$\frac{1}{c\sqrt{1-v^2/c^2}}\frac{dp^0}{dt} = \frac{e}{c^2}F^{0i}u_i = \frac{e}{c^2}\frac{E_i v_i}{\sqrt{1-v^2/c^2}}.$$

Now recall that $p^0 = \mathscr{E}/c$ where $\mathscr{E}$ is the energy. Thus, the above equation is equivalent to

$$\frac{d\mathscr{E}}{dt} = e\boldsymbol{E}\boldsymbol{v}.$$

Throughout this section we use calligraphic $\mathscr{E}$ for the energy and normal $\boldsymbol{E}$ for the electric field.

The meaning of this equation is obvious: the rate of the energy change is equal to the work performed by the Lorentz force per unit time. Note that only the electric field does the work, the reason being that the magnetic force is always perpendicular to the velocity and thus perpendicular to the displacement. We see that the time component of eq. (3.2) is a consequence of the three space equations, and therefore eq. (3.2) is completely equivalent to eq. (3.1).

3.2. Motion in constant fields

Constant $\boldsymbol{E}$

Consider first the case when the magnetic field is absent, $\boldsymbol{B} = 0$, and the electric field is constant. Let us chose the coordinate system is such a way that the only non-zero component of the electric field is the x-component, $\boldsymbol{E} = (E, 0, 0)$. Once this is done, we can still rotate the axes around the x-axis to ensure that the initial momentum of a particle lies in the in the plane (xy). Then there is no force in the z-direction and the motion is entirely in the plane (xy). In components, eq. (3.1) reduces to the following equations

$$\begin{aligned}\frac{dp_x}{dt} &= eE,\\ \frac{dp_y}{dt} &= 0,\end{aligned}$$

which can be easily integrated:

$$\begin{aligned}p_x &= eEt,\\ p_y &= p_0 = \text{const}.\end{aligned}$$

In the first equation we have absorbed an arbitrary integration constant into redefinition of t, or in other words, have chosen the moment $t = 0$ so that at this moment $p_x = 0$. We can always reintroduce this constant by replacing $t \to t - t_0$ in our solution, then p_x will be zero at $t = t_0$.

Once we know the momentum, we can determine the particle energy as a function of time from the standard relativistic relation:

$$\mathscr{E} = \sqrt{m^2c^4 + c^2p^2} = \sqrt{m^2c^4 + c^2p_0^2 + (ceEt)^2} = \sqrt{\mathscr{E}_0^2 + (ceEt)^2},$$

where

$$\mathscr{E}_0 \equiv \sqrt{m^2c^4 + c^2p_0^2}$$

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is the particle energy at $t = 0$.

To find the particle position as a function of time one may use the relativistic relation between momentum and velocity (1.59),

$$\boldsymbol{v} = \frac{c^2 \boldsymbol{p}}{\mathscr{E}}. \tag{3.3}$$

For the x-component we have

$$v_x = \frac{dx}{dt} = \frac{c^2 p_x}{\mathscr{E}} = \frac{c^2 eEt}{\sqrt{\mathscr{E}_0^2 + (ceEt)^2}}.$$

It is straightforward to integrate this equation. One finds

$$x(t) = x_0 - \frac{\mathscr{E}_0}{eE} + \frac{1}{eE}\sqrt{\mathscr{E}_0^2 + (ceEt)^2}. \tag{3.4}$$

We have chosen the integration constant in such a way that $x(0) = x_0$.

The projection of eq. (3.3) onto y-axis gives the equation

$$\frac{dy}{dt} = \frac{c^2 p_0}{\mathscr{E}} = \frac{c^2 p_0}{\sqrt{\mathscr{E}_0^2 + (ceEt)^2}}.$$

Integrating this equation one finds

$$y(t) = y_0 + \frac{p_0 c}{eE}\operatorname{arcsh}\frac{ceEt}{\mathscr{E}_0}. \tag{3.5}$$

Again, we have chosen the integration constant so that $y(0) = y_0$.

One may exclude t from eqs. (3.4) and (3.5) and find the trajectory $x(y)$. From eq. (3.5) we have

$$ceEt = \mathscr{E}_0 \operatorname{sh}\left[\frac{eE}{cp_0}(y - y_0)\right].$$

This has to be substituted into eq. (3.4), which gives

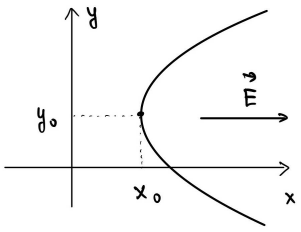
$$x - x_0 = \frac{\mathscr{E}}{eE}\left\{\operatorname{ch}\left[\frac{eE}{cp_0}(y - y_0)\right] - 1\right\}.$$

This determines the explicit dependence $x(y)$.

Recall that

$$\operatorname{ch}^2(x) - \operatorname{sh}^2(x) = 1$$
$$\operatorname{sh}(0) = 0$$
$$\operatorname{ch}(0) = 1$$

The entire trajectory is shown on the plot. The particle starts at some initial moment of time t somewhere on the curve and then moves either towards positive y in the case $p_0 > 0$, or towards negative y in the opposite case. The complementary part of the trajectory represents either the past motion at times before t, or the motion from the same starting point with the initial momentum $-p_0$. In all cases, as $t \to \infty$ both momentum and energy grow infinitely, while the velocity



$$|v| = c\left\{\frac{c^2 p_0^2 + (ceEt)^2}{m^2 c^4 + c^2 p_0^2 + (ceEt)^2}\right\}^{1/2}$$

tends to the velocity of light $v \to c$ while always staying smaller.

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As a final comment we note that if we take eq. (3.4) and multiply it by eE, we find

$$eE(x(t) - x_0) = \sqrt{\mathscr{E}_0^2 + (ceEt)^2} - \mathscr{E}_0 = \mathscr{E} - \mathscr{E}_0.$$

This is nothing but the energy conservation: the work of the electric force $eE(x - x_0)$ equals to the energy gain $\mathscr{E} - \mathscr{E}_0$ by the particle.

Constant B

• In the case of zero electric field eq. (3.1) becomes

$$\frac{d\boldsymbol{p}}{dt} = e\boldsymbol{v} \times \boldsymbol{B}. \tag{3.6}$$

In this case the particle energy does not depend on time,

$$\frac{d\mathscr{E}}{dt} = 0.$$

One may see this directly from eq. (3.6). Indeed, using the relativistic expression for the energy (1.58) we have

$$\frac{d\mathscr{E}}{dt} = \frac{c^2}{\mathscr{E}}\boldsymbol{p}\frac{d\boldsymbol{p}}{dt} = \frac{c^2 me}{\mathscr{E}\sqrt{1 - v^2/c^2}}\boldsymbol{v} \cdot (\boldsymbol{v} \times \boldsymbol{B}) = 0.$$

Here we expressed $\boldsymbol{p}$ in terms of velocity and substituted $d\boldsymbol{p}/dt$ from eq. (3.6).

Since the energy depends only on the absolute value of particle velocity $v = |\boldsymbol{v}|$, constant energy implies that v is also constant in time. Note that the velocity $\boldsymbol{v}$ itself *is not constant*: its absolute value is conserved, but the direction changes with time. This simplifies eq. (3.6) enormously.

To simplify this equation further recall that

$$\boldsymbol{p} = \frac{m\boldsymbol{v}}{\sqrt{1 - v^2/c^2}} = m\gamma\boldsymbol{v}, \qquad \text{where} \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}.$$

Since $|\boldsymbol{v}| = \text{const}$, the gamma-factor γ is also constant, and we may take it outside of the time derivative in eq. (3.6). The equation then becomes

$$m\gamma\frac{d\boldsymbol{v}}{dt} = e\boldsymbol{v} \times \boldsymbol{B}.$$

Note that this trick only works at $\boldsymbol{E} = 0$, otherwise the energy is not conserved and the gamma-factor depends on time.

Dividing both sides by $m\gamma$ we finally rewrite this equation in the form

$$\frac{d\boldsymbol{v}}{dt} = \boldsymbol{v} \times \boldsymbol{\omega}_B \tag{3.7}$$

where

$$\boldsymbol{\omega}_B = \frac{e\boldsymbol{B}}{m\gamma} = \frac{c^2 e\boldsymbol{B}}{\mathscr{E}} \tag{3.8}$$

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is the vector directed along $\boldsymbol{B}$ which has units of $1/s$ and is interpreted as the *angular velocity*. As we will see shortly, the motion of the particle is circular with the angular velocity given by ω_B called *gyration frequency*.

• In order to solve this equation we first rewrite it in components. Let us choose the coordinates in such a way that the magnetic field has the z-direction, in which case the vector $\boldsymbol{\omega}_B$ has only the z-component,

$$\boldsymbol{\omega}_B = (0, 0, \omega_B).$$

In these coordinates vector eq. (3.7) is equivalent to the following three equation,

$$\frac{dv_x}{dt} = v_y\omega_B, \tag{3.9}$$
$$\frac{dv_y}{dt} = -v_x\omega_B, \tag{3.10}$$
$$\frac{dv_z}{dt} = 0. \tag{3.11}$$

The first two equations form a coupled pair. The last equation decouples and gives

$$v_z = \text{const.}$$

The motion along the z axis is therefore simply $z = z_0 + v_z t$.

There are many ways to solve the remaining two equations. For instance, one may exclude one of the velocity components and obtain a closed second-order equation for the remaining one, which turns out to be the harmonic oscillator equation. For educational purposes, let us use a different method. If we multiply the second equation by i and add to the first one we get

$$\frac{d}{dt}(v_x + iv_y) = -i\omega_B(v_x + iv_y),$$

which is a closed first order differential equation for a complex variable $v_x + iv_y$. The solution is obvious:

$$v_x + iv_y = a\mathrm{e}^{-i\omega_B t}$$

where a is a complex integration constant. We can write this constant in terms of two real parameters, modulus and phase, as $a = v_0 \exp(-i\alpha)$. Then the solution becomes

$$v_x + iv_y = v_0\mathrm{e}^{-i(\omega_B t+\alpha)}.$$

Separating imaginary and real parts we find

$$\begin{aligned} v_x &= v_0\cos(\omega_B t + \alpha), \\ v_y &= -v_0\sin(\omega_B t + \alpha). \end{aligned} \tag{3.12}$$

The total velocity of the particle obviously satisfies

$$\boldsymbol{v}^2 = v_x^2 + v_y^2 + v_z^2 = v_0^2 + v_z^2$$

and is independent of time.

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We can integrate eqs. (3.12) and find the trajectory of the particle,

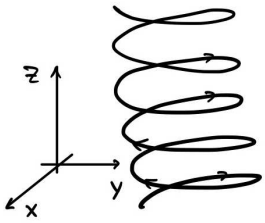
$$\begin{aligned} x(t) &= x_0 + R\sin(\omega_B t + \alpha), \\ y(t) &= y_0 + R\cos(\omega_B t + \alpha), \end{aligned} \tag{3.13}$$

where the constant R is

$$R = \frac{v_0}{\omega_B} = \frac{v_0 \mathscr{E}}{c^2 eB} = \frac{p_\perp}{eB}. \tag{3.14}$$

Here $p_\perp$ is the projection of the particle momentum onto the plane (xy) perpendicular to the magnetic field.

In the absence of the z-component of velocity the motion of the particle is circular, with the orbit centered on the point (x_0, y_0) and having radius R called *gyration radius*. The gyration radius decreases with the increasing magnetic field, and increases with increasing particle momentum (and energy). If the z-component of particle velocity is non-zero, it remains constant, and the trajectory becomes a helix with the axis along z-direction. The radius of the helix is determined by the projection of particle momentum onto the plane perpendicular to the magnetic field.



• The deflection of particles in the magnetic field has many applications, and is at the origin of a variety of astrophysical phenomena. To get an idea of numbers, consider a few examples.

The Earth possesses magnetic field. At the Earth surface its value is of the order of $0.3\,\mathrm{G} = 3\cdot 10^{-5}\mathrm{T}$. In space around the Earth the field decreases with distance from the center roughly like a dipole field, that is $\propto 1/r^3$. At double the Earth radius (6400 km above the ground) the field is therefore ~ 8 times weaker than at the surface. So let us take as an example $B = 10^{-6}$ T, which is the field at about 10'000 km above the Earth surface. Consider cosmic ray electrons of energy $E = 1$ MeV. In the same units the rest energy of an electron is $m_e c^2 \simeq 0.5$ MeV, so these electrons are relativistic with the gamma-factor $\gamma = 2$. Their momentum is

$$p = \frac{1}{c}\sqrt{E^2 - m_e^2 c^4} = m_e c\sqrt{3} \simeq 10^{-30}\mathrm{kg}\cdot 3\cdot 10^8 \frac{\mathrm{m}}{\mathrm{s}}\sqrt{3} = 5\cdot 10^{-22}\frac{\mathrm{kg\,m}}{\mathrm{s}}.$$

In order to determine whether these cosmic ray electrons can directly bombard the Earth surface, or get trapped by the magnetic field, let us calculate the gyration radius:

$$R = \frac{p}{eB} = \frac{5\cdot 10^{-22}\frac{\mathrm{kg\,m}}{\mathrm{s}}}{1.6\cdot 10^{-19}\mathrm{C}\;10^{-6}\mathrm{T}} \simeq 3\cdot 10^3\mathrm{m} = 3\;\mathrm{km}.$$

This is much smaller than the Earth radius, so the electrons make circles around the magnetic field lines and are trapped for a long time. This gives rise to radiation belts (Van Allen belts) around the Earth.

• As a second example consider an accelerator. Suppose we want to accelerate protons to energy ~ 6 TeV. The maximum magnetic field that can be currently achieved in a sufficiently large volume is about 8 Tesla (the magnetic field of the Large Hadron Collider). What is the minimum size of the ring that is required?

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The rest energy of a proton is $m_p c^2 \simeq 1$ GeV, so the 6 TeV protons have the gamma-factor $\gamma = 6000$, and the momentum

$$p \simeq m_p c\gamma = 3 \cdot 10^{-15} \frac{\text{kg m}}{\text{s}}.$$

For the gyro-radius we therefore find

$$R = \frac{3 \cdot 10^{-15} \text{kg m/s}}{1.6 \cdot 10^{-19} \text{C } 8\text{T}} = 2.3\text{km}.$$

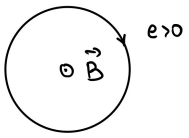
This is close to the actual LHC radius of 4.3 km.

why the difference by about factor 2?

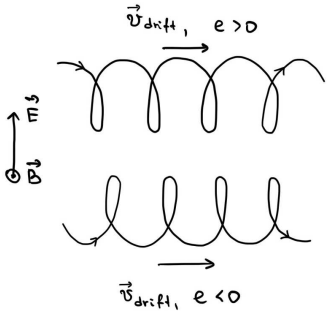
• The third example is a mass spectrometer.

add an example

Constant E and B, non-relativistic case



• Before discussing a general case consider qualitatively the motion in *perpendicular* electric and magnetic fields. As we have seen above, if $\boldsymbol{E} = 0$ the motion of a particle is circular, with the radius of the orbit depending on the particle momentum un the plane perpendicular to $\boldsymbol{B}$. What happens if we switch on the electric field perpendicular to the magnetic field?



Let us assume for simplicity that the electric field is small so that it can be viewed as a correction. When the particle circles around the magnetic field direction, part of the trajectory it moves along the electric field, and another part against the electric field. Assuming its charge is positive, at the first part it accelerates, while at the second decelerates. In the first case the radius of the orbit becomes larger, in the second – smaller. As a result, the orbit drifts (is systematically displaced) *in the direction perpendicular to* $\boldsymbol{E}$. If the same argument is repeated for a negatively charged particle, one finds that it drifts *in the same direction*.

Let us now go back to a general case when $\boldsymbol{E}$ is not necessarily perpendicular to $\boldsymbol{B}$, and calculate the drift velocity $\boldsymbol{v}_{\text{drift}}$. For simplicity we will assume that the particle is non-relativistic, $|\boldsymbol{v}| \ll c$ and $\boldsymbol{p} \simeq m\boldsymbol{v}$. Then the equation (3.1) reads

$$m\frac{d\boldsymbol{v}}{dt} = e\boldsymbol{E} + e\boldsymbol{v} \times \boldsymbol{B}.$$

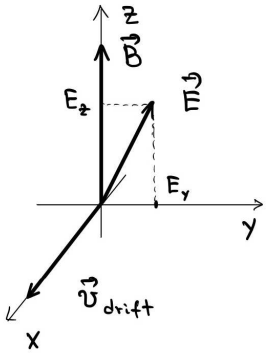
Without loss of generality, we may assume that the magnetic field is directed along the z-axis. In general, the electric field is not parallel to the magnetic field and thus has a non-zero projection onto the (xy)-plane. By rotating coordinates around the z-axis we can make the x-component of the electric field vanish, so that in the resulting frame we have

$$\begin{aligned} \boldsymbol{B} &= (0, 0, B), \\ \boldsymbol{E} &= (0, E_y, E_z). \end{aligned}$$

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In this frame the equations take the form

$$\begin{aligned} m\frac{dv_x}{dt} &= ev_yB, \\ m\frac{dv_y}{dt} &= -ev_xB + eE_y, \\ m\frac{dv_z}{dt} &= eE_z, \end{aligned}$$



The third equation decouples (note that this only happens in the non-relativistic limit) and can be easily solved,

$$\begin{aligned} v_z &= \frac{eE_z}{m}t + v_{0z}, \\ z &= z_0 + v_{0z}t + \frac{eE_z}{2m}t^2, \end{aligned}$$

where z_0 and v_{0z} are two integration constants. The motion in the z-direction is a uniformly accelerated motion. Obviously, this solution is only valid as long as $v_z \ll c$.

Consider the other two equations. We may use the same method as in the absence of the electric field, that is, we multiply the second equation by i and add to the first one:

$$m\frac{d}{dt}(v_x + iv_y) = -ieB(v_x + iv_y) + ieE_y.$$

If we introduce the variable $u = v_x + iv_y$, we may rewrite this equation as

$$\frac{du}{dt} = -i\omega_B u + i\frac{eE_y}{m}.$$

This is an inhomogeneous linear equation; its general solution is a sum of a general solution to the homogeneous part and a particular solution to the inhomogeneous equation. The former we have found above in sect. 3.2. The latter is easy to guess: obviously, a constant function $u = eE_y/(m\omega_B)$ satisfies the equation. Therefore, a general solution is

$$u(t) = v_0 \mathrm{e}^{-i\omega_B t - i\alpha} + \frac{eE_y}{m\omega_B}.$$

Separating the imaginary and real parts we finally find

$$\begin{aligned} v_x &= v_0 \cos(\omega_B t + \alpha) + \frac{E_y}{B}, \\ v_y &= -v_0 \sin(\omega_B t + \alpha). \end{aligned}$$

Here we have substituted the expression for ω_B, eq. (3.8) with $\gamma = 1$.

We see that the particle velocity in the plane (xy) has a constant contribution E_y/B in the x-direction. This is the drift velocity. For the particle trajectory we get

$$\begin{aligned} x(t) &= x_0 + \frac{E_y}{B}t + R\sin(\omega_B t + \alpha), \\ y(t) &= y_0 + R\cos(\omega_B t + \alpha), \end{aligned}$$

where, as in sect. 3.2, $R = v_0/\omega_B$. In the (xy)-plane, the particle makes

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circles of radius R around the point $(x_0 + (E_y/B)t, y_0)$ which is moving with the velocity

$$v_{\text{drift}} = \frac{E_y}{B}.$$

We may write this relation in the vector form as

$$\boldsymbol{v}_{\text{drift}} = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2} \tag{3.15}$$

which makes no reference to a particular frame. Interestingly, the drift velocity is independent of e and m.

Constant $\boldsymbol{E}$ and $\boldsymbol{B}$, general case

In the preceding sections, we have considered several examples of a particle motion in constant electric and magnetic fields of different configurations. At this point it is worth recalling that the Lorentz equation (3.1) is a relativistic equation, so that one may solve it in any frame and then transform into the frame of interest. It may happen, therefore, that some of the considered cases are related by a Lorentz transformation and thus are not, in fact, independent. In order to address this question, we have to consider in more detail how the electric and magnetic fields $\boldsymbol{E}$ and $\boldsymbol{B}$ change upon Lorentz transformations.

First, we recall that while the fields themselves are not invariant, there exist (see sect. 2.3) two combinations of the fields which are. These invariant combinations are

$$\begin{aligned} I_1 &= c^2\boldsymbol{B}^2 - \boldsymbol{E}^2, \\ I_2 &= \boldsymbol{E}\boldsymbol{B}. \end{aligned}$$

The cases in which these invariants are different are certainly not related by any change of frame.

For the discussion of this section we will need the actual transformation rules for the fields themselves. It is straightforward to derive transformations for $\boldsymbol{E}$ and $\boldsymbol{B}$ directly from the general transformation rule in the 4-vector form,

$$F'^{\mu\nu} = \Lambda^{\mu}{}_{\sigma}\Lambda^{\nu}{}_{\rho}F^{\sigma\rho}$$

where $\Lambda^{\mu}{}_{\sigma}$ is the usual Lorentz transformation matrix. For a boost in the x-direction it is given by eq. (1.38). As an example, consider the x-component of the electric field which corresponds to the component F^{10} of the field stress tensor. After the boost in the x-direction the new value of the x-component of the electric field is

$$E'_x = F'^{10} = \Lambda^1{}_{\sigma}\Lambda^0{}_{\rho}F^{\sigma\rho} = \Lambda^1{}_0\Lambda^0{}_1F^{01} + \Lambda^1{}_1\Lambda^0{}_0F^{10} =$$

$$= -\text{sh}^2\theta F^{10} + \text{ch}^2\theta F^{10} = F^{10} = E_x.$$

We see that the x-component of $\boldsymbol{E}$ does not change. In a similar way one may determine the transformation laws of other components of $\boldsymbol{E}$ and $\boldsymbol{B}$. Together all these relations may be written in terms of the 3-vectors $\boldsymbol{E}$

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and $\boldsymbol{B}$ as follows (see Jackson),

$$\boldsymbol{E}' = \gamma(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) - \frac{\gamma^2}{1+\gamma}\frac{\boldsymbol{v}}{c^2}(\boldsymbol{v}\boldsymbol{E}), \tag{3.16}$$

$$\boldsymbol{B}' = \gamma(\boldsymbol{B} - \frac{\boldsymbol{v}}{c^2} \times \boldsymbol{E}) - \frac{\gamma^2}{1+\gamma}\frac{\boldsymbol{v}}{c^2}(\boldsymbol{v}\boldsymbol{B}), \tag{3.17}$$

where $\boldsymbol{v}$ is the relative velocity of the frames and $\gamma = 1/\sqrt{1 - v^2/c^2}$ is the corresponding gamma-factor.

Using these expressions one may show (see Appendix A) that if either $I_1 \neq 0$ or $I_2 \neq 0$, then one can make a boost in the direction perpendicular to both $\boldsymbol{E}$ and $\boldsymbol{B}$ to a frame where the electric and magnetic fields are parallel, $\boldsymbol{E}'||\boldsymbol{B}'$. Therefore, if $I_2 = 0$ and $I_1 \neq 0$, then either $\boldsymbol{E}'$ or $\boldsymbol{B}'$ will vanish in the new frame, depending on the sign of I_1. The particle motion then reduces to the case of pure electric, or pure magnetic field. In all other cases, including the one when $I_1 = I_2 = 0$, both electric and magnetic fields are non-zero in any frame. The solution in this case may be found analytically in the parametric form.

• With this new insight, we may now generalize our treatment of the perpendicular fields in sect. 3.2 to a fully relativistic case. If the electric and magnetic fields are perpendicular to each other and $c|\boldsymbol{B}| > |\boldsymbol{E}|$, one can go to a frame where the electric field *is zero* by making a boost in the direction perpendicular to both $\boldsymbol{E}$ and $\boldsymbol{B}$. In this frame the motion is circular in the plane perpendicular to the magnetic field. Obviously, the velocity $\boldsymbol{v}$ of this new frame relative to the old one is nothing but the drift velocity that we introduced in sect. 3.2.

Let us find the boost velocity at which the new electric field vanishes. Eq. (3.16) simplifies when $\boldsymbol{v} \perp \boldsymbol{E}$ and becomes

$$\boldsymbol{E}' = \gamma(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}).$$

To have $\boldsymbol{E}' = 0$ we therefore must require

$$\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B} = 0.$$

When $\boldsymbol{B} \perp \boldsymbol{E}$ this equation is solved by

$$\boldsymbol{v} = \frac{\boldsymbol{E} \times \boldsymbol{B}}{B^2}.$$

Thus we recover our result of sect. 3.2, eq. (3.15), but now without the non-relativistic approximation. Note that from eq. (3.17) we have in the new frame

$$\boldsymbol{B}' = \frac{1}{\gamma}\boldsymbol{B},$$

so that the magnetic field there is weaker because $\gamma > 1$.

3.3. Slowly varying magnetic field

So far we have considered static electromagnetic fields which are constant over the whole space. In any realistic situation this is an approximation. How does the motion change if the fields vary in space and/or in time?

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This problem is not analytically solvable in general. However, if the fields change slowly, one might expect that the solutions that we have found in the previous section can serve as a reasonable first approximation.

We will be primarily interested in motion in the magnetic field. In the case of constant magnetic field the motion of charged particles is circular in the plane perpendicular to the magnetic field lines, and is characterized by the frequency ω_B and the radius of the orbit R. How does this motion change if the field varies slowly? More precisely, "slowly" means that the relative change of the field $\Delta B/B$ has to be much smaller than one over times of order $1/\omega_B$ and distances of order R:

$$\frac{1}{B\omega_B}\frac{\partial B}{\partial t} \ll 1, \qquad \frac{R}{B}\frac{\partial B}{\partial x} \ll 1.$$

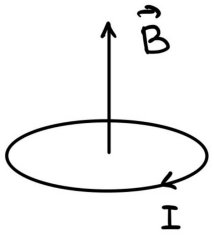
The presence of these small parameters makes the problem tractable analytically.

Adiabatic invariance of magnetic flux through particle orbit

It is a general property of mechanical systems with periodic motion that there exist certain combinations of variables which are approximately preserved when the external parameters of the system change slowly. These combinations are called *adiabatic invariants*.

Consider again a particle of charge e moving in a uniform magnetic field with the velocity perpendicular to it. Such a particle rotates around a circle of radius R given by eq. (3.14) with the angular velocity ω_B, eq. (3.8), and the period $T/(2\omega_B)$. On the time scales much longer than the period, this motion can be viewed as a circular loop of current with the area πR^2 and the current

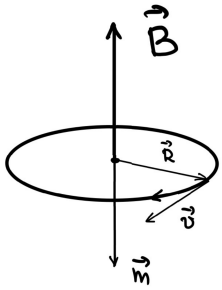
$$I = \frac{e}{T} = \frac{1}{2\pi} e\omega_B.$$



As we have seen in sect. 2.3, a flat loop of current has a magnetic moment whose value is

$$m = I \cdot \pi R^2 = \frac{1}{2} e\omega_B R^2 = \frac{1}{2}\frac{ec^2}{\mathscr{E}} \cdot BR^2$$

where $\mathscr{E}$ is the particle energy. The direction of the magnetic moment is perpendicular to the plane of the loop. Moreover, one may easily see that it is anti-parallel to the magnetic field $\boldsymbol{B}$.



The product BR^2 is, up to a factor π, the flux Φ of the magnetic field through the loop of current,

$$\Phi = \pi BR^2.$$

Let us show that this quantity is an adiabatic invariant, that is, it is preserved under slow changes of the magnetic field.

Consider first the case when the magnetic field slowly varies with time. In this case the flux through the particle orbit changes, and therefore by the Faraday law there is an induced "electromotive force" or f.e.m. As we have already discussed in sect. 2.3, this is actually an electric

field with non-zero circulation along the contour. By definition of the "electromotive force", it equals the work that one has to perform to move a unit electric charge a full circle along the contour. Therefore, as the particle makes a complete circle, its energy changes by

$$\Delta\mathscr{E} = e\frac{d\Phi}{dt} = e\pi R^2\frac{dB}{dt}.$$

If the magnetic field varies slowly, one may approximate the derivative $d\mathscr{E}/dt$ by the ratio $\Delta\mathscr{E}/T$, so

$$\frac{d\mathscr{E}}{dt} \simeq \frac{\Delta\mathscr{E}}{T} = \frac{1}{2}e\omega_B R^2\frac{dB}{dt}. \tag{3.18}$$

Since the motion along the magnetic field completely decouples, this relation is also applicable when the particle has a non-zero momentum along the magnetic field.

Consider now the change with time of the quantity

$$\frac{e^2}{\pi}\Phi = e^2BR^2 = \frac{p_\perp^2}{B},$$

where $p_\perp$ is the particle momentum in the plane perpendicular to $\boldsymbol{B}$. We have

$$\frac{d}{dt}\left(\frac{p_\perp^2}{B}\right) = \frac{1}{B}\frac{d(p_\perp^2)}{dt} - \frac{p_\perp^2}{B^2}\frac{dB}{dt}. \tag{3.19}$$

To progress further we need to determine $dp_\perp^2/dt$. Using the general expression for the relativistic energy we may relate it to the rate of energy change,

$$\mathscr{E}^2 = m^2c^4 + c^2p_\perp^2 + c^2p_\parallel^2 \qquad \Longrightarrow \qquad \frac{dp_\perp^2}{dt} = \frac{1}{c^2}\frac{d\mathscr{E}^2}{dt} = \frac{2\mathscr{E}}{c^2}\frac{d\mathscr{E}}{dt}.$$

Substituting this expression into eq. (3.19) we find

$$\frac{d}{dt}\left(\frac{p_\perp^2}{B}\right) = \frac{2\mathscr{E}}{c^2B}\frac{d\mathscr{E}}{dt} - \frac{p_\perp^2}{B^2}\frac{dB}{dt} =$$

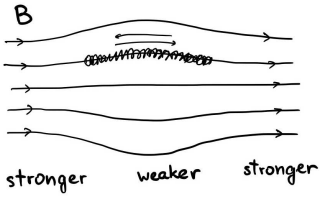
$$= \frac{e\mathscr{E}}{c^2B}\omega_B R^2\frac{dB}{dt} - \frac{p_\perp^2}{B^2}\frac{dB}{dt} =$$

$$= \left\{\frac{e\mathscr{E}}{c^2B}\frac{ec^2B}{\mathscr{E}}\frac{p_\perp^2}{e^2B^2} - \frac{p_\perp^2}{B^2}\right\}\frac{dB}{dt} = 0.$$

In the last equation we have substituted explicit expressions for ω_B and R.

• When we derived this result we only used the Faraday induction law. We know that this law is applicable not only when the magnetic field itself changes, but also when the contour moves in the magnetic field which is constant in time, but varying in space. Therefore, our conclusion that the magnetic flux through the orbit remains approximately constant should be valid in that case as well.

This leads to an interesting consequence for the motion of a charged particle in a static, but space-dependent field — magnetic trapping of charged particles. Consider a field which is almost uniform, but is somewhat weaker in a localized region, see the plot. One may imagine,



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for instance, the field in a gap between two identical solenoids placed one after another along the same line.

Since the magnetic field is static, there is no f.e.m. induced, and the particle energy is conserved. In general, the particle momentum can have both components — perpendicular and parallel to $\boldsymbol{B}$. In the uniform magnetic field they wold be separately conserved, but in a non-uniform field this is not the case. First, we note that if the parallel momentum $p_{||}$ is not too large, the adiabatic approximation holds and the flux through the particle orbit is (approximately) conserved. This implies that as the particle orbit moves along the magnetic lines and the local value of B changes, the perpendicular momentum $p_\perp$ must also change to keep the flux constant:

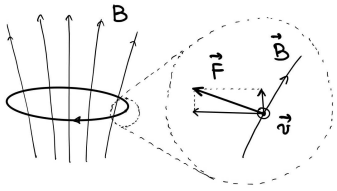
Quantitatively, we require that the relative change of the magnetic field value after a full gyration circle is much smaller than one, $\Delta B/B \ll 1$.

$$\Phi = \pi B R^2 \propto \frac{p_\perp^2}{B} = \text{const} \qquad \Longrightarrow \qquad p_\perp \propto \sqrt{B}.$$

Thus, when the particle is moving so that the magnetic field increases, $p_\perp$ also increases like $\sqrt{B}$. But since the particle energy is constant (because the magnetic field is static),

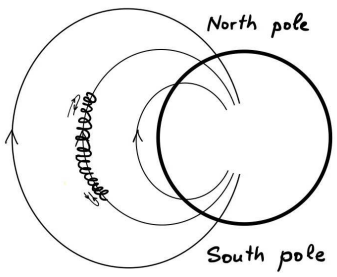
$$\frac{\mathscr{E}^2}{c^2} = m^2c^2 + p_\perp^2 + p_{||}^2 = \text{const},$$

when $p_\perp$ increases the parallel momentum $p_{||}$ must decrease. Thus, when particle is moving towards the region of stronger field it slows down. At some point it stops (the parallel momentum becomes zero), turns back and goes in the opposite direction until it reaches another region of stronger field where the story repeats. The particle is therefore trapped between the two regions of stronger B.



The physical origin of the force in the direction of weaker field, which stops the particle and makes it turn back, is clear from the picture. In a slightly inhomogeneous magnetic field, the force which makes the particle turn in circles is not directed exactly to the center of the orbit, but has a small projection perpendicular to the plane of the orbit. This projection is always directed towards the region of the weaker field independently of the phase of the gyration, so it does not disappear when averaged over many circles.

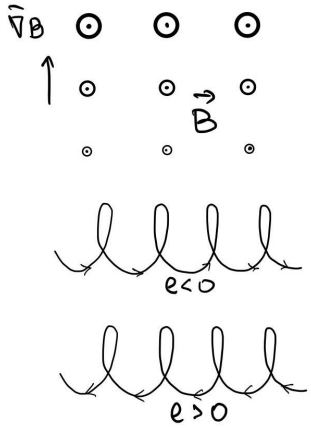
This result is, of course, in agreement with the force acting on a magnetic moment in a slowly varying field which we have derived in sect. 2.3, eq. (2.85). Recall that a particle circling in a magnetic field is equivalent to a magnetic moment that is directed *against* the field, so the scalar product in eq. (2.85) is negative, and therefore the force acts in the direction opposite to the gradient of B, i.e., towards the region of a weaker field.



The magnetic field of the Earth acts as a natural magnetic trap for cosmic ray particles. Around the equatorial plane the magnetic field lines are roughly parallel to the Earth surface (perpendicular to the plane). They become progressively denser (therefore, the field becomes stronger) as one goes along the magnetic lines towards the poles. Particles that circle around the magnetic field lines in the equatorial region experience a force that repels them from the poles. They are, therefore, confined in this region if their parallel momentum is not sufficient to overcome this force. This is the reason for the existence of the Van Allen radiation belts around the Earth.

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Drift in a non-uniform magnetic field



Consider now the motion in the magnetic field which has a constant direction, but changing magnitude in the direction perpendicular to the magnetic field, see the plot. As before, we will assume that this change is slow: the *relative* change of the magnetic field magnitude at distances of order of the particle orbit is much smaller than one. The corresponding small dimensionless parameter is

$$\epsilon = \frac{R}{B}\frac{dB}{dx} \ll 1. \tag{3.20}$$

Qualitatively, the situation is similar to that in the case of electric field: as the particle moves around its circular orbit, in the region where the field is stronger it turns faster, while where the field is weaker — slower. We therefore expect that the orbit will *drift* in the direction perpendicular to both the magnetic field and its gradient. The opposite charges will drift in the opposite directions. We want to calculate the drift velocity.

Since the electric field is zero, the motion is described by the equation

$$\frac{d\boldsymbol{p}}{dt} = e\boldsymbol{v} \times \boldsymbol{B},$$

and the particle energy is conserved. We may therefore rewrite this equation as

$$\frac{d\boldsymbol{v}}{dt} = \boldsymbol{v} \times \boldsymbol{\omega}_B \qquad \boldsymbol{\omega}_B = \frac{e\boldsymbol{B}}{m\gamma}$$

with $\boldsymbol{\omega}_B$ given by eq. (3.8).

Let us choose the coordinates in such a way that the magnetic field is directed along the z-axis,

$$\boldsymbol{B} = (0, 0, B(x)).$$

Then there is no force in the z-direction, and the motion in this direction decouples. We may assume that the velocity along $\boldsymbol{B}$ is zero and concentrate on the motion in the plane perpendicular to $\boldsymbol{B}$.

We will solve the resulting equation of motion using the perturbation theory. In the leading approximation we can neglect the variation of $\boldsymbol{B}$. Then the motion of particle is circular as described in sect. 3.2. The leading order particle trajectory $\boldsymbol{x}_0(t)$ and velocity $\boldsymbol{v}_0(t)$ are given by eqs. (3.13) and (3.12), respectively. In the next approximation we have to take into account the change of the magnetic field with coordinates. This results in a correction to the particle trajectory, so that we can write $\boldsymbol{x}(t) = \boldsymbol{x}_0(t) + \boldsymbol{x}_1(t)$, where the correction $\boldsymbol{x}_1(t)$ is small as compared to the leading term. Because the correction is small, we only need to take it into account in the linear order; the equations for $\boldsymbol{x}_1(t)$ will therefore be linear, hence the simplification.

In a systematic way this strategy is implemented as follows. Let us choose the origin of coordinates in the center of the unperturbed orbit and expand the magnetic field in the Taylor series,

$$B(x) = B_0 + \boldsymbol{\nabla}B \cdot \boldsymbol{x} + \ldots = B_0\left(1 + \frac{\boldsymbol{\nabla}B \cdot \boldsymbol{x}}{B_0} + \ldots\right).$$

Note that the gradient $\boldsymbol{\nabla}B$ lies in the plane (xy) because the magnetic field must have zero divergence, so that

$$\boldsymbol{\nabla}\boldsymbol{B} = \frac{\partial B}{\partial z} = 0.$$

Therefore, B must be independent of z.

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In the last equation the second term is of order ϵ for $|\boldsymbol{x}|$ of order R, the radius of the unperturbed orbit. So, this term is small as compared to 1. The higher order terms in the expansion are suppressed by even higher powers of ϵ, and we will neglect them.

When we substitute this into the equation of motion we get

$$\frac{d\boldsymbol{v}}{dt} = \boldsymbol{v} \times \boldsymbol{\omega}_0 \left(1 + \frac{\boldsymbol{\nabla} B \cdot \boldsymbol{x}}{B_0} + \dots\right), \tag{3.21}$$

where $\omega_0 = eB_0/(m\gamma)$ is the unperturbed frequency. The second term on the right is a correction; if we neglect it we recover, of course, the equation for the unperturbed circular orbit.

This equation is too complicated to solve exactly because it is non-linear. So let us use the perturbation technique outlined above. To do so, we write the trajectory as a sum of the unperturbed part and the correction,

$$\begin{aligned} \boldsymbol{x}(t) &= \boldsymbol{x}_0(t) + \boldsymbol{x}_1(t) + \dots \\ \boldsymbol{v}(t) &= \boldsymbol{v}_0(t) + \boldsymbol{v}_1(t) + \dots \end{aligned}$$

where the corrections $\boldsymbol{x}_1(t)$ and $\boldsymbol{v}_1(t)$ are supposed to be $O(\epsilon)$ and dots stand for higher order terms. We now substitute these expressions into eq. (3.21):

$$\frac{d\boldsymbol{v}_0}{dt} + \frac{d\boldsymbol{v}_1}{dt} + \dots = (\boldsymbol{v}_0 + \boldsymbol{v}_1 + \dots) \times \boldsymbol{\omega}_0 \left(1 + \frac{1}{B_0} \boldsymbol{\nabla} B \cdot (\boldsymbol{x}_0 + \boldsymbol{x}_1 + \dots) + \dots\right).$$

This equation is still equivalent to the original eq. (3.21). The idea now is to expand the r.h.s. in powers of ϵ and equate terms of the same order.

Leading order. In the leading order we only keep the first terms from both sides and get

$$\frac{d\boldsymbol{v}_0}{dt} = \boldsymbol{v}_0 \times \boldsymbol{\omega}_0.$$

This is the equation describing the motion in the constant field $\boldsymbol{B}_0$, the one we have solved in sect. 3.2. Its solution is $\boldsymbol{x}_0(t) = (x_{01}, x_{02}, 0)$ where

$$x_{01}(t) = R \sin(\omega_0 t + \alpha), \tag{3.22}$$

$$x_{02}(t) = R \cos(\omega_0 t + \alpha), \tag{3.23}$$

and α is an arbitrary phase. We also have the relations

$$\begin{aligned} \boldsymbol{v}_0 &= \boldsymbol{x}_0 \times \boldsymbol{\omega}_0, \\ \boldsymbol{\omega}_0 \times \boldsymbol{v}_0 &= \omega_0^2 \boldsymbol{x}_0, \end{aligned}$$

and so on. These are the standard relations for a circular motion.

First order. On the left there is only one term of order ϵ: $d\boldsymbol{v}_1/dt$. On the right we have two contributions that are linear in the correction, so in this order we have

$$\frac{d\boldsymbol{v}_1}{dt} = \boldsymbol{v}_1 \times \boldsymbol{\omega}_0 + \boldsymbol{v}_0 \times \boldsymbol{\omega}_0 \frac{(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)}{B_0} =$$

$$= \left\{ \boldsymbol{v}_1 + \boldsymbol{v}_0 \frac{(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)}{B_0} \right\} \times \boldsymbol{\omega}_0. \tag{3.24}$$

Note that the second term proportional to $(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)/B_0$ is already of order ϵ, so we do not need to include further corrections to it since they will be $O(\epsilon^2)$.

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This equation determines $\boldsymbol{v}_1(t)$ once $\boldsymbol{x}_0(t)$ and $\boldsymbol{v}_0(t)$ are known. In mathematical terms, this is a linear inhomogeneous equation for $\boldsymbol{v}_1(t)$. The fact that the equation for the perturbation is *linear* is generic. It is guaranteed by the method itself, and is not specific to our particular problem.

Even though there exist general methods to solve linear inhomogeneous equations (e.g., the Green's function method we have discussed in sect. 2.2), the solution in our case will be too complicated to be useful. From the qualitative discussion in the beginning of this section we expect that the motion will be approximately circular, while the center of the orbit will slowly drift with the velocity that we want to determine. This drift velocity is contained in the velocity $\boldsymbol{v}_1$.

In order to separate the drift velocity from periodic corrections to the circular motion we may *average* the particle velocity over many periods. The circular velocity and the periodic corrections will average to zero, while the systematic drift will remain. Mathematically, the average of a time-dependent function $f(t)$ is defined as

$$\langle f \rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T f(t)dt.$$

Averaging is linear, that is for two functions $f(t)$ and $g(t)$

$$\langle \alpha f + \beta g \rangle = \alpha \langle f \rangle + \beta \langle g \rangle$$

where α and β are two constants.

Obviously, a constant averages to itself. Also note that the averaging is a linear operation.

The oscillating circular velocity averages to zero, $\langle \boldsymbol{v}_0 \rangle = 0$. Therefore, the average velocity of our particle will come only from $\boldsymbol{v}_1$,

$$\langle \boldsymbol{v} \rangle = \langle \boldsymbol{v}_0 \rangle + \langle \boldsymbol{v}_1 \rangle = \langle \boldsymbol{v}_1 \rangle.$$

Let us determine $\langle \boldsymbol{v}_1 \rangle$ directly from eq. (3.24) without actually solving it. To this end we take average of both sides:

$$\langle \frac{d\boldsymbol{v}_1}{dt} \rangle = \left\{ \langle \boldsymbol{v}_1 \rangle + \langle \boldsymbol{v}_0 \frac{(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)}{B_0} \rangle \right\} \times \boldsymbol{\omega}_0. \tag{3.25}$$

We have taken $\boldsymbol{\omega}_0$ out of the average because it is constant. Let us have a look at the l.h.s.:

$$\langle \frac{d\boldsymbol{v}_1}{dt} \rangle = \lim_{T\to\infty} \frac{1}{T} \int_0^T \frac{d\boldsymbol{v}_1}{dt} dt = \lim_{T\to\infty} \frac{\boldsymbol{v}_1(T) - \boldsymbol{v}_1(0)}{T} = 0.$$

In the last equality we have used the fact that $\boldsymbol{v}_1(t)$ stays limited while T goes to infinity. Thus, we have proven a general statement: the average of a time derivative of a bounded function is zero. Therefore, eq. (3.25) gives

$$\langle \boldsymbol{v}_1 \rangle = -\langle \boldsymbol{v}_0 \frac{(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)}{B_0} \rangle = \boldsymbol{\omega}_0 \times \langle \boldsymbol{x}_0 \frac{(\boldsymbol{\nabla} B \cdot \boldsymbol{x}_0)}{B_0} \rangle$$

where we have replaced $\boldsymbol{v}_0$ by its expression in terms of $\boldsymbol{x}_0$ in order to simplify subsequent calculations.

It remains to calculate the average. Let us first rewrite this equation in components. Since the vector $\boldsymbol{\omega}_0 = (0, 0, \omega_0)$ only has the z-component, the velocity $\boldsymbol{v}_1$ lies in the plane (xy). For its two components we have the

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following equation,

$$\langle v_1 \rangle_a = \omega_0 \epsilon_{a3b} \frac{\nabla_c B}{B_0} \langle x_{0b} x_{0c} \rangle,$$

where $a, b, c = 1, 2$. Both components of $\boldsymbol{x}_0$ are harmonic functions, eqs.(3.22) and (3.23). Their averages are

$$\begin{aligned} \langle x_{01} x_{01} \rangle &= R^2 \langle \sin^2(\omega_0 t + \alpha) \rangle = \frac{1}{2} R^2, \\ \langle x_{02} x_{02} \rangle &= R^2 \langle \cos^2(\omega_0 t + \alpha) \rangle = \frac{1}{2} R^2, \\ \langle x_{01} x_{02} \rangle &= R^2 \langle \cos(\omega_0 t + \alpha) \sin(\omega_0 t + \alpha) \rangle = 0. \end{aligned}$$

Together we may write these relations as

$$\langle x_{0a} x_{0b} \rangle = \frac{1}{2} R^2 \delta_{ab}.$$

For the drift velocity we have, therefore,

$$\langle v_1 \rangle_a = \frac{R^2}{2B_0} \omega_0 \epsilon_{a3b} \nabla_b B$$

or, in vector notations,

$$\boldsymbol{v}_{\text{drift}} = \langle \boldsymbol{v}_1 \rangle = \frac{R^2}{2B_0} \boldsymbol{\omega}_0 \times \boldsymbol{\nabla} B. \tag{3.26}$$

This is our final result.

From eq. (3.26), the magnitude of the drift velocity can be written as (recall that we consider the case when $\boldsymbol{\nabla} B$ is perpendicular to $\boldsymbol{B}$ and thus to $\boldsymbol{\omega}_0$)

$$|\boldsymbol{v}_{\text{drift}}| = \frac{1}{2} R\omega_0 \frac{R|\boldsymbol{\nabla} B|}{B_0} = \frac{1}{2} \epsilon v_0.$$

We see therefore that in the approximation $\epsilon \ll 1$ the drift velocity of the orbit as a whole is much smaller than the velocity of the particle, $|\boldsymbol{v}_{\text{drift}}| \ll v_0$.

• As an example, consider again the magnetic field of the Earth. To a good approximation it can be represented as a dipole with the value

$$m_E = 8 \cdot 10^{22} \text{A m}^2$$

in the direction from the North to the South pole, $\boldsymbol{m}_E = (0, 0, -m_E)$. We can therefore write the magnetic field as follows (cf. multipole expansion of the magnetic field in Sect. 2.3),

$$\boldsymbol{B} = \frac{\mu_0}{4\pi} \frac{3\boldsymbol{x}(\boldsymbol{x}\boldsymbol{m}_E) - \boldsymbol{m}_E x^2}{x^5}.$$

In the equatorial plane one has $\boldsymbol{x} \perp \boldsymbol{m}_E$, so only the last term contributes. The field is then

$$\boldsymbol{B} = -\frac{\mu_0}{4\pi} \frac{\boldsymbol{m}_E}{x^3}.$$

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If we substitute here the numbers we get the field at the equator

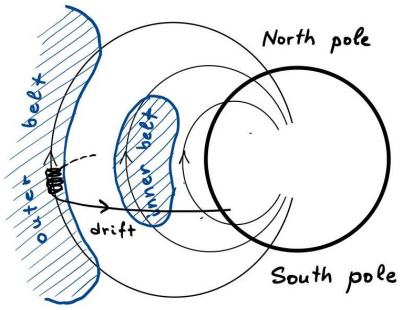
$$\begin{aligned} \frac{\mu_0}{4\pi} &= 10^{-7}\frac{\mathrm{N}}{\mathrm{A}^2}, \\ x &= R_E = 6400\,\mathrm{km}, \\ \Longrightarrow\ B &= 3.1\cdot 10^{-5}\mathrm{T} = 0.31\,\mathrm{Gauss} \end{aligned} \tag{3.27}$$

which is the correct value.

Since the field is proportional to $1/x^3$, at a distance from the center equal twice the Earth radius the field is 8 times weaker and equals

$$B = 0.39\cdot 10^{-5}\mathrm{T}.$$

Let us take this as a characteristic value. The Earth actually has two radiation belts. The inner one extends from $\sim (0.2-2)R_E$ above the Earth surface, while the outer one is in the range $\sim (3-10)R_E$. The value above is characteristic of the inner belt.



Consider as an example protons of energy 100 MeV ($\gamma = 1.1$) and electrons of energy 100 keV ($\gamma = 1.2$). Both values are typical for the inner belt. They corresponds to velocities

$$v_p = 0.42c,$$

$$v_e = 0.55c.$$

In the above field protons and electrons have gyration frequencies

$$\omega_p = \frac{eB}{m_p\gamma_p} \simeq 350\,\mathrm{s}^{-1},$$

$$\omega_e = \frac{eB}{m_e\gamma_e} \simeq 6.4\times 10^5\,\mathrm{s}^{-1},$$

respectively. From the frequency and velocity one finds the gyration radius $R = v/\omega$ for each particle,

$$R_p = 360\,\mathrm{km},$$

$$R_e = 0.26\,\mathrm{km}.$$

Both are much smaller than the characteristic length scale, the Earth radius R_E.

In the equatorial plane the field is everywhere perpendicular to the plane, but varies as a function of r. We should expect, therefore, that both protons and electrons will experience the gradient drift. The gradient of the magnetic field is in the radial direction towards the center. Knowing that the dipole field decreases with distance as $\propto 1/r^3$, we can write its gradient as

$$\partial_r B = -\frac{3B}{r}.$$

The minus sign indicates that its direction is towards the center. We can now calculate the expansion parameter of the perturbation theory for both particles, as well as their drift velocities. From eq. (3.20) we find,

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substituting $r = 2R_E$,

$$\epsilon_p = \frac{3R_p}{2R_E} = 0.08,$$

$$\epsilon_e = \frac{3R_e}{2R_E} = 6 \times 10^{-5}.$$

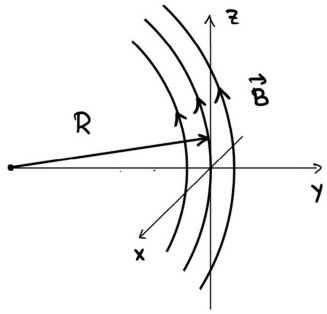
For the drift velocities we may use eq. (3.26), or directly the relation $v_{\rm drift} = \epsilon v/2$. We find

$$v_{\rm drift,p} = 5000\,{\rm km/s},$$

$$v_{\rm drift,e} = 5\,{\rm km/s}.$$

with this velocity, it takes protons a few seconds only to make a full turn around the Earth.

• There remains one more type of magnetic field variation to consider, namely the one where magnetic field slowly changes the direction, so that the field lines are curved, see the figure where R represents the radius of curvature of the field lines. In general, a particle may have the velocity component $v_{||}$ parallel to the field, as well as the perpendicular component $v_\perp$. In this case, in addition to the gyration it will move along the field line with the velocity $v_{||}$. Since the lines are curved, it will experience the centrifugal force $\boldsymbol{F} = mv_{||}^2\boldsymbol{R}/R^2$ which will result in the drift of the particle orbit in the direction perpendicular to the force.

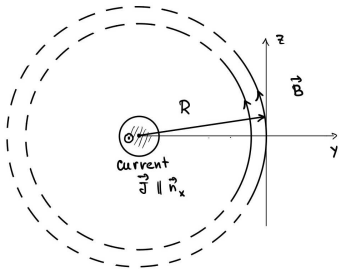


In order to find the drift velocity one may notice that exactly the same force would act on the particle if there were a radial electric field of the value $\boldsymbol{E}_{\rm eff} = \boldsymbol{F}/e = mv_{||}^2\boldsymbol{R}/(eR^2)$. Since forces are the same, the motion is the same, and we can use the expression for the drift in the electric field, where we have to substitute $\boldsymbol{E}_{\rm eff}$. Thus, we find the drift velocity

$$\boldsymbol{v}_C = \frac{\boldsymbol{E}_{\rm eff} \times \boldsymbol{B}}{B^2} = \frac{v_{||}^2}{\omega_0 R}\frac{\boldsymbol{R} \times \boldsymbol{B}}{RB}. \tag{3.28}$$

Here ω_0 is the gyration frequency in the field B.

This is not, however, the full story. In many applications (including the Earth magnetic field) one may assume, to a good approximation, that the region of interest is free from currents. In this case the magnetic field, apart from zero divergence $\boldsymbol{\nabla B} = 0$ also has zero curl, $\boldsymbol{\nabla} \times \boldsymbol{B} = 0$. As a consequence, in this case the magnitude of the field cannot be constant in the radial direction if the magnetic field lines are circular. The reason is easy to understand. Instead of explicitely calculating the curl, let us imagine that the field in the region of interest is a part of an *axially-symmetric* azimuthal magnetic field. By the Ampere law, the field at the distance r from the axis satisfies $2\pi r B(r) = I$, where I is the total current passing through the circle of radius r. But since there is no current in the region of interest, and therefore, by axial symmetry, nowhere at distances $\sim r$ from the axis, the enclosed current I is the same for different values of r in this range. Therefore, we must have



$$B(r) \propto \frac{1}{r},$$

and

$$\boldsymbol{\nabla} B = -\boldsymbol{n}_r \frac{B_0}{R},$$

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where $\boldsymbol{n}_r$ is the unit vector in the radial direction and B_0 the amplitude the field at the origin.

The gradient of the magnetic field gives rise to the gradient drift in addition to the curvature one, with the drift velocity given by eq. (3.26),

$$\boldsymbol{v}_G = \frac{R_g^2}{2B}\boldsymbol{\omega}_0 \times \boldsymbol{\nabla} B,$$

where $\boldsymbol{\omega}_0$ and R_g are the gyration frequency and radius, respectively. Using explicit expressions for these quantities we can rewrite the gradient drift velocity as follows,

$$\boldsymbol{v}_G = \frac{1}{2B}\left(\frac{m\gamma v_\perp}{qB}\right)^2 \frac{q}{m\gamma}\boldsymbol{B} \times \left(-\frac{\boldsymbol{R}B}{R^2}\right) = \frac{1}{2}\frac{v_\perp^2}{\omega_0 R}\frac{\boldsymbol{R}\times\boldsymbol{B}}{RB}.$$

The combined drift velocity equals

$$\boldsymbol{v}_{\text{drift}} = \boldsymbol{v}_C + \boldsymbol{v}_G = \frac{1}{\omega_0 R}\left(\frac{1}{2}v_\perp^2 + v_{||}^2\right)\frac{\boldsymbol{R}\times\boldsymbol{B}}{RB}.$$

If one assumes that the velocities $v_\perp \sim v_{||}$ are due to thermal motion with temperature T, the numerical value of the drift velocity is estimated as

$$\boldsymbol{v}_{\text{drift}} \simeq 2\frac{\text{m}}{\text{s}}\left(\frac{T}{1^\circ\,\text{K}}\right)\left(\frac{1\text{m}}{R}\right)\left(\frac{1\text{G}}{B}\right).$$

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4. Electromagnetic waves

We will discuss in this chapter a particular class of solutions to the Maxwell equations — the electromagnetic waves. These are general solutions to the *homogeneous* part of Maxwell equations. They exist even in the absence of charges and currents.

4.1. Wave equation

In Chapter 2 we have considered static electric and magnetic fields produced by a time-independent distribution of charges and currents. When time dependence is allowed, the Maxwell equations admit another type of solutions that exist even in the absence of charges and currents. These solutions are *electromagnetic waves*. They may be viewed as oscillations of $\boldsymbol{E}$ and $\boldsymbol{B}$ about the equilibrium state $\boldsymbol{E} = \boldsymbol{B} = 0$. These oscillations propagate in space and carry energy and momentum.

We start from the Maxwell equations in the vacuum, that is, in the equations (2.95)-(2.98) of Chapter 2 we set the charge density and the current density to zero. In the absence of charges and currents the Maxwell equations depend only on one constant – the product $\epsilon_0\mu_0$. Using the definitions of ϵ_0 and μ_0 this constant can be written as

$$\frac{1}{\epsilon_0\mu_0} = 8.988 \cdot 10^{16}\frac{\mathrm{m}^2}{\mathrm{s}^2} = \left(2.998 \cdot 10^8 \frac{\mathrm{m}}{\mathrm{s}}\right)^2 \equiv c^2.$$

As we will see below, Maxwell equations describe propagation of waves, and the constant c is the velocity of these waves. In terms of c the Maxwell equations read

$$\boldsymbol{\nabla}\boldsymbol{E} = 0 \tag{4.1}$$

$$\boldsymbol{\nabla}\times\boldsymbol{B} - \frac{1}{c^2}\frac{\partial \boldsymbol{E}}{\partial t} = 0 \tag{4.2}$$

$$\boldsymbol{\nabla}\times\boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = 0 \tag{4.3}$$

$$\boldsymbol{\nabla}\boldsymbol{B} = 0 \tag{4.4}$$

where the last two equations are the homogeneous Maxwell equations which are valid even in the presence of charges and currents.

Our first step is to derive a closed equation for the components of $\boldsymbol{E}$ and $\boldsymbol{B}$. In order to exclude one of the variables let us calculate the curl of eq. (4.3):

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{E}) + \frac{\partial}{\partial t}\boldsymbol{\nabla}\times\boldsymbol{B} = 0.$$

For the first term, component i, we have

$$\epsilon_{ijk}\partial_j\epsilon_{kmn}\partial_m E_n = (\delta_{im}\delta_{jn} - \delta_{in}\delta_{jm})\partial_j\partial_m E_n = \partial_i(\boldsymbol{\nabla}\boldsymbol{E}) - \Delta E_i = -\Delta E_i$$

where we have used eq. (4.1). Thus, in the vector form,

$$\boldsymbol{\nabla}\times(\boldsymbol{\nabla}\times\boldsymbol{E}) = -\Delta\boldsymbol{E}.$$

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For the second term we use eq. (4.2) to express $\nabla \times \boldsymbol{B}$ in terms of $\boldsymbol{E}$. Adding the two terms we get a closed equation for $\boldsymbol{E}$:

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{E} = 0. \tag{4.5}$$

This is called the *d'Alambert equation*, and the operator in brackets the d'Alambert operator. In mathematical terms, this equation is a *consequence* of eqs. (4.1)–(4.4), meaning any solution of eqs. (4.1)–(4.4) must satisfy the d'Alambert equation, but not the other way around.

Repeating the same steps starting from eq. (4.2) gives the d'Alambert equation for $\boldsymbol{B}$,

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{B} = 0. \tag{4.6}$$

We see that, as a consequence of the Maxwell equations, each of the six components of $\boldsymbol{E}$ and $\boldsymbol{B}$ must individually satisfy the d'Alambert equation.

Digression. Before discussing the solutions, let us show that the same d'Alambert equation is satisfied by the vector and scalar potentials. We have seen before that the potentials are not defined uniquely — they can be changed by a gauge transformation without changing $\boldsymbol{E}$ and $\boldsymbol{B}$. Explicitely, this transformation reads:

$$\begin{aligned} \phi &\rightarrow \phi + \partial_t \lambda(t, \boldsymbol{x}), \\ \boldsymbol{A} &\rightarrow \boldsymbol{A} - \nabla\lambda(t, \boldsymbol{x}), \end{aligned}$$

where $\lambda(t, \boldsymbol{x})$ is an arbitrary function. We can choose this function as we wish; let us choose it in such a way that the scalar potential vanishes, $\phi = 0$. Note that this condition does not fix the gauge freedom completely: making an additional transformation with $\lambda_1(\boldsymbol{x})$ which *does not* depend on t leaves ϕ unchanged. In the gauge $\phi = 0$ the electric and magnetic fields are expressed in terms of $\boldsymbol{A}$ as follows,

$$\begin{aligned} \boldsymbol{E} &= -\frac{\partial \boldsymbol{A}}{\partial t} - \nabla\phi = -\frac{\partial \boldsymbol{A}}{\partial t}, \\ \boldsymbol{B} &= \nabla \times \boldsymbol{A}. \end{aligned} \tag{4.7}$$

If we substitute $\boldsymbol{E}$ in this gauge into the first of the Maxwell equations we find

$$\frac{\partial}{\partial t}\nabla\boldsymbol{A} = 0$$

which tells us that the divergence $\nabla\boldsymbol{A}$ is time-independent. We therefore may use the residual gauge transformations with a time-independent $\lambda_1(\boldsymbol{x})$ to set $\nabla\boldsymbol{A} = 0$.

In the resulting gauge where $\phi = 0$ and $\nabla\boldsymbol{A} = 0$ eq. (4.2) gives:

$$\nabla \times (\nabla \times \boldsymbol{A}) + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\boldsymbol{A} = 0$$

The first term is (by the same calculation as above)

$$\nabla \times (\nabla \times \boldsymbol{A}) = \nabla(\nabla\boldsymbol{A}) - \Delta\boldsymbol{A} = -\Delta\boldsymbol{A},$$

so that finally we get the d'Alambert equation for $\boldsymbol{A}$,

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{A} = 0.$$

It is important to note that, starting from general ϕ and $\boldsymbol{A}$, we have found that any solution to the Maxwell equations can be constructed from scalar and vector potentials which are, in turn, written in terms of only two functions (two components of $\boldsymbol{A}$ in the Coulomb gauge), both satisfying the d'Alambert equation. One says that electromagnetic field has *two degrees of freedom*. We will see below how the same fact manifests itself in terms of $\boldsymbol{E}$ and $\boldsymbol{B}$.

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4.2. Plane wave

Let us now discuss particular solutions to Maxwell equations — the plane waves. Since the components of $\boldsymbol{E}$ and $\boldsymbol{B}$ must individually satisfy the d'Alambert equation, consider the solutions to this equation first, and then use the results to construct solutions to the full set of Maxwell equations (4.1)– (4.4).

For a general function $f(t, \boldsymbol{x})$ the d'Alambert equation reads

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2} - \frac{\partial^2}{\partial y^2} - \frac{\partial^2}{\partial z^2}\right) f(t, x, y, z) = 0, \tag{4.8}$$

where we have written all the variables explicitely. The *plane wave* is a configuration that is constant along a plane, that is, independent of the two out of three space coordinates. Assuming these two coordinates are y and z, we have therefore $f = f(t, x)$. The d'Alambert equation then becomes

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial x^2}\right) f(t, x) = 0. \tag{4.9}$$

To find a general solution of this equation, let us make the following change of variables:

$$\begin{aligned} \xi &= ct - x, \\ \eta &= ct + x. \end{aligned}$$

For the derivatives we have then

$$\begin{aligned} \partial_x &= \frac{\partial \eta}{\partial x}\partial_\eta + \frac{\partial \xi}{\partial x}\partial_\xi = \partial_\eta - \partial_\xi \\ \frac{1}{c}\partial_t &= \frac{1}{c}\frac{\partial \eta}{\partial t}\partial_\eta + \frac{1}{c}\frac{\partial \xi}{\partial t}\partial_\xi = \partial_\eta + \partial_\xi. \end{aligned}$$

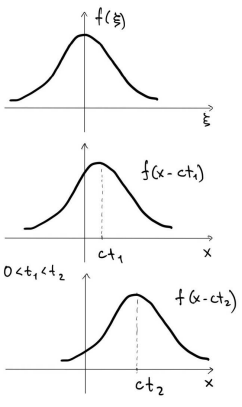
The d'Alambert equation becomes

$$\left\{(\partial_\eta + \partial_\xi)^2 - (\partial_\eta - \partial_\xi)^2\right\} f(\xi, \eta) = 4\partial_\xi \partial_\eta f(\xi, \eta) = 0.$$

This equation is trivial to solve. The general solution consists of the sum of two arbitrary functions, one depending only on ξ and another only on η,

$$f = f_1(\xi) + f_2(\eta) = f_1(ct - x) + f_2(ct + x).$$

It is straightforward to check by direct substitution that this solution satisfies the original eq. (4.9).



The two functions $f_1(ct-x)$ and $f_2(ct+x)$ describe a wave that propagates, without changing its shape, in the *positive* and *negative* direction of x, respectively. For instance, consider f_1. If at $t = 0$ it has a value f_* at the point x_0, so that $f_1(-x_0) = f_*$, at time t it will have the same value f_* at $x = x_0 + ct$ which is shifted by ct towards positive x. The propagation velocity obviously equals c.

These solutions are easy to generalize to the case of an arbitrary direction of propagation. Take

$$f(t, \boldsymbol{x}) = f(ct - \boldsymbol{n}\boldsymbol{x})$$

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where $\boldsymbol{n}$ is a unit vector. First, this is a plane wave because it takes the same value at any point of a plane perpendicular to $\boldsymbol{n}$. The points on such a plane can be parameterized as $\boldsymbol{x} + \boldsymbol{a}$ where $\boldsymbol{a}$ is an arbitrary vector perpendicular to $\boldsymbol{n}$. For all these points $f(ct - \boldsymbol{n}(\boldsymbol{x} + \boldsymbol{a})) = f(ct - \boldsymbol{n}\boldsymbol{x})$. By the same argument as above, this function describes a wave propagating in the direction $\boldsymbol{n}$. Finally, it is straightforward to check that this function satisfies the d'Alambert equation (4.8) for any direction of $\boldsymbol{n}$,

$$\left(\frac{1}{c^2}\frac{\partial^2}{\partial t^2} - \partial_i\partial_i\right) f(ct - \boldsymbol{n}\boldsymbol{x}) = f'' - n_i n_i f'' = 0,$$

where f'' denotes the second derivative of f with respect to its argument.

The d'Alambert equations for $\boldsymbol{E}$ and $\boldsymbol{B}$ are solved in exactly the same way. Let us use the plane waves to construct the solution to the full set of Maxwell equations (4.1)– (4.4). Since these equations mix the components of $\boldsymbol{E}$ and $\boldsymbol{B}$, we should take the waves propagating in the same direction for all of them,

$$\boldsymbol{E} = \boldsymbol{E}(ct - \boldsymbol{n}\boldsymbol{x}), \tag{4.10}$$

$$\boldsymbol{B} = \boldsymbol{B}(ct - \boldsymbol{n}\boldsymbol{x}). \tag{4.11}$$

At this point we have 6 independent functions. We will see, however, that Maxwell equations impose further relations among them, so that at the end only two remain independent.

Before we proceed let us note that the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ enter the Maxwell equations only through their derivatives. Therefore, any fields that are constant in space and time are solutions. These solutions are trivial and we disregard them in what follows.

Let us start with the first eq. (4.1). Denote the argument of the field by $\xi \equiv ct - \boldsymbol{n}\boldsymbol{x}$. Substituting the electric field (4.10) we get

$$0 = \boldsymbol{\nabla}\boldsymbol{E} = -n_i\partial_\xi E_i = -\partial_\xi(\boldsymbol{n}\boldsymbol{E})$$

Thus,

$$\boldsymbol{n}\boldsymbol{E} = \text{const},$$

and since we agreed to ignore constant fields we get

$$\boldsymbol{n}\boldsymbol{E} = 0. \tag{4.12}$$

Repeating the argument for the last equation (4.4) we find in a similar way

$$\boldsymbol{n}\boldsymbol{B} = 0. \tag{4.13}$$

We see that in the plane wave the directions of both $\boldsymbol{E}$ and $\boldsymbol{B}$ must be orthogonal to the propagation direction $\boldsymbol{n}$. The field components parallel to $\boldsymbol{n}$ must be zero, so we are down to 4 independent components out of original 6.

Consider now the remaining two equations. From eq. (4.2) we find

$$\boldsymbol{\nabla}\times\boldsymbol{B} - \frac{1}{c^2}\frac{\partial\boldsymbol{E}}{\partial t} = -\boldsymbol{n}\times\partial_\xi\boldsymbol{B} - \frac{1}{c}\partial_\xi\boldsymbol{E} = -\partial_\xi\left(\boldsymbol{n}\times\boldsymbol{B} + \frac{1}{c}\boldsymbol{E}\right) = 0,$$

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from which we conclude that the combination in the brackets is constant. Ignoring again constant fields we find

$$\boldsymbol{n} \times c\boldsymbol{B} + \boldsymbol{E} = 0. \tag{4.14}$$

Repeating the arguments, eq. (4.3) gives

$$\boldsymbol{n} \times \boldsymbol{E} - c\boldsymbol{B} = 0. \tag{4.15}$$

Note that these are vector equations, so formally we have 3 + 3 = 6 more equations on the components of $\boldsymbol{E}$ and $\boldsymbol{B}$.

However, not all of these equations are independent. One of eqs. (4.14), (4.15) is a consequence of the other. For instance, one may express $c\boldsymbol{B}$ from eq. (4.15) and substitute it into eq. (4.14):

$$\boldsymbol{n} \times (\boldsymbol{n} \times \boldsymbol{E}) + \boldsymbol{E} = 0$$

The l.h.s. of this equation is

$$\epsilon_{ijk} n_j \epsilon_{kpq} n_p E_q - E_i = (\delta_{ip}\delta_{jq} - \delta_{iq}\delta_{jp}) n_j n_p E_q - E_i = E_i - n_i(\boldsymbol{nE}) - E_i = 0$$

in view of eq. (4.12). Therefore, once eqs. (4.12), (4.13) and (4.15) are satisfied, eq. (4.14) is also satisfied, so it gives nothing new. (Alternatively, we could have started from eq. (4.14), in which case eq. (4.15) would not be independent.)

So, how many independent components do we finally have? Eqs. (4.12) and (4.13) imply that both $\boldsymbol{E}$ and $\boldsymbol{B}$ lie in the plane perpendicular to $\boldsymbol{n}$. In this plane we may choose, say, $\boldsymbol{E}$ arbitrarily — this makes 2 independent components. After that eq. (4.15) determines $\boldsymbol{B}$ *uniquely*. This $\boldsymbol{B}$ is perpendicular to both $\boldsymbol{n}$ and $\boldsymbol{E}$, and satisfies

$$c|\boldsymbol{B}| = |\boldsymbol{E}|.$$

All the equations (4.12)–(4.15) are satisfied.

We conclude that a plane wave propagating in a given direction $\boldsymbol{n}$ is completely characterized by 2 arbitrary independent functions of $ct - \boldsymbol{nx}$. They may be chosen to be two components of the electric field in the plane orthogonal to $\boldsymbol{n}$. Then eq. (4.15) determines the value of $\boldsymbol{B}$.

For instance, let us choose our coordinates in such a way that $\boldsymbol{n} = (0, 0, 1)$. In this frame take $E_1 = f_1(ct - \boldsymbol{nx})$, $E_2 = f_2(ct - \boldsymbol{nx})$ where f_1 and f_2 are two arbitrary functions, so that $\boldsymbol{E} = (f_1, f_2, 0)$. Then eq. (4.15) gives $\boldsymbol{B} = (1/c)(-f_2, f_1, 0)$. All the Maxwell equations are satisfied. The fields $\boldsymbol{E}$ and $\boldsymbol{B}$ are both orthogonal to $\boldsymbol{n}$ and orthogonal to each other, and also $|c\boldsymbol{B}| = |\boldsymbol{E}|$ as illustrated in Fig. 4.1.

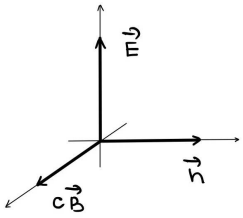


Figure 4.1.: Mutual orientation of $\boldsymbol{n}$, $\boldsymbol{E}$ $\boldsymbol{E}$ in a plane wave propagating in the direction $\boldsymbol{n}$.

4.3. Monochromatic plane wave

Consider now a particular type of plane waves called *monochromatic* plane waves in which both $\boldsymbol{E}$ and $\boldsymbol{B}$ oscillate with the same frequency ω,

$$\boldsymbol{E}, \boldsymbol{B} \propto \mathrm{e}^{-i\omega t}.$$

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The term "monochromatic" originates from the fact that for the electromagnetic waves describing the light the frequency determines the color.

If the plane wave is monochromatic, its dependence on t and $\boldsymbol{x}$ is fixed, because in a plane wave the fields depend on these variables in the combination $ct - \boldsymbol{n}\boldsymbol{x}$:

$$\boldsymbol{E}, \boldsymbol{B} \propto \mathrm{e}^{-i\omega(ct-\boldsymbol{n}\boldsymbol{x})/c} = \mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}} \tag{4.16}$$

where

$$\boldsymbol{k} = \frac{\omega}{c}\boldsymbol{n}, \quad |\boldsymbol{k}| = \frac{\omega}{c} \tag{4.17}$$

is called the *wave vector*.

Some general properties of a monochromatic plane wave can be understood already from its dependence on t and $\boldsymbol{x}$ in eq. (4.16). If we consider *a given point in space*, the fields there change periodically as a function of time. After a time interval

$$T = \frac{2\pi}{\omega}$$

the exponential in eq. (4.16) returns to its original value, and so do the fields $\boldsymbol{E}$ and $\boldsymbol{B}$. Therefore, T is a period of a wave and $f = 1/T$ its frequency. If instead we consider the space dependence of fields *in a given moment of time*, we see from eq. (4.16) that:

(i) in every plane perpendicular to $\boldsymbol{k}$ the fields are the same, as they should for a plane wave

(ii) if $\boldsymbol{x}$ changes by the distance

$$\lambda = \frac{2\pi}{k} = 2\pi\frac{c}{\omega}$$

in the direction of $\boldsymbol{k}$, the phase of the exponential in eq. (4.16) changes by $i\boldsymbol{k}\boldsymbol{n}\lambda = i\boldsymbol{k}\boldsymbol{n}(2\pi/k) = 2\pi i$ and the fields return to the original values.

At a given moment of time the fields are thus *periodic in space* in the direction of $\boldsymbol{k}$ with the period λ. The distance λ is the *wavelength*.

How do we write a general monochromatic plane wave? Since Maxwell equations are linear, it is a common practice to use complex functions. It is assumed that the physical electric and magnetic fields are obtained by taking the real part. If complex fields satisfy Maxwell equations, their real and imaginary parts independently satisfy them as well. So we can write, in general,

$$\boldsymbol{E} = \mathrm{Re}\left[\mathbf{E}\,\mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}\right], \tag{4.18}$$

$$\boldsymbol{B} = \mathrm{Re}\left[\mathbf{B}\,\mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}\right], \tag{4.19}$$

where $\mathbf{E}$ and $\mathbf{B}$ are two constant complex vectors. From the general plane wave relations (4.12) and (4.13) they must be both orthogonal to $\boldsymbol{k}$, while the other two relations (4.14) and (4.15) imply

$$\boldsymbol{k} \times \mathbf{B} + \frac{\omega}{c^2}\mathbf{E} = 0, \tag{4.20}$$

$$\boldsymbol{k} \times \mathbf{E} - \omega\mathbf{B} = 0. \tag{4.21}$$

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To derive these relations we have used eq. (4.17).

This is in fact a most general form. We have seen above that we need 2 functions to characterize a general plane wave — for instance, 2 components of the electric field in the plane perpendicular to $\boldsymbol{n}$. For a monochromatic wave both functions must be of the form (4.16). Each one is characterized by the amplitude and the phase (equivalently, by the coefficients in a linear combination of sine and cosine), so we should have 4 real arbitrary parameters in total. A complex constant vector $\mathbf{E}$ perpendicular to $\boldsymbol{n}$ has exactly 4 real parameters. The magnetic field vector $\mathbf{B}$ is then uniquely determined from eq. (4.21).

4.4. Polarization

The exact behavior of the electric field in a general monochromatic plane wave depends on the constant complex vector $\mathbf{E}$. Different behaviors are referred to as different *polarizations* of the wave. Let us see which polarizations a monochromatic plane wave can have.

A constant complex vector $\mathbf{E}$ in the plane orthogonal to $\boldsymbol{k}$ can be written as a linear combination of two mutually orthogonal unit basis vectors $\boldsymbol{\epsilon}_1$ and $\boldsymbol{\epsilon}_2$ such that

$$\boldsymbol{\epsilon}_1 \boldsymbol{n} = \boldsymbol{\epsilon}_2 \boldsymbol{n} = \boldsymbol{\epsilon}_1 \boldsymbol{\epsilon}_2 = 0,$$

$$\boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2 = \boldsymbol{n}.$$

The two vectors $\boldsymbol{\epsilon}_1$ and $\boldsymbol{\epsilon}_2$ form an orthonormal basis in the plane perpendicular to $\boldsymbol{k}$. We may therefore write

$$\mathbf{E} = A_1 \boldsymbol{\epsilon}_1 + A_2 \boldsymbol{\epsilon}_2,$$

where A_1 and A_2 are two *complex* coefficients having a modulus and a phase each, so that $A_1 = a_1 \mathrm{e}^{i\phi_1}$, $A_2 = a_2 \mathrm{e}^{i\phi_2}$. Four real parameters a_1, a_2, ϕ_1 and ϕ_2 completely characterize the vector $\mathbf{E}$,

$$\mathbf{E} = a_1 \mathrm{e}^{i\phi_1} \boldsymbol{\epsilon}_1 + a_2 \mathrm{e}^{i\phi_2} \boldsymbol{\epsilon}_2. \tag{4.22}$$

and, therefore, determine the polarization of a monochromatic plane wave.

Consider first the simplest case when one of the amplitudes is zero, say $a_2 = 0$. Then

$$\mathbf{E} = a_1 \mathrm{e}^{i\phi_1} \boldsymbol{\epsilon}_1$$

and from eq. (4.18) we have for the electric field:

$$\boldsymbol{E} = a_1 \boldsymbol{\epsilon}_1 \mathrm{Re}\left[\mathrm{e}^{i\phi_1} \mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}\right] = a_1 \boldsymbol{\epsilon}_1 \cos(-\omega t + \boldsymbol{k}\boldsymbol{x} + \phi_1).$$

We see that in this case the electric field of the wave is always parallel to the direction of $\boldsymbol{\epsilon}_1$. The value of $\boldsymbol{E}$ oscillates with the amplitude a_1. Such waves are called *linearly polarized* in the direction $\boldsymbol{\epsilon}_1$. Note that in such a wave the magnetic field is always parallel to $\boldsymbol{\epsilon}_2$, as follows from eq. (4.21).

It is clear that the wave corresponding to $a_1 = 0$ is also linearly polarized, but now in the direction $\boldsymbol{\epsilon}_2$. Moreover, a wave that has both $a_1, a_2 \neq 0$

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but $\phi_1 = \phi_2$ is also linearly polarized. Indeed, in this case

$$\boldsymbol{E} = (a_1\boldsymbol{\epsilon}_1 + a_2\boldsymbol{\epsilon}_2)\mathrm{Re}\left[\mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x} + i\phi_1}\right] = (a_1\boldsymbol{\epsilon}_1 + a_2\boldsymbol{\epsilon}_2)\cos(-\omega t + \boldsymbol{k}\boldsymbol{x} + \phi_1).$$

This is again a wave in which the electric field is always parallel to the same direction, that of the vector $a_1\boldsymbol{\epsilon}_1 + a_2\boldsymbol{\epsilon}_2$. If we define $a = \sqrt{a_1^2 + a_2^2}$, the oscillations of $\boldsymbol{E}$ have the amplitude a, and the direction of $\boldsymbol{E}$ is set by the unit vector $(a_1\boldsymbol{\epsilon}_1 + a_2\boldsymbol{\epsilon}_2)/a$. We can reduce this case to the previous ones by changing the basis in the plane perpendicular to $\boldsymbol{k}$.

• It remains only one situation to consider, the one when both $a_1, a_2 \neq 0$ and $\phi_1 \neq \phi_2$. Denote α the difference of two phases, so that $\phi_2 = \phi_1 + \alpha$. The common part of the phase can then be factored out and eliminated by the change of time t by a constant. So, without loss of generality we can write

$$\mathbf{E} = a_1\boldsymbol{\epsilon}_1 + \mathrm{e}^{i\alpha}a_2\boldsymbol{\epsilon}_2.$$

When α is non-zero, the direction of the electric field is no more constant. To understand what happens, let us start again by considering a particular case $a_1 = a_2$ and $\alpha = \pm\pi/2$. In this case

$$\mathbf{E} = a_1(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2).$$

The corresponding electric field reads

$$\boldsymbol{E} = a_1\mathrm{Re}\left[(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)\mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}\right]$$

$$= a_1\left[\boldsymbol{\epsilon}_1\cos(\omega t - \boldsymbol{k}\boldsymbol{x}) \pm \boldsymbol{\epsilon}_2\sin(\omega t - \boldsymbol{k}\boldsymbol{x})\right].$$

It is easy to understand that this expression describes, in each point in space, a vector of constant magnitude a_1 rotating in the counter-clockwise direction for the "+" sign and in the clockwise direction for the "−" sign. Such polarizations are called *circular*. There are two circular polarizations: *left* for the "+" sign (counter-clockwise) and *right* for the "−" sign (clockwise).

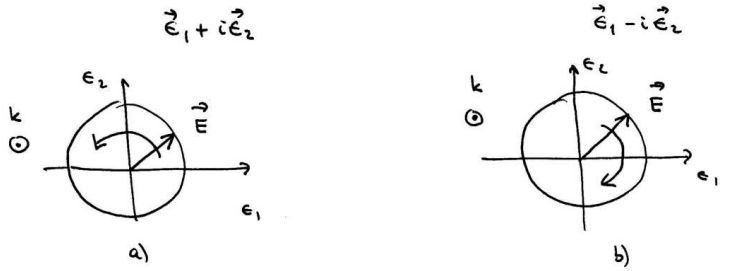


Figure 4.2.: Circular polarizations: (a) – left, (b) – right.

The two circular polarizations may be used as a basis to decompose an arbitrary monochromatic plane wave instead of linearly polarized waves as in eq. (4.22). One defines two normalized complex basis vectors

$$\boldsymbol{\epsilon}_\pm = \frac{1}{\sqrt{2}}(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2) \tag{4.23}$$

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such that

$$\begin{aligned} \boldsymbol{\epsilon}_{\pm}^{*}\boldsymbol{\epsilon}_{\mp} &= 0, \\ \boldsymbol{\epsilon}_{\pm}^{*}\boldsymbol{\epsilon}_{\pm} &= 1. \end{aligned}$$

Then the complex vector of a general monochromatic plane wave can be written as follows:

$$\mathbf{E} = A_{+}\boldsymbol{\epsilon}_{+} + A_{-}\boldsymbol{\epsilon}_{-} \tag{4.24}$$

where again A_+ and A_- are two arbitrary complex coefficients. Without loss of generality one may assume A_+ to be real. Then we can write

$$\begin{aligned} A_{+} &= \text{real positive} \\ A_{-} &= \mathrm{e}^{i\alpha} r A_{+}, \end{aligned}$$

so we have 3 real parameters: amplitude A_+, relative phase α and r,

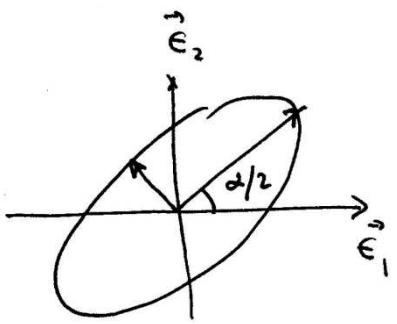


Figure 4.3.: Elliptic polarization: the vector of electric field traces an ellipse in the plane perpendicular to $\boldsymbol{K}$.

the ratio of the magnitudes of the two coefficients. For general values of these parameters the wave has an *elliptical* polarization: at a given pont in space, the vector of the electric field sweeps an ellipse whose semi-axes are $A_+|(1+r)|/\sqrt{2}$ and $A_+|(1-r)|/\sqrt{2}$ and whose semi-major axis is turned with respect to $\boldsymbol{\epsilon}_1$ by an angle $\alpha/2$, see fig. 4.3. In the case $r = 1$ we recover the linear polarization, while in the case $r = 0$ the circular one.

Digression. Let us see how the parameters of the ellipse are expressed in terms of A_+, α and r. To this end we substitute eq. (4.24) into eq. (4.18) and calculate the electric field by separating the real part. We find

$$\boldsymbol{E} = \frac{A_+}{\sqrt{2}}\left[(\boldsymbol{\epsilon}_1 + r\cos\alpha\boldsymbol{\epsilon}_1 + r\sin\alpha\boldsymbol{\epsilon}_2)\cos\theta - (\boldsymbol{\epsilon}_2 + r\sin\alpha\boldsymbol{\epsilon}_1 - r\cos\alpha\boldsymbol{\epsilon}_2)\sin\theta\right]$$

where for shortness we have denoted $\theta \equiv -\omega t + \boldsymbol{kx}$. Since we expect an ellipse turned with respect to the basis vectors, let us rotate the basis by an angle β

$$\begin{aligned} \boldsymbol{\epsilon}_1 &= \boldsymbol{e}_1\cos\beta + \boldsymbol{e}_2\sin\beta \\ \boldsymbol{\epsilon}_2 &= -\boldsymbol{e}_1\sin\beta + \boldsymbol{e}_2\cos\beta \end{aligned}$$

and adjust β in such a way that in the new frame the ellipse is aligned with the axes. In terms of new basis vectors the electric field becomes

$$\boldsymbol{E} = \frac{A_+}{\sqrt{2}}\Big\{\boldsymbol{e}_1(\cos\beta + r\cos(\alpha+\beta))\cos\theta + \boldsymbol{e}_2(\sin\beta + r\sin(\alpha+\beta))\cos\theta$$

$$+\boldsymbol{e}_1(\sin\beta - r\sin(\alpha+\beta))\sin\theta + \boldsymbol{e}_2(-\cos\beta + r\cos(\alpha+\beta))\sin\theta\Big\}.$$

We see that if we choose $\beta = -\alpha/2$ the coefficients in front of the basis vectors simplify, and we get

$$\boldsymbol{E} = \frac{A_+}{\sqrt{2}}\left\{[(1+r)\cos\frac{\alpha}{2}\boldsymbol{e}_1 + (r-1)\sin\frac{\alpha}{2}\boldsymbol{e}_2]\cos\theta\right.$$

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$$-[(1+r)\sin\frac{\alpha}{2}\boldsymbol{e}_1-(r-1)\cos\frac{\alpha}{2}\boldsymbol{e}_2]\sin\theta\Big\}.$$

Re-grouping the terms we finally obtain

$$\boldsymbol{E}=\frac{A_+}{\sqrt{2}}\Big\{(1+r)\cos(\theta+\frac{\alpha}{2})\boldsymbol{e}_1+(r-1)\sin(\theta+\frac{\alpha}{2})\boldsymbol{e}_2\Big\}.$$

When the phase $\theta+\alpha/2=-\omega t+\boldsymbol{k}\boldsymbol{x}+\alpha/2$ changes by 2π the vector $\boldsymbol{E}$ sweeps the ellipse oriented along the axes $\boldsymbol{e}_1$ and $\boldsymbol{e}_2$.

Stokes parameters

How does one measure the polarization of EM waves in practice? In case of light, the easiest quantity to measure is the intensity which is proportional to the square of the amplitude. To determine the polarization parameters one passes the light through different polarization filters and measures intensities of transmitted waves, from which one may then recover the polarization parameters.

• *Stokes parameters* Following Stokes' 1852 one defines four easily measurable quantities completely describing the polarization: I, Q, U and V (in some books the same parameters are denoted S_0, S_1 S_2 and S_3). All these parameters are quadratic in amplitude. To measure them two filters are commonly used: the one that only transmits the component linearly polarized in the direction of the filter polarization axis, and another one that only transmits one of the two circular polarizations. Suppose one measures the following intensities of the transmitted light

no filter	I
linear, axis at 0°	F_0
linear, axis at 90°	F_{90}
linear, axis at 45°	F_{45}
linear, axis at 135°	F_{135}
circular	F_C

Then the Stokes parameters are defined as follows:

$$\begin{aligned}Q&=F_0-F_{90},\\U&=F_{45}-F_{135},\\V&=2F_C-I.\end{aligned}$$

Note that all three Stokes parameters may be positive and negative.

The Stokes parameters may be expressed in terms of the complex electric field vector $\boldsymbol{E}$ used in the previous section. One may decompose this vector either in the linear polarization basis $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, or in the circular polarization basis $\boldsymbol{\epsilon}_\pm$. In terms of the linear polarization basis one has

$$\boldsymbol{E}=a_1\mathrm{e}^{i\phi_1}\boldsymbol{\epsilon}_1+a_2\mathrm{e}^{i\phi_2}\boldsymbol{\epsilon}_2.$$

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The Stokes parameters are expressed in terms of $a_{1,2}$ and $\phi_{1,2}$ as follows:

$$\begin{aligned} I = S_0 = &\quad |\boldsymbol{\epsilon}_1 \boldsymbol{E}|^2 + |\boldsymbol{\epsilon}_2 \boldsymbol{E}|^2 = &\quad a_1^2 + a_2^2 \\ Q = S_1 = &\quad |\boldsymbol{\epsilon}_1 \boldsymbol{E}|^2 - |\boldsymbol{\epsilon}_2 \boldsymbol{E}|^2 = &\quad a_1^2 - a_2^2 \\ U = S_2 = &\quad 2\mathrm{Re}\,[(\boldsymbol{\epsilon}_1 \boldsymbol{E})^*(\boldsymbol{\epsilon}_2 \boldsymbol{E})] = &\quad 2a_1 a_2 \cos(\phi_2 - \phi_1) \\ V = S_3 = &\quad 2\mathrm{Im}\,[(\boldsymbol{\epsilon}_1 \boldsymbol{E})^*(\boldsymbol{\epsilon}_2 \boldsymbol{E})] = &\quad 2a_1 a_2 \sin(\phi_2 - \phi_1) \end{aligned}$$

A purely linear polarization corresponds to $\phi_2 - \phi_1 = 0$, in which case $V = 0$. The direction of polarization is then controlled by Q and U.

Alternatively, in terms of circular polarization basis $\boldsymbol{\epsilon}_\pm$ one has

$$\mathscr{E} = a_+ \mathrm{e}^{i\phi_+} \boldsymbol{\epsilon}_+ + a_- \mathrm{e}^{i\phi_-} \boldsymbol{\epsilon}_- .$$

In these terms

$$\begin{aligned} I = S_0 = &\quad |\boldsymbol{\epsilon}_+^* \boldsymbol{E}|^2 + |\boldsymbol{\epsilon}_-^* \boldsymbol{E}|^2 = &\quad a_+^2 + a_-^2 \\ Q = S_1 = &\quad 2\mathrm{Re}\,[(\boldsymbol{\epsilon}_+^* \boldsymbol{E})^*(\boldsymbol{\epsilon}_-^* \boldsymbol{E})] = &\quad 2a_+ a_- \cos(\phi_- - \phi_+) \\ U = S_2 = &\quad 2\mathrm{Im}\,[(\boldsymbol{\epsilon}_+^* \boldsymbol{E})^*(\boldsymbol{\epsilon}_-^* \boldsymbol{E})] = &\quad 2a_+ a_- \sin(\phi_- - \phi_+) \\ V = S_3 = &\quad |\boldsymbol{\epsilon}_+^* \boldsymbol{E}|^2 - |\boldsymbol{\epsilon}_-^* \boldsymbol{E}|^2 = &\quad a_+^2 - a_-^2 \end{aligned}$$

Purely circularly polarized wave has either $a_+ = 0$ or $a_- = 0$, in which case both $Q = U = 0$.

• *Stokes sphere.* In both parameterizations only the difference of the phases enters in the above expressions. This means that four Stokes parameters depend on three quantities, so there must be a relation among them. Indeed, it is easy to see from explicit expressions that they satisfy

$$I^2 = Q^2 + U^2 + V^2. \tag{4.25}$$

If one imagines the 3-dimensional space of Q, U and V. The values that satisfy eq. (4.25) lie on the sphere of radius I in this space. Each point on this sphere corresponds to a different polarization of the wave. The points on the circle $V = 0$ correspond to linearly polarized wave, each point to a different direction of polarization. The two poles $V = \pm I$ correspond to two circularly polarized states. All other points correspond to an elliptical polarization.

• *Non-coherent waves.* What has been said so far concerns a single EM wave. The real light is a superposition of many waves with different frequencies, phases, polarization states etc. The Stokes parameters which are measured in this case are averages over time and space (the size of the detector). If the individual waves are completely random, Q, U and V average to zero as they do not have a definite sign. If there is a preference for some particular polarization, they may be different from zero, so in general

$$I^2 \geq Q^2 + U^2 + V^2.$$

Such a polarization state is called partially polarized; it is characterized by the point in the QUV-space lying inside the Stokes sphere (4.25).

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4.5. Energy of an electromagnetic wave. Poynting vector

Consider how does energy conservation work for a system consisting of charges and electromagnetic field. A collection of point charges q_i having velocities $\boldsymbol{v}_i$ in an electric field $\boldsymbol{E}(\boldsymbol{x})$ changes its total energy at a rate

$$\sum_i q_i \boldsymbol{v}_i \boldsymbol{E}(\boldsymbol{x}_i)$$

where $\boldsymbol{x}_i$ are position of charges. In the limit of continuous charge distribution we replace $q_i \boldsymbol{v}_i$ by $d^3x \boldsymbol{J}(\boldsymbol{x})$,

$$\sum_i q_i \boldsymbol{v}_i \boldsymbol{E}(\boldsymbol{x}_i) \to \int d^3x \boldsymbol{J}(\boldsymbol{x}) \boldsymbol{E}(\boldsymbol{x}).$$

Let us transform this expression making use of the Maxwell equations. We will see that the resulting equation can be interpreted as the energy conservation for the whole system including the electromagnetic field.

We first express the current from Maxwell equations,

$$\boldsymbol{J} = \frac{1}{\mu_0} \boldsymbol{\nabla} \times \boldsymbol{B} - \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t}.$$

This gives

$$\int_V d^3x \boldsymbol{J}\boldsymbol{E} = \int_V d^3x \boldsymbol{E} \left\{ \frac{1}{\mu_0} \boldsymbol{\nabla} \times \boldsymbol{B} - \epsilon_0 \frac{\partial \boldsymbol{E}}{\partial t} \right\}.$$

The second term is a time derivative of the electric energy density $\epsilon_0 \boldsymbol{E}^2/2$, so we leave it as is. The first term can be rewritten as follows,

$$\boldsymbol{E} \cdot (\boldsymbol{\nabla} \times \boldsymbol{B}) = \epsilon_{ijk} E_i \partial_j B_k = \partial_j (\epsilon_{ijk} E_i B_k) - \epsilon_{ijk} B_k \partial_j E_i =$$

$$= -\boldsymbol{\nabla} \cdot (\boldsymbol{E} \times \boldsymbol{B}) + \boldsymbol{B} \cdot (\boldsymbol{\nabla} \times \boldsymbol{E}) = -\boldsymbol{\nabla} \cdot (\boldsymbol{E} \times \boldsymbol{B}) - \boldsymbol{B} \frac{\partial \boldsymbol{B}}{\partial t}.$$

In the last equality we have used Maxwell equations to replace $\boldsymbol{\nabla} \times \boldsymbol{E}$.

Substituting this into the integral we get

$$\int_V d^3x \boldsymbol{J}\boldsymbol{E} = -\int_V d^3x \left\{ \frac{1}{\mu_0} \boldsymbol{\nabla} \cdot (\boldsymbol{E} \times \boldsymbol{B}) + \frac{1}{\mu_0} \boldsymbol{B} \frac{\partial \boldsymbol{B}}{\partial t} + \epsilon_0 \boldsymbol{E} \frac{\partial \boldsymbol{E}}{\partial t} \right\}. \quad (4.26)$$

Now recall that the energy density of the electromagnetic field is

$$w = \frac{1}{2\mu_0} \boldsymbol{B}^2 + \frac{\epsilon_0}{2} \boldsymbol{E}^2.$$

Thus, we may write eq. (4.26) as follows:

$$\int_V d^3x \left\{ \boldsymbol{J}\boldsymbol{E} + \frac{\partial w}{\partial t} \right\} = -\int_{\partial V} d\boldsymbol{s} \frac{1}{\mu_0} \boldsymbol{E} \times \boldsymbol{B} \equiv -\int_{\partial V} d\boldsymbol{s}\, \boldsymbol{S},$$

where we have defined

$$\boldsymbol{S} = \frac{1}{\mu_0} \boldsymbol{E} \times \boldsymbol{B}. \quad (4.27)$$

This quantity is called **Poynting vector** (Poynting'1884).

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The interpretation of the above relation is as follows: the rate of change of energy contained in the volume V, including charges and electromagnetic field, equals minus the flux of energy through the boundary of the volume V. The Poynting vector, therefore, describes the *energy flux* associated with the electromagnetic field.

Let us see what are the units of $\boldsymbol{S}$. We have

$$\begin{aligned} [E] &= \frac{\mathrm{N}}{\mathrm{C}} \\ [B] &= \frac{\mathrm{N\,s}}{\mathrm{C\,m}} \\ [\mu_0] &= \frac{\mathrm{N\,s^2}}{\mathrm{C^2}} \\ [S] &= [EB/\mu_0] = \frac{\mathrm{N}}{\mathrm{m\,s}} = \frac{\mathrm{J}}{\mathrm{s\,m^2}}. \end{aligned}$$

We see therefore that $\boldsymbol{S}$ has units of energy per area per time, as the energy flux should have.

• Let us calculate Poynting vector of a plane electromagnetic wave. In the case of a plane wave, the magnetic field is expressed in terms of the electric field and the propagation direction $\boldsymbol{n}$ as follows,

$$\boldsymbol{B} = \frac{1}{c}\boldsymbol{n} \times \boldsymbol{E}.$$

The Poynting vector therefore equals

$$\boldsymbol{S} = \frac{1}{c\mu_0}\boldsymbol{E} \times (\boldsymbol{n} \times \boldsymbol{E}) = \sqrt{\frac{\epsilon_0}{\mu_0}} E^2\, \boldsymbol{n}. \tag{4.28}$$

On the other hand, the energy density is

$$w = \frac{1}{2\mu_0}\boldsymbol{B}^2 + \frac{\epsilon_0}{2}\boldsymbol{E}^2 = \frac{1}{2c^2\mu_0}\boldsymbol{E}^2 + \frac{\epsilon_0}{2}\boldsymbol{E}^2 = \epsilon_0 E^2.$$

Comparing the two expressions we see that the Poynting vector can be written as

$$\boldsymbol{S} = w\, c\, \boldsymbol{n}.$$

As expected, the energy flux equals the energy density times the velocity (speed of light).

For a monochromatic plane wave one may express the Poynting vector in terms of two amplitudes of polarizations. In the linear polarization basis, a general monochromatic wave is written as

$$\boldsymbol{E} = \mathrm{Re}\left\{(a_1\boldsymbol{\epsilon}_1 + a_2\boldsymbol{\epsilon}_2 \mathrm{e}^{i\phi})\mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}\right\} =$$

$$= a_1 \cos\theta \boldsymbol{\epsilon}_1 + a_2 \cos(\theta + \phi)\boldsymbol{\epsilon}_2,$$

where we have denoted for shortness $\theta = -\omega t + \boldsymbol{k}\boldsymbol{x}$. We find, therefore, for the average of $\boldsymbol{E}^2$ over a long period of time:

$$\langle \boldsymbol{E}^2 \rangle = \frac{1}{2}(a_1^2 + a_2^2),$$

which may be combined with eq. (4.28) to give the Poynting vector.

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4.6. Electromagnetic waves in media

In the absence of external sources and currents, the Maxwell equations in a medium read

$$\begin{aligned} \boldsymbol{\nabla} \boldsymbol{B} &= 0, \\ \boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} &= 0, \\ \boldsymbol{\nabla} \boldsymbol{D} &= 0, \\ \boldsymbol{\nabla} \times \boldsymbol{H} - \frac{\partial \boldsymbol{D}}{\partial t} &= 0. \end{aligned}$$

If the medium is linear and isotropic, and if its response time is much faster than the frequency of field oscillations, we may assume linear relations

$$\begin{aligned} \boldsymbol{D} &= \epsilon \boldsymbol{E}, \\ \boldsymbol{H} &= \frac{1}{\mu} \boldsymbol{B}. \end{aligned}$$

In this case the last two Maxwell equations become

$$\begin{aligned} \boldsymbol{\nabla} \boldsymbol{E} &= 0, \\ \boldsymbol{\nabla} \times \boldsymbol{B} - \epsilon \mu \frac{\partial \boldsymbol{E}}{\partial t} &= 0. \end{aligned}$$

Together with the first two these equations form a closed system which contains only one constant

$$v^2 = \frac{1}{\epsilon \mu}.$$

This constant has units of velocity squared and determines the speed of electromagnetic waves in the medium.

Since μ is typically very close to μ_0 and ϵ is larger than ϵ_0, the wave propagation speed in medium is smaller than the speed of light in the vacuum. The ratio

$$n \equiv \frac{c}{v} \tag{4.29}$$

is called the **refraction index** of the medium. Obviously, it satisfies $n \geq 1$.

If the response of the medium is not much faster than the frequency of the wave, but is still linear, one may describe the medium by frequency-dependent electric and magnetic constants. In this case, the wave propagation velocity, as well as the refraction index, may depend on the frequency of wave.

• Given that the equations in a linear medium are obtained from those in the vacuum by changing $c \to v$, their solutions are related in the same way. Repeating the same steps as in the vacuum, we find that all

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components of $\boldsymbol{E}$ and $\boldsymbol{B}$ must satisfy a d'Alambert-type equation,

$$\begin{aligned}\left(\frac{1}{v^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{E} &= 0,\\ \left(\frac{1}{v^2}\frac{\partial^2}{\partial t^2} - \Delta\right)\boldsymbol{B} &= 0.\end{aligned} \tag{4.30}$$

Correspondingly, a general plane wave solution is given by arbitrary functions of a single argument,

$$\boldsymbol{E}(vt - \boldsymbol{n}\boldsymbol{x}), \quad \boldsymbol{B}(vt - \boldsymbol{n}\boldsymbol{x}),$$

where the unit vector $\boldsymbol{n}$ specifies the propagation direction. The velocity of propagation of these waves is v.

In order to satisfy the full system of Maxwell equations, the components of $\boldsymbol{E}$ and $\boldsymbol{B}$ must be further constrained. In a way similar to the vacuum case one finds that they must satisfy the equations

$$\begin{aligned}\boldsymbol{n}\boldsymbol{E} &= 0,\\ \boldsymbol{n}\boldsymbol{B} &= 0,\\ -\boldsymbol{n}\times\boldsymbol{E} + v\boldsymbol{B} &= 0,\\ -\boldsymbol{n}\times\boldsymbol{B} - \frac{1}{v}\boldsymbol{E} &= 0.\end{aligned}$$

Like in the vacuum case, only one of the last two equations is linearly independent. The total number of arbitrary functions needed to characterize a plane wave equals 2 as in the vacuum case.

• In the case of a monochromatic wave, all the quantities are periodic functions of time with the frequency ω. In a monochromatic plane wave the dependence on time and space is, therefore, fixed:

$$\boldsymbol{E}, \boldsymbol{B} \propto \mathrm{e}^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}}$$

where the wave vector has to satisfy the relation

$$\boldsymbol{k} = \frac{\omega}{v}\boldsymbol{n}.$$

Note that this equation may be obtained directly by substituting the exponential behavior into eqs. (4.30). The fields of a monochromatic plane wave, apart from being orthogonal to $\boldsymbol{k}$, must obey the relations

$$\boldsymbol{k}\times\boldsymbol{E} - \omega\boldsymbol{B} = 0, \tag{4.31}$$

$$\boldsymbol{k}\times\boldsymbol{B} + \frac{\omega}{v^2}\boldsymbol{E} = 0. \tag{4.32}$$

• *Reflection and refraction* When the properties of medium change as a function of space, the propagation of waves may no longer be straight. The simplest example of this phenomenon is reflection and refraction of waves at the interface between two media with different refraction indices. Fig. 4.4 shows such a situation where two media with refraction indices n' at $z > 0$ and n at $z < 0$ are separated by the plane $z = 0$. The incident plane wave falls along the direction $\boldsymbol{k}$. At the interface it splits into reflected (direction $\boldsymbol{k}''$) and refracted (direction $\boldsymbol{k}'$) plane waves.

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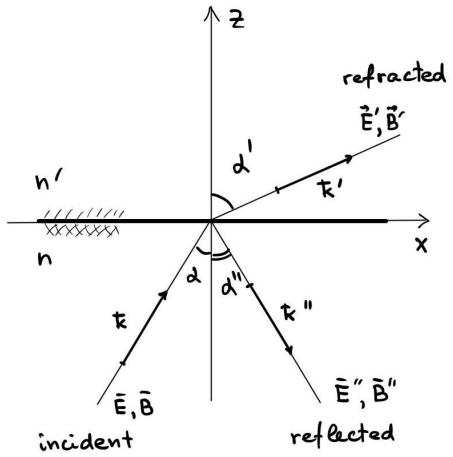


Figure 4.4.: Incident, reflected and refracted plane waves. Only propagation directions are shown on the plot.

The incident and reflected waves occupy the half-space $z < 0$, while the refracted one occupies the part $z > 0$.

The parameters of the incident, reflected and refracted waves are related by the field matching conditions eqs. (2.110), (2.111), (2.123) and (2.124). Since these matching conditions at the interface should be satisfied *at all times*, it is clear that the frequencies of all three waves must be the same. The wave vectors then have the following magnitudes when expressed in terms of the common frequency ω:

$$|\boldsymbol{k}| = |\boldsymbol{k}''| \equiv k = \frac{\omega}{v} = \frac{\omega}{c} n = \omega\sqrt{\epsilon\mu},$$

$$|\boldsymbol{k}'| \equiv k' = \frac{\omega}{v'} = \frac{\omega}{c} n' = \omega\sqrt{\epsilon'\mu'}.$$

We will write the electric fields these three waves as follows:

$$\begin{aligned} \text{incident:} \quad & \boldsymbol{E} e^{-i\omega t + i\boldsymbol{k}\boldsymbol{x}} \\ \text{reflected:} \quad & \boldsymbol{E}'' e^{-i\omega t + i\boldsymbol{k}''\boldsymbol{x}} \\ \text{refracted:} \quad & \boldsymbol{E}' e^{-i\omega t + i\boldsymbol{k}'\boldsymbol{x}}. \end{aligned}$$

where $\boldsymbol{E}$, $\boldsymbol{E}''$ and $\boldsymbol{E}'$ are constant vectors. In all three waves the magnetic field is expressed in terms of the electric field and the wave vector by means of eq. (4.31).

The directions of propagation of the reflected and refracted waves, as well as their amplitudes, are determined by the continuity conditions (2.110), (2.111), (2.123) and (2.124) at the interface of the two media $z = 0$. In terms of the electric and magnetic fields they read:

$$\boldsymbol{n} \times (\boldsymbol{E}_2 - \boldsymbol{E}_1) = 0, \tag{4.33}$$

$$\boldsymbol{n} \cdot (\epsilon_2 \boldsymbol{E}_2 - \epsilon_1 \boldsymbol{E}_1) = 0, \tag{4.34}$$

$$\boldsymbol{n} \cdot (\boldsymbol{B}_2 - \boldsymbol{B}_1) = 0, \tag{4.35}$$

$$\boldsymbol{n} \times \left(\frac{\boldsymbol{B}_2}{\mu_2} - \frac{\boldsymbol{B}_1}{\mu_1}\right) = 0. \tag{4.36}$$

Here subscripts 1,2 indicate the fields on one or the other side of the interface. In our case $\boldsymbol{E}_1 = \boldsymbol{E} + \boldsymbol{E}''$ and $\boldsymbol{E}_2 = \boldsymbol{E}'$, and similarly for the magnetic fields.

Field matching conditions:

$$\begin{aligned} \boldsymbol{n} \times (\boldsymbol{E}_2 - \boldsymbol{E}_1) &= 0 \\ \boldsymbol{n} \cdot (\boldsymbol{D}_2 - \boldsymbol{D}_1) &= \sigma_{\text{ext}} \\ \boldsymbol{n} \cdot (\boldsymbol{B}_2 - \boldsymbol{B}_1) &= 0 \\ \boldsymbol{n} \times (\boldsymbol{H}_2 - \boldsymbol{H}_1) &= j_{\text{ext}} \end{aligned}$$

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The continuity equations have to be satisfied everywhere on the interface plane $z = 0$. Because of the oscillatory behavior of exponentials, this is only possible if the phase factors are the same for all three waves along the whole plane. In other words, the continuity conditions require

$$(\boldsymbol{kx})|_{z=0} = (\boldsymbol{k'x})|_{z=0} = (\boldsymbol{k''x})|_{z=0} \tag{4.37}$$

for all $\boldsymbol{x}$. Geometrically this means that projections of the three wave vectors $\boldsymbol{k}$, $\boldsymbol{k'}$ and $\boldsymbol{k''}$ onto the plane $z = 0$ must be equal. It follows then that the three vectors themselves must lie in the same plane perpendicular to the plane $z = 0$. This plane is called *plane of incidence* and is defined by the vectors $\boldsymbol{k}$ and $\boldsymbol{n}$. Also, the magnitudes of the wave vectors must satisfy the relation

$$k \sin\alpha = k'' \sin\alpha'' = k' \sin\alpha'$$

where the angles are defined relative to the normal to the plane of interface, see Fig. 4.4. Since the incident and reflected waves propagate in *the same* medium, $k = k''$ and therefore the first equality implies

$$\alpha = \alpha''. \tag{4.38}$$

In other words, the angle of reflection equals the angle of incidence.

The second equality reads

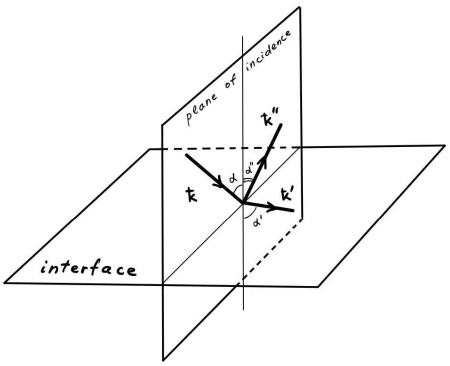
$$\sin\alpha' = \frac{k}{k'} \sin\alpha.$$

Taking into account that $k/k' = v'/v = n/n'$, one can rewrite this equation as

$$\sin\alpha' = \frac{n}{n'} \sin\alpha. \tag{4.39}$$

This relation is known as the **Snell law**; it determines the direction of propagation of the refracted wave which is, in general, different from the incident one.

One may immediately notice from eq. (4.39) that when the ratio n/n' is larger than one, which is typically the case when the wave propagates from more dense to a less dense medium, the solution for the refracted wave does not exist for all incidence angles. When α approaches a value such that $\sin\alpha = n'/n$, the refracted angle approaches $\pi/2$, meaning the refracted wave becomes parallel to the interface. If the incidence angle increases even further, the refracted wave does not exist, and everything is reflected. This phenomenon is known as *total internal reflection*. For the visible light, it manifest itself in many everyday phenomena like shining of diamonds, shining of cracks in the glass, mirages, etc. It is also the basis of the optical fiber technology.



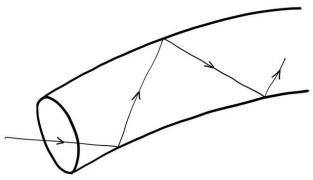


Figure 4.5.: An optical fiber. The refraction index of the fiber material is larger than that of the air.

• Let us derive the relation between the parameters of the incident, reflected and refracted waves. These relations depend on the wave polarization. Consider first the case when the electric field is perpendicular to the plane of incidence. The orientations of electric and magnetic fields in this case are shown in Fig. 4.9. For each of the three waves they follow eq. (4.31). From this equation we also have the relation between the

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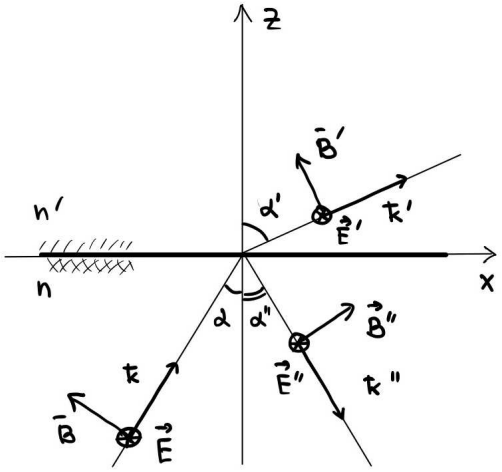


Figure 4.6.: Electric and magnetic fields of incident, reflected and refracted waves in the case when the electric field is parallel to the interface and is perpendicular to the plane of incidence.

amplitudes of $\boldsymbol{E}$ and $\boldsymbol{B}$ for each wave:

$$B = \frac{k}{\omega}E = \frac{n}{c}E, \tag{4.40}$$

$$B' = \frac{k'}{\omega}E' = \frac{n'}{c}E', \tag{4.41}$$

$$B'' = \frac{k''}{\omega}E'' = \frac{n}{c}E''. \tag{4.42}$$

We can now write the continuity conditions (4.33)- (4.36) for this polarization. There are two conditions for the components of $\boldsymbol{E}$, eqs. (4.33) and (4.34). The first of these equations gives

$$E' - E - E'' = 0, \tag{4.43}$$

while the second reduces to a trivial equation $0 = 0$. The continuity of the magnetic field components give two equations:

$$nE\sin\alpha + nE''\sin\alpha'' - n'E'\sin\alpha' = 0,$$

$$\frac{n}{\mu}E\cos\alpha - \frac{n}{\mu}E''\cos\alpha'' - \frac{n'}{\mu'}E'\cos\alpha' = 0.$$

If we take into account that $\alpha'' = \alpha$ and the Snell law, the first of these equations is equivalent to eq. (4.43) and gives nothing new.

It has already been mentioned in Sect. 2.4, in most cases of interest the value of μ is very close to μ_0. Let us assume for simplicity that $\mu = \mu' = \mu_0$, in which case the second equation becomes

$$n(E - E'')\cos\alpha - n'E'\cos\alpha' = 0. \tag{4.44}$$

So, we have finally two eqs. (4.43) and (4.44) which determine the amplitudes of the reflected and refracted waves given the amplitude of the incident wave.

Since the equations are linear in the amplitudes of electric fields, one can exclude one of the variables by dividing both equations by E and changing the variables to $x_{1,2} = E_{1,2}/E$. In the new variables the equations

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read

$$x_1 = 1 + x_2, \tag{4.45}$$

$$(1 - x_2)\cos\alpha = \frac{n'}{n} x_1 \cos\alpha'. \tag{4.46}$$

From these equations we find

$$x_2 = \frac{\cos\alpha - (n'/n)\cos\alpha'}{\cos\alpha + (n'/n)\cos\alpha'},$$

while the value of x_1 is determined from eq. (4.45).

It remains to express $\cos\alpha'$ in terms of α by using the Snell law. One finds

$$\cos\alpha' = \frac{n}{n'}\sqrt{\frac{n'^2}{n^2} - \sin^2\alpha},$$

and finally,

$$x_1 = \frac{E'}{E} = \frac{2\cos\alpha}{\cos\alpha + \sqrt{(n'/n)^2 - \sin^2\alpha}}, \tag{4.47}$$

$$x_2 = \frac{E''}{E} = \frac{\cos\alpha - \sqrt{(n'/n)^2 - \sin^2\alpha}}{\cos\alpha + \sqrt{(n'/n)^2 - \sin^2\alpha}}. \tag{4.48}$$

Let us have a closer look at the behavior of the electric field as a function of the incidence angle α. We have to distinguish two cases depending on the relative value of refraction indices. Consider first the case $n' > n$ (transition from smaller to larger refraction index). In this case eq. (4.47)-(4.48) are valid for the whole range of incidence angles $0 \le \alpha \le \pi/2$. In the two limits we have:

$\alpha \to 0$:

$$\frac{E'}{E} \to \frac{2n}{n+n'} < 1 \tag{4.49}$$

$$\frac{E''}{E} \to \frac{n-n'}{n+n'} < 0 \tag{4.50}$$

$\alpha \to \frac{\pi}{2}$:

$$\frac{E'}{E} \to 0 \tag{4.51}$$

$$\frac{E''}{E} \to -1 \tag{4.52}$$

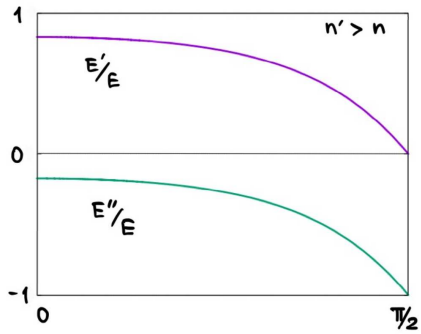


Figure 4.7.: The amplitudes of refracted and reflected waves, E' and E'', polarized in the direction perpendicular to the plane of incidence, for the case $n' > n$ (transition from less refractive to more refractive medium).

At intermediate values of α the amplitudes interpolate between these two limiting cases as shown on the plot. Note that, since E''/E is negative, the electric fields of the incident and reflected wave have opposite directions.

Consider now the case $n' < n$, which corresponds to the transition from a more refractive to a less refractive medium. In this case the solution only exists for incidence angles satisfying

$$\alpha < \arcsin(n'/n).$$

For larger values of α there is no transmitted wave, only the reflected one (total internal reflection). Equations (4.47)-(4.48) have in this case the

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following limits:

$$\alpha \to 0:$$

$$\frac{E'}{E} \to \frac{2n}{n+n'} > 1 \tag{4.53}$$

$$\frac{E''}{E} \to \frac{n-n'}{n+n'} > 0 \tag{4.54}$$

$$\alpha \to \arcsin(n'/n):$$

$$\frac{E'}{E} \to 2 \tag{4.55}$$

$$\frac{E''}{E} \to 1 \tag{4.56}$$

The limit $\alpha \to 0$ is formally the same, but now $n' < n$, so the amplitude of the refracted wave is larger than that of the incident one. The amplitude of the reflected wave now has the same sign as in the incident wave. The behavior at intermediate angles is shown in Fig. 4.8.

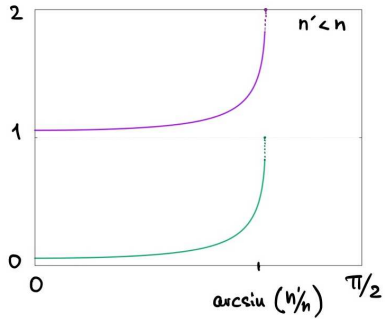


Figure 4.8.: The amplitudes of refracted and reflected waves, E' and E'', polarized in the direction perpendicular to the plane of incidence, for the case $n' < n$ (transition from more refractive to less refractive medium). For $\alpha > \arcsin(n'/n)$ there occurs total internal reflection.

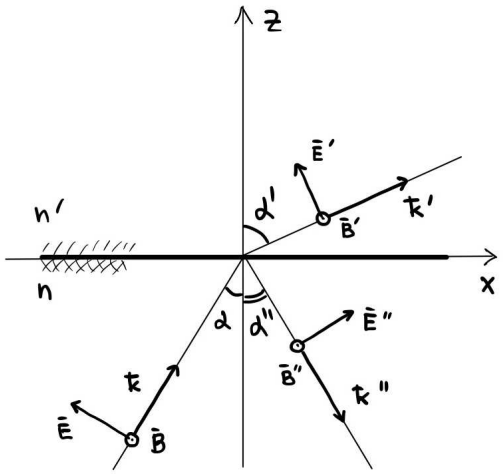


Figure 4.9.: Electric and magnetic fields of incident, reflected and refracted waves in the case when the electric field lies in the plane of incidence.

• *Incident wave polarized in the plane of incidence.* Consider now the other polarization when the electric field lies in the plane of incidence. Eqs. (4.33)-(4.36) in this case, upon projection onto the plane of the interface and on the normal vector $\boldsymbol{n}$, translate into the following equations:

$$\begin{aligned} E\cos\alpha - E''\cos\alpha'' - E'\cos\alpha' &= 0, && (4.57)\\ \epsilon E\sin\alpha - \epsilon E''\sin\alpha'' - \epsilon' E'\sin\alpha' &= 0, && (4.58)\\ 0 &= 0, \\ nE + nE'' - n'E' &= 0. && (4.59) \end{aligned}$$

In the last equation we have expressed B in terms of E by means of eqs. (4.40)-(4.42), and assumed, as before, $\mu = \mu' = \mu_0$.

The second equation is equivalent to the fourth one. Indeed, multiplying it by μ and using $\mu = \mu'$ we may rewrite it as

$$n^2 \sin\alpha(E + E'') - n'^2 E' \sin\alpha' = 0.$$

Using the Snell law $n\sin\alpha = n'\sin\alpha'$, this equation is equivalent to

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eq. (4.59). Thus, we have to solve eqs. (4.57) and (4.59).

As before, denote

$$x_1 \equiv \frac{E'}{E}, \quad x_2 \equiv \frac{E''}{E}.$$

In these notations the equations become

$$\begin{aligned}(1 - x_2)\cos\alpha - x_1 \cos\alpha' &= 0,\\ n(1 + x_2) - n' x_1 &= 0.\end{aligned}$$

Let us use the second equation to express x_1,

$$x_1 = \frac{n}{n'}(1 + x_2).$$

From the first equation we have then

$$(1 - x_2)\cos\alpha = \frac{n}{n'}(1 + x_2)\cos\alpha'.$$

Solving this equation for x_2 and substituting into the expression for x_1 we obtain:

$$x_1 = \frac{E'}{E} = \frac{2(n/n')\cos\alpha}{\cos\alpha + (n/n')^2\sqrt{(n'/n)^2 - \sin^2\alpha}}, \tag{4.60}$$

$$x_2 = \frac{E''}{E} = \frac{\cos\alpha - (n/n')^2\sqrt{(n'/n)^2 - \sin^2\alpha}}{\cos\alpha + (n/n')^2\sqrt{(n'/n)^2 - \sin^2\alpha}}. \tag{4.61}$$

As for another polarization, we should consider separately two cases. In the case $n < n'$, which corresponds to propagation from a less refractive to a more refractive medium, α may change from 0 to $\pi/2$ without having a negative argument under the square root in eqs. (4.60)-(4.61). The limiting values of the amplitudes are then as follows:

$\alpha \to 0$:

$$\frac{E'}{E} \to \frac{2n}{n + n'} < 1 \tag{4.62}$$

$$\frac{E''}{E} \to \frac{n' - n}{n + n'} > 0 \tag{4.63}$$

$\alpha \to \frac{\pi}{2}$:

$$\frac{E'}{E} \to 0 \tag{4.64}$$

$$\frac{E''}{E} \to -1 \tag{4.65}$$

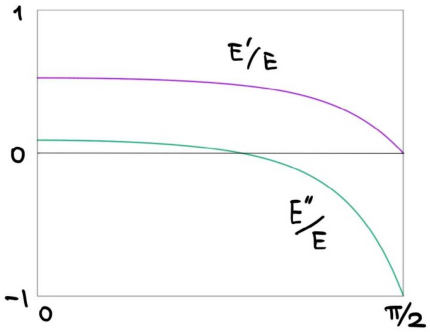


Figure 4.10.: The amplitudes of refracted and reflected waves, E' and E'', polarized in the plane of incidence, for the case $n' > n$ (propagation from less refractive to more refractive medium).

Note the change of sign of E''/E in the limit $\alpha \to 0$ compared to another polarization. The full behavior of amplitudes is shown in Fig. 4.10.

Finally, consider the case $n > n'$. As for another polarization, in this case the solution only exists for $\alpha \le \arcsin(n'/n)$, while at larger angles the total internal reflection takes place and there is no refracted wave. The

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amplitudes have the following asymptotic values:

$$\alpha \to 0:$$

$$\frac{E'}{E} \to \frac{2n}{n+n'} > 1 \tag{4.66}$$

$$\frac{E''}{E} \to \frac{n'-n}{n+n'} < 0 \tag{4.67}$$

$$\alpha \to \arcsin(n'/n):$$

$$\frac{E'}{E} \to 2\frac{n}{n'} \tag{4.68}$$

$$\frac{E''}{E} \to 1 \tag{4.69}$$

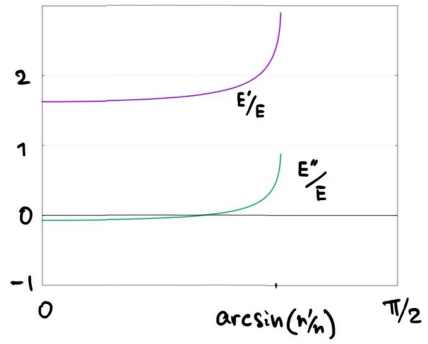


Figure 4.11.: The amplitudes of refracted and reflected waves, E' and E'', polarized in the plane of incidence, for the case $n' < n$ (propagation from more refractive to less refractive medium).

The behavior of the amplitudes as a function of the incidence angle is shown in Fig. 4.11.

As a consistency check, let us compare the two polarizations in the limit $\alpha \to 0$ corresponding to the incident wave falling perpendicular to the plane of the interface. In this case the plane of incidence is not defined and one polarization is equivalent to the other. The amplitudes of the reflected and refracted waves in this case must have the same value independently of the polarization. This is indeed what follows from our equations.

• We see from these plots that the case of polarization in the plane of incidence is rather peculiar: the amplitude of the reflected wave passes through zero at some angle α. This is not an artifact of the values of n and n' that were chosen to make the plots. Our analysis of the asymptotics shows that independently of whether $n > n'$ or $n' > n$, the ratio E''/E has different signs in the two limiting cases, and thus by continuity must pass through zero somewhere in between.

The angle at which this happens can be found from eq. (4.61) by requiring that the numerator vanishes,

$$\cos\alpha - (n/n')^2\sqrt{(n'/n)^2 - \sin^2\alpha} = 0.$$

If we denote the ratio $n'/n \equiv r$, this equations becomes

$$r^2\cos\alpha = \sqrt{r^2 - \sin^2\alpha}$$

Taking square of both sides in order to get rid of the square root, we find

$$\cos^2\alpha = \frac{1}{1+r^2}, \quad \sin^2\alpha = \frac{r^2}{1+r^2}.$$

From these relations,

$$\tan\alpha = r = \frac{n'}{n}. \tag{4.70}$$

This angle is called **Brewster's angle**. When the angle of incidence equals the Brewster's angle, one of the polarizations is not reflected, and therefore the reflected wave is *completely linearly polarized* in the direction perpendicular to the plane of incidence.

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Emission of electromagnetic waves

5.

The behavior of electric and magnetic fields is described by the Maxwell equations (2.95)–(2.98). So far we have considered their solution in two special cases: when the charge and the current distributions are static, in which case the resulting electric and magnetic fields are also static, and when the charges and currents are absent. In the latter case the solutions are propagating electromagnetic waves. How electromagnetic waves are created? To answer this question we have to solve Maxwell equations for time-dependent charge and current distributions.

From mathematical point of view the Maxwell equations are linear inhomogeneous equations. Their general solution is a sum of a general solution of the homogeneous equations — that is, an arbitrary electromagnetic wave — plus a particular solution to the inhomogeneous equations. That latter we now have to find.

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5.1. Retarded potentials

Since the Maxwell equations are linear, this problem can be solved in general terms. It is convenient to work with the scalar and vector potentials ϕ and $\boldsymbol{A}$. As we have seen in Sect. 2.3, in the non-static case the electric and magnetic fields are expressed in terms of these potentials as follows,

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \tag{5.1}$$

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}. \tag{5.2}$$

The second equation becomes the standard electrostatic definition of the scalar potential $\boldsymbol{E} = -\boldsymbol{\nabla}\phi$ when there is no time dependence.

• These two definitions of the potentials *automaticaly* solve the homogeneous Maxwell equations (2.97) and (2.98). Indeed, for the first equation this is evident,

$$\boldsymbol{\nabla B} = \boldsymbol{\nabla}(\boldsymbol{\nabla} \times \boldsymbol{A}) = 0.$$

Homogeneous Maxwell equations:

$$\begin{aligned} \boldsymbol{\nabla B} &= 0 \\ \boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} &= 0 \end{aligned}$$

For the second equation we have

$$\boldsymbol{\nabla} \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = \boldsymbol{\nabla} \times \left(-\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t} \right) + \frac{\partial}{\partial t} \boldsymbol{\nabla} \times \boldsymbol{A} = 0.$$

Thus, we only have to solve the inhomogeneous equations.

• Consider first the vector equation (2.96). Substituting the expressions for the electric and magnetic fields in terms of the potentials we get the equation

$$\boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \boldsymbol{A}) - \frac{1}{c^2} \frac{\partial}{\partial t} \left(-\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t} \right) = \mu_0 \boldsymbol{J}.$$

Inomogeneous Maxwell equations:

$$\begin{aligned} \boldsymbol{\nabla E} &= \frac{\rho}{\epsilon_0} \\ \boldsymbol{\nabla} \times \boldsymbol{B} - \frac{1}{c^2} \frac{\partial \boldsymbol{E}}{\partial t} &= \mu_0 \boldsymbol{J} \end{aligned}$$

After working out the double curl and regrouping the terms this becomes

$$\left(-\Delta + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} + \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t}\right) = \mu_0 \boldsymbol{J}. \tag{5.3}$$

This equation is complicated because all the components of $\boldsymbol{A}$ and ϕ are mixed together in each of its three components. However, recall that the potentials are not defined uniquely and we have some freedom to chose them — the gauge freedom. Namely, if we change simultaneously both potentials as follows:

$$\begin{aligned} \phi &\rightarrow \phi + \frac{\partial}{\partial t}\lambda(\boldsymbol{x}, t), \\ \boldsymbol{A} &\rightarrow \boldsymbol{A} - \boldsymbol{\nabla}\lambda(\boldsymbol{x}, t), \end{aligned}$$

with an arbitrary function $\lambda(\boldsymbol{x}, t)$, the fields $\boldsymbol{E}$ and $\boldsymbol{B}$ do not change. In 4-vector form this transformation is written as

$$A_\mu \rightarrow A_\mu + \partial_\mu \lambda.$$

Recall that

$$\begin{aligned} \partial_\mu &= (\tfrac{1}{c}\partial_t, \boldsymbol{\nabla}) \\ A^\mu &= (\phi, c\boldsymbol{A}). \end{aligned}$$

Thus, we can impose one condition on the vector potential. By a suitable choice of λ we may ensure that the potentials satisfy the following condition (this choice is called *Lorentz gauge*):

$$\partial_\mu A^\mu = 0,$$

which in terms of ϕ and $\boldsymbol{A}$ reads

$$\boldsymbol{\nabla}\boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t} = 0. \tag{5.4}$$

If we choose the potentials in this way the second term in eq. (5.3) is zero and we have simply

$$\left(-\Delta + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} = \mu_0 \boldsymbol{J}. \tag{5.5}$$

This equation is much simpler than eq. (5.3): every component satisfies the same inhomogeneous d'Alambert equation, but with its own source term.

Consider now the scalar equation (2.95). Substituting there eq. (5.2) we find

$$\boldsymbol{\nabla}\left(-\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}\right) = \frac{1}{\epsilon_0}\rho.$$

The first term in this equation is $-\Delta\phi$. If we add and subtract the term that completes this to the full d'Alambert operator we get

$$-\Delta\phi + \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \frac{1}{c^2}\frac{\partial^2 \phi}{\partial t^2} - \frac{\partial}{\partial t}\boldsymbol{\nabla}\boldsymbol{A} = \frac{1}{\epsilon_0}\rho,$$

or, grouping the terms,

$$\left(-\Delta + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\phi - \frac{\partial}{\partial t}\left(\boldsymbol{\nabla}\boldsymbol{A} + \frac{1}{c^2}\frac{\partial \phi}{\partial t}\right) = \frac{1}{\epsilon_0}\rho.$$

The second parenthesis contains exactly the combination that is set to zero

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by our gauge condition (5.4), so this reduces again to the inhomogeneous d'Alambert equation,

$$\left(-\Delta + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\right)\phi = \frac{1}{\epsilon_0}\rho. \tag{5.6}$$

In summary, in the Lorentz gauge the Maxwell equations reduce to two inhomogeneous d'Alambert equations (5.5) and (5.6). As the Maxwell equations themselves, these are linear inhomogeneous equations, so the general solution is a sum of a general solution to the homogeneous equation (electromagnetic waves considered before), and a *particular* solution to the inhomogeneous equations.

• This derivation is even more straightforward in the 4-vector notations. Recall that in these notations the Maxwell equations are written in terms of $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$ have the form

$$\partial_\mu F^{\mu\nu} = \frac{1}{c\epsilon_0} J^\nu.$$

In terms of the 4-vector potential this equation reads

$$\partial_\mu \partial^\mu A^\nu - \partial^\nu(\partial_\mu A^\mu) = \frac{1}{c\epsilon_0} J^\nu.$$

If we impose the gauge condition $\partial_\mu A^\mu = 0$, the second term on the left side vanishes, and we get the equation

$$\partial_\mu \partial^\mu A^\nu = \frac{1}{c\epsilon_0} J^\nu.$$

If now we recall that four components of the 4-current are $J^\mu = (c\rho, \boldsymbol{J})$, we see that this equation is equivalent to eqs. (5.5), (5.6).

• Consider first eq. (5.6) for the scalar potential. We have to find any solution to this equation. For that we will essentially use the Green's function method, but we will formulate it in a more intuitive way.

Since this equation is linear, we may decompose the charge density on the r.h.s. into a sum of contributions, solve the equation for each of them and then sum the solutions. It is convenient to represent ρ as a sum of charge densities each having a support only in a volume dV around each point in space. In other words, we divide the whole space into infinitesimal volumes dV each having a charge

$$dq(t) = \rho(\boldsymbol{x}, t) d^3x.$$

Note that since ρ is time-dependent, the individual charges $dq(t)$ also depend on time.

Each of these individual charges can be considered point-like, so that for a single contribution into the total solution we have to solve the equation

$$-\Delta\phi + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi = \frac{dq(t)}{\epsilon_0}\delta^3(\boldsymbol{x} - \boldsymbol{x}'), \tag{5.7}$$

where $\boldsymbol{x}'$ is a position of the corresponding infinitesimal volume. Now the parallel with the Green's function method becomes transparent.

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To solve the above equation we first note that the operator on the l.h.s. does not depend on the choice of the origin of the coordinates, so that choosing it in the point $\boldsymbol{x}'$ will eliminate this variable. Therefore we can solve the equation for $\boldsymbol{x}' = 0$ and obtain the solution for other values of $\boldsymbol{x}'$ simply replacing $\boldsymbol{x}$ by $\boldsymbol{x} - \boldsymbol{x}'$.

Furthermore, after setting $\boldsymbol{x}' = 0$ the r.h.s. becomes spherically symmetric, so we may look for the solutions that only depend on $r = |\boldsymbol{x}|$, that is, we may take $\phi = \phi(r, t)$. In this case only the radial part of the spherical Laplacian (2.57) contributes. The equation therefore becomes

$$-\frac{1}{r^2}\partial_r\left(r^2\partial_r\phi\right) + \frac{1}{c^2}\partial_t^2\phi = \frac{dq(t)}{\epsilon_0}\delta^3(\boldsymbol{x}). \tag{5.8}$$

Everywhere except the point $\boldsymbol{x} = 0$ this equation reduces to its homogeneous part,

$$-\frac{1}{r^2}\partial_r\left(r^2\partial_r\phi\right) + \frac{1}{c^2}\partial_t^2\phi = 0.$$

To solve the homogeneous equation let us make the change of variables

$$\phi(r, t) = \frac{\chi(r, t)}{r}.$$

If we substitute this into the homogeneous equation we see that the first derivative of χ cancel out, and the resulting equation is

$$-\partial_r^2\chi + \frac{1}{c^2}\partial_t^2\chi = 0.$$

This eqaution we have already solved in the context of plane waves. Its general solution is a combination of two arbitrary functions

$$\chi(r, t) = f_1(ct - r) + f_2(ct + r).$$

The first one describes an expanding spherical wave, the second — a contracting one. Since we ony need a particular solution, we may take the first function. Thus, the solution is an arbitrary function of $ct - r$,

$$\chi = \chi(ct - r), \qquad \phi(r, t) = \frac{\chi(ct - r)}{r}, \tag{5.9}$$

where $\chi(ct - r)$ is an arbitrary function of one argument.

Now we have to choose the function $\chi(ct - r)$ in such a way that the full inhomogeneous equation (5.7) is satisfied everywhere, including the point $\boldsymbol{x} = 0$. To this end let us integrate this equation over a spherical volume of radius ϵ centered at the origin. On the right the integral will give simply $dq(t)/\epsilon_0$ because of the delta-function. Thus we get the equation

$$\int_{|\boldsymbol{x}|\le\epsilon} d^3x \left\{-\Delta\phi + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi\right\} = \frac{dq(t)}{\epsilon_0}. \tag{5.10}$$

In the second term on the r.h.s. we can immediately substitute the solution (5.9):

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi = \frac{1}{c^2}c^2\frac{\chi''(ct - r)}{r} = \frac{\chi''(ct - r)}{r}$$

where primes on χ denote the derivatives over its whole argument, so

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this term gives, using the spherical symmetry,

$$\int_{|\boldsymbol{x}|\le\epsilon} d^3x \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi = 4\pi \int_0^\epsilon r dr \chi''(ct-r).$$

We can evaluate this integral by performing integration by parts several times as follows,

$$\int_0^\epsilon r dr \chi''(ct-r) = -\int_0^\epsilon r d\left[\chi'(ct-r)\right] =$$

$$= -\Big[r\chi'(ct-r)\Big]_0^\epsilon + \int_0^\epsilon \chi'(ct-r)dr =$$

$$= -\epsilon\chi'(ct-\epsilon) - \chi(ct-r)\Big|_0^\epsilon =$$

$$= -\epsilon\chi'(ct-\epsilon) - \chi(ct-\epsilon) + \chi(ct).$$

The integration gives the volume of the sphere and we find

$$\int_{|\boldsymbol{x}|\le\epsilon} d^3x \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi = 4\pi[-\epsilon\chi'(ct-\epsilon) - \chi(ct-\epsilon) + \chi(ct)]. \qquad (5.11)$$

Now consider the first term. The integrand is not well defined at the origin where $1/r$ is singular. To progress let us assume, following the logic of Sect. 2.2, that we have regularized it by making finite and smooth in the very vicinity of the point $\boldsymbol{x} = 0$ which is even much smaller than ϵ. Then we can use the Stokes theorem to transform the volume integral into a surface one,

$$-\int_{|\boldsymbol{x}|\le\epsilon} d^3x \Delta\phi = -\int_{|\boldsymbol{x}|\le\epsilon} d^3x \boldsymbol{\nabla}\boldsymbol{\nabla}\phi = -\int_{|\boldsymbol{x}|=\epsilon} d\boldsymbol{s}\boldsymbol{\nabla}\phi.$$

Importantly, the resulting integral is taken over the *surface* of the sphere $|\boldsymbol{x}| = \epsilon$, so it does not depend on our regularization at the origin, which we can forget about from this point on. Since ϕ only depends on r, its gradient only has a radial component, so the scalar product with $d\boldsymbol{s}$ gives

$$d\boldsymbol{s}\boldsymbol{\nabla}\phi = ds\partial_r\phi = ds\left(-\frac{\chi'}{r} - \frac{\chi}{r^2}\right).$$

Therefore, we get

$$-\int_{|\boldsymbol{x}|\le\epsilon} d^3x \Delta\phi = 4\pi\epsilon^2\left(\frac{\chi'}{r} + \frac{\chi}{r^2}\right)\Big|_{r=\epsilon} = 4\pi\epsilon\chi'(ct-\epsilon) + 4\pi\chi(ct-\epsilon).$$

Combining the two contributions we find

$$\int_{|\boldsymbol{x}|\le\epsilon} d^3x \left\{-\Delta\phi + \frac{1}{c^2}\frac{\partial^2}{\partial t^2}\phi\right\} = 4\pi\chi(ct).$$

Note that the ϵ-dependent terms have canceled as expected, because outside of the origin the integrand is zero.

To summarize, eq. (5.10) is a valid equation provided the function χ satisfies

$$4\pi\chi(ct) = \frac{dq(t)}{\epsilon_0}.$$

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We have, therefore,

$$\chi(r,t) = \chi(ct-r) = \frac{dq(t-r/c)}{4\pi\epsilon_0},$$

and thus the solution to eq. (5.7) is

$$\phi(\boldsymbol{x},t) = \frac{dq(t-|\boldsymbol{x}-\boldsymbol{x}'|/c)}{4\pi\epsilon_0|\boldsymbol{x}-\boldsymbol{x}'|}.$$

Here we have re-introduced an arbitrary position $\boldsymbol{x}'$ of the infinitesimal volume by replacing $r \to |\boldsymbol{x}-\boldsymbol{x}'|$.

Finally, we may replace back

$$dq(t-|\boldsymbol{x}-\boldsymbol{x}'|/c) = d^3x'\rho(\boldsymbol{x}', t-|\boldsymbol{x}-\boldsymbol{x}'|/c),$$

and sum up (integrate) contributions of individual volumes. The result is

$$\phi(\boldsymbol{x},t) = \frac{1}{4\pi\epsilon_0}\int d^3x' \frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|}\rho(\boldsymbol{x}', t-|\boldsymbol{x}-\boldsymbol{x}'|/c). \tag{5.12}$$

Repeating the same steps for the components of the vector potential $\boldsymbol{A}$ we find

$$\boldsymbol{A}(\boldsymbol{x},t) = \frac{\mu_0}{4\pi}\int d^3x' \frac{1}{|\boldsymbol{x}-\boldsymbol{x}'|}\boldsymbol{J}(\boldsymbol{x}', t-|\boldsymbol{x}-\boldsymbol{x}'|/c). \tag{5.13}$$

These solutions to the d'Alambert equations (5.5)and (5.6) are called *retarded potentials*.

What is remarkable in these expressions is the time dependence on the r.h.s. The volume d^3x' around the point $\boldsymbol{x}'$ contributes to the potentials in the point $\boldsymbol{x}$ at time t according to the charge and current densities *it had in the past moment* t' where $t' = t-|\boldsymbol{x}-\boldsymbol{x}'|/c$. The time difference $t-t' = |\boldsymbol{x}-\boldsymbol{x}'|/c$ is precisely the time it takes a light signal to propagate from $\boldsymbol{x}'$ to $\boldsymbol{x}$. These expressions automatically take into account the finite speed propagation of an electromagnetic signal. Otherwise they are quite similar to the static expressions for the same quantities, cf. eqs. (2.13), (2.79).

5.2. Electromagnetic field of a moving point charge

The retarded potential give a formal solution to the d'Alabmert equation for arbitrary time-dependent charge and density distributions. They may be simplified in some concrete cases. As a first example we consider just a single moving charge. The potentials in this case take a remarkably simple form which is known as the Liénard-Viechert potentials. However, even in this simple case the calculation of the electric and magnetic fields is rather lengthy.

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Liénard-Viechert potentials

Consider an electric charge e moving along the trajectory $\boldsymbol{r}(t)$. Denote the velocity of the charge $\boldsymbol{v}(t)$. We may represent the charge density of this charge by a delta-function that at each moment of time is centered on the charge, so that

$$\rho(\boldsymbol{x}, t) = e\delta^3(\boldsymbol{x} - \boldsymbol{r}(t)).$$

The corresponding current density is

$$\boldsymbol{J}(\boldsymbol{x}, t) = e\boldsymbol{v}(t)\delta^3(\boldsymbol{x} - \boldsymbol{r}(t)).$$

We may now use the general expressions for the retarded potentials (5.12) and (5.13) to write down the scalar potential created by the point charge as

$$\phi(\boldsymbol{x}, t) = \frac{e}{4\pi\epsilon_0}\int \frac{d^3x'}{|\boldsymbol{x} - \boldsymbol{x}'|}\delta^3(\boldsymbol{x}' - \boldsymbol{r}(t - |\boldsymbol{x} - \boldsymbol{x}'|/c)),$$

and for the vector potential

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{e\mu_0}{4\pi}\int \frac{d^3x'}{|\boldsymbol{x} - \boldsymbol{x}'|}\boldsymbol{v}(t - |\boldsymbol{x} - \boldsymbol{x}'|/c)\delta^3(\boldsymbol{x}' - \boldsymbol{r}(t - |\boldsymbol{x} - \boldsymbol{x}'|/c)).$$

• Both integrals contain a delta-function, so in principle should be easy to evaluate. However, this delta-function depends on the integration variables $\boldsymbol{x}'$ in a complicated way, so in practice direct calculation is complicated.

To simplify the calculation we use the standard trick to simplify the argument of the delta-function and thus evaluate the integral over d^3x'. consider the identity

$$f(t - |\boldsymbol{x} - \boldsymbol{x}'|/c) = \int d\tau f(\tau)\delta(\tau - t + |\boldsymbol{x} - \boldsymbol{x}'|/c).$$

Using this relation we may write the scalar potential as

$$\phi(\boldsymbol{x}, t) = \frac{e}{4\pi\epsilon_0}\int \frac{d^3x'}{|\boldsymbol{x} - \boldsymbol{x}'|}\int d\tau\delta(\tau - t + |\boldsymbol{x} - \boldsymbol{x}'|/c)\delta^3(\boldsymbol{x}' - \boldsymbol{r}(\tau)).$$

Note that now the 3-dimensional delta-function depends on $\boldsymbol{x}'$ in a simple way, so we may first evaluate the integral over d^3x' by simply replacing $\boldsymbol{x}'$ by $\boldsymbol{r}(\tau)$. This gives

$$\phi(\boldsymbol{x}, t) = \frac{e}{4\pi\epsilon_0}\int d\tau \frac{1}{|\boldsymbol{x} - \boldsymbol{r}(\tau)|}\delta(\tau - t + |\boldsymbol{x} - \boldsymbol{r}(\tau)|/c). \tag{5.14}$$

The remaining integral still contains a delta-function of a complicated argument, but only a 1-dimensional one.

• We now have to evaluate the remaining integral over τ. The rule how to calculate the integral of a δ-function of a complicated argument has been derived in Sect. 2.2. In our case

$$f(\tau) = \frac{1}{|\boldsymbol{x} - \boldsymbol{r}(\tau)|}; \qquad g(\tau) = \tau - t + |\boldsymbol{x} - \boldsymbol{r}(\tau)|/c.$$

Delta-function of a complicated argument:

$$\int dx f(x)\delta(g(x)) = \sum_i \frac{f(a_i)}{|g'(a_i)|}$$

where a_i satisfy $g(a_i) = 0$.

The derivative g' is

$$\frac{\partial g(\tau)}{\partial \tau} = 1 - \frac{\boldsymbol{v}(\tau)(\boldsymbol{x} - \boldsymbol{r}(\tau))}{c|\boldsymbol{x} - \boldsymbol{r}(\tau)|}.$$

Denote

$$\boldsymbol{R} = \boldsymbol{R}(\tau) \equiv \boldsymbol{x} - \boldsymbol{r}(\tau); \qquad R = |\boldsymbol{R}|.$$

Then the above relation is written as

$$\frac{\partial g(\tau)}{\partial \tau} = 1 - \frac{\boldsymbol{v}\boldsymbol{R}}{cR}$$

where both $\boldsymbol{R}$ and $\boldsymbol{v}$ are functions of τ. Using the integration formula we therefore get

$$\phi(\boldsymbol{x}, t) = \frac{ec}{4\pi\epsilon_0} \frac{1}{cR - \boldsymbol{v}\boldsymbol{R}}. \tag{5.15}$$

In a similar way, for the vector potential one finds

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{ec\mu_0}{4\pi} \frac{\boldsymbol{v}}{cR - \boldsymbol{v}\boldsymbol{R}}. \tag{5.16}$$

These are the *Liénard-Viechert potentials* of a point moving charge.

One should keep in mind that despite their simple appearance, these expressions contain a hidden complication: on the r.h.s. the quantities $\boldsymbol{R}(\tau)$ and $\boldsymbol{v}(\tau)$ should be evaluated not at time t but at the earlier moment τ which satisfies the equation

$$t - \tau = \frac{1}{c}|\boldsymbol{x} - \boldsymbol{r}(\tau)|. \tag{5.17}$$

This is the moment before t at which a light signal should be emitted at the current charge position $\boldsymbol{r}(\tau)$ in order to reach the point $\boldsymbol{x}$ exactly at time t. In other words, this moment is determined by the intersection of the particle world line with the past light cone drawn from the observation point $(\boldsymbol{x}, t)$, as illustrated in Fig. 5.1. In our derivation this moment is fixed by the zero of the argument of the delta-function in eq. (5.14), which is precisely eq. (5.17). The solutions to eq. (5.17) depend on $\boldsymbol{x}$ and t, $\tau = \tau(\boldsymbol{x}, t)$, and in this way the dependence on these variables enters the expressions for the potentials.

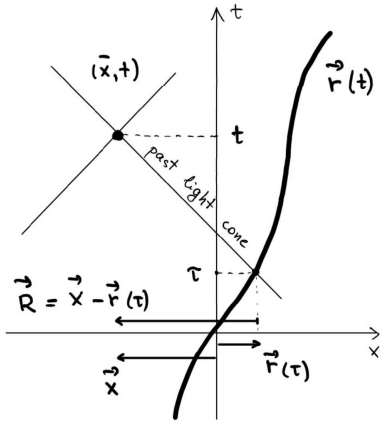
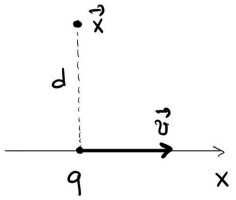


Figure 5.1.: Calculation of the Liénard-Viechert potentials. The contribution at the point $(\boldsymbol{x}, t)$ comes from the point of the particle world line $\boldsymbol{r}(t)$ where it intersects with the past light cone drawn from $(\boldsymbol{x}, t)$.

• Consider, as an example, a uniformly moving charge. Take its velocity v to be in the x-direction, so that the trajectory is written as $\boldsymbol{r}(t) = (vt, 0, 0)$. Let us calculate the potentials at a distance d from the charge in the point $(0, d, 0)$ at the moment $t = 0$. In summary:

$$\begin{aligned} \boldsymbol{r}(t) &= (vt, 0, 0), \\ \boldsymbol{v}(t) &= (v, 0, 0), \\ \boldsymbol{x} &= (0, d, 0). \end{aligned} \tag{5.18}$$



The Liénard-Viechert potentials are determined through the quantity $\boldsymbol{R}$

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for which we have

$$\begin{aligned} \boldsymbol{R}(\tau) &= \boldsymbol{x} - \boldsymbol{r}(\tau) = (-v\tau, d, 0) \\ R &= |\boldsymbol{x} - \boldsymbol{r}(\tau)| = \sqrt{v^2\tau^2 + d^2}. \end{aligned}$$

The retarded moment of time at which all the quantities should be evaluated is determined from eq. (5.14) which in our case gives

$$-\tau = \frac{1}{c}\sqrt{v^2\tau^2 + d^2}.$$

The solution to this equation reads

$$\tau = -\frac{d}{c}\frac{1}{\sqrt{1 - v^2/c^2}}.$$

From these expressions we find:

$$\begin{aligned} R &= -c\tau = \frac{d}{\sqrt{1 - v^2/c^2}}, \\ \boldsymbol{v}\boldsymbol{R} &= \frac{v^2}{c}\frac{d}{\sqrt{1 - v^2/c^2}}, \\ cR - \boldsymbol{v}\boldsymbol{R} &= cd\sqrt{1 - v^2/c^2}. \end{aligned}$$

Substituting this into equations for the Liénard-Viechert potentials this gives

$$\begin{aligned} \phi &= \frac{1}{4\pi\epsilon_0}\frac{e}{d\sqrt{1 - v^2/c^2}}, \\ \boldsymbol{A} &= \frac{\mu_0}{4\pi}\frac{e\boldsymbol{v}}{d\sqrt{1 - v^2/c^2}} = \frac{1}{4\pi\epsilon_0}\left(\frac{e}{d}\frac{v/c^2}{\sqrt{1 - v^2/c^2}}, 0, 0\right). \end{aligned}$$

Obviously, this is a Lorentz transformation of the potentials at a distance d from a static point charge,

$$\begin{aligned} \phi &= \frac{1}{4\pi\epsilon_0}\frac{e}{d}, \\ \boldsymbol{A} &= 0. \end{aligned}$$

This is precisely what is expected.

Calculation of electric and magnetic fields

The electric and magnetic fields are determined by the potentials ϕ and $\boldsymbol{A}$ through eqs. (5.1), (5.2) which contain derivatives of the potentials with respect to $\boldsymbol{x}$ and t. The potentials depend on these variables in two ways: on $\boldsymbol{x}$ directly (cf. definition of $\boldsymbol{R}$) and through τ, and on t through τ. The moment τ in turn depends on $\boldsymbol{x}$ and t through eq. (5.17). To calculate the derivatives of the potentials we will need, therefore, the derivatives of τ with respect to $\boldsymbol{x}$ and t.

Electric field:

$$\boldsymbol{E} = -\boldsymbol{\nabla}\phi - \frac{\partial \boldsymbol{A}}{\partial t}$$

Magnetic field:

$$\boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}$$

• *Calculation of the derivatives of* τ. It is not possible to solve eq. (5.17) for an arbitrary trajectory and find the explicit dependence $\tau(\boldsymbol{x}, t)$. So, in

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order to find the derivatives of τ we have to differentiate eq. (5.17) and then solve the resulting equation for the derivative. Consider first the time derivative. We assume that $\tau = \tau(\boldsymbol{x}, t)$ and differentiate eq. (5.17) with respect to t using the chain rule. This gives

$$1 - \frac{\partial \tau}{\partial t} = \frac{1}{c}\frac{\partial}{\partial t}|\boldsymbol{x} - \boldsymbol{r}(\tau)| = -\frac{1}{c}\frac{\boldsymbol{x} - \boldsymbol{r}(\tau)}{|\boldsymbol{x} - \boldsymbol{r}(\tau)|}\frac{\partial \boldsymbol{r}}{\partial \tau}\frac{\partial \tau}{\partial t}.$$

The derivative $\partial \boldsymbol{r}(\tau)/\partial \tau = \boldsymbol{v}(\tau)$ is the velocity of charge at the moment τ. Recalling the definition of $\boldsymbol{R}$ we can rewrite this equation as follows:

$$1 - \frac{\partial \tau}{\partial t} = -\frac{\boldsymbol{R}\boldsymbol{v}}{cR}\frac{\partial \tau}{\partial t}.$$

From this equation we find

$$\frac{\partial \tau}{\partial t} = \left(1 - \frac{\boldsymbol{R}\boldsymbol{v}}{cR}\right)^{-1} = \frac{cR}{cR - \boldsymbol{v}\boldsymbol{R}}. \tag{5.19}$$

The quantities on the r.h.s. are calculated at the moment τ.

Consider now the derivatives with respect to $\boldsymbol{x}$. Following the same logic we differentiate eq. (5.17) with respect to x_i, considering t an independent variable, and τ a function of $\boldsymbol{x}$,

$$-\frac{\partial \tau}{\partial x_i} = \frac{x_i - r_i(\tau)}{c|\boldsymbol{x} - \boldsymbol{r}(\tau)|} - \frac{x_j - r_j(\tau)}{c|\boldsymbol{x} - \boldsymbol{r}(\tau)|}\frac{\partial r_j}{\partial \tau}\frac{\partial \tau}{\partial x_i} = \frac{R_i}{cR} - \frac{\boldsymbol{R}\boldsymbol{v}}{cR}\frac{\partial \tau}{\partial x_i}.$$

Isolating $\partial \tau/\partial x_i$ we find

$$\frac{\partial \tau}{\partial x_i} = -\frac{R_i}{cR - \boldsymbol{v}\boldsymbol{R}}. \tag{5.20}$$

• *Calculation of* $\boldsymbol{E}$ *and* $\boldsymbol{B}$. Now that we have calculated the derivatives of τ we can calculate electric and magnetic fields. We will consider first the electric field, which is expressed in terms of the potentials as follows,

$$E_i = -\partial_i \phi - \frac{\partial A_i}{\partial t}.$$

The potentials are given by eqs. (5.15) and (5.16).

It is convenient to calculate first the derivatives of $R_i = x_i - r_i(\tau)$. We have

$$\partial_i R_j = \delta_{ij} - v_j \partial_i \tau = \delta_{ij} + \frac{v_j R_i}{cR - \boldsymbol{v}\boldsymbol{R}}.$$

Using this equation,

$$\partial_i R = \frac{R_j}{R}\partial_i R_j = \frac{R_i}{R} + \frac{(\boldsymbol{v}\boldsymbol{R})R_i}{R(cR - \boldsymbol{v}\boldsymbol{R})} = \frac{cR_i}{cR - \boldsymbol{v}\boldsymbol{R}}.$$

The contribution of the scalar potential then equals

$$-\partial_i \phi = \frac{ce}{4\pi\epsilon_0}\frac{1}{(cR - \boldsymbol{v}\boldsymbol{R})^2}(c\partial_i R - R_j \partial_i v_j - v_j \partial_i R_j) =$$

$$= \frac{ce}{4\pi\epsilon_0}\frac{1}{(cR - \boldsymbol{v}\boldsymbol{R})^2}\left[\frac{c^2 R_i}{cR - \boldsymbol{v}\boldsymbol{R}} + R_j \dot{v}_j \frac{R_i}{cR - \boldsymbol{v}\boldsymbol{R}} - v_j\left(\delta_{ij} + \frac{v_j R_i}{cR - \boldsymbol{v}\boldsymbol{R}}\right)\right] =$$

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$$= \frac{ce}{4\pi\epsilon_0}\frac{1}{(cR-\boldsymbol{v}\boldsymbol{R})^2}\left[(c^2-v^2+\boldsymbol{R}\dot{\boldsymbol{v}})\frac{R_i}{cR-\boldsymbol{v}\boldsymbol{R}}-v_i\right].$$

Note the appearance of the term proportional to the *charge acceleration* $\dot{\boldsymbol{v}} = \partial \boldsymbol{v}/\partial\tau$.

In a similar way we start the calculation of the second term by the time derivatives of $\boldsymbol{R}$. We have

$$\partial_t R_i = -v_i\frac{\partial\tau}{\partial t} = -\frac{v_i cR}{cR-\boldsymbol{v}\boldsymbol{R}},$$

$$\partial_t R = -\frac{R_i}{R}\frac{v_i cR}{cR-\boldsymbol{v}\boldsymbol{R}} = -\frac{c\boldsymbol{v}\boldsymbol{R}}{cR-\boldsymbol{v}\boldsymbol{R}},$$

$$\partial_t(cR-\boldsymbol{v}\boldsymbol{R}) = \frac{1}{cR-\boldsymbol{v}\boldsymbol{R}}(-c^2\boldsymbol{v}\boldsymbol{R}+v^2cR-\dot{\boldsymbol{v}}\boldsymbol{R}cR).$$

Using these expressions we have for the contribution of the vector potential:

$$-\frac{\partial A_i}{\partial t} = -\frac{e\mu_0 c}{4\pi}\frac{\partial}{\partial t}\left(\frac{v_i}{cR-\boldsymbol{v}\boldsymbol{R}}\right) =$$

$$= -\frac{e}{4\pi c\epsilon_0}\left\{\frac{\partial_t v_i}{cR-\boldsymbol{v}\boldsymbol{R}}-\frac{v_i}{(cR-\boldsymbol{v}\boldsymbol{R})^2}\partial_t(cR-\boldsymbol{v}\boldsymbol{R})\right\} =$$

$$= \frac{e}{4\pi\epsilon_0}\frac{1}{(cR-\boldsymbol{v}\boldsymbol{R})^2}\left\{-\dot{v}_i R+\frac{v_i}{cR-\boldsymbol{v}\boldsymbol{R}}[-c\boldsymbol{v}\boldsymbol{R}+v^2R-\dot{\boldsymbol{v}}\boldsymbol{R}R]\right\}.$$

We now add the two contributions together and after some rearrangement of terms finally obtain

$$E_i(\boldsymbol{x},t) = \frac{ec}{4\pi\epsilon_0}\frac{1}{(cR-\boldsymbol{v}\boldsymbol{R})^2}\left\{\frac{(c^2-v^2+\dot{\boldsymbol{v}}\boldsymbol{R})}{cR-\boldsymbol{v}\boldsymbol{R}}\left(R_i-\frac{v_i}{c}R\right)-\frac{\dot{v}_i}{c}R\right\}. \quad (5.21)$$

This is the electric field of a moving charge. As in the expression for the potentials, here in the r.h.s. we have to take all the time-dependent quantities not at the moment t, but at a previous moment τ defined in eq. (5.17).

The magnetic field can be calculated in a similar way. We do not present this calculation here. It can be shown that $\boldsymbol{B}$ can be expressed in terms of the electric field as follows:

$$\boldsymbol{B} = \frac{1}{cR}\boldsymbol{R}\times\boldsymbol{E}. \quad (5.22)$$

Thus, the magnetic field of a point charge is always perpendicular to the electric field, like in the case of a plane electromagnetic wave considered previously.

• If the charge is non-relativistic so that $v \ll c$, this expression may be simplified. In this limit we have

$$\boldsymbol{E} = \frac{e}{4\pi\epsilon_0}\left\{\frac{\boldsymbol{R}}{R^3}+\frac{(\dot{\boldsymbol{v}}\boldsymbol{R})}{c^2R^3}\boldsymbol{R}-\frac{\dot{\boldsymbol{v}}}{c^2R}\right\}.$$

Note that when obtaining this expression we cannot, in general, drop the acceleration terms (the ones containing $\dot{\boldsymbol{v}}$) as we do not know their relative strength compared to other terms.

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The first contribution in this expression — the only one that survives at zero acceleration $\dot{\boldsymbol{v}} = 0$ — is just the Coulomb law giving the electric field of a charge (static in this approximation) at a distance R. In view of eq. (5.22) the magnetic field corresponding to this part is zero.

• In the general case, the full expression (5.21) can be separated into two parts. The first part is composed of the terms that do not contain the particle acceleration (equivalently, these are the terms that remain when one sets $\dot{\boldsymbol{v}} = 0$). These terms decrease at large distances as $1/R^2$. One may check that this contribution is nothing but the Lorentz transformation of a Coulomb electric field of a static point charge to the reference frame moving with the velocity $-\boldsymbol{v}$.

The terms proportional to the acceleration $\dot{\boldsymbol{v}}$ have a different behavior: they decrease at large distances as $1/R$. This means that at large distances form the moving charge these terms give the dominant contribution to the electric field, while the Lorentz transform of the Coulomb field can be neglected. These terms represent electromagnetic waves emitted by the moving charge. They are determined by the *charge acceleration*, and not by the velocity.

Dipole approximation

In principle, the expressions (5.21) and (5.22) give the electric and magnetic fields of an arbitrary moving charge and thus are sufficient to determine the electromagnetic fields created by any charge distribution. In practice, however, they are often too complicated to use. An approximation is therefore needed.

In most of practical applications one does not need to know the fields at distances close to the moving charge where the Coulomb field is still strong. So, in the same spirit as it has been done in the static case, consider the field at distances that are much larger than the size of the system of charges and/or currents. We are looking therefore for the analog of the multipole expansion, but in a non-static case.

• It is convenient to start directly from the expressions (5.12), (5.13) for the retarded potentials. By our assumption , in these expressions the charge and current densities are only non-zero in a compact region of space and therefore $\boldsymbol{x}'$ varies within this region, while we want to know the fields at large distances $\boldsymbol{x}$. We take therefore $|\boldsymbol{x}| \gg |\boldsymbol{x}'|$, and expand these expressions in powers of a small parameter

$$\frac{|\boldsymbol{x}'|}{|\boldsymbol{x}|} \ll 1.$$

Both potentials depend on $|\boldsymbol{x}|$ through the combination $|\boldsymbol{x} - \boldsymbol{x}'|$ for which we have

$$|\boldsymbol{x} - \boldsymbol{x}'| = \sqrt{\boldsymbol{x}^2 - 2\boldsymbol{x}\boldsymbol{x}' + \boldsymbol{x}'^2} \simeq x - \boldsymbol{n}\boldsymbol{x}' + \dots,$$

where $x \equiv |\boldsymbol{x}|$ and $\boldsymbol{n} = \boldsymbol{x}/x$ is a unit vector in the direction of $\boldsymbol{x}$.

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Neglecting the higher order corrections we thus may write

$$\phi(\boldsymbol{x},t) \simeq \frac{1}{4\pi\epsilon_0}\int d^3x' \frac{1}{x-\boldsymbol{n}\boldsymbol{x}'}\rho[\boldsymbol{x}', t-(x-\boldsymbol{n}\boldsymbol{x}')/c], \quad (5.23)$$

$$\boldsymbol{A}(\boldsymbol{x},t) \simeq \frac{\mu_0}{4\pi}\int d^3x' \frac{1}{x-\boldsymbol{n}\boldsymbol{x}'}\boldsymbol{J}[\boldsymbol{x}', t-(x-\boldsymbol{n}\boldsymbol{x}')/c]. \quad (5.24)$$

• As discussed above, at large distances from the system of charges only the contribution corresponding to the electromagnetic wave survives. This wave has some characteristic wavelength. At distances much larger than this wavelength we may consider the wave as plane. For a plane wave the electric and magnetic fields are related by

$$\boldsymbol{E} = c\boldsymbol{B}\times\boldsymbol{n}.$$

Therefore, it is sufficient to calculate the magnetic field.

• The magnetic field is given by the curl of the vector potential. If we are only interested in the leading term, we may neglect the correction in the denominator and write

$$B_i = \epsilon_{ijk}\frac{\partial}{\partial x_j}\left\{\frac{\mu_0}{4\pi x}\int d^3x' \boldsymbol{J}[\boldsymbol{x}', t-(x-\boldsymbol{n}\boldsymbol{x}')/c]\right\}.$$

In principle, when calculating the derivative we have to differentiate both x and $\boldsymbol{n}$. However, if the motion of charges is non-relativistic we may also neglect the $\boldsymbol{n}\boldsymbol{x}'/c$ correction in the argument of the current. Indeed, this term corresponds to the time interval $\Delta t \sim x'/c$. During this time interval the displacement of a charge moving with velocity v is $\Delta x \sim v\Delta t = vx'/c$. We may neglect this correction if this displacement is much smaller than the size of the system $\sim x'$. So,

$$\Delta x \ll x' \quad \Longrightarrow \quad \frac{v}{c} \ll 1.$$

The expression for the field then simplifies even further,

$$B_i = \epsilon_{ijk}\frac{\partial}{\partial x_j}\left\{\frac{\mu_0}{4\pi x}\int d^3x' J_k[\boldsymbol{x}', t-x/c]\right\} =$$

$$= -\frac{\mu_0}{4\pi}\epsilon_{ijk}n_j\left\{\frac{1}{x^2}\int d^3x' J_k[\boldsymbol{x}', t-x/c] + \frac{1}{cx}\int d^3x' \dot{J}_k[\boldsymbol{x}', t-x/c]\right\}.$$

Note the appearance of the time derivative of J_k in the second term.

Something very interesting has happened in this expression. Recall the multipole expansion in the static case. The leading term there appeared in the order $1/x^3$, the magnetic dipole contribution. The 'monopole' term was zero because of the equation $\nabla\boldsymbol{J} = 0$ in the static case. In the non-static case the monopole term does not have to be zero; this is the first term in the above expression which behaves like $1/x^2$.

The second term, however, decays even slower like $1/x$. This term is proportional to the time derivative of the current. The current in turn contains velocities of charges, so this term depends on *charge acceleraions*. This is the term describing the electromagnetic waves.

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Keeping only the second term which is dominant at large distances we have

$$B_i = -\frac{\mu_0}{4\pi c x}\epsilon_{ijk} n_j \int d^3x' \dot{j}_k[\boldsymbol{x}', t - x/c].$$

In the vector form we can write this as $\dot{\boldsymbol{B}}$

$$\boldsymbol{B} = \frac{1}{c}\dot{\boldsymbol{A}} \times \boldsymbol{n}$$

where

$$\dot{\boldsymbol{A}} = \frac{\mu_0}{4\pi x}\int d^3x' \dot{\boldsymbol{j}}[\boldsymbol{x}', t - x/c] \tag{5.25}$$

is the time derivative of the vector potential in this approximation. Correspondingly, for the electric field we get

$$\boldsymbol{E} = (\dot{\boldsymbol{A}} \times \boldsymbol{n}) \times \boldsymbol{n}.$$

• The expression (5.25) can be re-written in terms of the electric dipole moment. To see this, let us imagine that our system consists of moving point charges (this is in fact the case in the real world). For a single charge q, the vector potential is given by the Lienard-Viechert expression (5.16),

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{\mu_0 q}{4\pi}\frac{\boldsymbol{v}}{R - \boldsymbol{v}\boldsymbol{R}/c}$$

where $\boldsymbol{R} = \boldsymbol{x} - \boldsymbol{r}(\tau)$. The charge trajectory $\boldsymbol{r}(t)$ lies within our system, and thus $|\boldsymbol{r}| \ll x$. If we also assume that the particle is non-relativistic, $v \ll c$, then we can approximate the potential as

$$\boldsymbol{A}(\boldsymbol{x}, t) \simeq \frac{\mu_0 q}{4\pi x}\boldsymbol{v}(t - x/c).$$

If there are many charges, the total vector potential is a sum of their contributions,

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{\mu_0}{4\pi x}\sum_a q_a \boldsymbol{v}_a(t - x/c) = \frac{\mu_0}{4\pi x}\frac{\partial}{\partial t}\sum_a q_a \boldsymbol{r}_a(t - x/c).$$

The quantity under the time derivative is nothing but the dipole moment

$$\boldsymbol{d} = \sum_a q_a \boldsymbol{r}_a(t - x/c)$$

of the system. We have therefore

$$\boldsymbol{A}(\boldsymbol{x}, t) = \frac{\mu_0}{4\pi x}\dot{\boldsymbol{d}}.$$

For the electric and magnetic fields we find, correspondingly,

$$\boldsymbol{B} = \frac{\mu_0}{4\pi c x}\ddot{\boldsymbol{d}} \times \boldsymbol{n}, \tag{5.26}$$

$$\boldsymbol{E} = \frac{\mu_0}{4\pi x}(\ddot{\boldsymbol{d}} \times \boldsymbol{n}) \times \boldsymbol{n}. \tag{5.27}$$

We see that in this approximation the emitted wave is determined by the second derivative of the electric dipole $\ddot{\boldsymbol{d}}(t - x/c)$ taken at the moment $t - x/c$.

• Let us calculate the energy carried away by the emitted electromagnetic

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wave, which is given by the Poynting vector (**??**),

$$\boldsymbol{S} = \frac{1}{\mu_0}\boldsymbol{E} \times \boldsymbol{B}.$$

From eqs. (5.26) and (5.27) we see that the electric and magnetic fields are mutually orthogonal, and both are orthogonal to the direction of propagation $\boldsymbol{n}$. Their vector product is therefor directed along $\boldsymbol{n}$. We get

$$\boldsymbol{S} = \frac{\boldsymbol{n}}{16\pi^2 x^2}\frac{\mu_0}{c}|\ddot{\boldsymbol{d}} \times \boldsymbol{n}|^2 = \frac{\boldsymbol{n}}{16\pi^2 x^2}\frac{\mu_0}{c}|\ddot{\boldsymbol{d}}|^2 \sin^2\theta,$$

where θ is the angle between $\ddot{\boldsymbol{d}}$ and $\boldsymbol{n}$. The energy flux decreases with distance as $1/x^2$. It is not isotropic; the maximum of energy is emitted in the directions perpendicular to $\ddot{\boldsymbol{d}}$, while in the direction parallel to $\ddot{\boldsymbol{d}}$ the energy flux is zero.

The total energy emission rate is obtained by integrating this expression over the sphere of radius x. This gives

$$I = \int \boldsymbol{S}d\boldsymbol{s} = \frac{\mu_0|\ddot{\boldsymbol{d}}|^2}{16\pi^2 c}\int \sin^2\theta \sin\theta d\theta d\phi = \frac{\mu_0}{6\pi c}|\ddot{\boldsymbol{d}}|^2.$$

Note that this quantity is independent of x, as it should be because of the energy conservation.

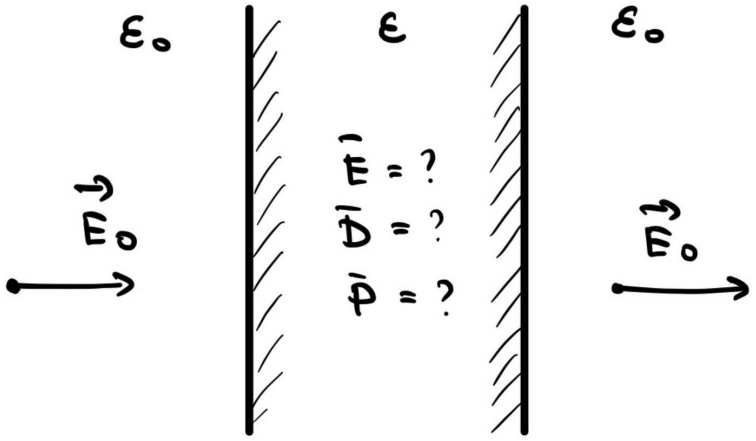
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Averaging is linear, that is for two functions $f(t)$ and $g(t)$

$$\langle \alpha f + \beta g \rangle = \alpha \langle f \rangle + \beta \langle g \rangle$$

where α and β are two constants.

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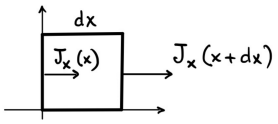


Figure 5.2.: Slice of dielectric of permittivity ϵ is embedded into a uniform electric field $\boldsymbol{E}_0$ perpendicular to the slice.

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Appendix

Appendix: Lorentz transformation to parallel fields

A.

We would like to demonstrate that if at least one of the field invariants is non-zero, there exists a frame in which the electric and magnetic fields are parallel, $\boldsymbol{E}||\boldsymbol{B}$. We will constrict such a frame by making a boost in the direction perpendicular to both $\boldsymbol{B}$ and $\boldsymbol{E}$. In order to improve readability of the calculations we use here units in which the speed of light is one, $c = 1$. In these units, the field invariants are

$$I_1 = \boldsymbol{B}^2 - \boldsymbol{E}^2, \tag{A.1}$$
$$I_2 = \boldsymbol{E}\boldsymbol{B}, \tag{A.2}$$

and, in the case of a boost in the direction perpendicular to both $\boldsymbol{B}$ and $\boldsymbol{E}$, the field transformations read

$$\boldsymbol{E}' = \gamma(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}), \tag{A.3}$$
$$\boldsymbol{B}' = \gamma(\boldsymbol{B} - \boldsymbol{v} \times \boldsymbol{E}). \tag{A.4}$$

We would like to show that there exists a frame in which $\boldsymbol{B}' \times \boldsymbol{E}' = 0$, which implies that the two fields are parallel.

To simplify calculations, let us choose spatial axes in such a way that $\boldsymbol{B}$ is in the direction z, $\boldsymbol{E}$ lies in the plane yz and the velocity is in the direction x. Denoting by $\boldsymbol{n}_1$, $\boldsymbol{n}_2$ and $\boldsymbol{n}_3$ three mutually orthogonal unit vectors in the direction of the axes, we have add figure

$$\boldsymbol{n}_i \times \boldsymbol{n}_j = \epsilon_{ijk}\boldsymbol{n}_k,$$

while the fields and velocity are written as

$$\boldsymbol{E} = E_\perp \boldsymbol{n}_2 + E_{||}\boldsymbol{n}_3, \tag{A.5}$$
$$\boldsymbol{B} = B\boldsymbol{n}_3, \tag{A.6}$$
$$\boldsymbol{v} = v\boldsymbol{n}_1. \tag{A.7}$$

Here v, B, $E_\perp$ and $E_{||}$ are magnitudes of the velocity, magnetic field and electric field perpendicular and parallel to $\boldsymbol{B}$, respectively. Without loss of generality we may assume that B and $E_\perp$ are positive. In terms of these magnitudes, the field invariants are

$$I_1 = B^2 - E_\perp^2 - E_{||}^2, \qquad I_2 = BE_{||}.$$

We can now calculate the transformed fields. We have

$$\boldsymbol{E}' = \gamma(E_\perp \boldsymbol{n}_2 + E_{||}\boldsymbol{n}_3 + vB\boldsymbol{n}_1 \times \boldsymbol{n}_3) = \gamma[(E_\perp - vB)\boldsymbol{n}_2 + E_{||}\boldsymbol{n}_3],$$

$$\boldsymbol{B}' = \gamma[B\boldsymbol{n}_2 - v\boldsymbol{n}_1 \times (E_\perp \boldsymbol{n}_3 + E_{||}\boldsymbol{n}_3)] = \gamma[(B - vE_\perp)\boldsymbol{n}_3 + vE_{||}\boldsymbol{n}_2].$$

Note that both fields lie in the plane yz. Making use of these expressions, the vector product of the transformed fields is

$$\boldsymbol{E}' \times \boldsymbol{B}' = \gamma^2 \boldsymbol{n}_1[(E_\perp - vB)(B - vE_\perp) - vE_{||}^2].$$

Requiring that $\boldsymbol{E}' \times \boldsymbol{B}' = 0$ gives the equation for the relative velocity of

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the two frames,

$$(E_\perp - vB)(B - vE_\perp) - vE_{||}^2 = 0.$$

Expanding the brackets we get

$$E_\perp B(1 + v^2) - v(B^2 + E^2) = 0, \tag{A.8}$$

where we have used $E^2 = E_\perp^2 + E_{||}^2$.

This is a quadratic equation for v; we are interested in solutions satisfying $-1 < v < 1$. The condition that the solutions to eq. (A.8) are real is

$$D \equiv (B^2 + E^2)^2 - 4B^2E_\perp^2 \geq 0.$$

The l.h.s. of this inequality can be expressed in terms of the field invariants:

$$D = (B^2 + E^2)^2 - 4B^2E^2 + 4B^2E_{||}^2 = (B^2 - E^2)^2 + 4B^2E_{||}^2 = I_1^2 + 4I_2^2.$$

The condition of reality then becomes

$$I_1^2 + 4I_2^2 \geq 0,$$

which is always satisfied. We have to examine the solutions further to see if they satisfy $|v| < 1$.

Let us start with a particular case $I_1 = I_2 = 0$, in which case $D = 0$. In this case $B = E_\perp = E$ and the equation becomes

$$v^2 - 2v + 1 = 0,$$

where we took into account that both B and $E_\perp$ are positive. The solution is thus $v = 1$, which in physical units becomes $v = c$. We see, therefore, that there are no physical solutions in this case.

Consider now the case $D > 0$. Let us rewrite eq. (A.8) in the following form,

$$v^2 - \frac{B^2 + E^2}{BE_\perp}v + 1 = 0.$$

Denoting the two roots of this equation as v_1 and v_2, we see that

$$v_1 v_2 = 1,$$

$$v_1 + v_2 = \frac{B^2 + E^2}{BE_\perp}.$$

Since the product and the sum of v_1 and v_2 are positive, they both must be positive. But since their product is 1, one must be larger and the other smaller than 1, except a special case which was considered above when both $v_1 = v_2 = 1$. We see, therefore, that at $D > 0$ there always exists a physical solution satisfying $v < 1$ ($v < c$ in physical units).

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